

FRM Part II Exam

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Questions with Answers - Liquidity and Treasury Risk
Measurement and Management

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Table of Contents

127 - Liquidity Risk	3
128 - Liquidity and Leverage	25
129 - Early Warning Indicators	59
130 - The Investment Function in Financial Services Management	79
131 - Liquidity and Reserves Management: Strategies and Policies	95
132 - Intraday Liquidity Risk Management	116
133 - Monitoring Liquidity	133
134 - The Failure Mechanics of Dealer Banks	146
135 - Liquidity Stress Testing	162
136 - Liquidity Risk Reporting and Stress Testing	179
137 - Contingency Funding Planning	195
138 - Managing and Pricing Deposit Services	211
139 - Managing Nondeposit Liabilities	227
140 - Repurchase Agreements and Financing	245
141 - Liquidity Transfer Pricing: A Guide to Better Practice	271
142 - The US Dollar Shortage in Global Banking and the International Policy Response	283
143 - Covered Interest Rate Parity Lost: Understanding the Cross-Currency Basis	300
144 - Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques	315
145 - Illiquid Assets	334

Reading 127: Liquidity Risk

Q.2255 Liquidity risk is best defined as:

- A. The risk that a counterparty in a transaction will delay meeting their financial obligation, thereby subjecting an institution to a shortage of funds.
- B. The risk that an institution will not be able to meet its financial needs at some future date.
- C. The risk that the amount of money in circulation within an economy is too low.
- D. Inability to meet short-term debt obligations without giving up capital or income.

The correct answer is **D**.

Liquidity risk is indeed the inability of a financial institution to meet its short-term debt obligations without sacrificing capital or income. This risk arises when an institution is unable to convert its assets into cash without incurring a significant loss. This inability could be due to various factors such as market conditions, the nature of the assets, or the institution's financial health. In a situation where the institution cannot liquidate its assets quickly enough to meet its obligations, it faces liquidity risk. This risk is a critical aspect of financial risk management as it directly impacts the institution's solvency and financial stability. Therefore, financial institutions employ various strategies and tools to manage and mitigate liquidity risk, such as maintaining a diversified portfolio, having adequate cash reserves, and using financial instruments that can be easily liquidated.

Choice A is incorrect. While it does involve a delay in meeting financial obligations, this definition more accurately describes counterparty or credit risk, not liquidity risk. Liquidity risk pertains to the institution's own ability to meet its short-term obligations.

Choice B is incorrect. This choice is too vague and could apply to several types of risks. Liquidity risk specifically refers to the inability of an institution to meet short-term debt obligations without sacrificing capital or income.

Choice C is incorrect. This choice describes a macroeconomic condition rather than a specific institutional risk. The amount of money in circulation within an economy can affect many aspects, but it doesn't directly define liquidity risk for a financial institution.

Things to Remember

- Market liquidity risk: The risk that an asset cannot be sold on short notice without impacting its price.
- Funding liquidity risk: The risk that a firm will not be able to meet its short-term obligations due to a lack of funding sources.

- Asset-liability management: Managing the balance between a firm's assets and liabilities to ensure liquidity and solvency.
 - Stress testing: Simulating extreme scenarios to assess the impact on liquidity and the ability to meet obligations.
 - Liquidity coverage ratio (LCR): A regulatory requirement that ensures institutions have enough high-quality liquid assets to cover short-term obligations.
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Q.2257 Joseph Bradley, FRM, works at I&M bank. As per the results of his intensive research, the US Treasury has issued bonds with a nominal value of approximately 800 billion dollars. I&M holds bonds worth 20 million dollars. This implies that:

- A. The bid-ask spread is exogenous to the bank.
- B. The bid-ask spread is endogenous to the bank.
- C. The bank is in a position to influence market price of Treasury bonds.
- D. The bank has too big an investment in gilts.

The correct answer is **A**.

The bid-ask spread is exogenous to the bank. The term 'exogenous' refers to factors or variables that are determined outside the system being studied and are often considered as given or constant. In this context, the bid-ask spread is determined by the larger bond market, not by I&M Bank's activities. Given the size of the bond market (\$800 billion) and the bank's relatively small position (\$20 million), the bank's trading activities would have an insignificant effect on the market price of bonds. Therefore, the bid-ask spread, which is the difference between the highest price that a buyer is willing to pay for an asset and the lowest price that a seller is willing to accept, is exogenous to the bank. It is determined by market forces beyond the bank's control.

Choice B is incorrect. The bid-ask spread is not endogenous to the bank. Endogenous variables are those that are influenced by other factors within the system. In this case, the bid-ask spread of Treasury bonds is determined by market forces and not directly controlled or influenced by I&M Bank's actions or decisions.

Choice C is incorrect. The bank does not have a position to influence market price of Treasury bonds. Given that the total face value of US Treasury bonds issued is approximately \$800 billion, I&M Bank's holding of \$20 million represents a very small fraction (0.0025%) of the total issuance, which would be insufficient to exert any significant influence on market prices.

Choice D is incorrect. It cannot be concluded that the bank has too big an investment in gilts

based on given information alone. Whether an investment size in gilts (UK government securities) or any other security for that matter, is 'too big' depends on various factors such as risk tolerance, investment strategy and portfolio diversification among others - none of which are provided in this question.

Things to Remember

- Exogenous variables are determined outside the system being studied and are considered as given or constant.
 - Endogenous variables are influenced by other factors within the system.
 - Bid-ask spread is the difference between the highest price a buyer is willing to pay and the lowest price a seller is willing to accept for an asset.
 - Market price of Treasury bonds is influenced by market forces and not directly controlled by individual banks.
 - Investment size being 'too big' depends on factors like risk tolerance, investment strategy, and portfolio diversification.
-

Q.2258 Which of the following statements is correct regarding the bid-ask spread of a financial instrument?

- A. The bid-ask spread is limited to 1%.
- B. The bid-ask spread is limited to 0.5%.
- C. The bid-ask spread ranges between 0.5% and 1%, depending on the type of security being traded.
- D. The bid-ask spread is not limited.

The correct answer is **D**.

The bid-ask spread is not limited. It is a dynamic variable that can fluctuate based on a variety of market factors. The bid-ask spread is essentially the difference between the highest price a buyer is willing to pay for an asset (the bid) and the lowest price a seller is willing to sell that asset for (the ask). This spread can vary widely depending on the liquidity of the asset, the volatility of the market, and the overall economic conditions. For instance, in a highly liquid market with many buyers and sellers, the bid-ask spread may be quite narrow. Conversely, in a less liquid market or during times of high market volatility, the bid-ask spread may widen.

significantly. Therefore, it is not accurate to state that the bid-ask spread is limited to a specific percentage or range. It is a variable that is influenced by market conditions and the characteristics of the specific asset being traded.

Choice A is incorrect. The bid-ask spread is not limited to 1%. The spread can be wider or narrower depending on various factors such as the liquidity of the asset, market volatility, and information asymmetry among market participants.

Choice B is incorrect. Similarly to Choice A, the bid-ask spread is not limited to 0.5%. It can vary widely based on a multitude of factors including but not limited to the ones mentioned above.

Choice C is incorrect. While it's true that the bid-ask spread can range between different percentages depending on the type of security being traded, it's misleading to suggest that this range is strictly between 0.5% and 1%. The actual range can be much broader or narrower depending on specific market conditions and characteristics of individual securities.

Things to Remember

- Bid-ask spread is a key concept in financial markets, representing the cost of trading an asset.
 - Market liquidity plays a significant role in determining the width of the bid-ask spread.
 - High market volatility can lead to wider bid-ask spreads, increasing trading costs.
 - Information asymmetry among market participants can also impact the bid-ask spread.
 - Different types of securities may have different typical bid-ask spread ranges.
-

Q.2262 Bilco Bank has implemented the LaR (liquidity at risk) method based on a 95% probability and a 1-day holding period. Suppose its calculations for the next day result in a figure of USD 15 million. What would that imply?

- A. The maximum loss over the next day is \$15 million with a probability of 95%.
- B. The maximum profit over the next day is \$15 million with a probability of 95%.
- C. The worst outcome over the next day is an inflow of cash of \$15 million with a probability of 95%.
- D. The worst outcome over the next day is an outflow of cash of \$15 million with a probability of 95%.

The correct answer is **D**.

The Liquidity at Risk (LaR) method is a risk management tool used by financial institutions to estimate the potential cash outflows or inflows that could occur over a specified holding period, given a certain level of confidence (in this case, 95%). In the context of this question, a LaR value of USD 15 million implies that the worst-case scenario for Bilco Bank over the next day is a cash outflow of USD 15 million. This means that, with a 95% level of confidence, the bank could experience a cash outflow of up to USD 15 million. This is considered the 'worst' outcome as it represents a decrease in the bank's liquidity, which could potentially impact its ability to meet its short-term obligations. Therefore, the bank needs to ensure that it has sufficient liquidity buffers in place to cover such a scenario.

Choice A is incorrect because it incorrectly interprets the LaR value as a potential loss. While a cash outflow could indeed lead to a decrease in the bank's liquidity, it is not the same as a loss. A loss would imply that the bank has incurred a negative return on its investments or operations, which is not what the LaR value represents. The LaR value is a measure of potential cash flows (either inflows or outflows), not profits or losses. Therefore, the statement that the maximum loss over the next day is USD 15 million with a probability of 95% is incorrect.

Choice B is incorrect because it incorrectly interprets the LaR value as a potential profit. The LaR method is not used to estimate potential profits; rather, it is used to estimate potential cash flows (either inflows or outflows) over a specified holding period, given a certain level of confidence. Therefore, the statement that the maximum profit over the next day is USD 15 million with a probability of 95% is incorrect. It is important to note that a cash inflow is not necessarily a profit, as it could be due to various factors such as the receipt of loan repayments or the sale of assets, which do not represent a return on investment.

Choice C is incorrect because it incorrectly interprets the LaR value as a potential cash inflow. In the context of this question, a positive LaR value represents a potential cash outflow, not an inflow. Therefore, the statement that the worst outcome over the next day is an inflow of cash of USD 15 million with a probability of 95% is incorrect. It is important to note that a cash inflow is not necessarily a 'worst' outcome, as it could increase the bank's liquidity and improve its ability to meet its short-term obligations.

Things to Remember

- **Liquidity Risk:** The risk that a company or financial institution may not be able to meet its short-term financial obligations due to an inability to convert assets into cash quickly enough.
- **Probability:** The likelihood of a specific event or outcome occurring, often expressed as a percentage.
- **Holding Period:** The length of time over which a financial institution assesses its risk exposure, typically used in risk management calculations.
- **Liquidity Buffer:** Reserves of cash or highly liquid assets that a financial institution holds to meet unexpected cash outflows or obligations.

Q.2266 Redding bank is in process of implementing a LaR framework. In the process, risk analysts realize that the bank has very similar hedging positions, albeit with different counterparties, in different sectors. In this scenario, which type of risk would be paramount?

- A. Credit risk
- B. Market risk
- C. Compliance risk
- D. Reputation risk

The correct answer is **A**.

Credit risk is the risk of loss that may occur from the failure of any party to abide by the terms and conditions of any financial contract, principally, the failure to make required payments on loans due to an inability to pay. In the given scenario, Redding bank has similar hedging positions with different counterparties in different sectors. This means that the bank is exposed to the risk that these counterparties may default on their obligations. This is a classic example of credit risk. The bank's exposure to credit risk is paramount in this scenario, as similar positions with different counterparties do not offset each other and increase the risk of simultaneous defaults. Therefore, the bank needs to manage this risk effectively to prevent potential losses.

Choice B is incorrect. Market risk refers to the potential for financial loss due to changes in the market conditions such as interest rates, exchange rates, equity prices or commodity prices. In this scenario, the bank has taken similar hedging positions across different sectors which does not necessarily increase its exposure to market risk. The main concern here is credit risk because of dealing with different counterparties.

Choice C is incorrect. Compliance risk pertains to the potential for legal penalties, financial forfeiture and material loss an organization faces when it fails to act in accordance with industry laws and regulations, internal policies or prescribed best practices. While compliance is always a concern for any bank, it's not specifically highlighted as a major issue in this scenario.

Choice D is incorrect. Reputation risk refers to the potential damage to a firm's reputation from any association, action or inaction which could be perceived negatively by others. Although reputation risk can have significant impacts on a bank's business operations and profitability, it isn't directly related to taking similar hedging positions across different sectors with various counterparties.

Things to Remember

- Counterparty Risk: The risk that the other party in a financial transaction may default or fail to fulfill their obligations.
- Diversification: Spreading investments across different assets to reduce risk.
- Netting: Offsetting the value of multiple positions or obligations in order to determine a

net position.

- Collateral: Assets pledged by a borrower to secure a loan or other credit, and subject to seizure in the event of default.
-

Q.2977 Suppose that XYZ Company has a current stock price of \$40 and a daily standard deviation of 1%. The current bid-ask spread is 2%. Calculate LVaR at the 95% confidence level. Assume a constant spread.

- A. 0.2452
- B. 1.3526
- C. 1.0981
- D. 1.06

The correct answer is **D**.

Using the constant spread approach,

$$\text{LVAR} = (V \times Z_{\alpha} \times \sigma) + (0.5 \times V \times \text{spread})$$

Where:

V = asset (or portfolio) value

Z_{α} = confidence parameter (Normal distribution)

σ = standard deviation of returns

$$\text{LVAR} = (40 \times 1.65 \times 0.01) + 0.5 \times 40 \times 0.02 = \$1.06$$

Q.3214 Tim Lauren is an analyst at a large commercial bank. He plans to invest in Grantson Automobile stock with bid and ask prices equal to \$53.70 and \$54.10, respectively. Given this information, the proportional bid-ask spread for Grantson Automobile stock is closest to?

A. 0.74%.

B. 1.48%.

C. 2.35%.

D. 1.45%.

The correct answer is **A**.

$$\begin{aligned}\text{Proportional Bid-ask spread} &= \frac{(\text{ask-price} - \text{bid-price})}{[(\text{ask-price} + \text{bid-price}) \times 0.5]} \\ &= \frac{(\$54.10 - \$53.70)}{[(\$54.10 + \$53.70) \times 0.5]} \\ &= 0.0074 \text{ or } 0.74\%\end{aligned}$$

Q.3940 Cathleen Wilson is the liquidity manager for CPR bank. She decides to invest in 50 million shares of one company and 20 million ounces of a commodity. Assume that the shares' bid price is \$80.4, offer price \$80.8, and the commodity's bid is \$30.6. However, she is not able to remember the commodity's offer price. Given the cost of liquidation for the portfolio to be \$12.0 million and the mid-market position in the commodity to be \$614 million, calculate the offer price for the commodities in a normal market.

- A. 30.8
- B. 31.6
- C. 34.6
- D. 28.9

The correct answer is **A**.

The mid-market price of shares is the average of bid and offer prices. Therefore, the total mid-market price for 50 million shares is:

$$\frac{80.4 + 80.8}{2} \times 50 = 80.6 \times 50 = \$4,030$$

The proportional bid-offer spread for the share is given by:

$$\frac{\text{Bid-offer spread}}{\text{Mid-market price}} = \frac{80.8 - 80.4}{80.6} = \frac{0.4}{80.6} = 0.004963$$

We let the proportional bid-offer for the share be x , and the offer price be w

Then

$$\begin{aligned} \frac{0.004963 \times 4,030}{2} + \frac{(x \times 614)}{2} &= 12.00 \\ \Rightarrow 1.99754 &= 307x \end{aligned}$$

$X=0.0065066$, which is the proportional bid-offer price for the commodity

$$\begin{aligned} \Rightarrow \frac{(w - 30.6)}{0.5 \times (w + 30.6)} &= 0.0065066 \\ \frac{(2w - 61.2)}{(w + 30.6)} &= 0.0065066 \end{aligned}$$

Thus, $w = 30.8$

Q.3941 Suppose that a bank invests in shares and a commodity whose mid-market position is \$1,400 million, and \$840 million respectively. You are also provided with the following information:

	Mean	Standard Deviation
Bid-offer spread for the shares	\$1.0 million	\$1.1 million
Bid-offer spread of the commodity	\$0.1 million	\$0.1 million
Proportional bid-offer spread for the shares	0.034	0.018
Proportional bid-offer spread for the commodity	0.0044	0.0044

Assuming the distribution of the spreads is normal, calculate the cost of liquidation in a stressed market at a 95% confidence level.

- A. \$40.10million
- B. \$50.23million
- C. \$49.42million
- D. \$107.93million

The correct answer is **C**.

$$\begin{aligned}
 \text{Cost of liquidation (stressed market)} &= \sum_{j=1}^n \frac{(\mu_j + \lambda\sigma_j)\alpha_j}{2} \\
 &= \frac{1,400 \times (0.034 + 1.645 \times 0.018)}{2} \\
 &\quad + \frac{840 \times (0.0044 + 1.645 \times 0.0044)}{2} \\
 &= \$49.415 \text{ million}
 \end{aligned}$$

In this question, we are considering the mean and standard deviation for the proportional bid-offer spread for both shares and the commodity, bearing in mind that $n=2$.

Additionally, α_j is the mid-market position, μ_j represents the mean for the proportional bid-offer, λ represents the constant factor given as 1.645 at 95% confidence level, and the σ represents the standard deviation for the propositional bid-offer spread.

Q.3942 Fatou James, the liquidity manager for CPQ bank, invests in shares and a commodity whose mid-market value of the positions are 1,426, and 814 respectively. The mean and standard deviation of the position in shares are \$1.12 and \$1.45, respectively. Suppose that the mean and standard deviation for the commodity are both \$0.63, and the mean for the proportional bid-offer spread for the shares is 0.0467, while the standard deviation for the proportional bid-offer spread for the shares is unknown. Given that the mean and standard deviation for the proportional bid-offer spread for the commodity are both 0.006857. Assume that the distribution of the spreads is normal. Further, the cost of liquidation under a stressed market condition at a 99% confidence level is 82.264. Calculate the standard deviation for the proportional bid-offer spread for shares.

A. 0.063

B. 0.048

C. 0.056

D. 0.0239

The correct answer is **D**.

Let the value of the standard deviation for the proportional bid-offer spread for shares be w .

The cost of liquidation under stressed market conditions at a 99% confidence level can be calculated using the following formula:

$$\begin{aligned}
 & \frac{\text{Value}_{\text{shares}} \times (\text{Mean}_{\text{shares}} + Z \times \text{SD}_{\text{shares}})}{2} + \frac{\text{Value}_{\text{commodity}} \times (\text{Mean}_{\text{commodity}} + Z \times \text{SD}_{\text{commodity}})}{2} \\
 &= \frac{1,426 \times (0.0467 + 2.326 \times w)}{2} + \frac{814 \times (0.006857 + 2.326 \times 0.006857)}{2} = \$82.264 \\
 &= 33.2971 + 1,658.438w + 9.2822 = \$82.264 \\
 &= 1,658.438w = \$39.6847 \\
 &= w = 0.02393
 \end{aligned}$$

Q.3943 A liquidity division for HTC bank invests in shares and a commodity. The mid-market value of the position in shares is \$W while the mid-market value of the position in the commodity is \$413. The mean and standard deviation of the bid-offer spread for the shares are \$1.24 and \$1.02, respectively. On the other hand, the mean and standard deviation of the bid-offer spread for the commodity are \$0.67 and \$0.34. Further, the mean and standard deviation of the proportional bid-offer spread for the shares is 0.0721 and 0.0675, respectively, while the mean and standard deviation of the proportional bid-offer spread for the commodity is 0.00524 and 0.00463, respectively. Assuming that the distribution of the spreads is normal, and the cost of liquidation at the 99% confidence level in a stressed market condition is \$95.062, calculate W, the mid-market value of the position in shares.

- A. \$905
- B. \$809
- C. \$801
- D. \$1,579

The correct answer is C.

$$\frac{W \times (0.0721 + 2.326 \times 0.0675)}{2} + \frac{413 \times (0.00524 + 2.326 \times 0.00463)}{2} = \$95.062\text{m}$$

$$0.5W(0.229105) + 3.3059 = \$95.062\text{m}$$

$$W = \frac{\$91.75606}{0.1145525}$$

$$= \$801$$

Note that 2.326 is the z-value at 99% confidence from the standard normal table.

Q.3944 A liquidity division for HTC bank invests in shares and a commodity whose mid-market position is \$801, and \$413, respectively. If the proportional bid-offer spread for the shares is 0.001879, and that for the commodity is 0.002488, What is the cost of liquidation in a normal market.

- A. 2.0982
- B. \$1.2663
- C. \$1.8990
- D. \$2.7660

The correct answer is **B**.

$$\frac{0.001879 \times 801}{2} + \frac{0.002488 \times 413}{2} = \$1.2663 \text{ million}$$

Q.3945 Richard Watson, the liquidity manager of RTC bank, has been finding ways of boosting the liquidity for the bank. He decides to invest in W shares of one company and 19 million ounces of a commodity. Assume that the shares are bid \$70.4, offer \$72.8, and the commodity is bid is \$46.6, offer \$47.2. The mid-market value of the position in shares is \$2,506 million. Calculate, W, the number of shares, and the cost of liquidation in a normal market.

- A. 35million and \$47.699
- B. 37million and \$42.076
- C. 20 million and \$65.000
- D. 17 million and \$34.985

The correct answer is **A**.

Since the mid-market value of the position in shares is \$2,506 million, W, the number of shares, can be calculated as follows:

$$\$2,506\text{m} = \frac{70.4 + 72.8}{2} \times W$$

Thus

$$W = 35 \text{ million}$$

The mid-market value of the position in the commodity = $46.9 \times 19 = \$891.1$ million

The proportional offer spread for the commodity is $\frac{47.2-46.6}{46.9} = 0.01279$, while the proportional bid-offer spread for the share is;

$$\frac{72.8 - 70.4}{71.6} = 0.03352$$

Then the cost of liquidation in a normal market;

$$\frac{0.03352 \times 2,506}{2} + \frac{0.01279 \times 891.1}{2} = \$47.699 \text{ million}$$

Q.3946 JCT bank has a position in two assets - shares of company X and a separate commodity Y. The following data has been extracted from the bank's liquidity division:

- The mid-market price of the shares is \$72,000,000 and the mid-market price of the commodity is \$47,000,000.
- The mean and standard deviation for the bid-offer spread for the shares are \$1.30 and \$1.03, respectively.
- The mean and standard deviation for the bid-offer spread for the commodity are \$1.32 and \$0.88, respectively.
- The bank holds 1 million units in each of the two positions.

Calculate the cost of liquidation at a 99% confidence level, assuming the spread distribution is approximately normal in a stressed market.

- A. \$2,500,000
- B. \$3,500,650
- C. \$3,531,536
- D. \$7,063,072

The correct answer is **C**.

Shares

Mean, proportional bid-offer spread: $\frac{\$1.3}{\$72} = 0.0181$

Standard deviation, proportional bid-offer spread: $\frac{\$1.03}{\$72} = 0.0143$

Commodity

Mean, proportional bid-offer spread: $\frac{\$1.32}{\$47} = 0.0281$

Standard deviation, proportional bid-offer spread: $\frac{\$0.88}{\$47} = 0.0187$

$$\begin{aligned} \text{Cost of liquidation} &= \frac{\$72,000,000 \times (0.0181 + 2.326 \times 0.0143)}{2} + \frac{\$47,000,000 \times (0.0281 + 2.326 \times 0.0187)}{2} \\ &= \$3,531,536 \end{aligned}$$

Q.3947 James Farouk, the liquidity manager of CRP financial institution, has been struggling with boosting liquidity in his institution. Farouk consults Richard Taylor, one of the staff in RCP seeking help on how he can source liquidity for the institution. Which of the following choices is not likely to be one of the choices offered by Taylor, assuming he was right?

- A. Liquidation of trading book positions
- B. Holdings of cash and treasury securities
- C. Increasing investments in real estate
- D. Ability to borrow money at short notice

The correct answer is C.

Increasing investments in real estate is not a viable strategy for boosting liquidity in a financial institution. While such investments may enhance the solvency status of the bank, they do not directly contribute to its liquidity. Liquidity refers to the ability of an institution to meet its short-term obligations, which requires readily available funds. Real estate investments, on the other hand, are typically long-term and illiquid, meaning they cannot be quickly converted into cash without potentially incurring a significant loss. Therefore, increasing investments in real estate would not effectively address the liquidity concerns of the institution.

Choice A is incorrect. Liquidation of trading book positions can indeed enhance the liquidity of a financial institution. Trading book positions are typically composed of financial instruments that can be easily sold or bought in the market, thus providing immediate liquidity.

Choice B is incorrect. Holding cash and treasury securities is another effective strategy for enhancing an institution's liquidity. Cash is the most liquid asset, and treasury securities are highly liquid due to their strong demand in the market.

Choice D is incorrect. The ability to borrow money at short notice also contributes to an institution's liquidity position. This strategy ensures that even if there are unexpected cash outflows, the institution can quickly secure funds to meet its obligations.

Things to Remember

- Liquidity management is crucial for financial institutions to ensure they can meet their short-term obligations.
- Real estate investments are typically long-term and illiquid, making them unsuitable for boosting liquidity.
- Trading book positions consist of easily tradable financial instruments that can enhance liquidity when liquidated.

- Holding cash and treasury securities are highly liquid assets that can improve an institution's liquidity position.
 - Borrowing money at short notice is a common strategy to address unexpected liquidity needs.
-

Q.3948 A credit downgrade of three notches (e.g., from AA+ to A+) is one of the acute stresses incorporated in the 30-day period considered in the calculation of the liquidity coverage ratio (LCR). Which of the following most accurately states the other stress events?

- A. Partial loss of deposits and decreased haircuts
- B. Drawdowns on lines of credit and partial loss of deposits
- C. Drawdowns on lines of deposits and decreased haircuts
- D. Reduced haircuts

The correct answer is **B**.

The Liquidity Coverage Ratio (LCR) is a regulatory requirement designed to ensure that a bank has an adequate stock of unencumbered high-quality liquid assets (HQLAs) that can be converted into cash to meet its liquidity needs for a 30 calendar day liquidity stress scenario. The LCR is calculated as the ratio of the stock of HQLAs to total net cash outflows over the next 30 calendar days. The stress scenario incorporated in the calculation of the LCR includes a credit downgrade of three notches (e.g., from AA+ to A+), drawdowns on lines of credit, and a partial loss of deposits. Drawdowns on lines of credit refer to the situation where a bank's customers, who have been granted a line of credit, decide to draw down on this facility, thereby increasing the bank's cash outflows. A partial loss of deposits refers to a situation where a bank's depositors decide to withdraw a portion of their deposits, thereby reducing the bank's available liquidity. These are realistic scenarios that could occur during a period of financial stress and are therefore included in the calculation of the LCR.

Choice A is incorrect. While partial loss of deposits is a stress event considered in the LCR, decreased haircuts are not. Haircuts refer to the difference between the market value of an asset and its collateral value; a decrease would imply an increase in liquidity, which contradicts the concept of a stress scenario.

Choice C is incorrect. The phrase "drawdowns on lines of deposits" seems to be a misinterpretation or typo as it doesn't make sense in this context. Banks do not draw down on deposits; instead, they face potential drawdowns on lines of credit extended to their customers.

Choice D is incorrect. Reduced haircuts alone do not constitute a comprehensive stress scenario for LCR calculation purposes. As mentioned earlier, reduced haircuts would imply increased liquidity rather than financial stress.

Things to Remember

- High-Quality Liquid Assets (HQLAs) are assets that can be easily and quickly converted into cash with little to no loss in value.
 - Net cash outflows are calculated by considering the expected cash outflows minus the expected cash inflows over the 30-day period.
 - Stress testing is a key component of risk management and regulatory requirements to assess a bank's ability to withstand adverse scenarios.
 - Lines of credit are pre-arranged borrowing facilities that allow customers to access funds up to a certain limit.
 - Deposit withdrawals can significantly impact a bank's liquidity position, especially during times of stress.
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Q.3949 Thomas Wong, one of the traders in the security market, has been a positive feedback trader. In his interview with a business analyst, Wong stated the factors that cause him to be a positive trader. Which choice among the following accurately states the reason as to why Wong practices positive trading?

- A. Urgent need for cash
- B. Need for liquidity
- C. Competition
- D. Predatory trading

The correct answer is **D**.

Predatory trading is a strategy where a trader anticipates that a stock's value will either increase or decrease and places trades accordingly. This strategy aligns with the concept of a 'positive feedback' strategy, where a trader buys more of a stock as its price increases and sells it as its price decreases. Thomas Wong, being a positive feedback trader, would likely engage in predatory trading as it aligns with his strategy of capitalizing on price movements. Therefore, predatory trading is the most accurate reason for Wong's positive feedback trading strategy.

Choice A is incorrect. An urgent need for cash does not necessarily dictate a positive feedback trading strategy. This strategy involves buying securities when prices are rising and selling when they are falling, which may not be the best approach if immediate liquidity is required.

Choice B is incorrect. The need for liquidity also does not directly influence the adoption of a positive feedback trading strategy. While this strategy can potentially provide liquidity through frequent trades, it's not specifically designed to enhance liquidity.

Choice C is incorrect. Competition in the market might influence a trader's overall approach but it doesn't specifically lead to the adoption of a positive feedback trading strategy. This type of strategy is more about capitalizing on price trends rather than outperforming competitors.

Things to Remember

- **Positive Feedback Trading:** Positive feedback trading is a strategy where a trader buys more of a stock as its price increases and sells it as its price decreases. This strategy is based on the belief that trends will continue in the same direction.
 - **Predatory Trading:** Predatory trading involves placing trades based on the anticipation of a stock's value either increasing or decreasing. It is a strategy that aims to capitalize on price movements in the market.
 - **Liquidity:** Liquidity refers to the ease with which an asset can be bought or sold in the market without causing a significant change in its price. Traders may consider liquidity when making trading decisions.
 - **Competition:** Competition in the market can influence a trader's decisions and strategies. Traders may need to consider competitive factors when developing their trading approaches.
-

Q.3950 The following statistics extracted from RTC bank's balance sheet reflect some of the underlying transactions conducted by the bank. The balance sheet has distinguished short-term and long-term liabilities and assets, according to Basel III. Use the data with the weighted factors to calculate the net stable funding ratio (NSFR) for the bank.

Assets	Short Term	Long Term	NSFR	LCR	Liabilities	Short Term	Long Term	NSFR	LCR
Cash		950	90%		Owners Equity		90		100%
Loans	4,200	3,000	100%		Deposits	500		100%	
Corporates									
Mortgages		3,200	100%	25%	Unsecured Debt	250	2,500	75%	45%
					Issuance				
Corporates	950	1,400	85%	15%	Deposits	540	5,000	10%	100%
					Financial Institution				
					Expenses	230	6,600	100%	
					Notes Payable	700	0	0	50%

A. 1.27

B. 1.33

C. 1.08

D. 0.75

The correct answer is **D**.

NSFR is defined as the ratio of the amount of available stable funding to the amount of required stable funding. Available stable funding represents the capital and liabilities that are expected to remain reliable over a 1-year horizon. On the other hand, required stable funding comprises the portion of assets and off-balance sheet exposures that need to be funded over the same 1-year horizon.

$$\begin{aligned}
 & \frac{\text{Amount of stable funding}}{\text{Required amount of stable funding}} \geq 100\% \\
 & = \frac{500 + 250 \times 0.75 + 540 \times 0.10 + 230 + 2,500 \times 0.75 + 5,000 \times 0.10 + 6,600}{950 \times 0.9 + 3,000 + 3,200 + 1,400 \times 0.85 + 4,200 + 950 \times 0.85} \\
 & = \frac{9,946.5}{13,252.5} \approx 0.75
 \end{aligned}$$

Q.3952 The following table represents a section extracted from ABC bank's balance sheet. Use the provided data to determine the Liquidity Coverage Ratio of the bank for 2018 and 2019, respectively.

Description	2018	2019
High Liquid assets	2100	1458
Average monthly withdrawals	1200	2458
Expected monthly outflows in a stressed scenario	1600	1700

- A. 126% and 145 %
- B. 131% and 86%
- C. 80% and 124%
- D. 129% and 134%

The correct answer is **B**.

The Liquidity Coverage ratio is equivalent to;

$$\frac{\text{High-quality liquid asset}}{\text{Net cash outflows in a 30-day period}} \geq 100\%$$

$$\text{LCR for 2018} = \frac{2100}{1600} = 1.312 = 131\%$$

$$\text{The LCR for 2019} = \frac{1458}{1700} = 0.857 = 86\%$$

Q.5383 While delivering a training session on liquidity risk to newly recruited individuals, Jane Doe, a risk manager at the national regulator, discusses the insights gained from the collapse of Metallgesellschaft. Which of the following lessons regarding liquidity risk failures would be most effectively exemplified by the case of Metallgesellschaft in 1993?

- A. Cash flow mismatches may occur when hedging liabilities by rolling forward futures contracts.
- B. Compared to forwards, futures offer a more effective hedge for mitigating commodities exposure.
- C. A bank run can be triggered by a negative public perception of emergency borrowing from the central bank.
- D. Excessive losses can arise from positive feedback trading in illiquid instruments.

The correct answer is **A**.

Hedging liabilities involves using futures contracts to protect against potential adverse movements in prices or interest rates. Rolling forward futures contracts refers to the practice of closing out existing contracts and simultaneously opening new contracts with extended maturity dates. It introduces the risk of cash flow mismatches. This occurs because the cash flows associated with the futures contracts may not align perfectly with the cash flows of the underlying liabilities being hedged.

B is incorrect. Both forwards and futures can serve as hedging tools for managing commodities exposure, but they can introduce liquidity risk when there is a cash flow mismatch.

C and D are incorrect. These were not issues in the case of Metallgesellschaft.

Things to Remember

- Cash flow mismatches can lead to liquidity risk in hedging strategies
 - Rolling forward futures contracts can introduce additional risks
 - Forwards and futures can both be used for hedging commodities exposure
 - Positive feedback trading in illiquid instruments can result in excessive losses
 - Bank runs can be triggered by negative public perception and emergency borrowing
-

Reading 128: Liquidity and Leverage

Q.2275 CRG Bank has a large portfolio of securities that could be quickly sold without significant price fluctuations. Such securities are said to be:

- A. Liquid
- B. Perfectly liquid
- C. Illiquid
- D. Perfectly illiquid

The correct answer is **A**.

Liquidity refers to the ease with which an asset, or security, can be converted into ready cash without affecting its market price. High liquidity is a desirable attribute for any asset or security because it means that the asset can be quickly sold without causing a significant change in its price and without incurring substantial transaction costs. In the context of the question, the securities in CRG Bank's portfolio are described as being able to be quickly sold without significant price fluctuations. This characteristic is indicative of high liquidity, hence, the term 'liquid' is the most appropriate description for such securities.

Choice B is incorrect. While "perfectly liquid" might seem like a stronger version of "liquid", it's not the term typically used in financial markets to describe securities that can be sold without significantly impacting their prices. Perfect liquidity is a theoretical concept where transactions do not affect the price and there are no transaction costs, which is rarely observed in real-world markets.

Choice C is incorrect. Illiquid securities are those that cannot be easily sold or exchanged for cash without a substantial loss in value. This description contradicts the scenario presented in the question, where CRG Bank's securities can be readily sold without causing significant changes in their prices.

Choice D is incorrect. Similar to choice C, "perfectly illiquid" refers to assets that cannot be converted into cash quickly or without significant loss of value. This does not align with the characteristics of CRG Bank's portfolio as described in the question.

Things to Remember

- Liquidity risk is the risk that an asset cannot be sold on short notice without incurring a loss.
- Market liquidity refers to the ability to buy or sell an asset in the market without causing a significant change in its price.
- Liquidity is important for financial institutions to meet short-term obligations and

manage cash flows effectively.

- Liquidity ratios, such as the current ratio and quick ratio, are used to assess a company's ability to meet its short-term liabilities.
-

Q.2276 Transaction liquidity risk can be best defined as:

- A. The risk of reducing the demand for an asset by increasing the number of transactions.
- B. The risk of moving the price of an asset adversely in the act of buying or selling it.
- C. The risk of moving the supply of an asset adversely in the act of buying or selling it.
- D. The risk that too many transactions on the trading floor could render the firm unable to meet its day-to-day operation costs.

The correct answer is **B**.

Transaction liquidity risk is indeed the risk of moving the price of an asset adversely in the act of buying or selling it. This risk arises when the market's liquidity is not sufficient to absorb the quantity being traded without impacting the price. In other words, if a large quantity of an asset is bought or sold, it could significantly move the price in an unfavorable direction. This is particularly relevant in illiquid markets where the lack of buyers or sellers at any given time can cause significant price movements. Therefore, managing transaction liquidity risk is crucial for financial institutions and traders to prevent substantial losses.

Choice A is incorrect. Transaction liquidity risk does not pertain to the reduction in demand for an asset due to an increase in the number of transactions. Rather, it refers to the potential adverse impact on the price of an asset during its buying or selling process.

Choice C is incorrect. This choice incorrectly defines transaction liquidity risk as a movement in supply of an asset during buying or selling activities. While supply and demand dynamics can influence prices, transaction liquidity risk specifically relates to how those transactions might negatively affect the price of that particular asset.

Choice D is incorrect. The statement describes operational risk rather than transaction liquidity risk. Operational risks are associated with failures in systems, processes, or controls, including those related to managing day-to-day operations and trading floor activities.

Things to Remember

- **Market Liquidity:** Refers to the ease with which an asset can be bought or sold in the market without significantly affecting its price.

- Illiquid Markets: Markets with low trading volumes and limited participants, making it difficult to buy or sell assets without impacting their prices.
 - Price Impact: The effect of a transaction on the price of an asset, which can be adverse if the market is illiquid.
 - Order Book: A list of buy and sell orders for a particular asset, showing the demand and supply at different price levels.
 - Market Depth: The ability of a market to absorb large orders without significant price movements.
-

Q.2277 Nathan James, a trader for Nathan Capital, works on the short-term funding desk at his firm. Over the past few months, markets have been highly volatile but Nathan Capital still enjoys a large capital base and is financially stable. In his monthly report to the liquidity subcommittee of the board of directors, James reports that in the last month, Nathan Capital's chief lender has been steadily increasing collateral requirements to roll over repo contracts. From the perspective of Nathan Capital, this represents:

- A. Transactions liquidity risk
- B. Balance sheet risk
- C. Systematic risk
- D. Maturity transformation risk

The correct answer is **B**.

Balance sheet risk, also known as funding liquidity risk, is the risk that creditors either withdraw credit or change the terms on which it is granted in such a way that the positions have to be unwound and/or are no longer profitable. This risk can arise due to perceived or actual deterioration in the borrower's credit quality, or due to overall deteriorating financial conditions. In the case of Nathan Capital, the lender is increasing the collateral requirements, thus changing the terms of credit. This is happening despite Nathan Capital being financially sound and showing no signs of financial deterioration. Therefore, this situation represents a balance sheet risk for Nathan Capital.

Choice A is incorrect. Transaction liquidity risk refers to the risk that a firm may not be able to execute a transaction at the prevailing market price because of the size of the transaction relative to normal market size. In this case, Nathan Capital is not facing any issues with executing transactions at prevailing market prices, but rather it's experiencing an increase in

collateral requirements from its chief lender.

Choice C is incorrect. Systematic risk refers to the overall impact of financial system or market and cannot be eliminated through diversification. It includes factors such as interest rates, inflation rates, and economic recessions that affect all participants in a market equally. The situation described for Nathan Capital does not indicate any systematic risks affecting all firms equally.

Choice D is incorrect. Maturity transformation risk arises when financial institutions borrow short-term while lending long-term, creating a mismatch between their assets and liabilities' maturities. This scenario doesn't mention anything about Nathan Capital's borrowing or lending practices related to different maturities.

Things to Remember

- Balance sheet risk is also known as funding liquidity risk and can lead to unwinding positions or unprofitable situations due to changes in credit terms.
 - Transaction liquidity risk refers to the risk of not being able to execute a transaction at prevailing market prices due to transaction size.
 - Systematic risk is the overall impact of financial system or market and includes factors like interest rates and economic recessions.
 - Maturity transformation risk arises from borrowing short-term while lending long-term, creating a maturity mismatch.
-

Q.2278 Yemi Bank was recently ordered to pay a hefty fine as punishment for engaging in certain illegal trades. Following this event, creditors have expressed fears over their investment and would want to introduce more conditions governing the use of funds lent to the bank. This scenario gives an example of:

- A. Liquidity credit risk
- B. Strategic risk
- C. Systemic risk
- D. Balance sheet risk

The correct answer is **D**.

Balance sheet risk, also known as funding liquidity risk, is a type of financial risk that arises

when a firm's creditors either withdraw their credit or alter the terms of credit in such a way that the firm's positions have to be unwound or are no longer profitable. In the given scenario, Yemi Bank has been involved in illegal trading activities, which has led to a significant fine. This has caused concern among the bank's creditors, who are now considering introducing more conditions on the use of the funds they have lent to the bank. This situation is a clear example of balance sheet risk, as the bank may face difficulties in meeting its financial obligations due to the potential withdrawal or alteration of credit terms by its creditors.

Choice A is incorrect. Liquidity credit risk refers to the risk that a party may not be able to meet its financial obligations due to an inability to convert assets into cash quickly. In this case, Yemi Bank's situation does not involve any issues with liquidity or converting assets into cash.

Choice B is incorrect. Strategic risk refers to the potential for loss due to poor business decisions, improper implementation of strategies, or inadequate response to changes in the business environment. While Yemi Bank's unlawful trading activities could be seen as a poor business decision, the question specifically relates more closely with balance sheet risk as it involves creditors considering additional conditions on their loans due to concerns about the bank's activities.

Choice C is incorrect. Systemic risk refers to the possibility that an event at company level could trigger severe instability or collapse an entire industry or economy. The situation described in this question pertains only Yemi Bank and its creditors and does not suggest a threat of widespread economic impact.

Things to Remember

- Liquidity risk: The risk that a firm may not be able to meet its short-term financial obligations due to an inability to convert assets into cash quickly.
- Strategic risk: The potential for loss due to poor business decisions, improper implementation of strategies, or inadequate response to changes in the business environment.
- Systemic risk: The risk that an event at the company level could trigger severe instability or collapse an entire industry or economy.

It is important for financial institutions to manage balance sheet risk effectively to ensure they can meet their financial obligations and maintain the confidence of their creditors.

Q.2279 Elipsa Bank has witnessed a dramatic deterioration of the credit quality of its borrowers in its loan portfolio. As a result, the bank's balance sheet position has worsened. The bank will most likely have to contend with:

A. Systemic risk

- B. Funding liquidity risk
- C. Reputation risk
- D. Strategic risk

The correct answer is **B**.

Funding liquidity risk, also known as balance sheet risk, arises when a borrower's credit position is deteriorating or is perceived by market participants to be deteriorating. In the case of Elipsa Bank, the credit quality of its borrowers has significantly declined, which has negatively impacted its loan portfolio and overall balance sheet position. This situation makes it difficult for the bank to negotiate new credit terms for its own funding needs. Potential lenders and investors may be hesitant to lend to the bank under the same terms as before due to the increased risk. As a result, any new funding for the bank may come with stricter credit terms, such as higher interest rates. This scenario exemplifies funding liquidity risk, as the bank's ability to meet its funding needs is compromised due to the deteriorating credit quality of its borrowers.

Choice A is incorrect. Systemic risk refers to the potential for a major disruption in the function of an entire market or financial system due to the failure of a single entity or group of entities. While Elipsa Bank's situation is concerning, it does not necessarily imply a systemic risk unless the bank's failure could cause significant disruption to the broader financial system.

Choice C is incorrect. Reputation risk involves damage to an organization's reputation which can lead to loss of revenue or legal, regulatory, or financial costs. In this scenario, there is no mention of any reputational damage suffered by Elipsa Bank due to its declining loan portfolio and balance sheet position.

Choice D is incorrect. Strategic risk arises from poor business decisions, improper implementation of decisions, inadequate resource allocation, or failure to respond well to changes in the business environment. The scenario does not provide any information suggesting that Elipsa Bank has made poor strategic decisions leading up its current predicament.

Things to Remember

- Systemic risk is the risk of a major disruption in the function of an entire market or financial system due to the failure of a single entity or group of entities.
- Reputation risk involves damage to an organization's reputation which can lead to loss of revenue or legal, regulatory, or financial costs.
- Strategic risk arises from poor business decisions, improper implementation of decisions, inadequate resource allocation, or failure to respond well to changes in the business environment.

Q.2280 Egda Bank has had a rough year in which its financial health and performance substantially deteriorated. An intensive study of the market by its risk management department has attributed the deterioration to bad market conditions. The study also found out that a majority of players in the banking sector have had almost identical problems. This scenario gives an example of:

- A. Maturity transformation risk
- B. Balance sheet risk
- C. Systemic risk
- D. Banking sector liquidity risk

The correct answer is **C**.

Systemic risk refers to the risk that could cause a widespread or major disruption in the financial system, potentially leading to severe financial stress or even a financial crisis. It is the risk that the failure of one financial institution could cause other financial institutions to fail, leading to a domino effect that could potentially destabilize the entire financial system. In the given scenario, Egda Bank and other banks in the sector are facing similar challenges due to unfavorable market conditions. This indicates that the risk is not specific to Egda Bank but is affecting the entire banking sector, which is a characteristic of systemic risk. Therefore, the situation described in the scenario is an example of systemic risk.

Choice A is incorrect. Maturity transformation risk refers to the risk that arises when financial institutions borrow short-term and lend long-term, creating a mismatch in the maturity of their assets and liabilities. This is not the case described in the scenario, where unfavorable market conditions are affecting all banks in the sector.

Choice B is incorrect. Balance sheet risk pertains to risks associated with assets, liabilities, and equity on a bank's balance sheet. While it's possible that Egda Bank may have balance sheet risks due to its declining financial performance, this does not explain why other banks in the sector are also facing challenges.

Choice D is incorrect. Banking sector liquidity risk refers to a situation where most or all banks within a specific banking system lack sufficient liquid assets to meet their obligations. The scenario does not provide any information suggesting that there's an industry-wide issue with liquidity.

Things to Remember

- Systemic risk is a crucial concept in risk management and refers to the risk of widespread financial instability or collapse due to interconnectedness and interdependencies within the financial system.
- Systemic risk can arise from various sources such as economic downturns, market

disruptions, regulatory changes, or the failure of a major financial institution.

- It is important for financial institutions to assess and manage systemic risk through measures such as stress testing, diversification, and contingency planning.
 - Maturity transformation risk is a specific type of risk related to the maturity mismatch between assets and liabilities on a bank's balance sheet.
 - Balance sheet risk encompasses risks associated with the composition and quality of assets, liabilities, and equity on a financial institution's balance sheet.
 - Banking sector liquidity risk refers to the risk of insufficient liquid assets within the banking system to meet short-term obligations, potentially leading to liquidity crises.
-

Q.2281 Aruba Commercial Bank's risk management team has recently expressed concerns about an escalating exposure to funding liquidity risk. This risk is associated with the bank's ability to meet its obligations without incurring unacceptable losses. Which of the following scenarios could potentially be the reason for this heightened exposure to funding liquidity risk?

- A. Financing short-term loans mostly with short-term deposits.
- B. Financing long-term loans mostly with long-term deposits.
- C. Financing long-term loans mostly with short-term deposits.
- D. Financing short-term loans mostly with long-term deposits.

The correct answer is C.

Many banks are largely short-term borrowers (via deposits), so their capacity to maintain long-term positions and their flexibility when circumstances or expectations change is limited. If the bank uses deposits to fund long-term loans, there's a big possibility that it will not have cash readily available as and when the depositor comes calling.

Things to Remember

- Funding liquidity risk is the risk that a company or financial institution will be unable to meet its short-term financial obligations due to an inability to liquidate assets or obtain funding at a reasonable cost.
- Short-term deposits are typically more volatile and can be withdrawn by depositors on short notice, leading to funding liquidity issues if they are used to fund long-term loans.
- Long-term deposits are more stable sources of funding as they have longer maturity periods, providing more certainty for the bank's funding needs.
- Matching the maturity of assets and liabilities is an important aspect of liquidity risk management to ensure that the bank can meet its obligations without incurring losses.
- Banks need to maintain a balance between short-term and long-term funding sources to manage liquidity risk effectively.

Q.2282 What do you understand by *matched funding* as used in the context of lending?

- A. Financing long-term loans with borrowed funds.

- B. Financing loans with reserve cash saved over a period of time.
- C. Financing long-term assets with long-term debt.
- D. Giving a particular facility to the beneficiary who qualifies for it, taking into consideration their full credit profile.

The correct answer is **C**.

Matched funding refers to the practice of financing long-term assets with long-term debt. This is an ideal scenario as it aligns the maturity of the assets (loans given out by the bank) with the maturity of the liabilities (debt taken on by the bank). This reduces the risk of a liquidity crunch, where the bank might have to pay off its liabilities before its assets have matured. However, in practice, this is not always possible due to the nature of banking operations. Banks often engage in maturity transformation, where they use short-term deposits to finance long-term loans. This can lead to a mismatch in the maturity of assets and liabilities, increasing the risk of a liquidity crisis. However, banks manage this risk through various risk management practices and regulatory requirements.

Choice A is incorrect. While it is true that financial institutions often finance long-term loans with borrowed funds, this does not accurately define the concept of 'matched funding'. Matched funding specifically refers to the practice of financing assets with debt that has a similar maturity or due date. Therefore, simply financing long-term loans with borrowed funds, without considering the matching of maturities, does not constitute matched funding.

Choice B is incorrect. Financing loans with reserve cash saved over a period of time can be part of an institution's overall financial strategy but it doesn't represent 'matched funding'. In matched funding, the focus is on aligning the maturities of assets and liabilities to mitigate risks associated with interest rate fluctuations and liquidity concerns.

Choice D is incorrect. Giving a particular facility to a beneficiary who qualifies for it based on their full credit profile pertains more to credit risk management and underwriting practices rather than matched funding. It doesn't involve matching the duration or maturity dates between assets and liabilities which is central to matched funding.

Things to Remember

- Matched funding helps reduce the risk of a liquidity crunch by aligning the maturity of assets with the maturity of liabilities.
- Banks often engage in maturity transformation, using short-term deposits to finance long-term loans, which can lead to a mismatch in asset and liability maturities.
- Risk management practices and regulatory requirements are used by banks to manage the risks associated with mismatched funding.
- Interest rate fluctuations and liquidity concerns are key considerations in matched

funding strategies.

Q.2283 Which of the following **best** explains why a bank may be incentivized to finance long-term loans with short-term deposits, despite the grave liquidity issues this could create?

- A. It is much easier to find long-term deposits than short-term deposits.
- B. Interest rates on long-term loans are usually higher than those on short-term loans.
- C. It is much easier to find clients for long-term financing.
- D. Long-term facilities are less risky than short-term ones.

The correct answer is **B**.

The primary reason why banks may be incentivized to finance long-term loans with short-term deposits is the difference in interest rates. Typically, interest rates on long-term loans are higher than those on short-term loans. This is primarily due to the yield curve, which is generally upward sloping. This means that the longer the maturity of a loan, the higher the interest rate that can be charged. Therefore, by financing long-term loans with short-term deposits, banks can earn a higher return on their investments. This practice, however, exposes the banks to rollover risk. Rollover risk is the risk that the short-term debt cannot be refinanced, or can be refinanced only on highly disadvantageous terms. Despite this risk, the potential for higher returns often incentivizes banks to engage in this practice.

Choice A is incorrect. It is not necessarily easier to find long-term deposits than short-term deposits. In fact, the opposite may be true as short-term deposits offer more flexibility and are therefore often more attractive to depositors.

Choice C is incorrect. The ease of finding clients for long-term financing does not directly relate to the decision of banks to finance long-term loans with short-term deposits. This choice does not address the risk-return trade-off that banks consider when making such decisions.

Choice D is incorrect. Long term facilities are not inherently less risky than short term ones. The risk associated with a loan or facility depends on a variety of factors including the creditworthiness of the borrower, interest rate fluctuations, and market conditions among others.

Things to Remember

- **Yield curve:** The yield curve is a graphical representation of the relationship between the interest rates and the time to maturity of debt securities. It typically slopes upward, indicating that longer-term securities have higher interest rates.
- **Rollover risk:** Rollover risk is the risk that a borrower will not be able to refinance or

roll over its debt obligations at maturity, or will only be able to do so at unfavorable terms.

- Interest rate risk: Banks face interest rate risk when the interest rates on their assets and liabilities are not perfectly matched. This can lead to fluctuations in their net interest income.
 - Liquidity risk: Liquidity risk is the risk that a bank may not be able to meet its short-term obligations due to a lack of liquid assets or the inability to sell assets quickly without incurring significant losses.
-

Q.2284 In a recent discussion between two senior risk managers at a Chinese bank, the topic of rollover risk was brought up as a growing concern for their institution. What does the term "rollover risk" most likely refer to?

- A. The bank was increasingly lending money to high-risk individuals, thereby increasing its exposure to credit risk.
- B. The amount recovered from non-performing loans was gradually declining, thereby increasing the bank's losses.
- C. The bank was increasingly financing long-term loans with short-term deposits, thereby making it hard to repay its own short-term debt.
- D. The bank's portfolio at risk was growing at a relatively higher rate.

The correct answer is C.

Funding long-term assets with short-term debt exposes an intermediary to rollover risk, the risk that the short-term debt cannot be refinanced, or can be refinanced only on highly disadvantageous terms.

Things to Remember

- Rollover risk is a type of liquidity risk that arises when an institution is unable to refinance its short-term debt obligations at or near their maturity.
- It is common for financial institutions to face rollover risk when they fund long-term assets with short-term liabilities.
- Rollover risk can lead to financial distress or even bankruptcy if an institution is unable to meet its short-term debt obligations.
- Managing rollover risk involves maintaining a well-balanced maturity profile of assets and liabilities to ensure that short-term obligations can be met.

Q.2285 A certain bank has a portfolio of CDOs (Collateralized Debt Obligations). In order to improve its liquidity position, its finance department has proposed an entry to secondary markets by selling the CDOs under a repo agreement. Is the proposal viable?

- A. No, because CDOs are ineligible for repo agreements.
- B. No, because liquidity positions can only be improved via direct sale of assets.

C. Yes, because CDOs are always eligible for repo transactions regardless of their credit quality.

D. Yes, because CDOs can be used as collateral in repo agreements provided their credit quality is high.

The correct answer is **D**.

Collateralized Debt Obligations (CDOs) can indeed be used as collateral in repurchase agreements, provided their credit quality is high. A repurchase agreement, or 'repo', is essentially a short-term loan where the borrower sells an asset to the lender with a promise to repurchase it at a later date at a predetermined price. The asset sold is considered as 'collateral' for the loan. In this context, the bank's CDOs can serve as the collateral. However, the key factor here is the 'credit quality' of the CDOs. High credit quality implies that the CDOs are less risky and more likely to meet their debt obligations. This makes them more acceptable as collateral in a repo agreement. Therefore, if the bank's CDOs have high credit quality, they can be used in a repo agreement to improve the bank's liquidity position.

Choice A is incorrect. CDOs are not ineligible for repo agreements. The eligibility of a CDO for a repo agreement depends on its credit quality and the specific terms of the agreement.

Choice B is incorrect. Improving liquidity positions does not solely depend on direct sale of assets. Other financial strategies, such as entering into repo agreements, can also be used to enhance liquidity.

Choice C is incorrect. It's not accurate to say that CDOs are always eligible for repo transactions regardless of their credit quality. The eligibility largely depends on the credit quality and other factors like market conditions.

Things to Remember

- **Repo Agreement:** A repurchase agreement, or 'repo', is a short-term loan where the borrower sells an asset to the lender with a promise to repurchase it at a later date at a predetermined price.
- **Credit Quality:** The credit quality of an asset, such as a CDO, is a measure of its riskiness and likelihood to meet its debt obligations. Higher credit quality assets are less risky and more acceptable as collateral in financial transactions.
- **Liquidity Management:** Improving liquidity positions involves managing the availability of cash and other liquid assets to meet short-term obligations. Various strategies, including repo agreements, can be used to enhance liquidity.
- **Market Conditions:** The eligibility of assets like CDOs for repo transactions can also be

influenced by market conditions and the specific terms of the agreement.

Q.2286 Baraba bank is in the process of calculating its leverage. An extract of its balance sheet is as follows:

Assets: \$50 million

Equity: \$20 million

Deposits: \$30 million

What is the level of leverage?

- A. 1
- B. 2.5
- C. 1.67
- D. 1.5

The correct answer is **B**.

Leverage should be calculated as (all \$ amounts in million):

$$\frac{\text{Total Assets}}{\text{Equity}} = \frac{50}{20} = 2.5$$

Q.2288 Barakuda is considering borrowing additional funds to in order to finance an ambitious expansion plan. In what circumstances would it be appropriate for the bank to borrow funds and go ahead with its plan?

- A. If its return on assets is equal to expenses of additional borrowing.
- B. If the cost of borrowing equals the expenses of the existing borrowing.
- C. If the cost of borrowing is less than the return on assets.
- D. If the total value of existing assets is less than the total value of existing debt.

The correct answer is **C**.

If the cost of borrowing is less than the return on assets, it is financially viable for the bank to borrow additional funds and proceed with its expansion plan. This is because the return on

assets (ROA) is a measure of how efficiently a bank is using its assets to generate earnings. If the ROA is higher than the cost of borrowing, it means that the bank is earning more from its assets than it is paying in interest on the borrowed funds. This would result in a net gain for the bank, making the borrowing and expansion plan financially justifiable. The bank would be able to repay the borrowed capital while still making a profit. This is a fundamental principle in finance known as the cost of capital, which states that a company should only invest in projects or expansions that are expected to generate a return that is higher than the cost of the capital used to finance them.

Choice A is incorrect. If the return on assets is equal to the expenses of additional borrowing, it means that all the returns generated by the assets are used to cover borrowing costs. This leaves no room for profit, which makes this option unviable for expansion.

Choice B is incorrect. The cost of new borrowing being equal to the expenses of existing borrowing does not necessarily justify an expansion plan. It's important that the return on assets exceeds both these costs in order to generate a profit and justify expansion.

Choice D is incorrect. If the total value of existing assets is less than total value of existing debt, it indicates a negative net worth for Barakuda which would make further borrowing risky and potentially unsustainable.

Things to Remember

- Return on Assets (ROA) is a financial ratio that shows the percentage of profit a company earns in relation to its overall assets.
 - Cost of borrowing refers to the interest expense incurred by a company when it borrows funds.
 - Cost of capital is the minimum return that a company must earn on its investments in order to maintain its market value and attract funds.
 - Net gain is the difference between the revenue generated from an investment and the total costs or expenses incurred.
 - Net worth is the difference between a company's total assets and total liabilities, representing the equity of the company's owners.
-

Q.2731 The shares of a company currently trade at a bid/ask rate of \$20.20 and \$20.35. If the sample standard deviation of the spread is \$0.0003, what will be the expected transaction cost assuming a 99% confidence interval on transaction cost?

- A. \$0.321
- B. \$0.075
- C. \$0.082
- D. \$0.821

The correct answer is **C**.

$$\text{Transaction costs} = \pm P \times \frac{1}{2}(S + 2.33\sigma)$$

Where

P = midprice

S = expected bid-ask spread

$$P = \frac{\text{Bid price} + \text{Ask price}}{2} = \frac{20.2 + 20.35}{2} = \$20.275$$

$$S = \frac{\text{Ask price} - \text{Bid price}}{\text{Midprice}} = \frac{20.35 - 20.20}{20.275} = \$0.0074$$

Thus,

$$\text{Transaction costs} = \pm 20.275 \times \frac{1}{2}(0.0074 + 2.33 \times \$0.0003) = \$0.0821$$

Q.2982 In the period of the global financial turmoil, the collapse of the financial institutions can largely be attributed to both illiquidity and insolvency. Which of the following does NOT describe the sequence of events in the collapse of an intermediary?

- A. Issues about the solvency of the company were raised by the reports of losses at the intermediary, or at other institutions.
- B. All companies, financial intermediaries, and non-financial companies are still willing to lend to the intermediary.
- C. To raise funds, the intermediary is forced to liquidate assets, which might lead to

losses in a distressed market.

D. Being aware that the challenges faced by the intermediary are now being compounded by the realized mark-to-market losses, the lenders become more and more reluctant to extend credit to the intermediary.

The correct answer is **B**.

This choice states that all companies, financial intermediaries, and non-financial companies are still willing to lend to the intermediary. However, this is not accurate. In the context of a financial crisis, firms, financial intermediaries, and non-financial companies would typically be very reluctant to lend to an intermediary that is facing solvency issues. This reluctance is often reflected in high credit spreads and a decrease in the volume of loan proceeds that the affected company can acquire. Therefore, this choice does not accurately describe a step in the sequence of events leading to the collapse of a financial intermediary during a financial crisis.

Choice A is incorrect. This statement accurately represents a step in the sequence leading to the collapse of a financial intermediary. During a financial crisis, reports of losses at an institution can indeed raise concerns about its solvency, which can trigger further negative events.

Choice C is incorrect. This statement accurately represents a step in the sequence leading to the collapse of a financial intermediary. When faced with liquidity issues and unable to secure loans from other institutions, an intermediary may be forced into liquidating assets at potentially unfavorable prices.

Choice D is incorrect. This statement also accurately represents part of this sequence; lenders often become more reluctant to extend credit as they realize that their risks are increasing due to mark-to-market losses at the borrowing institution.

Things to Remember

- Illiquidity and insolvency are two key factors that can lead to the collapse of financial institutions during a crisis.
- Mark-to-market losses can significantly impact the financial health of an intermediary during a crisis.
- High credit spreads and decreased loan proceeds are common indicators of reluctance to lend to a financially distressed institution.
- Reports of losses at an institution can raise concerns about its solvency, triggering further negative events.
- Liquidating assets in a distressed market can lead to additional losses for a struggling

intermediary.

Q.3135 A firm has a return on asset of 2.5% and a value at risk of 2.45% at 95% confidence. Its cost of debt is 2%.

What should be its debt equity ratio if its desired return on equity is twice its 95% VaR.

A. 2.4

B. 3.4

C. 5.8

D. 4.8

The correct answer is **D**.

An entity's return on equity can be computed using the following formulas:

$$r_E = Lr_A - (L - 1)r_D$$

Where:

r_A = return on assets

r_E = return on equity

r_D = cost of debt

$$L = \text{leverage ratio} = \frac{(\text{Equity} + \text{Debt})}{\text{Equity}} = 1 + \frac{\text{Debt}}{\text{Equity}}$$

You may also want to express this in words as:

$$\begin{aligned} \text{ROE} &= (\text{leverage ratio} \times \text{ROA}) - [(\text{leverage ratio} - 1) \times \text{cost of debt}] \\ \text{ROE} &= 2 \times \text{VaR} = 2 \times 0.0245 = 0.049 \end{aligned}$$

Thus,

$$\begin{aligned} 0.049 &= L \times 0.025 - (L - 1)0.02 \\ 0.049 &= 0.025L - 0.02L + 0.02 \\ 0.049 &= 0.005L + 0.02 \\ L &= \frac{0.029}{0.005} = 5.8 \\ L &= 1 + \frac{D}{E} \\ \frac{D}{E} &= 5.8 - 1 = 4.8 \end{aligned}$$

Q.3225 Mark Sanders is a chief investment officer at Kremlin Pension Fund managing defined benefit plan for state employees. His fund manager has calculated the 1-day value at risk (VaR) of the position at \$62 million. However, given the magnitude of the position it is most likely that any liquidation will take place over four trading days. In this scenario, what will be the liquidity-adjusted VaR for Kremlin?

- A. \$116,250,000
- B. \$84,897,000
- C. \$22,897,000
- D. \$54,250,000

The correct answer is **B**.

$$\text{VaR}_i \left(\alpha, \frac{1}{252} \right) (X) \times \sqrt{\frac{(1 + T)(1 + 2T)}{6T}}$$

$$\text{liquidity-adjusted VaR} = \$62,000,000 \times \sqrt{\frac{(1 + 4 \times 1 + 8)}{6 \times 4}} = \$84,897,000.$$

Q.3226 Samar Sarkar Brokerage specializes in providing margin loans to U.S. hedge funds that intend to buy securities on margin. One of the brokerage's clients, Xenon Hedge Fund, wants to take a \$3,000,000 equity position while putting up the minimum equity amount required by the Federal Reserve. Determine the margin loan amount and the leverage ratio of this position.

- A. Margin Loan \$3,000,000; Leverage Ratio 1.0
- B. Margin Loan \$1,500,000; Leverage Ratio 1.0
- C. Margin Loan \$1,500,000; Leverage Ratio 2.0
- D. Margin Loan \$3,000,000; Leverage Ratio 0.0

The correct answer is **C**.

According to Regulation T of the Federal Reserve Board, one may borrow up to 50 percent of the purchase price of securities that can be purchased on margin. Therefore, the margin loan will be \$1,500,000 ($= \$3,000,000 \times 50\%$). The remaining \$1,500,000 has to be financed by equity. The leverage ratio is calculated as total assets divided by equity. Therefore, the hedge fund's leverage ratio is 2.0 ($= \frac{\$3,000,000}{\$1,500,000}$).

Note: By buying on margin, you borrow money from a broker. Margin accounts increase an investor's purchasing power and allow them to use other people's money to increase financial leverage.

Things to Remember

- Margin loans allow investors to borrow money from a broker to purchase securities.
 - Regulation T of the Federal Reserve Board sets the maximum amount that can be borrowed on margin at 50% of the purchase price of securities.
 - Leverage ratio is calculated as total assets divided by equity, indicating the level of financial leverage used by an investor.
 - Buying on margin increases an investor's purchasing power but also increases the risk due to the use of borrowed funds.
-

Q.3228 Tom Daniel is a portfolio manager at XinWin pension fund investing in common stocks. XinWin has a policy around liquidity risk measure to limit each of its holdings to a maximum of 45% of its 30-day average value traded. If the fund size is \$7 billion, what is the maximum allocation that the fund can hold in a stock with a 30-day average value traded of \$390 million?

A. 17.56%

B. 2.25%

C. 2.51%

D. 5.57%

The correct answer is **C**.

$$390 \times 0.45 = \$175.5 \text{ million}$$

$$\% \text{ of allocation in fund} = \frac{175.5}{7000} = 0.0251 \text{ or } 2.51\%$$

Q.3230 Abraham Maslow is an equity strategist at FinteseCapital. He intends to use leverage to increase the returns on a convertible arbitrage strategy. The expected return on assets of the strategy is 11%. The fund has \$16 million invested in the strategy and will finance the investment with 60% borrowed funds. The cost of borrowing is 7%. Using this information, the return on equity (ROE) is closest to:

- A. 17%
- B. 13.66%
- C. 10.66%
- D. 11%

The correct answer is **A**.

$$\begin{aligned} \text{Debt} &= \$16 \times 0.60 = \$9.6 \text{ million} \\ \text{Leverage ratio} &= \frac{\text{total assets}}{\text{equity}} \\ \text{leverage ratio} &= \text{Leverage ratio} = \frac{16}{(16 - 9.6)} = 2.5 \text{ times} \\ r^e &= Lr^a - (L - 1)r^d \end{aligned}$$

Where:

r^a = return on assets

r^e = return on equity

r^d = cost of debt

L = leverage ratio

$$\text{Return on equity} = 2.5 \times 11\% - [(2.5 - 1)(7\%)] = 27.5\% - 10.5\% = 17\%$$

Q.3875 There's always a risk that the price of an asset will move adversely in the act of buying or selling it. This risk is low if assets can be liquidated or a position can be covered quickly, cheaply, and without moving the price too much. The risk described here is:

- A. Balance sheet risk
- B. Transactions liquidity risk
- C. Systematic risk
- D. funding liquidity risk

The correct answer is **B**.

Transaction liquidity risk is the risk associated with an adverse movement in the price of an asset during the process of buying or selling it. This risk is low if the asset can be liquidated or if a position can be covered quickly, inexpensively, and without causing a significant shift in the price. An asset is considered liquid if it is a good substitute for cash or if it is 'near' cash. An asset is said to carry a liquidity premium if its price is lower and its expected return is higher because it is not perfectly liquid. A market is considered liquid if market participants can quickly and without excessive transaction costs or price deterioration, put on or unwind positions.

Choice A is incorrect. Balance sheet risk refers to the possibility that a company's balance sheet can portray a misleading picture of its financial strength. It does not directly relate to the risk associated with asset price movement during transactions.

Choice C is incorrect. Systematic risk, also known as market risk, refers to the overall impact of market movements and cannot be eliminated through diversification. It does not specifically refer to the potential for an asset's price to move unfavorably during a transaction.

Choice D is incorrect. Funding liquidity risk pertains to a situation where an entity is unable to obtain funds at normal prices or quantities, thereby affecting its ability to meet its obligations when they come due. This type of risk doesn't directly deal with the potential for an asset's price moving unfavorably during transactions.

Things to Remember

- Transaction liquidity risk is also known as market liquidity risk.
- Liquidity risk can be managed through diversification, maintaining ample cash reserves, and using hedging strategies.
- Market liquidity risk can be affected by factors such as trading volume, bid-ask spreads, and market depth.
- Asset liquidity is crucial for investors who may need to sell assets quickly to meet financial obligations or take advantage of investment opportunities.
- Illiquid assets may have higher liquidity risk and may require a higher liquidity premium to compensate investors for the risk.

Q.3876 Convertible arbitrage hedge funds experienced significant losses during the 2007-2009 subprime crisis. These losses were most likely due to:

A. Increase of the target's stock price due to stock market corrections

- B. Unavailability of financing due to market conditions
- C. The decline of the acquirer's price as they filed for bankruptcy during the crisis
- D. Selling of the highly liquid assets

The correct answer is **B**.

Convertible arbitrage hedge funds rely heavily on leverage to boost returns. When market conditions make financing unavailable, as was the case during the subprime crisis, the value of convertible bonds can plummet. This situation was further exacerbated by increasing redemptions, which further strained funding liquidity. Convertible bonds also have a limited 'clientele' among investors. When these investors develop an aversion to the product during a period of market stress, it becomes difficult to sell the product without significant price declines. The gap between the prices of convertible bonds and their replicating portfolios widened significantly, with no-arbitrage capital being brought into the market. The theoretical price is the value of the replicating portfolio, taking into account credit, risk-free rates, and the embedded option.

Choice A is incorrect. An increase in the target's stock price due to market corrections would not have caused significant losses for convertible arbitrage hedge funds during the subprime crisis. Convertible arbitrage strategies involve buying a company's convertible securities and short selling its common stock. If the stock price increases, it would offset losses on the short position with gains on the convertible securities.

Choice C is incorrect. The decline of an acquirer's price as they filed for bankruptcy during the crisis could cause losses for some hedge funds, but it was not a widespread issue that affected most convertible arbitrage hedge funds during this period. Moreover, these types of funds typically do not take large positions in companies that are at high risk of bankruptcy.

Choice D is incorrect. Selling highly liquid assets wouldn't necessarily lead to significant losses unless those assets were sold at a discount or below their intrinsic value which was not generally observed during subprime crisis specifically for convertible arbitrage hedge funds.

Things to Remember

- Convertible arbitrage hedge funds involve buying convertible securities and short selling the underlying common stock to exploit pricing inefficiencies.
- These funds rely on leverage to boost returns, making them susceptible to market conditions that affect financing availability.
- Convertible bonds can experience significant price declines during market stress, leading to losses for arbitrageurs.
- Investor aversion to convertible bonds during crises can further exacerbate price

declines and liquidity issues.

- Theoretical pricing of convertible bonds considers factors like credit risk, risk-free rates, and embedded options.
-

Q.3877 ABC Ltd. is a US-based food processing firm. The firm has a ROA of 10%, total assets equal to \$5, equity capital equal to \$2. The firm's cost of debt is 3%. Calculate the firm's ROE.

- A. 19.8
- B. 20.1
- C. 20.5
- D. 21.2

The correct answer is **C**.

$$\begin{aligned}\text{The firm's leverage} &= \frac{A}{E} = \frac{E + D}{E} = 1 + \frac{D}{E} \\ L &= \frac{5}{2} = 2.5\end{aligned}$$

Its

$$\begin{aligned}\text{ROE} &= (\text{ROA} \times \text{Leverage ratio}) - [(\text{Leverage ratio} - 1) \times \text{cost of debt}] \\ &= (0.10 \times 2.5) - [(2.5 - 1) \times 0.03] = 0.205 \text{ or } 20.5\%\end{aligned}$$

Q.3878 ABC Ltd. is a US-based food processing firm. The firm has a ROA of 10%, total assets equal to \$7, equity capital equal to \$2. The firm's cost of debt is 5%. Calculate the firm's ROE.

- A. 25.5%
- B. 27.5%
- C. 22.5%
- D. 33.5%

The correct answer is **C**.

$$\text{Leverage ratio} = \frac{7}{2} = 3.5$$

$$\begin{aligned}\text{ROE} &= (\text{ROA} \times \text{Leverage ratio}) - [(\text{Leverage ratio} - 1) \times \text{cost of debt}] \\ &= (0.10 \times 3.5) - [(3.5 - 1) \times 0.05] = 0.225 \text{ or } 22.5\%\end{aligned}$$

Q.3879 Suppose that an investor borrows \$200. He then invests collateral of \$225. Calculate the haircut of this transaction.

- A. 25.0
- B. 45.0
- C. 212.5
- D. 225.0

The correct answer is **A**.

A haircut is equivalent to the value of the collateral less the amount borrowed. The \$25 difference is the one referred to as the haircut. The lender aims to ensure that the loan amount is less than the collateral. From the lender's perspective, the haircut is the extent to which collateral value can fall and still be fully collateralized.

Things to Remember

- Collateral: Assets provided by a borrower to secure a loan or credit. In case of default, the lender can seize the collateral to recover the loan amount.
 - Loan-to-Value Ratio: The ratio of the loan amount to the value of the collateral. A lower ratio indicates a lower risk for the lender.
 - Margin Call: A demand by a lender for an investor to deposit additional money or securities to cover potential losses in case the value of the collateral falls below a certain level.
 - Securities Lending: The practice of loaning securities to earn additional income. Haircuts are often applied in securities lending transactions to protect the lender.
-

Q.3880 A hedge fund has \$1,000,000 in cash, corresponding to an initial placement of \$1,000,000 in equity by its owners. The hedge fund wants to finance a long position \$1,000,000 worth of stocks at the Reg T margin requirement of 50 percent. A broker facilitates the trade by lending the hedge fund \$500,000. Determine the new leverage ratio for the hedge fund.

A. 0.5

B. 1.0

C. 2.0

D. 1.5

The correct answer is **D**.

Before the trade, this is how the hedge fund's balance sheet looks like:

Assets

Cash: 1,000,000

Liabilities

Equity: 1,000,000

After the trade, here's the hedge fund's balance sheet:

Assets

Cash: 500,000

Stocks: 1,000,000

Liabilities

Equity: 1,000,000

Margin loan: 500,000

The leverage ratio (LR) is equal to total assets divided by equity.

$$LR = 1,500,000 / 1,000,000 = 1.5$$

Things to Remember

Regulation T specifies the amount of credit brokers-dealers may extend to investors for the purchase of securities and also sets forth regulations for the use of cash accounts. A margin account is required for investors who wish to purchase securities with broker-dealer credit. Under Reg T, investors may borrow a maximum of 50% of the purchase price and must pay the remaining balance in cash.

Q.3881 Suppose a company traded at an ask price of \$335 and a bid price of \$331. The sample standard deviation of the spread is 0.00019. Calculate the expected transaction cost following the zero mean normality assumption and 95% confidence interval on the transaction cost.

- A. 2.03
- B. 2.05
- C. 1.06
- D. 0.06

The correct answer is **B**.

$$\text{Expected transaction cost} = P \times \frac{1}{2}(s + \lambda (\text{standard deviation of spread}))$$

$$\text{Midprice (P)} = \frac{335 + 331}{2} = 333$$

$$s = \frac{\text{ask price} - \text{bid price}}{\text{midprice}} = \frac{335 - 331}{333} = 0.0120$$

$$\text{Expected transactions cost} = 333 \times \frac{1}{2}(0.0120 + 1.65(0.00019)) = 2.05$$

Q.3882 Suppose Delight Inc. traded at an ask price of \$200 and a bid price of \$199. The sample standard deviation of the spread is 0.0004. Calculate the 99% spread risk factor for the transaction.

- A. 0.00297
- B. 0.00325
- C. 0.00353
- D. 0.00381

The correct answer is **A**.

The 99% confidence interval of transaction cost in dollars, i.e., 99% spread risk factor is expressed as:

$$\pm P \times \frac{1}{2}(s + 2.33\sigma_s)$$

and

$$99\% \text{ spread risk factor} = \frac{1}{2}(s + 2.33\sigma_s)$$

$$\text{mid market price} = \frac{\text{ask} + \text{bid}}{2} = \frac{200 + 199}{2} = 199.5$$

and,

$$s = \frac{\text{ask} - \text{bid}}{\text{mid market price}} = \frac{200 - 199}{199.5} = 0.00501$$

Thus,

$$99\% \text{ spread risk factor} = \frac{1}{2}(0.00501 + 2.33(0.0004)) = 0.00297$$

Q.3883 A trader estimates that the number of trading days (T) required for the orderly liquidation of a position is eight trading days (T = 8). By what percentage is the trader most likely to adjust the VaR?

- A. 79%
- B. 179%
- C. 18.75%
- D. 118.75%

The correct answer is **A**.

To adjust for the fact that the position could be liquidated over several days, the following formula can be used:

$$\text{VaR}_t \times \sqrt{\frac{(1 + T)(1 + 2T)}{6T}}$$

The adjustment to the overnight VaR of the position is

$$\sqrt{\frac{(1 + 8)(1 + (2 \times 8))}{6 \times 8}} = 1.7854$$

This means that the trader should increase VaR by 79%.

Q.3884 Velma Jones placed an order to sell a stock when the market price was \$50. Due to market volatility, by the time Jones's broker sold the stock, the price had fallen to \$48. In the market, this phenomenon is known as:

- A. Bid-ask spread
- B. Adverse price impact
- C. Slippage
- D. Leverage effect

The correct answer is **C**.

Slippage is a term used in finance to describe a situation where the execution price of a trade is

different from the expected price. This usually occurs due to market volatility and the time delay between when an order is placed and when it is actually executed. In the case of Velma Jones, she intended to sell her stock at \$50, but due to market fluctuations and the time it took for her broker to execute the trade, the price had fallen to \$48. This difference in price is referred to as slippage. It is a common occurrence in financial markets, particularly in volatile markets where prices can change rapidly within a short period of time.

Choice A is incorrect. The bid-ask spread refers to the difference between the highest price that a buyer is willing to pay for an asset and the lowest price that a seller is willing to accept. It does not refer to the difference between an expected execution price and actual execution price due to market volatility.

Choice B is incorrect. Adverse price impact refers to a situation where large trades cause significant changes in prices, which can negatively affect traders' profits. While this concept involves changes in prices, it doesn't specifically refer to the discrepancy between expected and actual trade execution prices due to market fluctuations.

Choice D is incorrect. The leverage effect refers to how debt can increase a company's return on equity or amplify potential losses, depending on whether the company's returns exceed its borrowing costs or not. This concept has no direct relation with differences in expected and actual trade execution prices caused by market volatility.

Things to Remember

- Slippage can occur in both buy and sell orders, and it is a common risk that traders and investors need to consider when executing trades.
 - Market volatility can increase the likelihood of slippage, as rapid price movements can result in significant differences between expected and actual execution prices.
 - Slippage can also be influenced by factors such as order size, liquidity of the market, and the speed of order execution.
 - Traders often use limit orders and other strategies to try to minimize the impact of slippage on their trades.
-

Reading 129: Early Warning Indicators

Q.3844 An indicator that provides information and significant potential stress before the occurrence of an actual event is most likely known as a:

- A. Sharp indicator
- B. Granular indicator
- C. Leading indicator
- D. Lagging indicator

The correct answer is **C**.

A leading indicator, also known as a forward-looking indicator, is designed to signal potential stress or risk before the actual occurrence of an event. This type of indicator is particularly crucial in preparing for systematic risks. It allows institutions to take proactive measures to mitigate potential losses or disruptions. Leading indicators can be based on a variety of factors, including economic data, market trends, and internal metrics. By identifying potential risks early, institutions can adjust their strategies and operations to better manage these risks.

Choice A is incorrect. Sharp indicators refer to the granularity and specificity of a signal as it relates to an institution's profile. They are signals that stand out clearly within large sets of data, such as detecting drops in deposit balances of high-net-worth or rate-sensitive customers. However, sharpness relates to the clarity of a signal, not its timing. Therefore, sharp indicators do not necessarily indicate stress before an event occurs.

Choice B is incorrect. Granular indicators provide detailed information about specific areas or aspects of risk but they do not necessarily signal potential stress before the actual occurrence of an event.

Choice D is incorrect. Lagging indicators are measures that change after the economy as a whole does, hence they do not provide information or signal potential stress before the actual occurrence of an event.

Things to Remember

- Leading indicators are crucial for proactive risk management and can help institutions prepare for systematic risks.
- These indicators can be based on economic data, market trends, or internal metrics.
- Granular indicators provide detailed information about specific areas or aspects of risk but may not signal potential stress before an event.
- Lagging indicators change after the economy as a whole does and do not provide early warning signals.

Q.3845 Early warning indicator triggers are used to initiate management discussions and actions that call for formal documentation. Which of the following actions is least likely a cost-benefit decision?

- A. Increasing highly liquid assets
- B. Continuing to comply with LCR requirements
- C. Setting a minimum standard for liquidity buffer according to the bank's risk appetite
- D. Using a set and forget approach to make cost-benefit decisions

The correct answer is **D**.

The 'set and forget' approach is least likely to be a cost-benefit decision. This approach does not allow for the strategic management of the potentially catastrophic impacts of insufficient liquidity in a dynamic market. It is a passive strategy that does not involve active decision-making based on cost-benefit analysis. Instead, it involves setting a strategy and then leaving it unchanged, regardless of changes in the market or the bank's financial situation. This approach can lead to significant risks, as it does not allow for adjustments based on changing market conditions or the bank's liquidity position. Therefore, it is least likely to be a cost-benefit decision compared to the other options.

Choice A is incorrect. Increasing high liquid assets is indeed a cost-benefit decision. The benefit of having more liquid assets is that it provides a safety net in case of financial distress, but the cost is the opportunity cost of not investing those funds in potentially higher yielding investments.

Choice B is incorrect. Continuing to comply with LCR (Liquidity Coverage Ratio) requirements also involves a cost-benefit decision. The benefit here lies in maintaining regulatory compliance and avoiding penalties, while the cost could be associated with holding high-quality liquid assets which may yield lower returns.

Choice C is incorrect. Setting a minimum standard for liquidity buffer according to the bank's risk appetite also involves a cost-benefit analysis. The benefit comes from ensuring sufficient liquidity to meet unexpected cash flow needs, while the potential costs are tied to holding unproductive assets that could otherwise be invested for higher returns.

Choice D (correct answer) does not involve any explicit or implicit trade-off between costs and benefits as it implies an automatic approach without any active management or decision-making process involved.

Things to Remember

- Cost-benefit analysis is a critical tool in risk management that helps organizations evaluate the potential costs and benefits of different decisions or strategies.
- Highly liquid assets refer to assets that can be quickly converted into cash without

significant loss of value.

- Liquidity Coverage Ratio (LCR) is a regulatory requirement that banks maintain an adequate level of high-quality liquid assets to withstand short-term liquidity stress.
 - Risk appetite is the level of risk that an organization is willing to accept in pursuit of its objectives.
 - Dynamic market conditions require active management and decision-making to adapt to changes and mitigate risks effectively.
-

Q.3846 Companies with substantial trading focus use intraday reporting because they are more exposed to external market conditions. Intraday liquidity monitoring indicators include the following, except:

- A. Intraday credit lines stretched out to financial institution customers
- B. Daily minimum liquidity requirement
- C. Daily maximum liquidity requirement
- D. Available intraday liquidity

The correct answer is **B**.

The 'Daily minimum liquidity requirement' is not listed as an intraday liquidity monitoring indicator in the BCBS-2012 fundamental supervisory guideline. This guideline is a key reference for financial institutions, particularly those with a substantial trading focus, as it outlines the indicators that should be monitored to manage exposure to external market conditions. While the guideline does include a variety of indicators, such as 'Daily maximum liquidity requirement', 'Available intraday liquidity', and 'Intraday credit lines stretched out to financial institution customers', it does not mention the 'Daily minimum liquidity requirement' as an indicator. This is likely because the minimum liquidity requirement is a baseline measure that must be maintained at all times, rather than an intraday measure that can fluctuate based on market conditions.

Choice A is incorrect. Intraday credit lines stretched out to financial institution customers are indeed recognized as an intraday liquidity monitoring indicator according to the BCBS-2012 fundamental supervisory guideline. These credit lines can provide a measure of the potential liquidity needs of a financial institution's customers, which in turn can impact the institution's own liquidity position.

Choice C is incorrect. Daily maximum liquidity requirement is also recognized as an intraday liquidity monitoring indicator according to the BCBS-2012 fundamental supervisory guideline.

This requirement sets an upper limit on the amount of liquid assets that a financial institution must hold at any point during the day, providing a buffer against unexpected market shocks or operational disruptions.

Choice D is incorrect. Available intraday liquidity, which refers to the amount of liquid assets that a financial institution has on hand at any given time during the day, is another recognized intraday liquidity monitoring indicator according to BCBS-2012 guidelines. This measure provides real-time information about an institution's ability to meet its immediate obligations without incurring unacceptable losses.

Things to Remember

- Monitoring intraday liquidity is crucial for companies with substantial trading focus to manage exposure to external market conditions.
 - Intraday liquidity indicators help in assessing the ability to meet immediate obligations without incurring losses.
 - Regulatory guidelines, such as BCBS-2012 fundamental supervisory guideline, provide a framework for monitoring intraday liquidity.
 - Minimum liquidity requirements are typically baseline measures that need to be maintained at all times, not fluctuating intraday indicators.
 - Maximum liquidity requirements set upper limits on liquid assets to provide a buffer against unexpected market shocks.
 - Intraday credit lines stretched out to financial institution customers can impact an institution's liquidity position.
-

Q.3847 Your supervisor asks you to prepare a list of sound early warning indicators for liquidity problems for your bank. Which of the following are sound early warning indicators of a potential liquidity problem?

- I. Negative publicity regarding an asset class owned by the institution
- II. Stock price declines
- III. Widening debt/credit-default-swap spreads
- IV. Increase in credit lines
- V. Significant deterioration in the bank's financial condition
- VI. Increase in the weighted average maturity of liabilities

- A. I, II, III, V
- B. I, II, IV, VI
- C. II, III, IV, V
- D. III, IV, V, VI

The correct answer is **A**.

The Basel Committee on Banking Supervision (BCBS) provided its “Principles of Sound Liquidity Management and Supervision”¹ (Sound Principles) in September 2008, following the global financial crisis of 2007-2008. The following figure is a non-exhaustive list of EWIs recommended by the BCBS.

- Rapid asset growth, especially when funded with probable volatile liabilities
- Growing concentrations in assets or liabilities
- Increases in currency mismatches
- Decrease of weighted average maturity of liabilities
- Recurring incidents of positions approaching or breaching internal or regulatory limits
- Negative trends associated with a particular product line
- Significant deterioration in the bank’s financial condition
- Negative publicity Credit rating downgrade
- Stock price declines
- Rising debt costs
- Widening debt/credit-default-swap spreads
- Rising wholesale/retail funding costs
- Counterparties requesting additional collateral or resisting entering into new transactions
- Drop-in credit lines

- Increasing retail deposit outflows
- Increasing redemptions of CDs before maturity
- Difficulty accessing longer-term funding
- Difficulty placing short-term liabilities

Things to Remember

- Basel Committee on Banking Supervision (BCBS) provides guidelines on sound liquidity management and supervision
 - EWIs (Early Warning Indicators) are crucial for identifying potential liquidity problems
 - Rapid asset growth funded with volatile liabilities can indicate liquidity issues
 - Concentrations in assets or liabilities can pose liquidity risks
 - Currency mismatches and changes in maturity of liabilities are important indicators
 - Monitoring credit lines, deposit outflows, and funding costs is essential for liquidity management
-

Q.3848 Fredric Pete is a risk manager at ABC Bank. Pete wants to forecast the bank's losses. He starts by assessing simple measures that indicate whether the bank's risks are changing over time (i.e., early warning signs). Fred then applies regression techniques, for example, to forecast the losses. The early warning signs that Pete assesses most likely include:

- I. Audit scores
 - II. Staff turnover
 - III. Trade volumes
 - IV. Stock prices
- A. I and IV
 - B. II and III
 - C. I, II and IV
 - D. All of the above

The correct answer is **D**.

Early warning indicators are crucial tools in risk management. They provide insights into potential problems before they become significant issues. In the context of a bank, these indicators can reveal whether the institution is on the right track or if there are reasons for concern. The indicators mentioned in the options - audit scores, staff turnover, trade volumes, and stock prices - all serve as early warning signs. A decline in audit scores could indicate potential issues with the bank's internal controls or compliance with regulations. An increase in staff turnover could suggest problems with employee satisfaction or retention, which could negatively impact productivity and efficiency. Changes in trade volumes could reflect shifts in the bank's market activity, while fluctuations in stock prices could indicate changes in the bank's perceived value or financial stability. Therefore, all these indicators would likely be assessed by a risk manager like Pete when forecasting the bank's losses.

Choice A is incorrect. While audit scores (I) and stock prices (IV) can serve as early warning signs of changing risks within a bank, this choice does not include other important measures such as staff turnover (II) and trade volumes (III). Staff turnover can indicate internal issues that may affect the bank's risk profile, while trade volumes can reflect changes in market activity that could impact the bank's potential losses.

Choice B is incorrect. Although staff turnover (II) and trade volumes (III) are relevant measures for predicting potential losses, this choice excludes audit scores (I) and stock prices (IV). Audit scores provide insights into the effectiveness of a bank's internal controls, which are crucial for managing risks. Stock prices reflect market perceptions of a company's riskiness and future prospects.

Choice C is incorrect. This option includes audit scores(I), staff turnover(II), and stock prices(IV), but it leaves out trade volumes(III). Trade volumes are an important measure because they can indicate changes in market activity that could potentially affect the bank's exposure to financial risks.

Things to Remember

- Early warning indicators help in identifying potential problems before they escalate into significant issues.
- Audit scores can indicate issues with internal controls or compliance with regulations.
- Staff turnover can reflect problems with employee satisfaction or retention, impacting productivity.
- Trade volumes can show shifts in market activity affecting the bank's risk exposure.
- Stock prices can indicate changes in the bank's perceived value or financial stability.

Q.3849 The primary purpose of putting in place early warning indicators is to:

- A. It is a measure used in management to indicate how risky an activity is.
- B. Ensure that the bank holds enough liquid assets to enable it to meet financial obligations
- C. Safeguard asset quality and ensure that liability limits are not breached
- D. Initiate management discussion and necessary corrective action

The correct answer is **D**.

Early warning indicators (EWIs) are essentially risk management tools that are designed to alert management about potential risks or issues that could lead to financial distress if not addressed promptly. The primary purpose of these indicators is to initiate management discussions and necessary corrective actions. This is akin to the warning lights on a car's dashboard, which alert the driver to potential issues that need to be addressed to prevent damage to the vehicle. Similarly, EWIs in a financial institution serve to alert management to potential risks that need to be addressed to prevent financial distress. These indicators can be related to various aspects of the institution's operations, including liquidity, credit risk, market risk, and operational risk. When an EWI is triggered, it serves as a signal for management to discuss the issue and take necessary corrective actions to mitigate the risk.

Choice A is incorrect. While early warning indicators can provide some insight into the riskiness of an activity, their primary purpose is not to serve as a measure of risk. They are more focused on identifying potential risks and threats that may require management intervention.

Choice B is incorrect. Early warning indicators are not specifically designed to ensure that a bank holds enough liquid assets to meet its financial obligations. This would be more related to liquidity management strategies and regulatory requirements, rather than the function of EWIs.

Choice C is incorrect. Safeguarding asset quality and ensuring liability limits are not breached are important aspects of risk management, but they do not represent the primary purpose of implementing early warning indicators in a financial institution. EWIs primarily serve as signals for potential risks or threats requiring corrective action by management.

Things to Remember

- Early warning indicators (EWIs) are crucial tools in risk management for financial institutions.
- EWIs are designed to alert management about potential risks or issues that could lead to financial distress if not addressed promptly.
- These indicators can cover various risk categories such as liquidity risk, credit risk, market risk, and operational risk.

- When an EWI is triggered, it signals the need for management to discuss the issue and take corrective actions to mitigate the risk.
 - EWIs are like warning lights on a car's dashboard, alerting management to potential risks that need attention.
-

Q.3850 What is a sharp early warning indicator?

- A. An indicator that highlights the past performance
- B. An indicator that depends on another performance measure
- C. A highly granular indicator, involving a subset of data
- D. An indicator that needs professional experience and quantitative skills to decode

The correct answer is C.

A sharp early warning indicator is a highly granular indicator that involves a subset of data. It is designed to detect subtle changes within a specific segment of data that might otherwise go unnoticed in the larger data set. This type of indicator is particularly useful in identifying potential risks or threats at an early stage, allowing for timely intervention and risk mitigation. For instance, a sharp indicator might detect a drop in deposit balances among high-net-worth customers, signaling that these crucial customers may be considering leaving the bank. This early warning allows the bank to take appropriate action to retain these customers and prevent potential losses.

Choice A is incorrect. An early warning indicator is not necessarily related to past performance. While historical data can be useful in identifying potential risks, a 'sharp' early warning indicator specifically refers to a highly granular and specific subset of data, not just any past performance metrics.

Choice B is incorrect. Although some indicators may depend on other performance measures, this does not define what a 'sharp' early warning indicator is. The dependency on another measure does not make an indicator more or less 'sharp'. Instead, the granularity and specificity of the data used are what characterize a 'sharp' indicator.

Choice D is incorrect. While professional experience and quantitative skills can certainly aid in interpreting complex indicators, these are not defining characteristics of a 'sharp' early warning indicator. Such an indicator refers specifically to the use of highly granular and specific subsets of data for risk identification.

Things to Remember

- Early Warning Indicators are crucial tools in risk management that help in identifying

potential risks or threats at an early stage.

- Granularity in data refers to the level of detail or specificity in the data being analyzed.
 - Early warning indicators can be leading indicators, which provide signals about potential future events or trends.
 - Quantitative skills are important in analyzing and interpreting data, especially in the context of risk management.
 - Professional experience can provide valuable insights in understanding the context and implications of different indicators.
-

Q.3852 Early warning signs can be identified using a set of parameters and processes that identify probable risks at a nascent stage. When potential risk events are recognized at an earlier stage, an investment bank is able to prepare itself as the event develops. Which one of the following is **LEAST** likely an early warning sign for such events?

- A. Narrowing of debt or credit default swap spreads
- B. Increases in currency mismatches
- C. Drop-in credit lines
- D. Increasing redemptions of CDs before maturity

The correct answer is **A**.

The narrowing of debt or credit default swap spreads is not typically considered an early warning sign for potential risk events in investment banking. In fact, it is usually seen as an indication that the investment is in a good position. Credit default swaps (CDS) are financial derivative contracts that allow an investor to 'swap' or offset their credit risk with that of another investor. When the spread, or difference, between the credit default swap rates of two different investments narrows, it suggests that the perceived risk of the investment is decreasing. This is because a narrower spread indicates that the market believes there is a lower chance of default on the debt. Therefore, rather than being a warning sign, a narrowing of debt or credit default swap spreads is often viewed as a positive signal in the investment banking world.

Choice B is incorrect. Increases in currency mismatches can serve as an early warning sign for potential risk events. Currency mismatches occur when the currency of a bank's liabilities differs from that of its assets. This can lead to exchange rate risk, which could potentially harm the bank's financial position.

Choice C is incorrect. A drop in credit lines can also be an early warning sign for potential risk events. If a bank sees a significant decrease in its credit lines, it may indicate that borrowers are facing financial difficulties and are less likely to repay their loans, leading to increased credit risk for the bank.

Choice D is incorrect. Increasing redemptions of CDs before maturity could be an early warning sign as well. When depositors start withdrawing their deposits before maturity, it might indicate a lack of confidence in the bank's stability or profitability, which could lead to liquidity risks.

Things to Remember

- Early warning signs are crucial for identifying potential risks at an early stage in order to prepare and mitigate the impact of these events.
 - Monitoring parameters such as changes in credit lines, currency mismatches, and redemptions can help in detecting potential risk events.
 - Credit default swaps (CDS) are financial derivative contracts used to manage credit risk, and narrowing spreads indicate lower perceived risk.
 - Currency mismatches can lead to exchange rate risk, impacting a bank's financial position.
 - A drop in credit lines may signal financial difficulties for borrowers and increased credit risk for the bank.
 - Increasing redemptions of CDs before maturity can indicate depositor concerns about the bank's stability and profitability, leading to liquidity risks.
-

Q.3853 In which of the given circumstances is it most optimal to track early warning indicator metrics?

- A. After an industrial-wide decline in performance
- B. During a business as usual environment
- C. During periods of institution-specific stress
- D. After the occurrence of a loss-causing event

The correct answer is **B**.

Early warning indicators (EWIs) are designed to provide signals or act as a heads-up in the lead-up to a potential disaster. They are most effective when used in a 'business as usual' environment, where everything appears to be in order, or at least going as per expectations. In such an environment, any deterioration in EWI metrics can be easily detected, setting in motion a series of steps to avert the impending danger. For instance, if a bank's liquidity coverage ratio decreases below a specified threshold, the management can immediately direct the relevant personnel to liquidate long-term assets and correct the situation. Therefore, tracking EWIs during a 'business as usual' environment allows for proactive risk management and disaster prevention.

Choice A is incorrect. Monitoring early warning indicator metrics after an industry-wide decline in performance may be too late to prevent significant losses or disasters. These metrics are designed to provide advance notice of potential risks, so they should ideally be monitored before such a decline occurs.

Choice C is incorrect. While it can be beneficial to monitor these metrics during periods of institution-specific stress, this is not the most optimal time. The purpose of EWI metrics is to identify potential risks and prevent disasters before they occur, not during the occurrence of stressful events.

Choice D is incorrect. Monitoring EWI metrics after the occurrence of a loss-causing event would not be as beneficial as monitoring them beforehand. The goal of these indicators is to provide early warnings about potential risks that could lead to losses, so their utility would be diminished if they were only monitored after a loss has already occurred.

Things to Remember

- Early warning indicators (EWIs) are crucial tools in risk management to detect potential risks before they escalate into disasters.
- EWIs are typically monitored in a 'business as usual' environment to proactively manage risks.
- Monitoring EWIs after an industry-wide decline in performance may be too late to prevent significant losses.
- During periods of institution-specific stress, monitoring EWIs can be beneficial but not the most optimal time.
- Tracking EWIs after the occurrence of a loss-causing event is not as effective as monitoring them beforehand.

Q.3854 To accurately monitor liquidity risk, firms are advised to use a spotlight system in representing and communicating their performance against the thresholds of the EWIs. Which of

the following indicators should a firm be most concerned with?

- A. Red
- B. Blue
- C. Amber
- D. Green

The correct answer is **A**.

The color 'Red' in the spotlight system is used to indicate a significant concern that may warrant an immediate response. This is because a red indicator implies that the measure has crossed the threshold of the Early Warning Indicators (EWIs), indicating a high level of liquidity risk. This could mean that the firm's liquidity position is deteriorating rapidly, and immediate action is required to prevent potential financial distress or insolvency. Therefore, a firm should be most concerned with a red indicator as it signals a critical situation that needs urgent attention and action.

Choice B is incorrect. Blue is not typically used in a spotlight system for liquidity risk management. The standard colors used are red, amber, and green, each representing different levels of risk or performance against thresholds.

Choice C is incorrect. Amber indicates that the firm's performance against the Early Warning Indicators (EWIs) thresholds is at a moderate level of risk. While it does require attention and possibly some action to mitigate potential risks, it does not represent an immediate threat like the red indicator.

Choice D is incorrect. Green indicates that the firm's performance against the EWIs thresholds is within acceptable limits and there are no significant risks identified at this time. Therefore, it does not require immediate action as compared to a red indicator.

Things to Remember

- Early Warning Indicators (EWIs) are specific metrics or thresholds used by firms to monitor and assess their liquidity risk.
- A spotlight system is a visual representation tool that categorizes performance indicators into different colors (such as red, amber, and green) to quickly communicate the level of risk or performance against thresholds.
- Red in the spotlight system typically indicates a critical situation where the measure has crossed the threshold of EWIs, signaling a high level of liquidity risk that requires immediate attention.

- Amber in the spotlight system indicates a moderate level of risk against the EWIs thresholds, requiring attention and potential action to mitigate risks.
 - Green in the spotlight system indicates that the firm's performance is within acceptable limits and there are no significant liquidity risks identified at that time.
 - Monitoring liquidity risk is crucial for firms to ensure their financial stability and ability to meet obligations in a timely manner.
-

Q.3855 Swift Investments, a U.S. based firm, is a clearing member at the Chicago Mercantile Exchange. The firm has put together a comprehensive liquidity risk management framework that involves the use of a set of early warning indicators to help the firm detect liquidity distress at a nascent stage. To prudently manage its liquidity risk, which of the following reporting schedules is most appropriate for the firm's EWI dashboard?

- A. Daily
- B. Weekly
- C. Intraday
- D. Monthly

The correct answer is **C**.

Intraday reporting is the most appropriate for Swift Investments, given its role as a clearing member at the Chicago Mercantile Exchange. Clearing members are exposed to significant market risk and need to be able to respond quickly to changes in market conditions. Intraday reporting allows the firm to monitor its liquidity risk on a real-time basis and take immediate action if necessary. This is particularly important for firms with a substantial trading focus, as they are more exposed to external market conditions. Intraday reporting provides the most timely information, allowing managers to adjust their strategies and actions in response to potential crises.

Choice A is incorrect. Daily reporting, while more frequent than weekly or monthly, may not be sufficient for a firm like Swift Investments that operates in the fast-paced environment of the Chicago Mercantile Exchange. Liquidity risks can arise and escalate within a single trading day, making daily reports potentially outdated by the time they are reviewed.

Choice B is incorrect. Weekly reporting would not provide Swift Investments with timely information about liquidity risks. Given the firm's exposure to market conditions and its role in clearing transactions, it needs to monitor its liquidity risk indicators more frequently to detect signs of distress at an early stage.

Choice D is incorrect. Monthly reporting would be even less suitable for Swift Investments than daily or weekly reporting schedules due to the delay in receiving critical information about potential liquidity risks. In a volatile market environment, such delays could lead to significant financial losses.

Things to Remember

- **Liquidity Risk Management:** The process of identifying, assessing, and managing liquidity risk to ensure a firm can meet its financial obligations as they come due.
 - **Early Warning Indicators (EWI):** Key metrics or signals that help firms detect potential liquidity distress at an early stage.
 - **Real-time Monitoring:** Continuous monitoring of liquidity risk indicators to enable quick decision-making and response to changing market conditions.
 - **Market Risk:** The risk of losses in positions arising from movements in market prices.
 - **Clearing Member:** A financial institution that acts as an intermediary in the clearing and settlement of trades in financial markets.
-

Q.5376 When evaluating ABC Bank's liquidity profile, what particular trends should John Doe, a risk consultant, prioritize as the most prominent warning sign indicating potential liquidity risk at the bank?

- A. The spreads on the bank's issued debt and credit default swap have widened.
- B. There has been substantial growth in assets, which has been financed by an expansion in stable liabilities.
- C. While the stock price of the bank's peers has declined, there has been no corresponding decrease in the stock price of the bank itself.
- D. The bank's hedge ratio has fallen below 1, indicating difficulties in meeting short-term obligations.

The correct answer is **A**.

The widening spreads reflect a diminished market confidence in the bank, leading to an increase in the cost of funding. Widening spreads indicate that investors are demanding higher yields or compensation for the perceived increase in risk associated with lending to the bank. This can potentially strain the bank's liquidity position as it becomes more expensive for the bank to borrow or raise funds.

B is incorrect. A rapid expansion of assets funded by unstable liabilities as opposed to stable liabilities would pose greater concerns.

C is incorrect. A bank-specific early warning indicator (EWI) that would be more relevant is a decline in the bank's stock price compared to that of its peers as opposed to no decline in the bank's stock price.

D is incorrect. In the context of banking and financial institutions, the hedge ratio is not the most appropriate indicator of liquidity risk.

Things to Remember

- Liquidity risk is the risk that a company or bank may be unable to meet its short-term financial obligations.
- Key liquidity ratios include the current ratio, quick ratio, and cash ratio.
- Market confidence plays a crucial role in determining a bank's liquidity position.
- Asset-liability management is essential for managing liquidity risk effectively.
- Monitoring trends in funding costs and spreads can provide valuable insights into a bank's liquidity risk.

Q.5380 ABC Bank employs a collection of early warning indicators to detect emerging risks and possible vulnerabilities in its liquidity position. Which of the following serves as an early warning indicator for a potential liquidity issue?

- A. A rapid rise in the leverage ratio and heavy reliance on short-term repo funding.
- B. A reduction in the collateral haircuts imposed on the bank's collateralized exposures.
- C. An upward revision in the credit rating.
- D. Increased diversification of assets

The correct answer is **A**.

A rapid expansion of leveraged assets accompanied by significant utilization of short-term repos serves as an early warning sign for a potential liquidity problem. The leverage ratio measures the level of debt relative to the bank's capital and indicates the potential strain on its liquidity position. Additionally, relying on short-term repo funding means the bank has a dependence on short-term borrowing to finance its activities, which can pose liquidity risks if the availability of funding becomes limited or costly.

B, C and D are incorrect. Decreased collateral haircuts, a credit rating upgrade, and increased asset diversification are generally positive developments and do not typically indicate an early warning of a potential liquidity problem.

Things to Remember

- Leverage ratio: Measures the level of debt relative to the bank's capital and indicates potential strain on liquidity position.
 - Short-term repo funding: Indicates a dependence on short-term borrowing which can pose liquidity risks.
 - Collateral haircuts: Reduction in collateral haircuts may not indicate liquidity issues but can affect risk management.
 - Credit rating: Upward revision in credit rating is a positive development but may not serve as an early warning indicator for liquidity issues.
 - Asset diversification: While important for risk management, increased diversification of assets may not signal liquidity problems.
-

Q.5422 Jane Doe is considering investing in Company X. She has optimistic expectations of the company despite the current poor global economic environment. Which of the following biases would most likely be present in Jane Doe's valuation of Company X and consequent investment decision?

- A. Forward-looking bias
- B. Survivorship bias
- C. Illiquidity bias
- D. Sample selection bias

The correct answer is **A**.

A forward-looking bias occurs when the analysis or decision-making process is influenced by information or expectations about future events. It can lead to overemphasis on future projections and neglect of historical data or current conditions.

B is incorrect. Survivorship bias refers to the error that arises when only successful or surviving entities are considered in a study or analysis, while ignoring those that failed or did not survive. It can lead to an overestimation of performance or outcomes since the unsuccessful entities are excluded from the analysis.

C is incorrect. Illiquidity bias refers to the distortion or bias that arises when analyzing or valuing illiquid assets or securities. Illiquid assets are more difficult to sell or trade, and their pricing may be less transparent or subject to higher transaction costs. This bias can lead to inaccuracies or underestimation of the true value or risk associated with illiquid assets.

D is incorrect. Sample selection bias occurs when the sample or data used for analysis is not representative of the entire population or is biased towards certain characteristics or outcomes. It can lead to erroneous conclusions or generalizations.

Things to Remember

- Confirmation bias: This bias occurs when individuals seek out information that confirms their preexisting beliefs or expectations, while ignoring or discounting contradictory information.
- Overconfidence bias: This bias refers to the tendency for individuals to overestimate their own abilities, knowledge, or the accuracy of their predictions.
- Anchoring bias: Anchoring bias occurs when individuals rely too heavily on initial information or the first piece of information they receive when making decisions or judgments.

- Availability bias: Availability bias is the tendency to rely on information that is readily available or easily recalled when making decisions, even if it may not be the most accurate or relevant information.
 - Regret aversion bias: This bias occurs when individuals make decisions to avoid potential regret or disappointment, rather than based on rational analysis of risks and rewards.
-

Reading 130: The Investment Function in Financial Services Management

Q.3858 To safeguard public funds, depository institutions in the United States cannot accept deposits from federal, state and local governments unless they post collateral acceptable to these government units. This is most likely known as:

- A. Statutory requirements
- B. Pledging requirement
- C. Investment requirements
- D. Operational requirements

The correct answer is **B**.

Pledging requirements refer to the legal obligations that require securities to be pledged as collateral for specific deposits. In the context of depository institutions in the United States, these requirements are particularly relevant when dealing with deposits from federal, state, and local governments. These institutions are not allowed to accept such deposits unless they post collateral that is deemed acceptable by these governmental units. This is done to safeguard public funds and ensure that the government's interests are protected in the event of a default. The nature and extent of these pledging requirements can vary widely from state to state. However, most states allow a combination of federal and municipal securities to meet these requirements. Therefore, the term 'pledging requirement' accurately encapsulates this regulatory obligation imposed on depository institutions.

Choice A is incorrect. Statutory requirements refer to the laws that a company or an institution must follow. These are broader in scope and not specifically related to the requirement of providing security in the form of pledged collateral for accepting deposits from government units.

Choice C is incorrect. Investment requirements pertain to rules and regulations regarding how institutions can invest their funds. This does not directly relate to the provision of security for accepting government deposits.

Choice D is incorrect. Operational requirements refer to the day-to-day functioning and management practices that an institution must adhere to, which does not necessarily include providing collateral as security for government deposits.

Things to Remember

- **Depository institutions:** These are financial institutions that accept deposits from individuals and entities and provide various financial services.
- **Collateral:** Assets that a borrower pledges to a lender to secure a loan or other credit,

and which can be seized by the lender in case of default.

- Government deposits: Funds deposited by federal, state, or local government entities in depository institutions for various purposes.
 - Regulatory compliance: The adherence to laws, rules, and regulations set forth by regulatory authorities to ensure the stability and integrity of the financial system.
 - Municipal securities: Debt securities issued by state and local governments or their agencies to finance public projects and operations.
-

Q.3859 Assume that an investor currently holds a bond whose par value is \$2,000. The bond is currently priced at \$1,600, matures in 5 years, and pays an annual coupon of 8%. Calculate the yield to maturity of the bond.

- A. 13.80%
- B. 16.38%
- C. 16.59%
- D. 17.34%

The correct answer is **A**.

The coupon amount is $0.08 \times 2,000 = \$160$

$$1,600 = \frac{160}{(1 + \text{YTM})^1} + \frac{160}{(1 + \text{YTM})^2} + \frac{160}{(1 + \text{YTM})^3} + \frac{160}{(1 + \text{YTM})^4} + \frac{2,160}{(1 + \text{YTM})^5}$$
$$\text{YTM} = 13.80\%$$

Q.3860 Any securities which reach maturity within one year is most likely regarded as:

- A. Money market securities
- B. Capital market securities
- C. Federal agency securities

D. Certificates of deposits

The correct answer is **A**.

Money market securities are short-term debt instruments that are highly liquid and have a maturity period of one year or less. They are considered low-risk investments and are issued by governments, financial institutions, and corporations. Examples of money market securities include Treasury bills, commercial paper, and certificates of deposit. These securities are typically used by investors for parking their funds temporarily until they find a more lucrative investment opportunity. The short maturity period of these securities allows investors to get their principal amount back within a short span, thereby reducing the risk of price fluctuations due to market volatility.

Choice B is incorrect. Capital market securities typically have a maturity period that exceeds one year. They include long-term instruments such as stocks and bonds, which are used by businesses to raise capital for their long-term needs.

Choice C is incorrect. Federal agency securities can have varying maturity periods, ranging from less than a year to several years. Therefore, they cannot be classified as securities that mature within one year.

Choice D is incorrect. Certificates of deposits (CDs) also have varying maturities and can range from a few months to several years depending on the terms set by the issuing bank. Hence, it's not accurate to categorize all CDs as maturing within one year.

Things to Remember

- Money market securities are highly liquid and have a maturity period of one year or less
- Examples of money market securities include Treasury bills, commercial paper, and certificates of deposit
- Capital market securities typically have a maturity period that exceeds one year
- Federal agency securities can have varying maturity periods, ranging from less than a year to several years
- Certificates of deposits (CDs) have varying maturities and can range from a few months to several years

Q.3863 Suppose that Aaa-rated corporate bonds have an average gross yield to maturity of 8%, the prime rate on top-quality corporate loans is 6%, and Aaa-rated municipal bonds have a 5% gross yield to maturity. Calculate the tax-equivalent yield (TEY) for the Aaa-rated municipal bonds for a taxed financial firm in the top 35% federal income tax bracket.

- A. 5.00%
- B. 6.52%
- C. 7.69%
- D. 14.29%

The correct answer is **C**.

$$\text{TEY} = \frac{\text{After-tax return on a tax-exempt investment}}{(1 - \text{Investing firm's marginal tax rate})}$$

Recall that municipal bonds are tax-exempt; thus, the expected after-tax gross returns for Aaa-rated municipal bonds will be:

$$\begin{aligned} 5\% \times (1 - 0) &= 5\% \\ \text{TEY} &= \frac{5\%}{(1 - 0.35)} = 7.69\% \end{aligned}$$

This measure means that the Aaa-rated corporate bond and the prime-rated loans would have to have a before-tax yield of 7.69% to match the Aaa-municipal bond's after-tax yield of 5.0%.

Q.3864 Which of the following statements is **not** true?

- A. Treasury bills are the long term debt obligations issued by the federal government
- B. Inflation risk is the likelihood that rising prices for goods and services will erode the purchasing power of interest income and repaid principal from security or loan
- C. Bankers' acceptances are considered to be among the safest of all money market instruments
- D. Investment securities are expected to help stabilize financial institutions' income

The correct answer is **A**.

U.S. Treasury bill is a debt obligation of the United States government that should mature in one year following the date of issue (short-term). T-bills are issued in weekly or monthly auctions, and their high degree of security makes them very attractive.

B is true: Rising prices of goods and services distort the purchasing power of interest income and repaid principal from security or loan. This is known as inflation risk.

C is true: Because they represent a bank's promise to pay the holder a designated amount of money on a designated future date, bankers' acceptances are considered to be among the safest of all money market instruments.

D is true: Investment security portfolios help to stabilize income, so that revenues level out over the business cycle—when loan revenues fall, income from investment securities may rise.

Things to Remember

- U.S. Treasury bills are considered one of the safest investments because they are backed by the full faith and credit of the U.S. government.
- Inflation risk is a key consideration for investors as it can erode the real value of their investments over time.
- Bankers' acceptances are commonly used in international trade transactions to provide a guarantee of payment to the seller.
- Investment securities can include a variety of assets such as stocks, bonds, and mutual funds, and are held by financial institutions to generate income and manage risk.

Q.3865 The risk that the bank must sell part of its investment portfolio before its maturity for a capital loss is known as:

- A. Credit risk
- B. Business risk
- C. Default risk
- D. Liquidity risk

The correct answer is **D**.

Liquidity risk refers to the risk that a bank or other financial institution may encounter when it is forced to sell part of its investment portfolio before the investments have reached their maturity. This premature sale could potentially result in a capital loss for the bank. Liquidity risk is a significant concern for financial institutions as they must always be prepared for the possibility of needing to liquidate assets quickly to meet their obligations or to respond to an unexpected event. This risk is inherent in investments that are not easily sold or converted into cash without a substantial loss in value. These include assets like real estate, certain types of securities, and other illiquid investments.

Choice A is incorrect. Credit risk refers to the potential loss that a bank may face when a borrower fails to meet their contractual obligations. It does not refer to the risk associated with selling investments prematurely.

Choice B is incorrect. Business risk pertains to the uncertainty in profits or danger of loss and the events that could pose a risk due to some unforeseen events in future, which causes business to fail. It does not specifically relate to premature sale of investments.

Choice C is incorrect. Default risk, similar to credit risk, refers to the possibility that a borrower will be unable to make required debt payments and thus default on their obligations. This type of risk does not encompass situations where banks are forced into premature sale of their investments.

Things to Remember

- Liquidity risk is a key consideration for banks and financial institutions as it can impact their ability to meet short-term obligations.
- Asset-liability management is crucial in managing liquidity risk effectively.
- Regulatory requirements often mandate that banks maintain a certain level of liquidity to mitigate liquidity risk.
- Market liquidity risk is another aspect to consider, which refers to the risk of not being able to execute large transactions without impacting the market price.
- Contingency funding plans are put in place by banks to address potential liquidity

shortfalls in times of stress.

Q.3866 Money market instruments which represent a bank's commitment to pay a specified amount of money on a specific future date under specific conditions, are known as:

- A. Eurocurrency deposits
- B. Government agency securities
- C. Bank qualified bonds
- D. Bankers' acceptances

The correct answer is **D**.

Bankers' acceptances are regarded as one of the most secure instruments in the money market because they represent a bank's commitment to pay a specified amount on a future date. These instruments commonly originate from financial institutions offering credit guarantees to customers engaged in activities such as exporting, importing, storing goods, or buying currency. Legally, the financial institution issuing the credit guarantee assumes the primary responsibility of repaying the customer's debt in exchange for a fee. By providing their name and creditworthiness, the issuing institution enables their customer to access credit from other sources at a reduced cost.

A is incorrect: International Eurocurrency deposits are typically short-term, fixed maturity deposits issued in million-dollar units by the world's largest banks headquartered in financial centers around the globe.

B is incorrect: Government agency securities are marketable notes and bonds sold by agencies owned by the government or sponsored by the government.

C is incorrect: Bank-qualified bonds are those issued by smaller local governments, i.e., governments issuing no more than \$10 million of public securities per year. Banks buying bank-qualified bonds are allowed to deduct 80% of any interest paid to fund these purchases. This tax advantage is not available for nonbank-qualified bonds.

Things to Remember

- Bankers' acceptances are commonly used in international trade transactions to facilitate the purchase of goods and services.
- Bankers' acceptances are typically used by importers and exporters to reduce the risk of non-payment in cross-border transactions.
- Bankers' acceptances are often traded in the secondary market, providing liquidity to

investors.

- Bankers' acceptances are considered to be low-risk instruments due to the involvement of a bank's credit guarantee.
-

Q.3867 The most aggressive investment maturity strategy that requires the bank to regularly shift the maturities of its securities in responses to fluctuations in interest rates called the:

- A. Front-End Load Maturity Policy
- B. Back-End Load Maturity Policy
- C. Barbell Strategy
- D. Rate Expectation Strategy

The correct answer is **D**.

The Rate Expectation Strategy is the most aggressive of all maturity strategies and is predominantly used by major financial institutions. This strategy involves the continuous adjustment of securities' maturities in accordance with current economic and interest rate forecasts. When a rise in interest rates is anticipated, investments are shifted towards the short end of the maturity spectrum. Conversely, when a fall in interest rates is expected, investments are moved towards the long end of the spectrum. This strategy requires a deep understanding of economic trends and interest rate movements, making it a complex but potentially rewarding approach for institutions with the necessary expertise and resources.

Choice A is incorrect. The Front-End Load Maturity Policy refers to a strategy where the majority of the investment's costs are paid at the beginning of the investment period. This strategy does not involve adjusting maturities in response to changes in interest rates.

Choice B is incorrect. The Back-End Load Maturity Policy, on the other hand, involves paying most of an investment's costs at its maturity or end. Similar to Choice A, this policy does not require continual adjustments of securities' maturities based on interest rate changes.

Choice C is incorrect. The Barbell Strategy involves investing in short-term and long-term bonds but avoiding intermediate-term bonds. While this strategy may involve some adjustment based on interest rates, it does not necessitate continuous adjustment as described in the question.

Things to Remember

- Interest Rate Risk: The risk that an investment's value will change due to changes in interest rates.

- Duration: A measure of a security's price sensitivity to changes in interest rates.
 - Yield Curve: A graphical representation of the relationship between the yield on bonds of the same credit quality but different maturities.
 - Convexity: A measure of the curvature of the price-yield relationship of a bond.
 - Immunization: A strategy used to protect a portfolio against interest rate movements.
-

Q.3868 Loans will be terminated or paid off ahead of schedule. This is a particular problem with residential home mortgages and other consumer loans that are pooled and used as collateral in securitized assets. This risk is known as:

- A. Liquidity risk
- B. Prepayment risk
- C. Call risk
- D. Business risk

The correct answer is **B**.

Prepayment risk refers to the risk associated with the early repayment of a loan by the borrower. This risk is particularly relevant to asset-backed securities, as the actual interest and principal payments received from a pool of securitized loans may differ from the originally expected cash flows. A common example of this can be seen in mortgage-backed securities, where the underlying asset is a pool of home loans. The prepayment risk arises when borrowers pay off their mortgages ahead of schedule, thereby altering the expected cash flows from these securities.

Choice A is incorrect. Liquidity risk refers to the risk that a firm may not be able to meet its short-term financial demands due to an inability to convert assets into cash without incurring a loss. It does not refer to the risk of loans being paid off ahead of their scheduled time.

Choice C is incorrect. Call risk pertains to the possibility that a bond issuer might retire a bond before its maturity date, something that is usually done when interest rates decline. This does not directly relate to the early termination or payoff of loans as described in the question.

Choice D is incorrect. Business risk involves risks associated with operating a business such as changes in market conditions, competition, and regulations among others. It does not specifically deal with risks associated with loan prepayments.

Things to Remember

- Prepayment risk is also known as refinancing risk, as borrowers may choose to refinance their loans at lower interest rates, leading to early repayment.
 - Prepayment risk can have a significant impact on the valuation and performance of mortgage-backed securities and other asset-backed securities.
 - Factors such as interest rate movements, economic conditions, and borrower behavior can influence the level of prepayment risk in a loan pool.
 - Prepayment risk is a key consideration for investors in securitized assets, as it can affect the expected cash flows and returns on these investments.
-

Q.3871 The practice of protecting securities purchased from loss of return, no matter which way interest rates go is regarded as:

- A. Portfolio shifting
- B. Duration
- C. Portfolio immunization
- D. Securitization

The correct answer is **C**.

Portfolio immunization is a strategy that is designed to protect the returns on purchased securities, regardless of the direction in which interest rates move. This strategy is primarily concerned with balancing the risks associated with price and reinvestment in an investment portfolio. The goal of portfolio immunization is to ensure that the portfolio is not adversely affected by changes in interest rates, thereby safeguarding the returns on the securities purchased. This is achieved by carefully selecting securities that have offsetting price and reinvestment risks. In other words, if the price of a security falls due to an increase in interest rates, the loss is offset by the higher reinvestment return, and vice versa. This ensures that the total return on the portfolio remains stable, irrespective of changes in interest rates.

Choice A is incorrect. Portfolio shifting involves changing the composition of a portfolio in response to changes in market conditions or investment goals. It does not necessarily aim to safeguard returns on purchased securities against interest rate fluctuations.

Choice B is incorrect. Duration is a measure of the sensitivity of the price of a bond or other debt instrument to a change in interest rates. While it can be used as part of an immunization strategy, duration itself does not describe a strategy that aims to safeguard returns on purchased securities irrespective of interest rate fluctuations.

Choice D is incorrect. Securitization refers to the process by which financial assets are pooled and sold as securities, it doesn't involve any strategy for mitigating risk from fluctuating interest rates.

Things to Remember

- Duration is a key concept in fixed income investing as it measures the sensitivity of a bond's price to changes in interest rates.
 - Convexity is another important concept in fixed income investing, as it measures the curvature of the price-yield relationship of a bond.
 - Interest rate risk is the risk that the value of a security will decline due to changes in interest rates.
 - Reinvestment risk is the risk that future cash flows from an investment will have to be reinvested at a lower rate than the original investment.
 - Yield curve risk is the risk of experiencing losses due to changes in the shape or slope of the yield curve.
-

Q.5407 Joshua is a portfolio manager at AlphaOmega Investments. He's considering diversifying the firm's short-term investment portfolio by including bankers' acceptances. He recognizes that bankers' acceptances are short-term instruments primarily used in trade credit transactions, where a bank guarantees the payment of a customer for an export/import transaction. Joshua also acknowledges that they offer higher yields than T-bills but lower than Eurocurrency deposits, and they can be discounted at the Federal Reserve Bank. Despite these advantages, Joshua is concerned about certain disadvantages associated with bankers' acceptances. Based on his understanding, which of the following concerns about bankers' acceptances is the most valid?

- A. The risk that the bank will default on its guarantee of the acceptance.
- B. The availability of bankers' acceptances for certain maturities.
- C. The possibility that bankers' acceptances could be subject to negative interest rates.
- D. The inability to trade bankers' acceptances in the secondary market

The correct answer is **B**.

Joshua's valid concern about bankers' acceptances is the lack of availability for certain

maturities. The demand and supply of bankers' acceptances may not always match for certain maturity periods, which can limit the investment opportunities for a portfolio manager.

A is incorrect. While a bank default is theoretically possible, it is highly unlikely. Banks are carefully regulated and monitored, and they are typically large, well-established institutions. The risk of a bank defaulting on its guarantee of a banker's acceptance is generally considered to be quite low.

C is incorrect. Negative interest rates are a phenomenon that typically occur in broader economic environments and are not specific to individual instruments such as bankers' acceptances. Moreover, bankers' acceptances are issued at a discount and mature at par value, which precludes the possibility of negative interest rates on these instruments.

D is incorrect. Bankers' acceptances actually have an active market for secondary trading, which means they can be sold before maturity. This adds to their attractiveness by providing liquidity for the investors.

Things to Remember

- Bankers' acceptances, being a promise from a bank to pay the holder a certain amount on a future date, are among the safest money market instruments.
- They often originate from a financial institution's decision to guarantee the credit of a customer involved in the export, import, or storage of goods or currency purchasing. The financial firm becomes the primary obligor in the transaction, committing to pay the customer's debt in return for a fee.
- Bankers' acceptances are discount instruments, sold below par and mature at par. The return for an investor comes from the acceptance rising in price as it nears its maturity.
- Their rates of return typically align closely with yields on Eurocurrency deposits and Treasury bills.
- Eligible bankers' acceptances, usually denominated in U.S. dollars, with a maturity of six months or less, and related to the export or import of goods or storage of marketable commodities, can be discounted at the Federal Reserve Banks.
- Despite the issuer's obligation to pay at maturity, the current holder of a banker's acceptance can sell it in the secondary market, thus increasing their reserves and transferring interest rate risk to another investor.

Q.5423 ABC Bank currently has been experiencing high loan revenues due to a recent surge in the demand for loans. The bank intends to sell lower-yielding securities at a loss in order to reduce its current taxable income. Which of the following tools is the bank most likely utilizing?

- A. Tax exposure
- B. Tax swap
- C. Tax shelter
- D. Tax exemption

The correct answer is **B**.

A tax swap refers to a financial strategy where a bank sells a lower-yielding security at a loss in order to reduce its current taxable income and simultaneously purchases a higher-yielding security to maintain their desired investment exposure. The purpose of a tax swap is to generate a tax benefit by offsetting the realized loss against capital gains or taxable income.

A is incorrect. Tax exposure refers to the financial risk and liability that an individual or entity faces as a result of their tax obligations. It represents the potential negative impact that taxes can have on one's financial position and profitability.

C is incorrect. A tax shelter is a legal strategy or investment vehicle that allows individuals or businesses to minimize their tax liability by reducing their taxable income or claiming deductions, credits, or exemptions provided by tax laws.

D is incorrect. Tax exemption refers to a legal provision that relieves certain individuals, organizations, or activities from the obligation to pay taxes on specific types of income or transactions. It grants an exemption or exclusion from taxation, effectively reducing or eliminating the tax liability for the eligible entity or activity.

Things to Remember

- **Tax Management:** Banks often use tax management strategies to optimize their tax liabilities and maximize their after-tax returns.
- **Capital Gains:** Capital gains are profits that result from the sale of assets such as stocks, bonds, or real estate. They are typically subject to capital gains tax.
- **Tax Loss Harvesting:** Tax loss harvesting involves selling investments that have experienced a loss to offset gains and reduce taxable income.
- **Investment Exposure:** Maintaining desired investment exposure is crucial for banks to achieve their financial objectives while managing tax implications.

Q.5424 Chris Colombus is a treasury manager at ABC Bank. Chris is training new interns on the various investment securities that the bank regularly purchases. Which of the following is a disadvantage of bankers' acceptances that Chris would highlight in his presentation?

- A. Limited resale market.
- B. Issued in odd denominations.
- C. Low yields relative to other financial instruments.
- D. More price risk than T-bills.

The correct answer is **B**.

Bankers' acceptances are financial instruments used in international trade transactions. They are essentially a guarantee from a bank to pay a specified amount at a future date. When a company needs to make an international payment, it may use a banker's acceptance to ensure the recipient receives the funds. A disadvantage of banker's acceptances is that they are issued in odd denominations. By "odd denominations," it means that bankers' acceptances are not issued in standard or commonly used denominations. This makes it challenging for investors or buyers to trade or resell these securities. Other disadvantages include the income being taxable and their limited availability at specific maturities.

A is incorrect. A limited resale market is a disadvantage of short-term municipal obligations.

C is incorrect. Bankers' acceptances are not characterized by "low yields relative to other financial instruments."

D is incorrect. Bankers' acceptances typically have less price risk compared to Treasury bills (T-bills).

Things to Remember

- Bankers' acceptances are commonly used in international trade transactions as a form of payment guarantee.
- Bankers' acceptances are typically issued by a bank on behalf of a customer for a specific amount and maturity date.
- Bankers' acceptances are considered to be low-risk investments due to the backing of a bank.
- Bankers' acceptances are often used to facilitate import/export transactions and can be discounted or sold in the secondary market.
- Investors should be aware of the credit risk associated with the bank issuing the

acceptance when considering investing in bankers' acceptances.

Q.5425 Peter Jones is the Chief Finance Officer at ABC Bank. He wants to reduce the Bank's overall portfolio risk by adding assets with different risk profiles and correlations to the Bank's current pool of investment securities. Which of the following securities are considered to have an implicit guarantee of the government?

- A. Commercial paper.
- B. Federal agency securities issued by government-sponsored enterprises
- C. International eurocurrency deposits.
- D. Short-term treasury notes.

The correct answer is **B**.

Federal agency securities issued by government-sponsored enterprises are debt securities issued by various government-sponsored entities (GSEs) or federal agencies in the United States. These entities are created by the U.S. government to fulfill specific purposes and support various sectors of the economy. While federal agency securities are not direct obligations of the U.S. government, they are considered to have an implicit guarantee or backing by the government.

A is incorrect. Commercial paper refers to short-term debt instruments issued by corporations and financial institutions to meet their short-term funding needs. It's important to note that while commercial paper is generally considered a relatively safe investment, it is not entirely risk-free.

C is incorrect. Eurocurrency deposits are large fixed-maturity time deposits issued in million-dollar denominations by major global banks located in financial centers worldwide, with London being a prominent hub.

D is incorrect. Short-term treasury notes are debt securities issued by the U.S. Department of the Treasury with maturities typically ranging from a few days to one year. They are considered low-risk investments as they are backed by the full faith and credit of the U.S. government.

Things to Remember

- Government-sponsored enterprises (GSEs) are privately owned corporations with public purposes created by the United States Congress to reduce the cost of capital for certain borrowing sectors of the economy.
- Implicit guarantee refers to the perception that certain securities or entities have the backing of the government, even though they are not direct obligations of the government.

- Portfolio risk reduction strategies often involve diversification by adding assets with different risk profiles and correlations to achieve a more balanced and less volatile overall portfolio.
 - Investors often seek securities with implicit government guarantees for added security and lower risk in their investment portfolios.
-

Reading 131: Liquidity and Reserves Management: Strategies and Policies

Q.3885 ABC Bank Limited predicts its cash inflows and outflows over the next one day to be as in the following table:

Predicted Cash Inflows and Outflows for ABC Bank Ltd.			
Deposit withdrawals	\$129	Sales of bank assets	61
Deposit inflows	\$119	Stockholder dividend payments	172
Scheduled loan repayments	\$122	Revenues from sale of non-deposit services	118
Acceptable loan requests	\$90	Repayments of bank borrowings	84
Borrowings from the money market	\$110	Operating expenses	70

Calculate the bank's projected net liquidity position within the next one day.

- A. -\$15
- B. -\$23
- C. \$15
- D. \$23

The correct answer is **A**.

Cash Inflows	Amount	Cash Outflow	Amount
Deposit inflows	\$119	Deposit withdrawals	\$129
Borrowings from the money market	\$110	Acceptable loan requests	\$90
Sales of bank assets	61	Stockholder dividend payments	172
Revenues from sale of non-deposit services	118	Repayments of bank borrowings	84
Scheduled loan repayments	\$122	Operating expenses	70
Total Cash Inflows	\$530	Total Cash Outflow	\$545

A financial firm's net liquidity position = Supplies of liquidity flowing into the financial firm - Demands on the financial firm for liquidity

$$\begin{aligned}
 &= \text{Total cash inflow} - \text{Total cash outflow} \\
 &= \$530 - \$545 = -\$15
 \end{aligned}$$

Therefore, ABC Bank Ltd. has a projected net liquidity deficit of \$15

Q.3886 Prime savings has a projected net liquidity surplus of \$31 million in the coming week. Given the following table showing Prime Savings' supply and demand for liquidity, calculate the total expected stockholder dividend payments for the coming week.

Supplies of Liquidity Flowing into ABC Bank Ltd. Savings

Deposit inflows	\$31
Revenues from nondeposit service sales	\$23
Scheduled repayments of previous	\$5
made customer loans	\$28
Asset sales	\$15
Money market borrowings	\$20
Total supply of liquidity	122

The demand for liquidity flowing from ABC Bank Ltd. savings

Expected quality loan demand	\$31
Necessary repayments of previous borrowings	\$26
Deposit withdrawals	\$22
Disbursements to cover operating expenses	\$10
Total demand for liquidity excluding stockholder dividend payments	\$89

- A. \$ 3 million
- B. \$2 million
- C. \$7 million
- D. \$11 million

The correct answer is **B**.

$$\begin{aligned}
 \text{Net Liquidity Surplus} &= \text{Total liquidity supply} - \text{Total liquidity demand} \\
 &\quad - \text{Stockholder dividend payments} \\
 \$31 \text{ million} &= \$122 \text{ million} - \$89 \text{ million} \\
 &\quad - \text{Stockholder dividend payments}
 \end{aligned}$$

Therefore, expected stockholder dividend payments must equal \$2 million for the coming week.

Q.3888 XYZ Bank has the following forecasts for its checkable deposits, time and savings deposits, commercial loans, and consumer loans over the next eight months as follows.

XYZ Bank Forecasts for Deposits and Loans Over the Next Eight Months in US\$

Month	Checkable Deposits	Time and Savings Deposits	Commercial Loans	Consumer Loans
January	219	639	729	209
February	216	591	731	267
March	203	593	783	265
April	189	580	785	213
May	210	562	797	211
June	187	589	789	234
July	189	626	791	232
August	209	618	768	230

Employ the sources and uses of funds approach to determine the month, which is likely to have a liquidity surplus.

- A. February
- B. March
- C. June
- D. August

The correct answer is **D**.

Total Deposits = Checkable Deposits + Time & Saving Deposits

Total Loans = Commercial Loans + Consumer Loans

Change in Previous Total Month's Deposits = Current month's total deposits - Previous months total deposits

Change in Previous Total Month's Loans = Current month's total loans - Previous month's total loans

Estimated Liquidity Deficit or Surplus = Change from previous month's total deposits - Change from previous month's total loans

Month	Total Deposits	Change from Previous Month's T.Deposits	Total Loans	Change from Previous Month's T.Loans	Estimated Liquidity Defecit or Surplus
January	858	0	938	0	0
February	807	−51	998	60	−111
March	796	−11	1048	50	−61
April	769	−27	998	−50	23
May	772	3	1008	10	−7
June	776	4	1023	15	−11
July	815	39	1023	0	39
August	827	12	998	−25	37

The bank has projected surpluses in April, July, and August. The surpluses should be invested profitably.

Q.3889 XYZ Bank has the following forecasts for its checkable deposits, time and savings deposits, commercial loans, and consumer loans over the next eight months as follows.

XYZ Bank Forecasts for Deposits and Loans Over the Next Eight Months in US\$

Month	Checkable Deposits	Time and Savings Deposits	Commercial Loans	Consumer Loans
January	219	639	729	209
February	216	591	731	267
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April	189	580	785	213
May	210	562	797	211
June	187	589	789	234
July	189	626	791	232
August	209	618	768	230

Using the sources and uses of funds approach, which of the following months are likely to result in a liquidity deficit?

- A. February
- B. April
- C. July
- D. August

The correct answer is **A**.

XYZ Bank has liquidity deficits in half of the months, i.e., February, March, May, and June as in the following table:

Month	Total Deposits	Change from Previous month	Total Liabilities	Change from Previous month	Estimated Liquidity Deficit or Surplus
January	858	0	938	0	0
February	807	-51	998	60	-111
March	796	-11	1048	50	-61
April	769	-27	998	-50	23
May	772	3	1008	10	-7
June	776	4	1023	15	-11
July	815	39	1023	0	39
August	827	12	998	-25	37

Q.3890 XYZ can use the following methods to meet its liquidity deficit. Which of the options does not apply?

- A. Aggressive advertising to attract NOW deposits
- B. Selling some of their securities
- C. Aggressively issuing new loans
- D. Selling securities under agreements to repurchase

The correct answer is **C**.

Aggressively issuing new loans is not a suitable strategy for addressing a liquidity deficit. In fact, it can exacerbate the problem. When a financial institution issues loans, it essentially provides funds to borrowers. This reduces the institution's available liquid assets. If the institution is already facing a liquidity deficit, issuing new loans can further deplete its liquid assets, making the deficit worse. Therefore, aggressively issuing new loans is not a viable strategy for addressing a liquidity deficit.

Choice A is incorrect. Aggressive advertising to attract NOW (Negotiable Order of Withdrawal) deposits can be a suitable strategy for addressing a liquidity deficit. NOW deposits are demand deposit accounts that allow the holder to write checks against the money deposited. This can increase the liquidity of XYZ by attracting more funds.

Choice B is incorrect. Selling some of their securities is also a viable strategy for addressing a liquidity deficit. By selling securities, XYZ can convert these assets into cash, thereby increasing its liquid resources.

Choice D is incorrect. Selling securities under agreements to repurchase (also known as repo transactions) can help address a liquidity deficit in the short term. In such transactions, XYZ sells securities and agrees to repurchase them at a later date at an agreed price. This provides immediate cash inflow and hence increases liquidity.

Things to Remember

- Liquidity deficit refers to a situation where a company does not have enough liquid assets to meet its short-term obligations.
- Aggressive advertising to attract NOW deposits can help increase the liquid assets of a financial institution.
- Selling securities can provide immediate cash inflow and help address liquidity deficits.
- Securities under agreements to repurchase (repo transactions) can be used to temporarily increase liquidity.

- Issuing new loans can worsen a liquidity deficit by reducing available liquid assets.
-

Q.3894 The following financial information pertains to Daylight Bank:

Assets:	Amount in US\$ in millions
Cash and due from depository institutions	546
U.S. Treasury securities	382
Other securities	550
Pledged securities	503
Federal funds sold and reverse repurchase agreements	396
Net loans and leases	2,374
Total Assets	3,431
Liabilities:	
Demand deposits	741
Savings deposits	976
Time deposits	1,351
Total deposits	2,686
Core deposits	1,111
Brokered deposits	214
Federal funds purchased and repurchase agreements	488
Other money market borrowings	186

Calculate Daylight Bank's net federal funds and repurchase agreements position.

- A. -11.54%
- B. -2.68%
- C. 2.68%
- D. 11.54%

The correct answer is **B**.

$$\begin{aligned}
 &\text{Net federal funds and repurchase agreements position} \\
 &= (\text{Federal funds sold and reverse repurchase agreements} \\
 &\quad - \text{Federal funds purchased and repurchase agreements}) \\
 &\quad \div \text{Total assets} \\
 &= \frac{396 - 488}{3431} = -2.68\%
 \end{aligned}$$

Q.3895 The following financial information pertains to Daylight Bank:

Assets:	Amount in US\$ in millions
Cash and due from depository institutions	546
U.S. Treasury securities	382
Other securities	550
Pledged securities	503
Federal funds sold and reverse repurchase agreements	396
Net loans and leases	2,374
Total Assets	3,431
Liabilities:	
Demand deposits	741
Savings deposits	976
Time deposits	1,351
Total deposits	2,686
Core deposits	1,111
Brokered deposits	214
Federal funds purchased and repurchase agreements	488
Other money market borrowings	186

Calculate Daylight Bank's cash position ratio.

- A. 7.97%
- B. 11.13%
- C. 15.91%
- D. 32.38%

The correct answer is **C**.

$$\begin{aligned}
 \text{Cash position indicator} &= \frac{\text{Cash and cash deposits due from depository institutions}}{\text{total assets}} \\
 &= \frac{546}{3431} = 15.91\%
 \end{aligned}$$

A higher proportion of cash implies that the institution is in a stronger position to handle immediate cash needs.

Q.3896 The following indicators are negative liquidity indicators. Which is the odd one out?

- A. Capacity ratio
- B. Pledged securities ratio
- C. Hot money ratio
- D. All of the above

The correct answer is C.

The Hot money ratio is not a negative liquidity indicator. This ratio is used to determine whether an institution has balanced its volatile liabilities with the money market instruments it holds. These instruments can be quickly sold to cover liabilities if necessary. Therefore, a high hot money ratio can indicate a strong liquidity position as the institution has a significant amount of liquid assets that can be quickly converted into cash to meet its short-term obligations. This is contrary to the concept of a negative liquidity indicator, which typically signifies potential liquidity risks or problems.

Choice A is incorrect. The Capacity ratio is indeed a negative liquidity indicator. It measures the proportion of an institution's assets that are financed by debt, and a higher ratio indicates greater financial risk due to potential difficulties in meeting debt obligations.

Choice B is incorrect. The Pledged securities ratio also falls under the category of negative liquidity indicators. This ratio represents the proportion of an institution's assets that are pledged as collateral for loans or other obligations, and a higher ratio suggests increased risk due to reduced flexibility in managing those assets.

Choice D is incorrect. As explained above, both Capacity Ratio and Pledged Securities Ratio are negative liquidity indicators, hence this option cannot be correct.

Things to Remember

- Liquidity risk is the risk that a company or financial institution may be unable to meet its short-term financial obligations due to an inability to convert assets into cash quickly enough.
- Capacity ratio is calculated as total liabilities divided by total assets, and a higher ratio indicates higher financial risk.
- Pledged securities ratio is calculated as pledged securities divided by total assets, and a higher ratio indicates reduced flexibility in managing assets.
- Hot money ratio is used to assess the balance between volatile liabilities and liquid

assets that can be quickly converted into cash.

Q.3897 No financial institution can confidently tell if it has sufficient liquidity until it has passed the market's test. Which of the following is a signal that the management should take note of

- A. Stock price behavior
- B. Public confidence
- C. Borrowings from the central bank
- D. All of the above

The correct answer is **D**.

All of the options listed are signals that management should take note of when assessing the liquidity of a financial institution.

Stock price behavior. The behavior of a firm's stock price can be a strong indicator of its liquidity. If investors perceive that the institution is experiencing a liquidity crisis, they may sell their shares, leading to a decline in the firm's stock prices. This is because investors are wary of the risks associated with a liquidity crisis, such as the firm's inability to meet its obligations.

Public confidence. Public confidence is another important factor that can signal liquidity issues. If customers lose confidence in the firm due to perceived risks, they may withdraw their investments or stop doing business with the firm. This can lead to a decrease in the firm's liquid assets, exacerbating the liquidity crisis.

Borrowings from the central bank. If a firm has been forced to borrow in larger volumes and more frequently from the central bank in its home territory, it could be a sign of liquidity problems. Central bank officials may also begin to question the institution's borrowings, which should be a red flag for the management. Therefore, all of the above are signals that management should take note of when assessing liquidity.

Choice A is incorrect. While stock price behavior can be an indicator of a company's financial health, it does not directly signal liquidity issues. Stock prices are influenced by a variety of factors, including market sentiment, economic indicators, and company-specific news. Therefore, relying solely on stock price behavior to gauge liquidity risk could lead to inaccurate conclusions.

Choice B is incorrect. Public confidence can impact a financial institution's ability to raise funds and maintain liquidity; however, it is not a direct measure of liquidity risk. Public confidence can fluctuate based on numerous factors that may or may not be related to the institution's actual financial condition.

Choice C is incorrect. Borrowings from the central bank are indeed an important source of emergency funding for banks facing severe liquidity problems; however, frequent or large-scale borrowings could indicate underlying issues with the bank's operations or risk management practices rather than being a direct signal of potential liquidity issues.

Things to Remember

- Liquidity risk: The risk that a company or financial institution may not be able to meet its short-term financial obligations due to a lack of liquid assets.
 - Market liquidity: The ability to buy or sell an asset in the market without causing a significant change in its price.
 - Central bank: A financial institution that is responsible for regulating a country's monetary policy and issuing currency.
 - Liquidity crisis: A situation where a company or financial institution does not have enough liquid assets to meet its short-term obligations.
 - Market sentiment: The overall attitude of investors towards a particular security or financial market.
-

Q.3898 One who is responsible for ensuring that the institution maintains an adequate level of legal reserves is known as?

- A. Chief finance officer
- B. Chief operational officer
- C. Money position manager
- D. None of the above

The correct answer is **C**.

Choice A is incorrect. The Chief Finance Officer (CFO) of a financial institution does play a significant role in the overall financial management, but they are not typically responsible for maintaining an adequate level of legal reserves. This task requires specialized knowledge and quick decision-making skills related to the institution's money position, which is usually the responsibility of a Money Position Manager.

Choice B is incorrect. The Chief Operational Officer (COO) oversees the day-to-day administrative and operational functions of a business. While they may be involved in strategic decisions about reserves, they do not directly manage or maintain legal reserves.

Choice D is incorrect. There is indeed a specific role within financial institutions that handles the maintenance of legal reserves - it's not 'None of the above'. As mentioned earlier, this role typically falls under the purview of a Money Position Manager.

Things to Remember

- Legal reserves are funds set aside by financial institutions to meet their legal obligations and regulatory requirements.
 - Money Position Manager is responsible for managing the institution's cash flow, liquidity, and reserves to ensure compliance with legal requirements.
 - Legal reserves are important for maintaining financial stability and protecting the institution from potential risks.
 - Regulatory authorities often set minimum requirements for legal reserves to ensure the safety and soundness of financial institutions.
-

Q.3899 A contractual agreement between bank and customer that permits the bank to move funds out of a customer checking account to generate higher returns for the customer and lower reserve requirements for the bank is known as:

- A. Clearing balance
- B. Sweep account
- C. Legal reserve
- D. Reserve Maintenance

The correct answer is **B**.

Choice A is incorrect. A clearing balance refers to the funds that a bank keeps in its account at a Federal Reserve Bank to meet its reserve requirement and clear financial transactions, not an agreement between a bank and its customer for fund transfers.

Choice C is incorrect. The legal reserve is the minimum amount of funds that a bank must hold as reserves, according to central banking regulations. It does not involve any contractual agreement with customers for transferring funds from their checking accounts.

Choice D is incorrect. Reserve maintenance refers to the process by which banks manage their required reserves, but it does not refer to any specific type of contractual agreement between a bank and its customer for fund transfers.

Things to Remember

- Sweep accounts are commonly used by banks to automatically transfer excess funds from a customer's checking account into a higher-interest investment option, such as a money market mutual fund.
 - These accounts help customers earn a higher return on their idle funds while also helping banks meet reserve requirements more efficiently.
 - Reserve requirements are set by central banks to ensure that banks have enough liquidity to cover withdrawals and other obligations.
 - By utilizing sweep accounts, banks can optimize their liquidity management and potentially earn higher returns on customer funds.
-

Q.3900 You are provided with a simple balance sheet for Loud bank. When the bank's depositors withdraw \$200 million, the bank faces a liquidity challenge.

Assets	Amount
Cash	\$48
Securities	\$940
Loans	\$3,932
Total assets	4,920
Liabilities and Equity	
Deposits	\$3,948
Other liabilities	\$440
Equity	\$532
Total Liabilities and equity	\$4,920

If Loud bank employs the asset conversion method and decides to sell its securities to cover the deposit drain, what happens to the size of the bank due to withdrawal?

- A. Increase
- B. Decrease
- C. No change
- D. Cannot establish from the information

The correct answer is **B**.

The size of the bank would decrease (shrink) by the amount of deposit withdrawal. Therefore, the total assets would decline to $\$4,920 - \$200 = \$4,720$ million

Q.3901 You are provided with a simplified balance sheet for Loud bank. When the bank's depositors withdraw \$200 million, the bank faces a liquidity challenge.

Assets	Amount
Cash	\$48
Securities	\$940
Loans	\$3,932
Total assets	4,920
Liabilities and Equity	
Deposits	\$3,948
Other liabilities	\$440
Equity	\$532
Total Liabilities and equity	\$4,920

Suppose that Loud bank's management depends on borrowed liquidity (liability management strategy) to cover the deposit drain. What happens to the size of the bank's balance sheet?

- A. Increase
- B. Decrease
- C. No change
- D. Cannot establish from the information

The correct answer is **C**.

Liability management, also known as purchased liquidity strategy, involves borrowing immediately spendable funds to cover all anticipated demands for liquidity. This strategy allows a firm to keep the volume and composition of its assets portfolio unchanged if it is satisfied with the assets it currently holds. In the case of Loud bank, the withdrawal of \$200 million by its depositors presents a liquidity challenge. However, if the bank's management decides to borrow an amount equal to the withdrawal, the size of the bank's balance sheet would remain unchanged. This is because the decrease in deposits due to the withdrawal would be offset by the increase in other liabilities due to the borrowed funds. Therefore, the total liabilities and equity of the bank would remain the same, resulting in no change in the size of the bank's balance sheet.

Choice A is incorrect. The size of the bank's balance sheet will not increase. This is because the bank is borrowing funds to cover a withdrawal, not to expand its assets or liabilities beyond their current levels.

Choice B is incorrect. The size of the bank's balance sheet will also not decrease. Although there has been a withdrawal, this has been offset by borrowing funds, so the overall size of the bank's balance sheet remains unchanged.

Choice D is incorrect. We can establish from the information provided that there would be no change in the size of Loud Bank if it borrows an amount equal to the withdrawal. This strategy allows Loud Bank to maintain its current asset portfolio and meet liquidity demands without altering its balance sheet's overall size.

Q.3902 You are provided with a simplified balance sheet for Loud bank. The bank faces liquidity challenge the moment the bank depositors withdraw \$200 million

Assets	Amount
Cash	\$48
Securities	\$940
Loans	\$3,932
Total assets	4,920
Liabilities and Equity	
Deposits	\$3,948
Other liabilities	\$440
Equity	\$532
Total Liabilities and equity	\$4,920

Loud Bank requires cash to cover its unexpected loan demand. The loan officer has \$400 million in loans that she wants to make. Using Loud Bank's balance sheet, what would happen to the size of the Loud bank if liability management strategy is used to provide funds for the loans?

- A. Increase
- B. Decrease
- C. No change
- D. Cannot establish from the information

The correct answer is **A**.

The size of the bank, in this context, refers to the total assets of the bank. When a bank uses a liability management strategy to fund loans, it essentially borrows money to lend it out. This increases the bank's liabilities (as it now owes money to the entity it borrowed from) and its assets (as it now has the loan it gave out as an asset). In this case, the loan officer wants to make \$400 million in loans. This would increase the bank's total assets by \$400 million, from \$4,920 million to \$5,320 million. Therefore, the size of the bank would increase.

Choice B is incorrect. A decrease in the overall size of Loud Bank would imply a reduction in its assets, liabilities, or both. However, implementing a liability management strategy to fund loans would involve increasing the bank's liabilities (through borrowing or issuing more equity), which would then be used to increase its assets (by providing more loans). Therefore, this strategy would not lead to a decrease in the bank's overall size.

Choice C is incorrect. No change in the overall size of Loud Bank implies that its total assets and total liabilities remain constant. However, as explained above, implementing a liability management strategy involves increasing both these components of the balance sheet - hence there will be an increase rather than no change.

Choice D is incorrect. The information provided about Loud Bank's current financial situation and proposed strategy is sufficient to establish that implementing a liability management strategy will result in an increase in the bank's overall size.

Q.3903 Suppose that a bank has a clearing balance averaging \$2.5 million during a certain two-week maintenance period and the Federal funds' interest rate over this same period averaged 8%. Calculate the Federal Reserve credit that it would earn:

- A. \$547.95
- B. \$6,575.34
- C. \$ 7,777.8
- D. \$15,342.47

The correct answer is **C**.

$$\begin{aligned}\text{Reserve credit} &= \text{Average clearing balance} \times \text{Annualized federal funds rate} \\ &\times \left(\frac{14 \text{ days}}{360 \text{ days}} \right) \\ &= \$2,500,000 \times 0.08 \times \frac{14}{360} = \$7,777.8\end{aligned}$$

Q.3904 Suppose Grand Bank's liquidity manager estimates that the bank experiences a liquidity deficit of \$500 million in the coming month with a probability of 45%, a \$575 million liquidity deficit with a probability of 20%, a liquidity surplus of \$350 million with a probability of 25%, and a \$100 million liquidity surplus with a probability of 10% What is the bank's expected liquidity requirement?

- A. -\$10.00
- B. -\$242.50
- C. \$22.75
- D. \$30.00

The correct answer is **B**.

Liquidity Deficit or Surplus in Millions	Probability	Expected Value
\$(500.00)	45%	\$(225.00)
\$(575.00)	20%	\$(115.00)
\$350.00	25%	\$87.50
\$100.00	10%	\$10.00
	100%	\$(17.50)

$$\begin{aligned}
 \text{Expected Liquidity Requirement} &= 0.45 \times (-\$500 \text{ million}) + 0.20 \times (-\$575 \text{ million}) \\
 &\quad + 0.25 \times (+\$350 \text{ million}) + 0.10 \times (+\$100 \text{ million}) \\
 &= -\$225 \text{ million} - \$115 \text{ million} \\
 &\quad + \$87.50 \text{ million} + \$10 \text{ million} \\
 &= -\$242.50 \text{ million}
 \end{aligned}$$

The bank is faced with an expected liquidity deficit of -\$242.5 million in the coming month.

Q.5379 The following are ABC Bank's estimates of cash inflows and outflows for the coming week.

Cash Flows	USD Million
Scheduled loan repayments	150
Acceptable loan requests	110
Borrowings from money market	120
Operating expenses	50
Deposit withdrawals	80
Deposit inflows	70
Stockholders dividend payments	30
Repayments of bank borrowings	50

What is ABC Bank's net liquidity position at the end of the coming week?

- A. 20
- B. -20
- C. 120
- D. 150

The correct answer is **A**.

Net liquidity position = Cash inflows – Cash outflows

Cash Inflows	Cash Outflows
Deposit inflows	Deposit withdrawals
Scheduled loan repayments	Acceptable loan requests
Borrowings from money market	Operating expenses
	Stockholders dividend payments
	Repayments of bank borrowings

Therefore, ABC Bank's net liquidity position at the end of the coming week is given by:

$$= (70 + 150 + 120) - (80 + 110 + 50 + 30 + 50) = \text{USD } 20 \text{ million}$$

Q.5382 John Doe is an investment analyst at ABC Investment Bank. ABC Investment Bank is the underwriter for a commercial paper that one of the country's national banks wants to issue. John Doe is performing an analysis of the national bank's financial statements and calculating its liquidity indicators over the past five years. Which of the following trends over this period should John Doe be most concerned about?

- A. The ratio of pledged securities at the bank has been decreasing.
- B. The loan commitments ratio for the bank has been decreasing.
- C. The average net position of the bank in repurchase agreements and government funds has increased.
- D. The capacity ratio of the bank has been rising.

The correct answer is **D**.

The capacity ratio, which compares net loans and leases to total assets, indicates a decrease in liquidity when net loans and leases increase in proportion to total assets. This is primarily due to the illiquid nature of loans and leases.

A is incorrect. Liquidity improves when the number of pledged or unavailable securities decreases in relation to the total number of securities.

B is incorrect. When loan commitments as a percentage of assets decline, liquidity rises.

C is incorrect. When overnight loans are more than overnight borrowing, liquidity rises.

Things to Remember

- Liquidity ratios are important indicators of a company's ability to meet its short-term obligations.
- Capacity ratio is a measure of a bank's ability to meet its financial obligations.
- Net loans and leases are considered less liquid assets compared to securities.
- Repurchase agreements involve the sale of securities with an agreement to repurchase them at a later date.
- Government funds refer to funds held by the bank on behalf of the government.

Reading 132: Intraday Liquidity Risk Management

Q.3905 Andera Jeremy is a bank treasurer in ABC bank; he realizes that the bank is experiencing liquidity shortage during the day, and it is unable to pay XYZ bank for its foreign transaction services owed. Jeremy suggests the use of intraday liquidity as a remedy. Which of the following definitions best applies to Jeremy's use of the funds?

- A. Asset purchasing and funding.
- B. Funding for Nostro accounts
- C. Collateral pledging
- D. Outgoing wire transfers

The correct answer is **B**.

In the given scenario, the primary purpose of acquiring the intraday fund is to settle the payment owed to XYZ bank. This purpose aligns with the definition of 'Funding for Nostro accounts'. Nostro accounts refer to an account that a bank holds in a foreign currency in another bank. Banks manage the cash they place in a correspondent bank account to a target average monthly balance as part of the return for providing banking services. The funds in these accounts are used to settle transactions with other banks, such as the payment owed to XYZ bank in this case. Therefore, the use of intraday liquidity suggested by Jeremy can be best described as 'Funding for Nostro accounts'.

Choice A is incorrect. Asset purchasing and funding refers to the acquisition of assets or investments for the purpose of generating returns. This does not align with Jeremy's proposed use of funds, which is to address a liquidity shortage issue during the day.

Choice C is incorrect. Collateral pledging involves using an asset as a guarantee for repayment of a loan, and it does not directly address liquidity shortages during the day. It's more related to securing long-term loans rather than managing intraday liquidity.

Choice D is incorrect. Outgoing wire transfers refer to electronic transfer of funds from one bank or credit union to another. While this could be part of managing intraday liquidity, it doesn't specifically address the issue Jeremy identified - delay in payment due to liquidity shortage.

Things to Remember

- Intraday liquidity management involves managing a bank's liquidity needs and obligations during the day, ensuring that it has enough funds to meet payment obligations as they arise.
- Nostro accounts are used by banks to facilitate foreign exchange and international transactions, holding funds in foreign currencies in accounts with other banks.

- Liquidity shortage can lead to operational disruptions, funding difficulties, and potential regulatory issues for banks.
 - Effective liquidity management is crucial for banks to maintain financial stability and meet their obligations in a timely manner.
-

Q.3907 A bank treasurer has been getting liquidity recovery during the day by borrowing Fed funds at London interbank offered rate, and Eurodollar deposits. Which of the following most accurately states the source under which the credit mentioned above falls?

- A. Other term funding
- B. Liquid assets
- C. Overnight borrowing
- D. Intraday credit

The correct answer is **C**.

Choice A is incorrect. Other term funding refers to longer-term borrowings, not the short-term liquidity management tools like Fed funds, LIBOR, and Eurodollar deposits that are used for managing liquidity on a daily basis.

Choice B is incorrect. Liquid assets are cash or assets that can be easily converted into cash. While Fed funds, LIBOR and Eurodollar deposits can provide liquidity, they themselves are not considered liquid assets.

Choice D is incorrect. Intraday credit refers to loans that must be paid back within the same trading day. Although these resources can help manage intraday liquidity needs, they do not fall under this category as they are not loans but rather interbank borrowing rates and deposit types.

Things to Remember

- Fed Funds: Funds that banks lend to each other overnight to meet reserve requirements.
 - London Interbank Offered Rate (LIBOR): The average interest rate at which major global banks borrow from one another.
 - Eurodollar Deposits: US dollar-denominated deposits at foreign banks or foreign branches of American banks.
 - Liquidity Risk: The risk that a company or bank may be unable to meet short-term financial demands.
 - Interbank Market: The market where banks lend and borrow funds from each other.
-

Q.3908 A bank treasurer working for FGH bank has been given an intraday credit whose use had

been determined according to the following characteristics, which he had considered and approved before the credit processing. The characteristics are listed below:

1. Have a single settlement per day, which is done majorly in the late afternoon timeframe.
2. May either serve as a source or use of funds reliant on the net position of a participant on any given day.
3. They can be forecast for securities that have multi-day settlements more difficult for same-day settlement activity

Which of the following choices most accurately outlines the use under which the intraday credit should be classified?

- A. Outgoing wire transfers
- B. Funding for Nostro accounts
- C. Collateral pledging
- D. Settlements at Payment, Clearing, and Settlement (PCS) Systems

The correct answer is **D**.

The characteristics outlined in the question are indicative of the operations of Payment, Clearing, and Settlement (PCS) Systems. PCS systems are integral to the financial markets, facilitating the exchange of payments, securities, and other financial transactions among financial institutions. They typically have a single settlement per day, often in the late afternoon. The net position of a participant on any given day can either serve as a source or use of funds, depending on the transactions that have taken place. Forecasting for securities with multi-day settlements can be more challenging for same-day settlement activity due to the complexity and volume of transactions. Therefore, the intraday credit in question should be classified under Settlements at PCS Systems.

Choice A is incorrect. Outgoing wire transfers are not the correct category for this intraday credit use. Wire transfers typically involve the electronic transfer of funds from one account to another, often across different financial institutions, and do not necessarily align with the characteristics described in the question such as a single settlement per day or forecasting for securities with multi-day settlements.

Choice B is incorrect. Funding for Nostro accounts does not fit with these characteristics either. Nostro accounts are held by a bank in a foreign country (in that country's currency), and while they can serve as a source or use of funds depending on the net position of a participant on any given day, they do not typically involve single daily settlements or forecasting for securities with multi-day settlements.

Choice C is incorrect. Collateral pledging involves using assets as security against a loan and does not match up with the characteristics described in this scenario such as primarily late afternoon settlements and more challenging forecasts for same-day settlement activity.

Things to Remember

- **Settlement Risk:** Intraday credit is often used to mitigate settlement risk, which is the risk that one party in a transaction may not deliver the security or payment as expected.
 - **Real-Time Gross Settlement (RTGS):** RTGS systems are commonly used for intraday credit to settle transactions individually and continuously in real-time, reducing credit and liquidity risks.
 - **Liquidity Management:** Intraday credit plays a crucial role in managing liquidity needs throughout the day, especially in the context of large-value payments and securities transactions.
 - **Central Counterparties (CCPs):** Intraday credit can also be provided to CCPs to facilitate the clearing and settlement of trades in financial markets.
-

Q.3910 A sound intraday risk structure for overseeing intraday liquidity is of the utmost importance in risk management. Which of the following accurately states the characteristics of a sound intraday risk structure?

- A. Active risk management
- B. Integration with risk governance
- C. Risk assessment
- D. All of the above

The correct answer is **D**.

A sound intraday risk structure for overseeing intraday liquidity risk is characterized by all the options listed. Active risk management involves the continuous monitoring and managing of risks. It is a proactive approach that seeks to identify potential risks before they materialize and take appropriate measures to mitigate them. Integration with risk governance refers to the alignment of the intraday risk structure with the overall risk governance framework of the organization. This ensures that the intraday risk structure is in sync with the organization's risk appetite, risk policies, and risk management strategies. Risk assessment is the process of identifying, analyzing, and evaluating risks. In the context of intraday liquidity risk, risk assessment involves the identification and evaluation of potential liquidity risks that could arise during the trading day. Therefore, all these elements - active risk management, integration with

risk governance, and risk assessment - are integral to a sound intraday risk structure.

Choice A is incorrect. While active risk management is a crucial feature of a robust intraday risk structure, it alone does not constitute the entirety of such a structure. Active risk management involves continuously monitoring and managing risks as they arise, but this needs to be complemented by other features like integration with risk governance and regular risk assessment for the overall structure to be effective.

Choice B is incorrect. Integration with risk governance ensures that the intraday risk structure aligns with the organization's overall approach to managing risks. However, this integration alone cannot ensure an effective intraday risk structure without active management of risks and regular assessment of these risks.

Choice C is incorrect. Risk assessment involves identifying potential risks and evaluating their possible impact on the organization's operations. While it forms an integral part of any sound intraday risk structure, it needs to be combined with active management of these identified risks and their integration into the broader framework of organizational governance for maximum effectiveness.

Things to Remember

- Sound intraday risk structure involves continuous monitoring and managing of risks
 - Integration with risk governance aligns the intraday risk structure with overall risk governance framework
 - Risk assessment is the process of identifying, analyzing, and evaluating risks
 - Active risk management, integration with risk governance, and risk assessment are integral to a sound intraday risk structure
-

Q.3911 FRM students were discussing the measures for quantifying and monitoring risk levels. They finally agreed to write down every two measures they think are accurate. Which of the following states the most accurate measures?

- A. Daily maximum intraday liquidity usage and client intraday credit usage
- B. Client intraday credit usage and total bank intraday credit lines available and usage
- C. Total bank intraday credit lines available and usage and total intraday credit lines to clients and counterparties.
- D. Time-sensitive obligations like settlement positions and payment throughput

The correct answer is **A**.

The measures mentioned in this option, namely 'Daily maximum intraday liquidity usage' and 'client intraday credit usage', are indeed accurate measures for quantifying risk levels. The 'Daily maximum intraday liquidity usage' refers to the highest amount of liquidity that a financial institution uses within a single day. This measure is crucial as it provides insights into the liquidity needs of the institution and can help in identifying potential liquidity shortfalls. On the other hand, 'client intraday credit usage' refers to the amount of credit that clients use within a single day. This measure is important as it provides an understanding of the credit risk exposure of the institution to its clients. Both these measures together provide a comprehensive view of the risk levels and are therefore the most accurate among the given options.

Choice B is incorrect. While client intraday credit usage and total bank intraday credit lines available and usage are important measures, they do not provide a complete picture of risk levels. They fail to account for the maximum liquidity usage which is crucial in quantifying risk levels.

Choice C is incorrect. Total bank intraday credit lines available and usage, along with total intraday credit lines to clients and counterparties, although significant, do not represent the most precise measures for quantifying risk levels. These metrics overlook daily maximum intraday liquidity usage which plays a vital role in assessing overall financial risk.

Choice D is incorrect. Time-sensitive obligations like settlement positions and payment throughput are important aspects of financial operations but they do not directly quantify risk levels as effectively as daily maximum intraday liquidity usage or client intraday credit usage would.

Things to Remember

- Quantifying risk levels is essential for financial institutions to manage their exposure to various risks effectively.
- Daily maximum intraday liquidity usage reflects the peak liquidity needs of an institution during a single day.
- Client intraday credit usage helps in understanding the credit risk exposure to clients.
- Total bank intraday credit lines available and usage provide insights into the credit capacity of the institution.
- Time-sensitive obligations like settlement positions and payment throughput are crucial for operational efficiency but may not directly quantify risk levels.

Q.3913 A bank has been keeping statistics necessary for understanding intraday flows. The statistics have been listed below: identify the pair that states the most accurate statistics for understanding intraday flows.

- A. Payment amounts and times for each processing step in the payment workflow.
- B. Time received or originated and credit status
- C. Past credit position and any suspensions of the payment
- D. Payer and payee and past credit position

The correct answer is **A**.

Payment amounts and times for each processing step in the payment workflow are indeed the most accurate statistics for understanding intraday flows. Intraday flows refer to the movement of funds within a single trading day. Therefore, knowing the payment amounts and the times for each processing step can provide a detailed picture of how funds are moving throughout the day. This information can help banks manage their liquidity, identify potential bottlenecks in the payment workflow, and make more informed decisions about their operations.

Choice B is incorrect. While the time a payment was received or originated and the credit status of the payer can provide some insight into intraday flows, they do not offer as comprehensive a view as payment amounts and times for each processing step. The latter allows banks to track funds throughout their entire journey within a single trading day, providing more accurate insights.

Choice C is incorrect. Past credit position and any suspensions of the payment may be useful for assessing credit risk, but they do not directly contribute to understanding intraday flows. These statistics are more backward-looking and do not necessarily reflect current fund movements within a single trading day.

Choice D is incorrect. Information about payer and payee along with past credit position can be useful in assessing counterparty risk but it does not provide detailed insights into intraday flows. It lacks information on timing or amount of payments which are crucial for understanding movement of funds within a single trading day.

Things to Remember

- Understanding intraday flows is crucial for banks to manage their liquidity effectively and make informed decisions about their operations.
- Payment amounts and times for each processing step in the payment workflow provide a detailed picture of how funds are moving throughout the day.
- Tracking funds throughout their entire journey within a single trading day helps identify potential bottlenecks in the payment workflow.
- Assessing credit risk and counterparty risk are important aspects of risk management in banking operations.

- Having accurate and timely data on intraday flows can help banks optimize their processes and improve efficiency.
-

Q.3914 A bank has been keeping statistics on payer and payee information as a way of understanding its intraday flows, payment system used, any suspensions of the payment, time received or originated, and routing information. Under which measure of understanding intraday flow does the above statistics most accurately lie?

- A. Total payments
- B. Other cash transactions
- C. Settlement positions
- D. Total intraday credit lines to clients and counterparties

The correct answer is **A**.

The statistics mentioned in the question are most accurately represented by the total payments measure for understanding intraday flows. This measure involves tracking all payments made and received by the bank during the intraday period. It includes details about the payer and payee, the payment system used, any suspensions of the payment, the time the payment was received or originated, and routing information. By analyzing these statistics, the bank can gain valuable insights into its intraday payment flows, which can help it manage its liquidity more effectively and make more informed decisions about its financial operations.

Choice B is incorrect. Other cash transactions refer to all other forms of cash inflows and outflows that are not related to payments. These could include investments, borrowings, or any other form of financial transaction. The data mentioned in the question specifically pertains to payment-related information and does not encompass all other cash transactions.

Choice C is incorrect. Settlement positions refer to the net obligations that a financial institution has at the end of a trading day. While payer and payee data can contribute towards understanding settlement positions, it does not provide a complete picture as it excludes other factors such as securities trades, foreign exchange deals etc.

Choice D is incorrect. Total intraday credit lines to clients and counterparties refers to the maximum amount that can be borrowed by clients or counterparties from the financial institution within a single day. This measure does not directly relate with payer/payee data or payment system details mentioned in the question.

Things to Remember

- Understanding intraday flows is crucial for managing liquidity and making informed financial decisions.

- Total payments measure involves tracking all payments made and received by the bank during the intraday period.
 - Payment system used, suspensions of payments, time received or originated, and routing information are important details in analyzing intraday flows.
 - Settlement positions reflect the net obligations of a financial institution at the end of a trading day.
 - Total intraday credit lines to clients and counterparties represent the maximum amount that can be borrowed within a single day.
-

Q.3915 Which of the following measures is derived by comparing a client's peak daily intraday overdraft to the established credit line?

- A. Client intraday credit usage
- B. Payment throughput
- C. Daily maximum intraday liquidity usage
- D. Total bank intraday credit lines available and usage

The correct answer is **A**.

Client intraday credit usage is the measure that is derived by comparing a client's peak daily intraday overdraft to the established credit line. This measure is crucial as it provides an indicator of the necessary liquidity required to support a client's business activities. It helps banks to understand the client's credit usage pattern and manage their credit lines effectively. This measure is particularly important in managing the risk associated with intraday credit and ensuring that the client does not exceed their credit limit.

Choice B is incorrect. Payment throughput refers to the total value of payments that can be processed by a system within a given time period. It does not involve comparing a client's highest daily intraday overdraft to their established credit line.

Choice C is incorrect. Daily maximum intraday liquidity usage refers to the maximum amount of liquidity used by an entity during a single day, which may include various transactions and not just overdrafts compared to credit lines.

Choice D is incorrect. Total bank intraday credit lines available and usage would refer to the overall capacity of the bank for providing intraday credit and how much of it has been utilized, rather than focusing on individual client's highest daily intraday overdraft in relation to their

established credit line.

Things to Remember

- Peak daily intraday overdraft: The highest amount by which a client's account is overdrawn during a single day.
 - Intraday liquidity: Refers to the availability of liquid assets to a bank or an entity during the trading day to meet payment obligations.
 - Credit risk management: The process of assessing and managing the risk of financial loss resulting from a client's failure to meet their credit obligations.
 - Liquidity risk: The risk that a bank may not have enough liquid assets to meet its payment obligations when they come due.
-

Q.3916 A bank XYZ usually monitors intraday and end-of-day settlement positions at every financial market utility in which it participates. It further maximizes the volume of the transaction-level. The details are taken and stored for further analysis. Which of the following indicates the most accurate measure for understanding intraday flow that the bank uses?

- A. Other cash transactions
- B. Total payments
- C. Total bank intraday credit lines available and usage
- D. Settlement positions

The correct answer is **A**.

Other cash transactions are the most accurate measure for understanding intraday flow that the bank uses. In the context of Bank XYZ, it monitors intraday and end-of-day settlement positions at every financial market utility in which it participates. This monitoring process involves other cash transactions, which are transactions that do not involve the exchange of physical cash but are instead conducted electronically or through other means. The bank also maximizes the volume of transaction-level detail taken and stored for further analysis. This detail includes data from other cash transactions, which can provide valuable insights into the bank's intraday flow. Therefore, other cash transactions are the most accurate measure for understanding the intraday flow that Bank XYZ uses.

Choice B is incorrect. Total payments do not provide a comprehensive understanding of the intraday flow as they only account for the outgoing funds and not the incoming funds or available

credit lines. They also lack transaction-level detail which is a priority for Bank XYZ.

Choice C is incorrect. While total bank intraday credit lines available and usage can give an idea about the bank's liquidity position, it does not necessarily reflect all transactions happening throughout the day, hence it does not fully capture the intraday flow.

Choice D is incorrect. Settlement positions are end-of-day balances and do not provide information on intraday activities or flows. Therefore, they cannot be used to understand the intraday flow that Bank XYZ employs.

Things to Remember

- Understanding intraday flow is crucial for banks to manage liquidity, monitor risks, and optimize their operations.
 - Monitoring intraday positions helps banks to ensure they have enough funds to meet their obligations throughout the day.
 - Transaction-level detail provides a granular view of the bank's activities and helps in identifying any potential issues or opportunities.
 - Settlement positions are important for end-of-day reconciliation but may not capture the full picture of intraday activities.
 - Cash transactions can include a wide range of activities such as wire transfers, electronic payments, and securities transactions.
-

Q.3917 Which of the following denotes the methods for tracking intraday flows and monitoring risk levels?

- A. Measures for understanding income flows
- B. Measures for quantifying and monitoring risk levels
- C. Measurement of credit level
- D. Measures for risk aversion

The correct answer is **B**.

Measures for quantifying and monitoring risk levels is a method for tracking intraday flows and monitoring risk levels.

A is incorrect: Measures for understanding income flows is not a method for tracking intraday flows and monitoring risk levels.

C is incorrect: Measurement of credit level does not apply in tracking intraday flows and monitoring risk levels.

D is incorrect: Measures for risk aversion is not a measure for tracking intraday flows and monitoring risk levels.

Things to Remember

- Tracking intraday flows involves monitoring the movement of funds within a financial institution or market throughout the trading day.
- Risk levels can be quantified using various metrics such as Value at Risk (VaR), Expected Shortfall (ES), and stress testing.
- Monitoring risk levels is crucial for financial institutions to ensure they are within their risk tolerance limits and regulatory requirements.
- Intraday risk monitoring helps institutions identify and address potential risks in real-time, reducing the likelihood of large losses.

Q.3918 ABC Bank has been a participant of FMU for a significant period. The bank has been actively-measuring and tracking the flow of outgoing payment transactions relative to total payments or time markers. Which one of the following is not a reason for the bank to perform that?

- A. To track its volume patterns to aid in ensuring that all the day's payments are processed on a timely basis.
- B. To meet FMU requirements for submitting a target percentage of payments by a deadline.
- C. To identify and track its peak periods over time and the correlation of this activity with its intraday liquidity on hand and intraday credit usage.
- D. To be able to know which means are best for acquiring intraday liquidity.

The correct answer is **D**.

The bank does not actively measure and track the flow of outgoing payment transactions relative to total payments or time markers to determine the best means of acquiring intraday liquidity. Intraday liquidity refers to funds that can be accessed during the course of a business day, usually to meet time-specific obligations. While the bank does monitor its payment transactions, this activity is not directly linked to the acquisition of intraday liquidity. The bank may use other methods, such as short-term borrowing or the sale of assets, to acquire intraday liquidity. Therefore, the assertion that the bank tracks its payment transactions to know the best means of acquiring intraday liquidity is incorrect.

Choice A is incorrect. Tracking volume patterns is a crucial part of ensuring that all payments are processed on time. By monitoring these patterns, the bank can anticipate high-volume periods and allocate resources accordingly to prevent any delays or issues in processing.

Choice B is incorrect. Meeting FMU requirements for submitting a target percentage of payments by a deadline is another valid reason for monitoring outgoing payment transactions. The FMU sets these targets to ensure smooth operation of the financial system, and banks must comply with them.

Choice C is incorrect. Identifying and tracking peak periods over time allows the bank to understand its intraday liquidity needs better and manage its intraday credit usage more effectively. This understanding helps in maintaining sufficient liquidity on hand during peak periods, thus preventing potential liquidity crises.

Things to Remember

- FMU stands for Financial Market Utility, which refers to a system that provides clearing, settlement, and recording services for financial transactions.
- Intraday liquidity management is crucial for banks to meet their payment obligations throughout the day and manage their liquidity risks effectively.
- Monitoring payment transactions helps banks in optimizing their liquidity management strategies and ensuring smooth operations.

- Short-term borrowing and asset sales are common methods used by banks to acquire intraday liquidity when needed.
 - Peak periods in payment transactions can have a significant impact on a bank's intraday liquidity needs and credit usage.
-

Q.3919 XYZ Bank Ltd. has been obtaining money from Fed funds to pay debts owed to ABC Bank Ltd. then servicing the foreign exchange services owed to fed fund providers the next morning. Which of the following set of choices consists of the source of funds for XYZ and the use for which the bank puts the funds into?

- A. Cash balances and Outgoing wire transfers
- B. Incoming funds flow and intraday credit and settlements at payment
- C. Overnight borrowings and funding for Nostro accounts
- D. Liquid assets and outgoing wire transfers.

The correct answer is **C**.

Overnight borrowings and funding for Nostro accounts. In the context of the given scenario, XYZ Bank Ltd. is obtaining its funds through overnight borrowings. This involves borrowing from Federal funds, the London Interbank Offered Rate (LIBOR), and Eurodollar deposits. The bank is then using these funds for funding Nostro accounts. Nostro accounts refer to accounts that a bank holds in a foreign currency in another bank. Banks manage the cash they place in a correspondent bank account to a target average monthly balance as part of the return for providing banking services. Therefore, the funds obtained through overnight borrowings are being used to transfer cash to a correspondent bank for foreign transaction services provided.

Choice A is incorrect. Cash balances are not the source of funds for XYZ Bank Ltd. in this scenario. The bank is procuring funds from Federal funds, not from its cash balances. Also, outgoing wire transfers are a method of transferring money, not a purpose for which these funds are being utilized.

Choice B is incorrect. Incoming funds flow and intraday credit do not accurately represent the source from which XYZ Bank Ltd. is obtaining its funds as it's clearly mentioned that the bank is procuring its fund through overnight borrowings (Federal Funds). Moreover, settlements at payment do not represent the purpose for which these funds are being utilized as they're used to service foreign exchange services owed to providers of Federal Funds.

Choice D is incorrect. Liquid assets aren't mentioned as a source of funding in this scenario; instead, it's specified that XYZ Bank Ltd obtains its funding through overnight borrowings (Federal Funds). Additionally, similar to choice A, outgoing wire transfers refer to a method of transferring money rather than representing the purpose for which these borrowed funds are

used.

Things to Remember

- Overnight borrowings are short-term loans that banks obtain to meet their funding needs for a short period, usually overnight.
 - Nostro accounts are accounts that a bank holds in a foreign currency in another bank to facilitate foreign exchange and international transactions.
 - Federal funds refer to unsecured borrowings between banks and other financial institutions overnight.
 - Liquid assets are assets that can be easily converted into cash without significant loss in value.
 - Intraday credit is a type of credit extended by a bank to its customers for a short period during the day to facilitate transactions.
 - Settlements at payment refer to the process of settling financial transactions, including the transfer of funds between parties.
-

Q.5426 Which of the following tools used by Financial Market Utilities (FMUs) to manage intraday settlement involves maintaining backup credit lines from commercial banks and central banks for liquidity?

- A. Contingent liquidity.
- B. Net debit cap.
- C. Settlement windows.
- D. Collateral.

The correct answer is **A**.

Contingent liquidity arrangements are credit lines or liquidity facilities established with commercial banks and central banks to ensure that the FMU has access to funds in case of unexpected liquidity shortfalls during the intraday settlement process. These arrangements are an essential part of risk management for FMUs to maintain smooth settlement operations.

B is incorrect. Net debit cap is a tool used by central banks like the Federal Reserve to limit the amount of intraday credit a depository institution (such as a bank) can obtain from the central bank. It is not a tool used by FMUs themselves to manage their intraday settlement processes or liquidity. It's more about regulating the credit extended by central banks to financial institutions.

C is incorrect. Settlement windows refer to specific time periods during the trading day when settlement of financial transactions occurs. They dictate when transactions will settle but do not directly involve maintaining backup credit lines or liquidity. While they are important for organizing settlement processes, they do not pertain to managing intraday liquidity risk through credit arrangements.

D is incorrect. Collateral is assets that a party pledges to secure a loan or other financial obligation. While collateral can be used to manage counterparty credit risk, it is not a tool specifically used by FMUs to manage intraday settlement liquidity. Collateral management is more related to risk mitigation and ensuring that obligations are met, but it's not a tool for obtaining additional liquidity during the day.

Things to Remember

- Financial Market Utilities (FMUs) are critical entities that provide clearing, settlement, and recording services for financial transactions.
- Intraday settlement refers to the process of settling financial transactions within the same trading day.
- Liquidity risk is the risk of not being able to meet financial obligations as they come due, which is a significant concern for FMUs.
- Contingent liquidity arrangements are crucial for FMUs to ensure they have access to funds in case of unexpected liquidity shortfalls.
- Net debit caps are limits set by central banks on the amount of intraday credit that depository institutions can obtain.
- Settlement windows are specific time periods during the trading day when financial transactions are settled.
- Collateral is assets pledged to secure a loan or financial obligation, which can be used to manage counterparty credit risk.

Reading 133: Monitoring Liquidity

Q.3920 In monitoring liquidity, it is essential to understand the identification and taxonomy of cash flows that occur during the business activities of a financial institution. A group of FRM students were asked to state the two perspectives of classifying cash flows. The students were then grouped into four sets, depending on their responses as follows. Group A: Time and amount Group B: Time and Credit scores Group C: Credit potential and amount Group D: Term structure and time Which group of students stated the most accurate perspectives?

- A. Group A
- B. Group B
- C. Group C
- D. Group D

The correct answer is **A**.

The two perspectives of classifying cash flows are indeed 'Time' and 'Amount'. When classifying based on 'Time', we distinguish between deterministic and stochastic cash flows. Deterministic cash flows are those that occur at future moments that are predictable or known with certainty at the reference time of their occurrence. Stochastic cash flows, on the other hand, are those that appear at random moments in the future in an unpredictable manner. This classification allows financial institutions to plan and manage their liquidity effectively.

When classifying based on 'Amount', again we distinguish between deterministic and stochastic cash flows. Deterministic cash flows occur in an amount known with certainty at the reference time, while stochastic cash flows are those whose amount cannot be fully determined at the reference time. This classification helps in understanding the potential variability in the cash flows and aids in effective risk management.

Choice B is incorrect. The classification of cash flows does not involve credit scores. Credit scores are used to assess the creditworthiness of an individual or a company, but they do not directly influence the categorization of cash flows in terms of time and amount.

Choice C is incorrect. Credit potential and amount are not the two perspectives for classifying cash flows. While the amount is a crucial factor, credit potential refers to a borrower's ability to repay debt which does not directly relate to how cash flows are classified.

Choice D is incorrect. Term structure and time do not represent the two perspectives for classifying cash flows. Term structure relates more to interest rates over different periods, while time indeed plays a role in classification, it alone with term structure doesn't provide complete perspective for classifying cashflows.

Things to Remember

- Classification of cash flows based on 'Time' can be further divided into deterministic

and stochastic cash flows.

- Deterministic cash flows are predictable or known with certainty at the reference time of their occurrence.
 - Stochastic cash flows occur at random moments in the future in an unpredictable manner.
 - Classification of cash flows based on 'Amount' also distinguishes between deterministic and stochastic cash flows.
 - Deterministic cash flows have an amount known with certainty at the reference time.
 - Stochastic cash flows have amounts that cannot be fully determined at the reference time.
 - Credit scores are used to assess creditworthiness but do not directly influence the classification of cash flows.
 - Credit potential refers to a borrower's ability to repay debt and is not a perspective for classifying cash flows.
 - Term structure relates more to interest rates over different periods and is not a perspective for classifying cash flows.
-

Q.3921 In the field of financial risk management, cash flows are often categorized into deterministic and stochastic types, each with distinct characteristics. During a group discussion for their Financial Risk Management (FRM) exam preparation, four students presented their understanding of deterministic cash flows. Which of the following statements accurately represents the characteristics of deterministic cash flows?

- A. Deterministic cash flows manifest themselves at some random instants in the future, and their amount cannot be fully determined
- B. Deterministic cash flows occur at future instants that are predictable and occur in an amount known with certainty
- C. Deterministic cash flows amount cannot be predicted, and they occur at future instants that are predictable.
- D. Deterministic cash flows are unpredictable, both in amount and time perspective.

The correct answer is **B**.

Deterministic cash flows are defined by their predictability both in terms of timing and amount. That's why they're termed "deterministic" - they're determined and known in advance.

A is incorrect. This statement describes stochastic cash flows, not deterministic ones. Stochastic cash flows are uncertain both in terms of when they will occur and in what amount.

C is incorrect. This statement is a mix of characteristics. While it correctly states that deterministic cash flows occur at predictable times, it incorrectly states that their amounts cannot be predicted.

D is incorrect. This again describes stochastic cash flows. Deterministic cash flows are, by definition, predictable in both timing and amount.

Things to Remember

- Stochastic cash flows are unpredictable in terms of both timing and amount, making them different from deterministic cash flows.
- Deterministic cash flows are used in financial risk management to calculate metrics such as net present value (NPV) and internal rate of return (IRR).
- Understanding the nature of cash flows, whether deterministic or stochastic, is crucial for risk assessment and decision-making in financial markets.
- Deterministic cash flows are often associated with fixed income securities, where the timing and amount of cash flows are known in advance.

Q.3923 An FRM tutor gave his students some characteristics of cash flows and told them to identify the cash flows. The characteristics were as follows;

- a. They are related to financial contracts such as fixed-rate bonds or fixed-rate mortgages or loans, bonds issued, and loans received by the bank held in its liabilities;
- b. They are produced by payments of periodic interests and periodic repayments of the capital installments if the asset is amortizing.

Which of the following most accurately states the correct cash flows under which the above characteristics fall?

- A. Deterministic cash flows
- B. Behavioral cash flows
- C. Stochastic cash flows
- D. Credit-related cash flows

The correct answer is **A**.

The characteristics provided in the question are indicative of deterministic cash flows. Deterministic cash flows are those that are predictable and can be calculated with certainty. They are often associated with fixed-rate financial contracts such as bonds, mortgages, or loans. These contracts have predetermined payment schedules, including periodic interest payments and capital repayments if the asset is amortizing. Therefore, the cash flows generated from these contracts can be calculated with certainty, making them deterministic.

Choice B (Behavioral cash flows) is incorrect. Behavioral cash flows are typically associated with the unpredictable behavior of customers, such as prepayments on mortgages or early withdrawals from deposit accounts. They are not directly linked to financial contracts like fixed-rate bonds or mortgages, nor are they generated by periodic interest payments and capital repayments.

Choice C (Stochastic cash flows) is incorrect. Stochastic cash flows refer to those that have a random probability distribution or pattern that may be analyzed statistically but may not be predicted precisely. While the described characteristics do involve some level of uncertainty, they do not necessarily imply randomness in the distribution of these cash flows.

Choice D (Credit-related cash flows) is incorrect. Credit-related cash flows primarily relate to the potential losses from a borrower's default on a loan repayment obligation. The characteristics described in the question pertain more towards deterministic nature of certain financial contracts rather than credit risk associated with them.

Things to Remember

- Cash flows are essential in financial risk management as they represent the movement of funds in and out of an organization.

- Deterministic cash flows are crucial for financial planning and analysis as they provide a clear picture of expected future cash inflows and outflows.
 - Stochastic cash flows are often used in risk modeling and simulation to account for uncertainty and variability in cash flow projections.
 - Credit-related cash flows are important in assessing the credit risk associated with financial contracts and the potential impact of default on cash flows.
 - Behavioral cash flows reflect the behavioral patterns of customers and their impact on cash flow dynamics, such as prepayments and early withdrawals.
-

Q.3924 The liquidity option impact is considered in two ways. A student was asked to give the ways, and the following choices were recorded as her responses after repeated trials. Which of the choices give the most accurate answer?

- A. Liquidity impact on the balance sheet, and a (positive or negative) financial impact on cash flows.
- B. A (positive or negative) financial impact and contract deposit rate.
- C. Contract deposit rate and liquidity impact on the balance sheet
- D. Available cash reserves and financial impact

The correct answer is **A**.

The impact of liquidity options is indeed considered in two ways: the impact on the balance sheet and the financial impact, which can be either positive or negative, on cash flows. The balance sheet impact refers to how liquidity options can affect the assets, liabilities, and equity of a company. The financial impact on cash flows refers to how liquidity options can influence the inflow and outflow of cash in a company. Therefore, choice A accurately identifies the two ways in which the impact of liquidity options is considered.

Choice B is incorrect. While a (positive or negative) financial impact is indeed one of the ways in which the impact of liquidity options is considered, contract deposit rate does not directly evaluate the impact of liquidity options. Contract deposit rate refers to the interest rate paid on deposits, which may be influenced by liquidity but it's not a direct measure of its impact.

Choice C is incorrect. Although liquidity's impact on the balance sheet is one way to evaluate its effect, contract deposit rate as explained above does not directly assess this effect.

Choice D is incorrect. Available cash reserves can be an outcome or result of managing

liquidity but it doesn't represent a method for evaluating the impact of liquidity options. The financial impact mentioned here could refer to either positive or negative effects on cash flows, but without specifying that it doesn't fully capture one of the two distinct ways in which these impacts are typically evaluated.

Things to Remember

- Liquidity risk: The risk that a company or individual may not be able to meet short-term financial obligations due to the inability to convert assets into cash quickly enough.
 - Balance sheet: A financial statement that provides a snapshot of a company's financial position at a specific point in time, including its assets, liabilities, and equity.
 - Cash flow: The movement of money into or out of a business, representing the operating, investing, and financing activities that impact the company's liquidity.
 - Contractual obligations: Legal agreements that bind parties to specific terms and conditions, including financial commitments that may impact liquidity.
 - Financial impact analysis: Evaluating the effects of certain decisions or events on a company's financial performance, including the impact on cash flows and overall financial health.
-

Q.3930 Which of the following statements is least accurate about TSECF based on financial risk management?

- A. In TSECF, cash flows are adjusted to include liquidity options, and thus behavioral models are used for typical banking products such as sight deposits, prepaying mortgages, and credit link usage.
- B. TSECF Includes the cash flows from all current contracts that include the assets and liabilities. Cash flows are stochastic in many cases because they link to market indices, such as Libor or Euribor fixings.
- C. Cash flows originated by new business increasing the assets are included; they are typically stochastic in both the amount and time dimensions, so they are treated employing models that consider all related risks.
- D. TSECF does not include the cash flows from all current contracts that include the assets and liabilities.

The correct answer is **D**.

The statement that TSECF does not include the cash flows from all current contracts that include the assets and liabilities is inaccurate. In fact, TSECF does include the cash flows from all current contracts. This includes both assets and liabilities. The cash flows are often stochastic, meaning they are subject to variability due to market indices such as Libor or Euribor fixings. Therefore, the assertion in choice D is incorrect, making it the least accurate statement about TSECF.

Choice A is incorrect. The statement accurately describes the TSECF concept, where cash flows are adjusted to include liquidity options. Behavioral models are indeed used for typical banking products such as sight deposits, prepaying mortgages, and credit link usage.

Choice B is incorrect. This statement correctly explains that TSECF includes cash flows from all current contracts involving assets and liabilities. These cash flows can be stochastic as they often link to market indices like Libor or Euribor fixings.

Choice C is incorrect. The statement correctly mentions that TSECF includes cash flows originated by new business increasing the assets which are typically stochastic in both amount and time dimensions. Models considering all related risks are used to treat these types of cash flows.

Things to Remember

- TSECF stands for Total Stock and Equity Cash Flow, which is a concept used in financial risk management to analyze and manage cash flows related to assets and liabilities.
 - Cash flows in TSECF can be adjusted to include liquidity options, and behavioral models are often used to analyze typical banking products.
 - Stochastic cash flows in TSECF are subject to variability and are often linked to market indices like Libor or Euribor fixings.
 - New business cash flows that increase assets are included in TSECF analysis, and they are typically stochastic in both amount and time dimensions.
 - Models that consider all related risks are used to treat stochastic cash flows in TSECF.
-

Q.3931 Suppose that you are given assets, liabilities, and their respective expiry terms of a financial institution, as shown in the table below. Given that the assets bear no default risk, no liquidity options are embedded within deposits, and the assets and the liabilities have been ordered according to their maturity, disregarding which kind of contract they are. Study the table to build the TSECF for the institution.

Expiry	Notional Positive Cashflows	Interest on Positive Cashflows	Notional on Negative Cashflows	Interest on Negative Cashflows
1	25	7	0	-5
2	0	5	-20	-6
3	0	5	0	-3
4	0	5	0	-3
5	60	8	-35	-3
6	0	2	0	-3
7	0	2	-20	-3
8	0	2	0	0
9	0	2	0	0
10	15	2	0	0
> 10	0	-	-25	0

Calculate the TSECCF for the assets and liabilities with a term of two years to expiry.

- A. -21
- B. 6
- C. 7
- D. 27

The correct answer is **B**.

The term structure of cumulated expected cash flows (TSECCF) is the collection of expected cumulated cash flows, starting at t_0 and ending at t_b ordered by date.

$$\text{TSECF for year 1} = 25 + 7 + 0 + -5 = 27$$

$$\text{TSECCF for year 2} = 27 + 5 - 20 - 6 = 6$$

Q.3932 Suppose that you are given assets, liabilities, and their respective expiry terms of a financial institution, as shown in the table below. Given that the assets bear no default risk, no liquidity options are embedded within deposits, and the assets and the liabilities have been

ordered according to their maturity, disregarding which kind of contract they are. Study the table to build the TSECF for the institution.

Expiry	Notional Positive Cashflows	Interest on Positive Cashflows	Notional on Negative Cashflows	Interest on Negative Cashflows
1	25	7	0	-5
2	0	5	-20	-6
3	0	5	0	-3
4	0	5	0	-3
5	60	8	-35	-3
6	0	2	0	-3
7	0	2	-20	-3
8	0	2	0	0
9	0	2	0	0
10	15	2	0	0
> 10	0	-	-25	0

Calculate the TSECCF for the assets and liabilities with a term of more than 10 years to expiry.

- A. 10
- B. 14
- C. 27
- D. 39

The correct answer is **B**.

In generating the TSECCF, we calculate the TSECF for the first year and then cumulate the amounts until the 10th year following TSECCF as follows,

$$\text{TSECCF}(t_0, t_{>10}) = \{CF(t_0, t_0, t_1), CF(t_0, t_0, t_2), \dots, CF(t_0, t_0, t_{>10})\}$$

This is as calculated in the following table.

Expiry	Notional Positive Cashflows	Interest on Positive Cashflows	Notional on Negative Cashflows	Interest on Negative Cashflows	TSECCF
1	25	7	0	-5	27
2	0	5	-20	-6	6
3	0	5	0	-3	8
4	0	5	0	-3	10
5	60	8	-35	-3	40
6	0	2	0	-3	39
7	0	2	-20	-3	18
8	0	2	0	0	20
9	0	2	0	0	22
10	15	2	0	0	39
> 10	0	-	-25	0	14

Q.3935 Assume that a bank decides to sell a quantity of the bond equal to a notional of 300,000 after 10 months (or 0.833 years) for 99.95. Keeping in mind that this trade can be dealt with to generate liquidity, calculate the term structure of cumulated liquidity generation capacity (TSCLGC) inflow resulting from this transaction given the interest rate is 10%.

- A. 324,840
- B. 324,890
- C. 494,675
- D. 543,785

The correct answer is **A**.

The calculations are as follows;

The term structure of cumulated liquidity generation capacity (TSCLGC) records an inflow equivalent to:

$$\text{Inflow} = \text{Amount} \times \text{price, including the accrued interest}$$

$$300,000 \times \left(\frac{99.95}{100} + 10\% \times 0.833 \right) = 324,840$$

Q.3936 A liquidity division in XYZ bank operates a buy/sell back operation. Assume that the bank buys a bond worth 200,000, 3 months (0.25 years) after it is issued with an interest rate of 10%

and sells it back after 6 months at the forward price. Assume that the purchase price is 99.50 of the par value. Calculate the amount the bank pays at the start of the contract.

- A. 204,000
- B. 309,400
- C. 403,600
- D. 199,000

The correct answer is **A**.

To calculate the amount the bank pays at the start of the contract, we need to consider the purchase price as a percentage of the par value and the accrued interest on the bond for the 3 months (0.25 years) since it was issued.

The purchase price is 99.50% of the par value, which is:

$$\text{Purchase Price} = \text{Par Value} \times \text{Percentage Price} = 200,000 \times 0.995 = \$199,000$$

The accrued interest on the bond for 3 months at an annual interest rate of 10% is:

$$\text{Accrued Interest} = \text{Par Value} \times \text{Annual Interest Rate} \times \text{Fraction of Year} = 200,000 \times 0.10 \times 0.25 =$$

Therefore, the total amount the bank pays at the start of the contract is the sum of the purchase price and the accrued interest:

$$\text{Total Amount Paid} = \text{Purchase Price} + \text{Accrued Interest} = \$199,000 + \$5,000 = \$204,000$$

So, the bank pays \$204,000 at the start of the contract.

Things to Remember

- **Buy/Sell Back Operations:** These are transactions where one party sells a security with a commitment to repurchase it at a future date. It is a common way for banks to manage their liquidity needs.
- **Par Value:** The face value of a bond, which is the amount the issuer agrees to repay the bondholder at maturity.
- **Accrued Interest:** The interest that has accumulated on a bond since the last interest payment date.

- Forward Price: The price at which a contract to buy or sell a security is agreed upon today for a future delivery date.
 - Percentage Price: The price of a bond expressed as a percentage of its par value.
-

Q.3938 A liquidity division in XYZ bank operates a buy/sell back operation. The bank buys a bond worth 200,000 with coupons of 10% per annum paid semi-annually and immediately gets into a forward contract to sell it back after six months. Assume that the purchase price is 99.50 of the par value and the six-month forward price is 99.90 of the par value. Calculate the total amount received by the bank after six months.

- A. 202,000
- B. 209,800
- C. 406,780
- D. D. 409,600

The correct answer is **B**.

Remember that the sum it receives includes accrued interest:

$$200,000 \times (99.90\% + 10\% \times 0.50) = 209,800$$

Q.3939 A liquidity division for RPQ bank decides to lend 600,000 of a bond after 3 months from purchase for a period of 6 months. In the process, they realize that The TSECF does not record any cash flow at the inception of the contract, whereas the TSAA shows a reduction of the available quantity of 600,000. After 6 months, the bond is returned to the bank whereby the bank receives a fee for the lending, which we assume to be equal to 3.5% p.a. Calculate the value of the received fee for the lending.

A. 4,800

B. 6,700

C. 10,500

D. 4,200

The correct answer is **C**.

The received fee for lending is calculated as follows:

$$600,000 \times (3.5\% \times 0.5) = 10,500$$

Note that 600,000 represents the notional, 3.5% is the interest rate, and 0.5 represents the maturity(half-year).

Reading 134: The Failure Mechanics of Dealer Banks

Q.2292 Coliseu Bank from Porto, Portugal, is a large dealer bank considered “too big to fail” and, therefore, can expect support from the government and central bank in case of distress. This support “guarantee” is associated with certain behaviors that may have potentially disastrous consequences. Which of the following gives an example of such behaviors?

- A. Manipulation of financial statements.
- B. Taking more inefficient risks.
- C. Offering products at exorbitant prices.
- D. Only doing business with high net-worth firms and individuals.

The correct answer is **B**.

The term 'too big to fail' refers to the concept that certain financial institutions are so large and interconnected that their failure would be disastrous to the economy, and they must therefore be supported by the government when they face potential failure. The implicit guarantee of government support can lead these institutions to take on more inefficient risks. This is because they may operate under the assumption that they will be rescued in the event of a crisis, thereby reducing their incentives to avoid risky behavior. This moral hazard problem is a major concern in financial regulation. The expectation of being bailed out if things go wrong can lead to excessive risk-taking, which can increase the likelihood of financial crises. In the case of Coliseu Bank, the expectation of government support in times of distress could potentially lead it to take on more inefficient risks, knowing that it will likely be bailed out if these risks do not pay off.

Choice A is incorrect. While manipulation of financial statements is a fraudulent activity and can lead to severe consequences, it's not directly related to the 'too big to fail' status of a bank. This behavior can occur in any organization, regardless of its size or status.

Choice C is incorrect. Offering products at exorbitant prices may be an unethical business practice but it doesn't necessarily result from the 'too big to fail' status. Banks might charge high prices due to various reasons such as high operational costs, market conditions or lack of competition but not specifically because they are considered too big to fail.

Choice D is incorrect. Only doing business with high net-worth firms and individuals does not necessarily lead to catastrophic outcomes nor does it stem from the 'too big to fail' status. It could be part of a bank's business strategy targeting premium segment but this doesn't inherently pose systemic risk associated with too-big-to-fail institutions.

Things to Remember

- **Moral Hazard:** The concept of moral hazard refers to the tendency of individuals or institutions to take on more risk because they are protected from the consequences of that risk. In the context of 'too big to fail' banks, the implicit guarantee of government

support can create a moral hazard problem where these institutions may engage in riskier behavior.

- **Systemic Risk:** Systemic risk is the risk that an event at the individual level can have widespread adverse effects on the entire financial system. 'Too big to fail' banks are often considered sources of systemic risk due to their size and interconnectedness.
 - **Financial Regulation:** Financial regulation refers to the rules and laws that govern the operation of financial institutions and markets. Regulators aim to maintain stability, protect consumers, and prevent systemic risks in the financial system.
 - **Market Discipline:** Market discipline is the idea that market forces, such as competition and transparency, can influence the behavior of financial institutions. It is seen as a complement to regulatory oversight in promoting sound risk management practices.
-

Q.2294 Bens bank from Brussels, Belgium, is a dealer bank. It operates in both primary and secondary markets. In the primary market, the bank most likely intermediates between:

- A. Issuers and investors of securities
- B. Sellers and buyers
- C. Other banks
- D. Corporate clients

The correct answer is **A**.

Dealer banks, such as Bens bank, play a crucial role in the primary market by acting as intermediaries between issuers and investors of securities. In this role, the dealer bank may sometimes act as an underwriter. This involves buying equities or bonds from an issuer and then selling them over time to investors. This process is essential for the functioning of the primary market as it facilitates the flow of capital from investors to issuers, who may need the funds for various purposes such as business expansion or debt repayment. The dealer bank earns a profit from the spread between the price at which it buys the securities from the issuer and the price at which it sells them to the investors.

Choice B is incorrect. While it's true that banks often act as intermediaries between sellers and buyers, this is a broad interpretation of the role of a bank. In the context of primary financial

markets, the bank specifically intermediates between issuers and investors of securities, not just any sellers and buyers.

Choice C is incorrect. Banks do interact with other banks in various capacities, but in the context of primary financial markets, their role as intermediaries isn't typically focused on other banks. Instead, they facilitate transactions between issuers who want to raise capital by selling securities and investors who are willing to buy these securities.

Choice D is incorrect. Although corporate clients can be either issuers or investors in the primary market scenario described here, this choice doesn't fully capture the intermediary role played by Bens Bank. The bank acts as an intermediary specifically between issuers (which could be corporations) and investors (which could also include corporations), not just with corporate clients in general.

Things to Remember

- **Primary Market:** The primary market is where new securities are issued and sold for the first time. It is where companies raise capital by issuing stocks or bonds to investors.
- **Secondary Market:** The secondary market is where existing securities are bought and sold among investors. It provides liquidity to investors who want to sell their securities.
- **Underwriting:** Underwriting is the process by which a dealer bank or financial institution agrees to buy a certain number of securities from an issuer at a specified price and then resell them to investors.
- **Spread:** The spread is the difference between the price at which a dealer bank buys securities from an issuer and the price at which it sells them to investors. It represents the dealer bank's profit.
- **Issuer:** An issuer is a company or entity that offers securities for sale to raise capital. Issuers can be corporations, governments, or other organizations.
- **Investors:** Investors are individuals or institutions that buy securities issued by companies or governments. They provide capital to issuers in exchange for the potential for returns.

Q.2295 Kralingen Bank from Rotterdam is a dealer bank. It operates in both primary and secondary markets. In the latter, the bank most likely intermediates between:

- A. Issuers and investors of securities
- B. Sellers and buyers
- C. Other intermediaries
- D. Corporate clients

The correct answer is **B**.

In the secondary market, dealer banks like Kralingen Bank act as intermediaries between sellers and buyers. This is because the secondary market is where previously issued securities are bought and sold. The dealer bank stands ready to buy securities at its bid prices from sellers and sell securities at its ask prices to buyers. This intermediation process facilitates the smooth functioning of the secondary market by providing liquidity and ensuring that securities can be bought and sold at any time. The dealer bank profits from the spread between the bid and ask prices.

Choice A is incorrect. Issuers and investors of securities are typically involved in primary markets, not secondary markets. In primary markets, new securities are issued and sold to investors. Kralingen Bank's role as an intermediary in the secondary market does not involve direct interaction with issuers of securities.

Choice C is incorrect. While it's possible for a dealer bank like Kralingen Bank to interact with other intermediaries, this isn't the most likely scenario in the context of its operations in the secondary market. The main function of a dealer bank in a secondary market is to facilitate transactions between buyers and sellers.

Choice D is incorrect. Corporate clients could be either buyers or sellers in the secondary market, but they do not represent both parties that Kralingen Bank would act as an intermediary between. The bank's role as an intermediary involves facilitating transactions between sellers and buyers specifically.

Things to Remember

- Primary Market: Where new securities are issued and sold to investors.
- Secondary Market: Where previously issued securities are bought and sold.
- Dealer Banks: Act as intermediaries in the secondary market between sellers and buyers.
- Liquidity: Provided by dealer banks in the secondary market to ensure smooth transactions.
- Spread: The difference between bid and ask prices that dealer banks profit from.

Q.2296 Arrenberg Bank is a large dealer bank performing many over-the-counter trades. What is the manner in which its over-the-counter trades are negotiated?

- A. Trades are negotiated both publicly or privately.
- B. Trades are negotiated publicly.
- C. Trades are negotiated privately.
- D. Trades are negotiated publicly with some portions of the contract kept secret.

The correct answer is **C**.

Over-the-counter trades are negotiated privately. Dealer banks, such as Arrenberg Bank, play a significant role in the intermediation of over-the-counter securities markets. These markets encompass bonds issued by corporations, municipalities, certain national governments, and securitized credit products. The nature of over-the-counter trades is such that they are privately negotiated between two parties without the oversight of an exchange. This allows for greater flexibility in terms of the terms and conditions of the trade, as well as the ability to tailor the trade to the specific needs and risk profiles of the parties involved. However, this also means that there is a higher degree of risk involved, as there is less transparency and regulatory oversight compared to trades conducted on an exchange.

Choice A is incorrect. Over-the-counter trades are not negotiated both publicly and privately. They are typically negotiated privately between two parties without the supervision of an exchange.

Choice B is incorrect. Over-the-counter trades are not negotiated publicly. These types of trades occur directly between two parties, away from the public eye and without the oversight of an exchange.

Choice D is incorrect. While it's true that some portions of a contract may be kept secret in over-the-counter trading, it's misleading to say that these trades are negotiated publicly with some portions kept secret. The negotiation process for over-the-counter trades typically occurs in private, away from public scrutiny or intervention.

Things to Remember

- Over-the-counter (OTC) trades are negotiated directly between two parties without the oversight of an exchange.
- Dealer banks play a significant role in intermediating OTC securities markets.
- OTC markets include bonds issued by corporations, municipalities, certain national governments, and securitized credit products.
- OTC trades offer flexibility in terms of trade terms and conditions, tailored to specific

needs and risk profiles.

- OTC trades involve higher risk due to less transparency and regulatory oversight compared to exchange-traded trades.
-

Q.2297 Grochowska Bank from Poznan, Poland, is intermediating for repurchase agreements (repos). In the Bank's repo agreements, what does the term "haircut" mean?

- A. "Haircut" is not used in repo agreements.
- B. A reduction in returns of a repo.
- C. An increase in returns of a repo.
- D. A form of risk mitigation for a repo.

The correct answer is **D**.

Securities dealers intermediate in the market for repurchase agreements or "repos." A repo is, in essence, a short-term cash loan collateralized by securities. One counterparty borrows cash from the other, and as collateral against performance on the loan, that counterparty posts government bonds, corporate bonds, securities from government-sponsored enterprises, or other securities such as collateralized debt obligations. The performance risk on a repo is typically mitigated by a "haircut" that reflects the risk or liquidity of the securities.

Things to Remember

1. Repurchase agreements (repos) are essentially short-term cash loans that are collateralized by securities. They are a common tool used in the financial industry for short-term borrowing.
 2. In a repo, one party borrows cash from the other and posts securities as collateral against the loan. The securities can be government bonds, corporate bonds, securities from government-sponsored enterprises, or other securities like collateralized debt obligations.
 3. The term 'haircut' in the context of repos refers to a reduction in the value of the collateral compared to the value of the loan. It is a risk mitigation measure designed to protect the lender from the risk of a decline in the market value of the collateral.
 4. The size of the 'haircut' reflects the perceived risk or liquidity of the securities being used as collateral. Riskier or less liquid securities would typically require a larger 'haircut'.
-

Q.2298 Which of the following best explains how dealer banks benefit from over-the-counter derivative trading?

- A. Charging transaction fees from investors in the form of the bid-ask spread.
- B. Taking almost equal but opposite trading positions.
- C. Charging a monthly premium with respect to every client.
- D. Selling the underlying commodities at a profit.

The correct answer is **A**.

Dealer banks primarily benefit from over-the-counter derivative trading by charging transaction fees from investors in the form of the bid-ask spread. The bid-ask spread is the difference between the highest price that a buyer is willing to pay for an asset and the lowest price that a seller is willing to accept. An individual looking to buy will find a sell order at the lowest ask, while a seller will find a buy order at the highest bid. This difference between the bid and ask prices is the transaction cost, which forms a significant part of the dealer banks' revenues. The larger the spread, the higher the potential profit for the dealer bank. This is a common practice in the financial markets and forms a significant part of the revenue for dealer banks involved in OTC derivative trading.

Choice B is incorrect. While dealer banks may sometimes take nearly equal but opposite trading positions, this is not the primary way they profit from OTC derivative trading. This strategy, known as a hedging strategy, is used to mitigate risk rather than generate profits.

Choice C is incorrect. Charging a monthly premium with respect to every client does not accurately describe how dealer banks primarily profit from OTC derivative trading. While some banks may charge fees or premiums for certain services, these are typically not the main source of income in OTC derivative trading.

Choice D is incorrect. Selling the underlying commodities at a profit does not represent the primary way in which dealer banks profit from their involvement in over-the-counter derivative trading. In fact, many derivatives do not involve physical delivery of any commodity and are settled financially instead.

Things to Remember

- Over-the-counter (OTC) derivative trading involves customized contracts between two parties, typically facilitated by dealer banks.
- Bid-ask spread is the difference between the buying price (bid) and selling price (ask) of a security or asset.
- Dealer banks act as intermediaries in OTC derivative trading, providing liquidity and market-making services.
- Hedging is a risk management strategy that involves taking offsetting positions to

reduce the impact of potential losses.

- OTC derivatives can be based on various underlying assets such as commodities, currencies, interest rates, or equities.
-

Q.2299 Utvin Bank from Timisoara, Romania, trades in over-the-counter derivatives as a dealer. What's the primary source of profit for the bank?

- A. Transaction commission
- B. Differences between bid and offer terms
- C. Interest rate
- D. Change in value of securities

The correct answer is **B**.

The primary source of profit for a dealer in the over-the-counter derivatives market, such as Utvin Bank, is the differences between bid and offer terms. This is often achieved by running a 'matched book'. In this strategy, the dealer aims to offset the risk of its client-initiated derivatives positions by seeking trades that counterbalance each other. Essentially, the dealer profits from the spread between the buying (bid) and selling (offer) prices of the derivatives. This is similar to how dealer banks conduct proprietary trading in over-the-counter derivatives markets, where they trade securities using their own funds, rather than the clients', to earn a profit for themselves.

Choice A is incorrect. While transaction commissions can be a source of profit for some financial institutions, it is not the main avenue through which a dealer in the over-the-counter derivatives market, like Utvin Bank, generates its profits. Dealers typically earn their profits from the spread between their buying and selling prices.

Choice C is incorrect. Interest rates can affect the profitability of financial institutions but they are not directly related to how a dealer in an over-the-counter derivatives market makes profits. The main profit comes from bid-offer spreads rather than changes in interest rates.

Choice D is incorrect. Changes in value of securities can lead to gains or losses for dealers but this isn't their primary source of income. Dealers primarily make money from bid-offer spreads, not from holding securities and benefiting (or suffering) from price changes.

Things to Remember

- Over-the-counter (OTC) derivatives are traded directly between two parties without going through an exchange or other intermediary.

- A 'matched book' strategy involves offsetting the risk of client-initiated derivatives positions by seeking trades that counterbalance each other.
 - Dealer banks conduct proprietary trading in OTC derivatives markets using their own funds to earn profits.
 - Transaction commissions can be a source of profit for some financial institutions, but not the primary source for OTC derivatives dealers.
 - Interest rates can impact the profitability of financial institutions, but they are not the main source of profit for OTC derivatives dealers.
 - Changes in the value of securities can lead to gains or losses for dealers, but it is not their primary source of income in OTC derivatives trading.
-

Q.2300 Which of the following derivatives are widely held by banks thanks to the nature of their business?

- A. Futures contracts
- B. Options contracts
- C. Forward contracts
- D. Interest-rate swaps

The correct answer is **D**.

Banks are heavily involved in the business of lending and borrowing money. They borrow money from depositors and lend it to borrowers. The interest rate they charge to borrowers is usually higher than the interest rate they pay to depositors. This difference in interest rates is a major source of profit for banks. However, changes in interest rates can affect this profit margin. To manage this risk, banks use interest-rate swaps. An interest-rate swap is a derivative in which two parties exchange interest payments. One party pays a fixed interest rate, and the other pays a variable interest rate. The variable rate is often based on a benchmark rate, such as the London Interbank Offered Rate (LIBOR). By using interest-rate swaps, banks can hedge against the risk of changes in interest rates. Therefore, due to the nature of their business, banks widely hold interest-rate swaps.

Choice A is incorrect. Futures contracts are not the most commonly held derivatives by banks. While futures can be used to hedge risk and speculate on future price movements, they are standardized contracts traded on an exchange, which limits their flexibility in terms of contract

size and maturity dates. This makes them less suitable for the specific needs of a bank's business operations.

Choice B is incorrect. Options contracts give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified price within a certain period of time. Although options can provide significant leverage and risk management benefits, they are not as commonly used by banks due to their complexity and higher costs compared to other derivatives.

Choice C is incorrect. Forward contracts are private agreements between two parties to buy or sell an asset at a specified future date for a price agreed upon today. While forwards can be customized according to the needs of each party, they carry counterparty risk since there's no clearinghouse guaranteeing performance of each party as in futures contracts. Therefore, forward contracts are also not as commonly held by banks.

Things to Remember

- Interest rate risk management is crucial for banks due to their business of borrowing and lending money.
 - Interest-rate swaps are commonly used by banks to hedge against changes in interest rates.
 - Futures contracts are standardized and traded on exchanges, limiting their flexibility for banks.
 - Options contracts give the holder the right, but not the obligation, to buy or sell an underlying asset.
 - Forward contracts are private agreements between two parties to buy or sell an asset at a specified future date.
 - Counterparty risk is a concern with forward contracts as there is no clearinghouse guaranteeing performance.
-

Q.2303 Bodrum bank has an exposure of \$20m with respect to Aurora bank. The latter has posted collateral worth \$15m. In case Aurora bank defaults, how much does Bodrum stand to lose?

- A. \$10m
- B. \$20m
- C. \$15m
- D. \$5m

The correct answer is **D**.

A useful gauge of counterparty risk in the over-the-counter market is the amount of exposure to default presented by the failure of counterparties to perform their contractual obligations. These exposures can be reduced through collateral.

In this example, Bodrum has covered only a part of its exposure to Aurora. In case the latter defaults, the former would only recover a maximum of \$15m by selling the collateral, thus would lose (\$20m - \$15m).

Things to Remember

- Counterparty risk is the risk that the other party in a financial transaction may default or fail to fulfill their obligations.
 - Exposure is the potential loss that an investor faces in a financial transaction.
 - Collateral is an asset or property that is pledged by a borrower to secure a loan or other credit, and is forfeited in the event of default.
 - Collateral can help reduce counterparty risk by providing a source of recovery in case of default.
 - Net exposure is the total exposure to a counterparty minus any collateral or hedges that have been posted.
-

Q.2305 Bank A is a dealer bank. It is headed in the direction of a solvency crisis. The bank's counterparties in over-the-counter are alarmed and are hastily harvesting assets from the bank to reduce their exposure. What can the bank do to regain the confidence of its counterparties?

- A. Refuse some trades with relatively smaller counterparties.
- B. Allow all trades with counterparties based on terms prevailing in the market.
- C. Borrow short-term.
- D. Hiding some transactions.

The correct answer is **B**.

If a dealer bank is perceived to have some risk of a solvency crisis, an over-the-counter derivatives counterparty would look for opportunities to reduce its exposure to that dealer bank. If the dealer wants to avoid an adverse signal of its weakness, the dealer cannot afford to refuse its counterparties the opportunity to make these trades at terms prevailing elsewhere in the market.

Therefore, the bank will be forced to allow all trades with the counterparties or risk triggering a death spiral.

Things to Remember

- Dealer Banks: These are financial institutions that engage in trading activities, especially in over-the-counter markets, acting as intermediaries between buyers and sellers.
- Solvency Crisis: A situation where a bank or financial institution is at risk of becoming insolvent, meaning it may not be able to meet its financial obligations.
- Counterparties: These are the parties on the other side of a financial transaction, such as trades in over-the-counter derivatives.
- Exposure: The amount of risk faced by a bank or financial institution due to its various transactions and obligations.
- Short-term Borrowing: A strategy where a bank borrows funds for a short period to meet its immediate liquidity needs.
- Transparency: The importance of being open and transparent in financial transactions to maintain trust and confidence among counterparties.

Q.2983 The following standard policy tools are applicable in the treating of the social costs of bank failures. Which one is NOT?

- A. Regulatory supervisions and the requirements for risk-based capital, which reduces the chance of a solvency threatening capital loss.
- B. Deposit insurance, a measure implemented in to protect bank depositors.
- C. Regulatory resolutions mechanisms, through which the authorities are given powers to liquidate or restructure a bank efficiently.
- D. Regulatory requirements guiding the departure of prime brokerage customers.

The correct answer is **D**.

Regulatory requirements guiding the departure of prime brokerage customers is not a standard policy tool used in treating the social costs of bank failures. This option refers to the rules and regulations that govern the exit of prime brokerage customers. While these regulations are important in the context of prime brokerage operations, they do not directly address the social costs of bank failures. The social costs of bank failures are typically addressed through measures aimed at preventing bank failures (such as regulatory supervision and risk-based capital requirements), protecting depositors (such as deposit insurance), and managing the resolution of failed banks (such as regulatory resolution mechanisms). Therefore, while regulatory requirements for prime brokerage customers are important in their own right, they do not directly address the social costs of bank failures and are not typically considered a standard policy tool in this context.

Choice A is incorrect. Regulatory supervisions and the requirements for risk-based capital are indeed standard policy tools used to mitigate the social costs associated with bank failures. They reduce the chance of a solvency threatening capital loss by ensuring that banks maintain adequate levels of capital based on their risk profile.

Choice B is incorrect. Deposit insurance is another common policy tool used in this context. It protects bank depositors from losses if a bank fails, thereby maintaining confidence in the banking system and preventing bank runs which can exacerbate financial crises.

Choice C is incorrect. Regulatory resolution mechanisms are also typically employed to manage the fallout from bank failures. These mechanisms give authorities powers to liquidate or restructure a failing bank efficiently, minimizing disruption to the broader economy and financial system.

Things to Remember

- **Regulatory Capital Requirements:** These are rules set by regulators that require banks to hold a certain amount of capital based on their risk profile to ensure they can absorb losses.
- **Bank Solvency:** Refers to a bank's ability to meet its financial obligations and maintain

a positive net worth, crucial for financial stability.

- Bank Liquidity: The ability of a bank to meet its short-term obligations and fund its operations without incurring significant losses.
 - Bank Resolution: The process of managing a failing bank, either through liquidation or restructuring, to minimize the impact on the financial system.
-

Q.2984 Dealer banks, like any other financial institutions, can face collapse due to a variety of factors. These factors can range from internal mismanagement to external market forces. However, not all mechanisms contribute to the collapse of a dealer bank. Which one of the following mechanisms is NOT typically associated with the collapse of a dealer bank?

- A. The flight of short-term creditors.
- B. Prime brokerage clients' departure.
- C. Various cash-draining undertakings by derivatives counterparties designed to lower their exposures to the dealer banks.
- D. Additional clearing-bank privileges.

The correct answer is **D**.

The failure of a dealer bank can be due to key mechanisms which are: the flight of short-term creditors, the departure of prime brokerage clients, various cash-draining undertakings by derivatives counterparties designed to lower their exposures to the dealer banks, and the loss of clearing-bank privileges.

Things to Remember

- Dealer banks play a crucial role in the financial system by providing liquidity, market-making services, and facilitating trading activities.
- Short-term creditors can quickly withdraw their funds from a dealer bank, leading to a liquidity crisis and potential collapse.
- Prime brokerage clients are important sources of revenue for dealer banks, and their departure can signal underlying issues within the institution.
- Cash-draining undertakings by derivatives counterparties can put pressure on a dealer bank's balance sheet and capital reserves.
- Clearing-bank privileges are essential for a dealer bank to settle transactions efficiently and maintain market confidence.
- Risk management practices, capital adequacy, and regulatory compliance are also critical factors in the stability of dealer banks.

Q.5381 Below is a table displaying the financing in the repo market at the end of the quarter for four broker-dealer banks with A ratings. The figures are in GBP billion.

Financial Security	Bank A	Bank B	Bank C	Bank D
Owned	240	189	320	280
Pledged as security	150	110	180	170
Not pledged	90	79	140	110

Which bank is the most susceptible to a liquidity crisis?

- A. Bank A
- B. Bank B
- C. Bank C
- D. Bank D

The correct answer is **A**.

If repo creditors grow apprehensive about a bank's solvency and opt not to renew their positions, it could trigger a liquidity crisis. Consequently, the bank might face difficulties in raising adequate cash, thereby exacerbating the crisis.

To manage the situation, the bank may be compelled to hastily sell its assets to buyers who are aware of its urgency, resulting in lower market valuations for unsold securities. Consequently, the amount of cash that could be raised through repurchase agreements collateralized by those securities may decrease.

This vulnerability is directly linked to the proportion of assets that a bank has pledged as collateral. Bank A would therefore be most susceptible to a liquidity crisis.

$$\begin{aligned} \text{Bank A} &= \frac{150}{240} = 62.5\% \\ \text{Bank B} &= \frac{110}{189} = 58.2\% \\ \text{Bank C} &= \frac{180}{320} = 56.2\% \\ \text{Bank D} &= \frac{170}{280} = 60.7\% \end{aligned}$$

Reading 135: Liquidity Stress Testing

Q.4125 The stress testing committee for the RTC bank is preparing to undertake a stress test for the bank. Which of the following factors is unlikely to be considered in the stress test process?

- A. Development of critical assumptions
- B. Scenario development
- C. The suitable scope and structure of the liquidity stress test
- D. Profit margins related to the process

The correct answer is **D**.

Profit margins related to the process are not typically considered during the stress testing process. Stress testing in banking is a simulation technique used to evaluate a bank's potential vulnerability to certain unlikely but severe events or market conditions. The primary focus of stress testing is to assess the bank's resilience in the face of adverse scenarios. It involves the evaluation of the bank's capital adequacy, exposure to risk, and the potential impact of various risk factors on the bank's financial condition. While profit margins are an important aspect of a bank's overall financial health, they are not directly related to the stress testing process. The stress test is more concerned with the bank's ability to withstand losses and continue its operations under stressful conditions, rather than its profitability.

Choice A is incorrect. The development of critical assumptions is a crucial part of stress test planning and execution process. These assumptions help in determining the potential impact of various risk factors on the bank's operations.

Choice B is incorrect. Scenario development is also an integral part of stress testing. It involves creating hypothetical situations that could potentially affect the bank's operations, and then assessing how well the bank would be able to handle these situations.

Choice C is incorrect. Determining the suitable scope and structure of the liquidity stress test forms a key component in planning a comprehensive stress test for any banking institution. This helps in understanding how much liquid assets are available to meet short-term obligations under different scenarios.

Things to Remember

- Stress testing is a key tool used by banks to assess their resilience to adverse scenarios and market conditions.
- Stress testing helps banks evaluate their capital adequacy, risk exposure, and potential impact of various risk factors on their financial condition.
- Development of critical assumptions is crucial in stress testing to determine the

potential impact of risk factors on the bank's operations.

- Scenario development involves creating hypothetical situations to assess the bank's ability to handle adverse conditions.
 - The scope and structure of liquidity stress tests are important in understanding a bank's ability to meet short-term obligations under different scenarios.
-

Q.4127 Axon, a fictional financial institution in Liverpool, has been using its available liquidity for funding of the business and doing orderly clearance of payment transactions. Under which category of liquidity use does this purpose fall?

- A. Contingent liquidity
- B. Operational liquidity
- C. Strategic liquidity
- D. Restricted liquidity

The correct answer is **B**.

Operational liquidity refers to the cash required for the daily funding of business operations and the orderly clearance of payment transactions. In the context of a financial institution, operational liquidity is crucial for maintaining the smooth functioning of the institution's daily activities. This includes, but is not limited to, the clearance of payment transactions, the funding of loans, and the management of deposits. Operational liquidity is a key component of a financial institution's liquidity management strategy, as it ensures that the institution has sufficient cash on hand to meet its immediate operational needs. In the case of Axon, the use of available liquidity for the funding of the business and the orderly clearance of payment transactions falls squarely within the definition of operational liquidity.

Choice A is incorrect. Contingent liquidity refers to the funds that a financial institution sets aside to meet unexpected cash flow needs or obligations, not for regular business operations or payment transactions.

Choice C is incorrect. Strategic liquidity is the cash and other liquid assets held by a financial institution for strategic purposes such as acquisitions, new ventures, or capital investments. It does not refer to the funds used for daily operational activities like clearing payment transactions.

Choice D is incorrect. Restricted liquidity refers to assets that are not readily convertible into cash due to legal or contractual constraints. It does not pertain to the use of available liquidity for funding business operations and ensuring smooth clearance of payment transactions.

Things to Remember

- Contingent liquidity is reserved for unexpected cash flow needs or obligations, not for regular business operations.
 - Strategic liquidity is held for strategic purposes like acquisitions or investments, not for daily operational activities.
 - Restricted liquidity refers to assets that are not easily converted into cash due to legal or contractual constraints.
 - Operational liquidity is crucial for daily funding of business operations and payment transactions.
 - Operational liquidity ensures smooth functioning of daily activities in a financial institution.
-

Q.4129 Which of the following choices is right about designing a stress testing model?

- A. Liquidity stress testing begins with identifying the risk and doing event analysis
- B. In liquidity stress testing the list of scenarios don't need to capture material liquidity appropriately
- C. Liquidity stress testing process begins with coming up with assumption
- D. The liquidity stress testing process should begin with developing a list of scenarios capturing material liquidity.

The correct answer is **A**.

Liquidity stress testing indeed begins with the identification of the risk and conducting an event analysis. This is a crucial first step in the process as it allows financial institutions to understand the potential risks they may face and the events that could trigger these risks. Event analysis involves examining past events that have led to liquidity stress and using this information to inform the design of the stress testing model. This step is critical in ensuring that the model is based on realistic and relevant scenarios, thereby enhancing its effectiveness in assessing the institution's liquidity risk.

Choice B is incorrect. This statement contradicts the fundamental purpose of liquidity stress testing. The list of scenarios in a liquidity stress test should indeed capture material liquidity risks appropriately, as this is crucial for assessing the financial institution's ability to maintain

sufficient liquidity under stress scenarios.

Choice C is incorrect. While assumptions are an integral part of any modeling process, they are not typically the initial step in designing a liquidity stress testing model. The process usually begins with identifying the risk and conducting an event analysis.

Choice D is incorrect. Although developing a list of scenarios capturing material liquidity risks forms part of the process, it does not constitute the initial steps involved in designing a liquidity stress testing model. As mentioned earlier, identifying risk and conducting event analysis precedes scenario development.

Things to Remember

- Stress testing is a risk management tool used to evaluate the potential impact of adverse events or scenarios on a financial institution's financial condition.
 - Liquidity stress testing focuses specifically on assessing an institution's ability to meet its short-term obligations in times of stress or crisis.
 - Scenario analysis is a key component of stress testing, involving the development of hypothetical scenarios that could impact the institution's liquidity position.
 - Stress testing models should be dynamic and regularly updated to reflect changes in the financial environment and the institution's risk profile.
 - Regulatory authorities often require financial institutions to conduct stress tests to ensure they have adequate liquidity and capital to withstand adverse events.
-

Q.4131 Afro Bank, a fictional financial institution in Turkey, has liquidity transfer restrictions. Which one of the following choices states the best initiative the bank should take in its liquidity stress testing?

- A. The bank should source help to emancipate from the transfer restrictions
- B. It should assess the effect of the restrictions on enterprise-level liquidity for both typical operating environment and stressed conditions
- C. The bank should stop the liquidity stress testing scenario until the end of the restrictions
- D. The bank should source other types of liquidity to include in the liquidity stress testing menu

The correct answer is **B**.

The bank should assess the effect of the restrictions on enterprise-level liquidity for both typical operating environment and stressed conditions. Liquidity stress testing is a critical tool for financial institutions to identify potential vulnerabilities in their liquidity risk management. It involves creating hypothetical scenarios to evaluate how an institution's liquidity position would fare under both normal and adverse conditions. In the case of Afro Bank, the liquidity transfer restrictions could significantly impact its liquidity position. Therefore, it is crucial for the bank to assess the effects of these restrictions under both typical and stressed conditions. This would provide the bank with a comprehensive understanding of its liquidity risk profile, enabling it to take appropriate measures to manage and mitigate this risk. Furthermore, this approach aligns with the principles of sound liquidity risk management as outlined by the Basel Committee on Banking Supervision, which emphasizes the importance of stress testing in identifying and measuring liquidity risk.

Choice A is incorrect. While sourcing help to emancipate from the transfer restrictions might be beneficial in the long run, it does not directly address the immediate need for managing and assessing liquidity risk under both normal and stressed conditions. This option does not provide a solution for liquidity stress testing which is crucial for Afro Bank's current situation.

Choice C is incorrect. Stopping the liquidity stress testing scenario until the end of restrictions would not be an effective way to manage and assess its liquidity risk. In fact, it could potentially exacerbate any existing or future risks as it would leave Afro Bank without a clear understanding of its financial position during this critical period.

Choice D is incorrect. Sourcing other types of liquidity to include in the liquidity stress testing menu may seem like a viable option, but it doesn't necessarily address how these transfer restrictions are affecting enterprise-level liquidity under both typical operating environment and stressed conditions. It's important that Afro Bank understands how these specific restrictions impact their overall financial health before seeking alternative sources of liquidity.

Things to Remember

- **Liquidity Risk Management:** The process of identifying, assessing, managing, and monitoring liquidity risk to ensure a financial institution can meet its obligations as they come due.
- **Stress Testing:** A tool used by financial institutions to assess their resilience to adverse scenarios by simulating the impact of various stress factors on their financial position.
- **Basel Committee on Banking Supervision:** An international regulatory body that sets global standards for the prudential regulation of banks, including guidelines on liquidity risk management.
- **Enterprise-Level Liquidity:** The overall liquidity position of a financial institution, taking into account all sources of liquidity and potential stress scenarios.

- Liquidity Transfer Restrictions: Limitations imposed on the movement of liquidity within a financial institution, which can impact its ability to manage liquidity risk effectively.
-

Q.4132 Among the following statements, which one is correct about the currency consideration in a consolidated liquidity stress testing process?

- A. The liquidity stress test is performed based on the available currency
- B. Liquidity impact of currency conversion is not necessary in consolidated stress testing as any currency can be used
- C. There should be considerations of liquidity impact of currency conversion requirements to avoid currency mismatch for offshore subsidiaries.
- D. In a consolidated liquidity stress test, the home currency can be used together with any other available currency.

The correct answer is C.

In a consolidated liquidity stress testing process, it is essential to consider the liquidity impact of currency conversion requirements. This is to avoid any potential currency mismatch for offshore subsidiaries. Currency mismatch can occur when the currency of the liabilities differs from the currency of the assets. This can lead to significant financial risks, especially in times of currency volatility. Therefore, it is crucial to consider the liquidity impact of currency conversion requirements in the stress testing process to ensure that the entity has sufficient liquidity to meet its obligations, regardless of currency fluctuations.

Choice A is incorrect. While the available currency does play a role in liquidity stress testing, it is not the sole factor to consider. The test also needs to account for potential currency mismatches and conversion requirements, especially for offshore subsidiaries.

Choice B is incorrect. This statement contradicts the principles of consolidated liquidity stress testing. It's crucial to consider the liquidity impact of currency conversion as using any currency without considering its impact can lead to inaccurate results and potential risks.

Choice D is incorrect. Using home currency along with any other available currency without considering their impacts on liquidity can lead to misleading results in a consolidated liquidity stress test. It's important to take into account the effects of different currencies and their conversion requirements.

Things to Remember

- Consolidated liquidity stress testing involves assessing the liquidity position of an

entire group of entities, including subsidiaries and affiliates.

- Currency conversion risk is a significant factor in liquidity stress testing, especially for multinational corporations with operations in different countries.
 - Offshore subsidiaries may face additional challenges related to currency conversion and liquidity management due to operating in different currencies.
 - Foreign exchange exposure should be considered when conducting liquidity stress testing to ensure that the entity can meet its obligations in different currency scenarios.
 - Using multiple currencies in a consolidated stress test requires careful consideration of the liquidity impact of currency conversion to provide accurate and reliable results.
-

Q.4134 FTC bank, a fictional financial institution operating in Tokyo, Japan, has been doing a liquidity stress test for a time Horizon of more than 12 months. Which of the following choices most accurately state the disadvantage related to such a time horizon?

- A. Banks usually continue to operate expensively indefinitely under stress without employing a recovery or resolution process.
- B. Longer-term projections may be affected by forecast error according to the baseline balance sheet and statement of income budgeting time horizon.
- C. Longer-term projections may lead to unnecessary confusion to those doing the liquidity stress testing program
- D. No institution would keep data for long-term liquidity stress testing

The correct answer is **B**.

The longer the time horizon for a liquidity stress test, the more susceptible the projections are to forecast errors. This is primarily due to the inherent uncertainty associated with predicting future events. The baseline balance sheet and income statement budgeting time horizon serve as the foundation for these projections. However, as the time horizon extends, the potential for deviations between the projected and actual figures increases. This is because numerous factors, such as market conditions, economic trends, and regulatory changes, can significantly influence a bank's financial performance. Therefore, while long-term liquidity stress tests can provide valuable insights into a bank's resilience under prolonged stress scenarios, they also carry the risk of forecast errors, which can lead to inaccurate assessments of the bank's liquidity risk.

Choice A is incorrect. Banks do not operate indefinitely under stress without employing a recovery or resolution process. It is a standard practice for banks to have contingency plans in place to manage and mitigate financial stress situations. Therefore, this statement does not accurately describe a potential disadvantage of adopting a lengthy time horizon for liquidity stress testing.

Choice C is incorrect. Longer-term projections do not necessarily lead to confusion in the liquidity stress testing program. While it may be challenging due to the increased complexity and uncertainty over longer periods, these challenges can be managed with proper planning, expertise, and use of sophisticated forecasting models.

Choice D is incorrect. The assertion that no institution would keep data for long-term liquidity stress testing is false. Financial institutions are required by regulatory bodies to maintain historical data which can be used for various purposes including long-term liquidity stress testing.

Things to Remember

- Forecast errors can arise due to various factors such as market conditions, economic trends, and regulatory changes.
 - Contingency plans are essential for banks to manage and mitigate financial stress situations.
 - Long-term liquidity stress testing can provide valuable insights into a bank's resilience under prolonged stress scenarios.
 - Regulatory bodies often require financial institutions to maintain historical data for various purposes, including stress testing.
 - Proper planning, expertise, and sophisticated forecasting models can help manage the challenges associated with longer-term projections in liquidity stress testing.
-

Q.4136 The treasury committee in a liquidity stress testing program for RTC bank were considering a hypothetical scenario in the process of liquidity stress testing. Which of the following characteristics is not considered in the scenario?

- A. Define the scenario
- B. Distinguish between systematic and idiosyncratic risk
- C. Distinguish between sources of liquidity
- D. Distinguish between levels of severity

The correct answer is **C**.

The sources of liquidity are not typically considered in the development of a hypothetical scenario in a liquidity stress testing program. This is because the focus of such a scenario is primarily on the potential risks and stressors that could impact the bank's liquidity, rather than the sources of liquidity themselves. The sources of liquidity are more relevant in the operational management of liquidity risk, where the bank needs to ensure it has access to sufficient funds to meet its obligations. In a stress testing scenario, the focus is more on understanding how different risk factors could impact the bank's liquidity position, and how severe these impacts could be under different conditions.

Choice A is incorrect. Defining the scenario is a crucial part of the stress testing process. It involves outlining the potential events or changes in market conditions that could impact the bank's liquidity position.

Choice B is incorrect. Distinguishing between systematic and idiosyncratic risk is also an important aspect of stress testing. Systematic risk refers to the risk that affects all entities in a market, while idiosyncratic risk pertains to risks specific to an individual entity or sector.

Choice D is incorrect. Distinguishing between levels of severity is another key characteristic considered in hypothetical scenarios for stress testing. This involves identifying and assessing different degrees of adverse events or conditions, from mild to severe, and their potential impact on liquidity.

Things to Remember

- **Liquidity Stress Testing:** A process used by banks and financial institutions to assess their ability to withstand liquidity stress events by analyzing potential risks and stressors that could impact their liquidity position.
- **Scenario Analysis:** In liquidity stress testing, scenario analysis involves creating hypothetical scenarios to simulate different market conditions and events that could impact the bank's liquidity.
- **Stress Testing:** A risk management technique used to evaluate the potential impact of adverse events or market conditions on a bank's financial stability, including liquidity risk.
- **Liquidity Risk:** The risk that a bank may not be able to meet its short-term obligations due to a lack of liquid assets or access to funding sources.
- **Operational Management of Liquidity Risk:** Involves managing the sources of liquidity available to the bank to ensure it can meet its obligations under normal operating

conditions.

Q.4137 In developing a liquidity stress test assumption, several essential factors are considered. Which of the following choices is not among those factors?

- A. Qualitative assessment of the expected liquidity behavior for each type of cash flow to identify where there is significant liquidity risk
- B. Quantitative modeling assumptions based on any existing historical data.
- C. Matrices of relative modeling assumptions based on scored risk levels and historical baseline data
- D. Matrices of relative modeling assumptions irrespective of scored risk levels and current data

The correct answer is **D**.

Matrices of relative modeling assumptions irrespective of scored risk levels and current data are not typically considered in the development of a liquidity stress test assumption. Liquidity stress testing involves making assumptions based on historical data and scored risk levels. Current data may not accurately reflect potential future liquidity scenarios, especially in times of stress. Therefore, it is not typically used in the development of these assumptions. Instead, historical data and scored risk levels are used to model potential future scenarios and prepare for adverse conditions.

Choice A is incorrect. Qualitative assessment of the expected liquidity behavior for each type of cash flow is indeed a critical component in developing liquidity stress test assumptions. It helps to identify areas where there might be significant liquidity risk, which can then be factored into the stress testing process.

Choice B is incorrect. Quantitative modeling assumptions based on any existing historical data are also typically considered when developing a liquidity stress test assumption. Historical data can provide valuable insights into past trends and patterns, which can then be used to make informed predictions about potential future scenarios.

Choice C is incorrect. Matrices of relative modeling assumptions based on scored risk levels and historical baseline data are often used in the development of a liquidity stress test assumption. These matrices allow for a more nuanced understanding of potential risks by taking into account both the severity and likelihood of different outcomes.

Things to Remember

- Liquidity stress testing is a crucial risk management tool used by financial institutions to assess their ability to withstand liquidity shocks.

- Qualitative assessment helps in understanding the nature of cash flows and identifying potential liquidity risks.
 - Quantitative modeling involves using historical data to create scenarios and assess potential liquidity needs.
 - Scoring risk levels can help prioritize areas of focus in liquidity stress testing.
 - Current data may not always accurately reflect future liquidity conditions, especially in times of stress.
-

Q.4139 A financial institution is attempting to develop assumptions for its liquidity stress testing. Which of the following assumptions is NOT considered?

- A. Deposit outflows
- B. Investment portfolio haircuts
- C. Collateral requirement
- D. Deposit inflows

The correct answer is **D**.

Deposit inflows are not typically considered in the assumptions for a liquidity stress test. Liquidity stress testing primarily focuses on potential scenarios that could lead to a liquidity crunch or shortfall. Therefore, the assumptions made during this process typically revolve around factors that could lead to a decrease in available liquidity. Deposit inflows, which represent an increase in available liquidity, are not typically considered in these assumptions. This is because the primary aim of liquidity stress testing is to assess the institution's ability to withstand scenarios where liquidity decreases, not increases.

Choice A is incorrect. Deposit outflows are a crucial assumption in liquidity stress testing. In times of financial stress, depositors may withdraw their funds, leading to a decrease in the institution's liquidity. Therefore, it is essential to consider potential deposit outflows during the process.

Choice B is incorrect. Investment portfolio haircuts are also typically considered during liquidity stress testing. A haircut refers to the reduction applied to the value of an institution's assets to account for potential losses that could occur if those assets had to be sold quickly (i.e., under stressed market conditions). This reduction can significantly impact an institution's available liquidity.

Choice C is incorrect. Collateral requirements are another important assumption in this

process. In times of financial stress, counterparties may demand more collateral as security against default risk, which can strain an institution's available liquid resources.

Things to Remember

- Liquidity stress testing is a critical risk management tool used by financial institutions to assess their ability to withstand liquidity shocks.
 - Assumptions made during liquidity stress testing help institutions identify potential vulnerabilities and develop contingency plans.
 - Factors such as market conditions, regulatory requirements, and customer behavior are key considerations in liquidity stress testing assumptions.
 - Deposit outflows, investment portfolio haircuts, and collateral requirements are common assumptions in liquidity stress testing scenarios.
 - Monitoring and managing liquidity risk is essential for the stability and resilience of financial institutions.
-

Q.4140 In liquidity stress testing, among the considered outputs are stress testing assumptions, liquidity position metrics, prospective liquidity position metrics, and capital and performance metrics. Which of the following choices is not among the four key stress testing assumptions?

- A. The assumption about the overall stress level represented by the scenario
- B. The assumption about the indication of the nature of the scenario, such as systemic, idiosyncratic, or both systemic and idiosyncratic
- C. Assumptions on the expected returns
- D. The assumption on the discussion of the impact of the scenario on cash flow

The correct answer is **C**.

Assumptions on the expected returns are not considered as one of the key stress testing assumptions in liquidity stress testing. Liquidity stress testing is primarily concerned with a firm's ability to meet its obligations as they come due without incurring unacceptable losses. Expected returns, while important in other aspects of financial management, do not directly impact a firm's liquidity position. Therefore, they are not typically included in the assumptions used in liquidity stress testing.

Choice A is incorrect. The assumption about the overall stress level represented by the scenario is indeed a key assumption in liquidity stress testing. It helps to determine the severity of the potential liquidity crisis and thus, plays a crucial role in risk management.

Choice B is incorrect. The assumption about the indication of the nature of the scenario, such as systemic, idiosyncratic, or both systemic and idiosyncratic is also an important part of liquidity stress testing. This helps to understand whether a potential crisis could be due to factors specific to an individual institution (idiosyncratic) or due to broader market-wide issues (systemic).

Choice D is incorrect. The assumption on the discussion of impact of scenario on cash flow forms an integral part of liquidity stress testing as it directly affects a firm's ability to meet its financial obligations during times of crisis.

Things to Remember

- Liquidity stress testing is a risk management technique used by financial institutions to assess their ability to withstand liquidity stress events.
 - Stress testing assumptions are crucial in determining the potential impact of adverse scenarios on a firm's liquidity position.
 - Prospective liquidity position metrics help in evaluating a firm's future liquidity needs and vulnerabilities.
 - Capital and performance metrics are important in assessing the overall financial health and resilience of a firm during liquidity stress events.
 - Expected returns are more relevant in the context of investment decision-making and portfolio management, rather than liquidity stress testing.
-

Q.4141 In the governance and control of the liquidity stress testing process, the treasury committee is considered to be instrumental in the whole process. Which among the following choices most accurately states one of the responsibilities undertaken by the treasury?

- A. Recommends stress testing scenarios
- B. Recruits members for the asset-liability committee (ALCO)
- C. Designs liquidity risk policy limits relative to stress test results and escalating exceptions
- D. Undertakes the liquidity stress testing framework periodical review, procedures, and controls to keep compliance with policy, regulatory, and control requirements.

The correct answer is **A**.

Treasury recommends stress testing scenarios.

B is incorrect: It is not the responsibility of the treasury to recruit members of ALCO. The treasury reviews and monitors components of the institution's assets and liabilities and recommend to the ALCO on matters relating to stress testing assumptions.

C is incorrect: Designing liquidity risk policy limits relative to stress test results and escalating exceptions is the responsibility of the asset-liability committee.

D is incorrect: Internal audit is responsible for undertaking liquidity stress testing framework periodical review, procedures, and controls to keep compliance with policy, regulatory, and control requirements.

Things to Remember

- Asset-Liability Committee (ALCO) is responsible for overseeing the institution's liquidity risk management framework and ensuring that it is in line with the organization's risk appetite and regulatory requirements.
- Liquidity stress testing is a key tool used by financial institutions to assess their ability to withstand liquidity shocks under different scenarios.
- Treasury plays a crucial role in liquidity management by managing the institution's cash flows, funding, and liquidity risk.
- Stress testing scenarios are designed to simulate adverse market conditions and assess the impact on the institution's liquidity position.
- Escalating exceptions in liquidity risk policy limits refer to situations where the institution's liquidity risk exceeds predefined thresholds, triggering a need for immediate action and reporting.

Q.4142 A group of FRM students was given the following characteristic to identify the committee of liquidity stress testing governance they belong to: The committee provides independent validation and control over the management governance of the liquidity stress testing model concerning the institution's model risk management policy. Which of the following committee meets the above description?

- A. The asset-liability committee
- B. Internal audit
- C. Model risk management
- D. The Treasury

The correct answer is **C**.

The Model Risk Management committee is responsible for providing independent validation and control over the management governance of the liquidity stress testing model. This is in line with the institution's model risk management policy. The committee's role is to ensure that the model used for liquidity stress testing is robust, reliable, and accurately represents the institution's liquidity risk. This involves reviewing the model's assumptions, methodologies, and outputs, and assessing the model's performance over time. The committee also ensures that the model is used appropriately within the institution and that any limitations or uncertainties associated with the model are properly communicated and understood.

Choice A is incorrect. The asset-liability committee (ALCO) is responsible for managing the balance sheet risk of a financial institution, including interest rate risk and liquidity risk. However, it does not provide independent validation and control over the management governance of the liquidity stress testing model.

Choice B is incorrect. While Internal Audit plays a crucial role in assessing the effectiveness of controls within an organization, its primary function is not to provide independent validation and control over the management governance of the liquidity stress testing model.

Choice D is incorrect. The Treasury department manages an institution's financial resources but does not typically oversee or validate models used in stress testing scenarios.

Things to Remember

- Model Risk Management (MRM) is a crucial function within financial institutions that focuses on identifying, assessing, and managing model risk.
- Liquidity stress testing is a key component of a financial institution's risk management framework, helping to assess the institution's ability to withstand liquidity shocks.
- The Asset-Liability Committee (ALCO) is responsible for overseeing the balance sheet management of a financial institution, including managing interest rate risk and liquidity risk.
- Internal Audit is an independent function within an organization that evaluates the effectiveness of risk management, control, and governance processes.

- The Treasury department is typically responsible for managing an institution's cash flow, investments, and funding activities.
-

Q.4143 Which of the following choices is correct about Liquidity stress testing and capital stress testing integration?

- A. To link liquidity stress testing and capital stress testing requires an institution to incorporate any essential capital infusions of subsidiary entities.
- B. For every liquidity stress test scenario, the institution should develop capital impact assumptions based only on the idiosyncratic conditions that occur under the scenario
- C. For every liquidity stress test scenario, the institution should develop capital impact assumptions based only on the overall market conditions that occur under the scenario
- D. For every liquidity stress test scenario, the institution should develop capital impact assumptions based only on the overall market conditions that occur under the scenario arise through investment portfolios.

The correct answer is **A**.

The integration of liquidity stress testing and capital stress testing necessitates that an institution includes any vital capital infusions from subsidiary entities. This is because the capital position of an institution can significantly influence its liquidity position. For instance, if a subsidiary entity is unable to provide the necessary capital infusion during a stress scenario, the parent institution may face a liquidity crunch. Therefore, it is crucial for institutions to consider the potential capital infusions from subsidiary entities when linking liquidity stress testing and capital stress testing.

Choice B is incorrect. While it's true that idiosyncratic conditions are important to consider in stress testing, they should not be the only factor considered when developing capital impact assumptions. A comprehensive risk assessment requires considering both idiosyncratic and market-wide conditions.

Choice C is incorrect. Similar to choice B, relying solely on overall market conditions for developing capital impact assumptions during a liquidity stress test scenario would not provide a complete picture of the potential risks. Both idiosyncratic and market-wide conditions need to be taken into account.

Choice D is incorrect. This option incorrectly suggests that overall market conditions arising through investment portfolios should be the only consideration for capital impact assumptions in every liquidity stress test scenario. This approach would neglect other important factors such as operational risks, credit risks, and specific business model vulnerabilities which are crucial for a comprehensive risk assessment.

Things to Remember

- Liquidity stress testing assesses an institution's ability to meet its short-term obligations under stressful conditions, while capital stress testing evaluates the impact of adverse scenarios on an institution's capital adequacy.
 - Integrating liquidity stress testing and capital stress testing helps institutions understand the interplay between liquidity and capital risks, providing a more holistic view of their risk profile.
 - Capital infusions from subsidiary entities can play a critical role in supporting the parent institution's liquidity position during stress scenarios.
 - Idiosyncratic conditions refer to institution-specific factors that can impact its financial health, while overall market conditions encompass broader economic and financial market factors.
 - Operational risks, credit risks, and business model vulnerabilities are important considerations in stress testing and risk assessment, in addition to liquidity and capital risks.
-

Reading 136: Liquidity Risk Reporting and Stress Testing

Q.4050 Which of the following details would not be obtained from a deposit tracker report?

- A. Month-end actuals for deposits by customer type
- B. Each month-end change
- C. Each month-end forecast for the position to the end of the year
- D. Annual loan to deposit ratio (LTD)

The correct answer is **D**.

The annual loan to deposit ratio (LTD) is not typically included in a deposit tracker report. The LTD ratio is a measure used by banks to assess their liquidity by dividing total loans by total deposits. This ratio is used to determine a bank's ability to cover loan losses and withdrawals by its depositors. However, the deposit tracker report is generated on a weekly or monthly basis and focuses on providing detailed insights into deposit trends and patterns. Therefore, it does not include the annual LTD ratio, which is a broader measure of a bank's financial health and liquidity.

Choice A is incorrect. Month-end actuals for deposits by customer type are typically included in a deposit tracker report. This information helps financial institutions understand the deposit trends and patterns of different customer types, which can be useful for strategic planning and decision-making.

Choice B is incorrect. Each month-end change is also usually included in a deposit tracker report. This data provides insights into the monthly fluctuations in deposits, enabling financial institutions to track their performance over time and make necessary adjustments.

Choice C is incorrect. Each month-end forecast for the position to the end of the year is another detail that's typically found in a deposit tracker report. These forecasts help financial institutions anticipate future trends and prepare accordingly.

Things to Remember

- **Deposit Tracker Report:** A report that provides detailed insights into deposit trends and patterns, typically generated on a weekly or monthly basis.
- **Loan to Deposit Ratio (LTD):** A measure used by banks to assess liquidity by dividing total loans by total deposits.
- **Customer Type Analysis:** Understanding deposit trends by customer type can help in strategic planning and decision-making.

- Monthly Changes: Tracking monthly fluctuations in deposits can help in performance evaluation and adjustments.
 - Forecasting: Using month-end forecasts to anticipate future trends and prepare accordingly.
-

Q.4051 VTR bank has been receiving most of its funding from current accounts and rolling deposits. Under what circumstance will the bank treat overnight balances as long-term according to regulation?

- A. If the overnight balances can be demonstrated to be acting as long-term in “behavioral” terms
- B. If the overnight deposits are predictable
- C. If the overnight deposits are the primary funding source for bank
- D. None of the above

The correct answer is **A**.

Things to Remember

1. The classification of deposits as short-term or long-term has significant implications for a bank's liquidity metrics. Long-term deposits are generally considered more stable and reliable, which can improve the bank's liquidity position and financial stability.
 2. The behavior of deposits is a key factor in determining their classification. Deposits that exhibit long-term behavior, such as consistency and stability, can be treated as long-term, even if they are technically overnight deposits.
 3. The size or significance of the deposits is not a determining factor in their classification. Even if overnight deposits form a significant part of the bank's funding, they cannot be treated as long-term unless they exhibit long-term behavior.
 4. Predictability of deposits is also not a determining factor in their classification. While predictability can help the bank manage its cash flows and liquidity, it does not imply that the deposits are stable or long-term.
-

Q.4053 MCT bank has been preparing a mismatch report after a stress testing process. Which among the following statements is accurate about mismatch reports?

- A. The mismatch report reflects the maturity gap for all assets and liabilities per time bucket and with an adjustment for liquid securities.
- B. The report doesn't apply in generating the horizon report for cash flow
- C. It presents the bank's marketable assets and liabilities
- D. It reports the end-of-day liquidity position

The correct answer is **A**.

The mismatch report usually comprises of the cumulative liquidity cash flow of both liquidity daily liquidity report and deposit tracker report

B is incorrect: The mismatch report is used to generate the survival horizon report for cashflows.

C is incorrect: The bank's marketable assets and liabilities are presented in a daily liquidity report

D is incorrect: The end of day liquidity position is presented in a daily liquidity report.

Things to Remember

- Asset Liability Management (ALM) is a strategic management tool to manage interest rate risk and liquidity risk faced by banks and other financial institutions.
- Mismatch reports help in identifying any gaps between the maturity profiles of assets and liabilities, which can help in managing liquidity risk.
- Liquidity risk is the risk that a bank may not be able to meet its obligations as they come due without incurring unacceptable losses.
- Survival horizon reports are used to assess the ability of a bank to survive a stress scenario by analyzing cash flows over a specified time horizon.

Q.4054 A financial institution is preparing a large depositor concentration report for a banking group. What is the most accurate meaning of the word “large” in this case?

- A. Someone that owns the majority of shares
- B. Someone that undertakes more than USD 50 million withdrawals
- C. Someone that deposits a deposit of 5% of total liabilities.
- D. None of the above

The correct answer is **C**.

In a large depositor concentration, the term “large” refers to someone that deposits a deposit of 5% of total liabilities.

A is incorrect: Someone that owns most shares is not referred to as “large” in deposit concentration

B is incorrect: Large deposit concentration is defined in terms of deposits, not withdrawals

Things to Remember

- Depositor concentration refers to the situation where a significant portion of a financial institution's deposits are held by a small number of depositors.
- Large depositor concentrations can pose risks to the financial institution, especially if one or a few large depositors decide to withdraw their funds.
- Regulatory authorities often monitor depositor concentration levels to ensure the stability of financial institutions.
- Financial institutions may use depositor concentration reports to assess their risk exposure and develop strategies to manage concentration risks.

Q.4055 Off-balance-sheet items serve as potential stress points for a bank’s funding. Which among the following is not an off-balance sheet item?

- A. Liquidity lines
- B. Letters of credit
- C. Revolving credit facilities

D. Liquid assets

The correct answer is **D**.

Liquid assets are not considered off-balance-sheet items. Instead, they are included in the asset portion of a bank's balance sheet. Liquid assets are those assets that can be easily converted into cash without significantly affecting their value. These include cash, marketable securities, and government bonds. These assets are crucial for a bank's liquidity management as they can be quickly sold to meet immediate and short-term obligations, thereby reducing the risk of insolvency. Therefore, liquid assets are on-balance-sheet items that contribute to a bank's liquidity and overall financial stability.

Choice A is incorrect. Liquidity lines are indeed classified as off-balance-sheet items. They represent commitments by a bank to lend funds to a client, and these commitments do not appear on the balance sheet until they are drawn upon by the client.

Choice B is incorrect. Letters of credit also fall under off-balance-sheet items. These are guarantees provided by a bank on behalf of its clients for making payments to third parties, and they do not appear on the balance sheet unless the client defaults and the bank has to make good on its guarantee.

Choice C is incorrect. Revolving credit facilities are another type of off-balance-sheet item. These facilities allow clients to borrow up to a certain limit as needed, and repay with flexibility over time. The undrawn portion of these facilities does not appear on the balance sheet but can create significant liabilities for banks if all their clients draw down their facilities at once.

Things to Remember

- Off-balance-sheet items are financial transactions or arrangements that are not recorded on a company's balance sheet, but still have an impact on its financial position and risk profile.
- Off-balance-sheet items can include items such as contingent liabilities, commitments, and certain types of financial derivatives.
- Off-balance-sheet items are important to consider when analyzing a company's financial health and risk exposure, as they can affect liquidity, solvency, and overall financial stability.
- Understanding off-balance-sheet items is crucial for risk management and regulatory compliance in the banking and financial industry.

Q.4056 The liability profile is a simple breakdown of the share of each type of liability at the bank. Which of the following is not a part of the liability profile?

- A. Customer individuals
- B. Highly liquid security repo
- C. Asset-backed securities
- D. Highly liquid assets

The correct answer is **D**.

Highly liquid assets are not part of the liability profile. The liability profile of a bank is a representation of the bank's financial obligations, i.e., what the bank owes to others. Highly liquid assets, on the other hand, are assets that can be quickly converted into cash without any significant loss in value. These assets are part of the bank's asset profile, not its liability profile. They represent resources that the bank owns and can use to meet its financial obligations. Therefore, highly liquid assets are not included in the liability profile.

Choice A is incorrect. Customer individuals are indeed a part of the bank's liability profile. They represent the deposits made by individual customers, which are liabilities for the bank as they are obligated to return these funds upon demand.

Choice B is incorrect. Highly liquid security repos (repurchase agreements) also form part of a bank's liability profile. These are short-term borrowing for dealers in government securities and the dealer sells the government securities to investors, usually on an overnight basis, and buys them back the following day at a slightly higher price.

Choice C is incorrect. Asset-backed securities can be part of a bank's liability profile if they have been issued by the bank itself. In this case, they represent obligations that must be met by generating sufficient returns from the underlying assets.

Things to Remember

- Liability profile is a breakdown of the share of each type of liability at the bank, representing the bank's financial obligations.
- Highly liquid assets are assets that can be quickly converted into cash without significant loss in value.
- Highly liquid assets are part of the bank's asset profile, not its liability profile.
- Customer individuals represent deposits made by individual customers, which are liabilities for the bank.

- Highly liquid security repos are short-term borrowings for dealers in government securities.
 - Asset-backed securities can be part of a bank's liability profile if they have been issued by the bank itself.
-

Q.4057 Which among the choices is not considered in regular liquidity qualitative reporting concerning individual liquidity adequacy standards (ILAS)?

- A. Details on any increase or decrease in retail deposits
- B. Details on average day-to-day opening cash position
- C. Details on shrinkage or growth of asset books
- D. Details on deposit month-end actuals by customer type

The correct answer is **D**.

Details on deposit month-end actuals by customer type are not typically included in regular liquidity qualitative reporting concerning individual liquidity adequacy standards (ILAS). These details are usually part of a separate report known as the deposit tracker report. This report is designed to provide a comprehensive view of the bank's deposit base, including the breakdown by customer type. It helps the bank to understand the composition of its deposit base and to manage its liquidity risk effectively. However, it is not a standard requirement in the regular qualitative reporting on liquidity under ILAS.

Choice A is incorrect. Details on any increase or decrease in retail deposits are typically included in the regular qualitative reporting. This information is crucial as it provides insights into the liquidity position of the financial institution and can indicate potential liquidity risks.

Choice B is incorrect. The average day-to-day opening cash position is also a typical component of regular qualitative reporting on liquidity. It helps to understand how much liquid assets the institution has at its disposal at the start of each business day, which can be critical for managing daily operations and meeting short-term obligations.

Choice C is incorrect. Information about shrinkage or growth of asset books forms part of this reporting too. Changes in asset books can significantly impact an institution's liquidity position, hence it's important to monitor and report these changes regularly.

Things to Remember

- Individual Liquidity Adequacy Standards (ILAS) are regulatory requirements that banks and financial institutions must adhere to in order to ensure they have adequate

liquidity to meet their obligations.

- Liquidity reporting is a key component of risk management for financial institutions, as it helps to assess and monitor the institution's ability to meet its short-term obligations.
 - Regular liquidity qualitative reporting typically includes information on various aspects such as funding sources, liquidity positions, and potential risks to liquidity.
 - Deposit tracker reports are used by banks to track and analyze their deposit base, including details on deposit composition by customer type. This helps in managing liquidity risk effectively.
 - Understanding the composition of retail deposits and their trends is important for assessing the stability and availability of funding sources for the financial institution.
-

Q.4058 The following is an excerpt from Bright Bank's Deposit Tracker from the End of the year 2018 to July 2019.

Deposit tracker	Month-end actuals 31/12/2018	Month-end actuals 31/01/2019	Month-end Forecasts 31/05/2019	Month-end Forecasts 30/06/2019
Total customer deposits	612,152	423,804	651,112	599,774
Total customer loans	547,478	373,381	520,829	480,122

Calculate the bank's loan-to-deposit ratio as at 31/12/2018

- A. 0.84
- B. 0.86
- C. 0.87
- D. 0.89

The correct answer is **D**.

$$\begin{aligned}\text{Loan-to-deposit (LTD) ratio} &= \frac{\text{Total customer loans}}{\text{Total customer deposits}} \\ &= \frac{547,478}{612,152} \\ &= 89\%\end{aligned}$$

Q.4059 The following is an excerpt from Bright Bank's Deposit Tracker from End of the year 2018 to July 2019.

Deposit tracker	Month-end actuals		Month-end Forecasts	
	31/12/2018	31/01/2019	31/05/2019	30/06/2019
Total customer deposits	612,152	423,804	651,112	599,774
Total customer loans	547,478	373,381	520,829	480,122

How much does the bank need to increase its total customer loans by to meet its targeted loan-to-deposit ratio of 85% on 30/06/2019?

- A. 22445
- B. 29685
- C. 31587
- D. 33465

The correct answer is **B**.

$$\begin{aligned}\text{Loan-to-deposit (LTD) ratio} &= \frac{\text{Total customer loans}}{\text{Total customer deposits}} \\ 85\% &= \frac{x}{599,774} \\ x &= 509,807 \\ \text{Increment} &= 509,807 - 480,122 = 29,685\end{aligned}$$

Q.4060 Which among the following choices is correct about derivatives value or notional treatment concerning UK regulations?

- A. For off-balance sheet items, derivative values are not included in the calculation of the liquidity ratio
- B. Derivatives are included to the extent of collateral payable or receivable under an ISDA/CSA agreement
- C. For off-balance sheet items, coupons receivable or payable are not included on their pay dates
- D. None of the above

The correct answer is **A**.

According to the UK regulations, derivative values or notionals are not included in the calculation of the liquidity ratio for off-balance sheet items. This means that the value of derivatives, which are financial instruments whose value is derived from the value of other assets, are not considered when calculating the liquidity ratio. The liquidity ratio is a measure of a company's ability to meet its short-term obligations, and it is calculated by dividing a company's liquid assets by its current liabilities. By excluding derivative values from this calculation, the UK regulations are effectively saying that these financial instruments do not contribute to a company's liquidity, or its ability to meet its short-term obligations.

Choice B is incorrect. In the context of UK regulations, derivatives are **not** included in the calculation of off-balance sheet items to the extent of their notional value, regardless of whether collateral is payable or receivable under an ISDA/CSA agreement.

Choice C is incorrect. Coupons receivable or payable on derivatives are indeed included in off-balance sheet calculations on their pay dates. This statement contradicts standard regulatory practices and thus, it's inaccurate.

Choice D is incorrect. As explained above, choices B and C contain inaccuracies regarding how derivatives are treated in relation to off-balance sheet items and liquidity ratios under UK regulations.

Things to Remember

- Derivatives are financial instruments whose value is derived from the value of an underlying asset, index, or rate.
- Liquidity ratio is a measure of a company's ability to meet its short-term obligations using its liquid assets.
- ISDA (International Swaps and Derivatives Association) and CSA (Credit Support Annex) agreements are common in derivative transactions to manage collateral.
- Off-balance sheet items refer to assets or liabilities that are not recorded on a company's balance sheet.
- Coupons on derivatives represent the periodic interest payments or cash flows associated with the derivative contract.

Q.4061 In preparation of a deposit tracker report, Cate Wilson, the manager of the bank preparing the report, realized that most of the retail bank deposits consisted of the current

accounts and rolling deposits. How should the bank treat these funds in the regulation process?

- A. As long-term liabilities
- B. As liquid assets
- C. As short-term liabilities
- D. None of the above

The correct answer is **C**.

Current accounts and rolling deposits are considered short-term liabilities in the context of banking regulations. This is because these types of deposits can be withdrawn by the customer at any time, making them a liability for the bank. The bank is obligated to return these funds upon the customer's request, hence they are classified as short-term liabilities. Regulatory liquidity metrics, which are used to assess a bank's ability to meet its short-term obligations, do not consider these funds as contributing to the bank's liquidity. This is because these funds can be withdrawn at any time, potentially leaving the bank without sufficient liquid assets to meet its obligations. However, there are exceptions to this rule. For instance, if the bank can demonstrate that these funds are behaving as long-term funds, the local regulator may allow the bank to treat them as such. This means that if the bank can show that these funds are not likely to be withdrawn in the short term, they may be classified as long-term liabilities.

Choice A is incorrect. Current accounts and rolling deposits cannot be classified as long-term liabilities. These are short-term in nature as they can be withdrawn by the depositor at any time, hence they do not represent a long-term obligation for the bank.

Choice B is incorrect. While current accounts and rolling deposits are liquid from the depositor's perspective, from the bank's perspective these are not assets but liabilities. The bank owes these funds to their depositors and therefore, they cannot be classified as liquid assets.

Choice D is incorrect. As explained above, current accounts and rolling deposits should indeed be classified in a specific way in regulatory processes - namely as short-term liabilities - so 'None of the above' is not an accurate answer.

Things to Remember

- Current accounts and rolling deposits are considered short-term liabilities due to their nature of being withdrawable by the customer at any time.
- Regulatory liquidity metrics do not consider these funds as contributing to a bank's liquidity as they can be withdrawn at any time.
- Exceptions may exist where these funds can be treated as long-term liabilities if the bank can demonstrate that they are not likely to be withdrawn in the short term.

- Short-term liabilities are important for assessing a bank's ability to meet its short-term obligations.
-

Q.4062 The regulatory authority for ABC bank generated a funding yield curve after conducting a stress test. They then compared the curve with the ones generated by their peers. The bank's management realized that their curve had risen significantly beyond the curves of its peers. What is the possible implication of these results?

- A. ABC is experiencing funding stress
- B. ABC is experiencing high liquidity levels
- C. ABC has more liabilities than assets
- D. ABC has a high level of solvency

The correct answer is **A**.

ABC is experiencing funding stress. A funding yield curve is a graphical representation of the interest rates on debt for a range of maturities. It shows the relationship between the interest rate (or cost of borrowing) and the time to maturity of the debt for a given borrower in a given currency. In this case, if ABC bank's funding yield curve rises significantly beyond the curves of its peers, it indicates that the bank is experiencing funding stress. This could be due to various factors such as increased borrowing costs, reduced access to funding sources, or a combination of these factors. The bank may need to take corrective measures to address this issue, such as improving its liquidity management, reducing its reliance on short-term funding, or increasing its capital buffers.

Choice B is incorrect. High liquidity levels would typically result in a lower funding yield curve, not a higher one. This is because when a bank has high liquidity, it means that it has more assets readily available to meet its short-term obligations, reducing the need for external funding and thus lowering the cost of such funding.

Choice C is incorrect. Having more liabilities than assets does not necessarily lead to a higher funding yield curve. While it's true that banks with more liabilities may face greater funding stress, this relationship isn't direct or guaranteed. Other factors such as the bank's creditworthiness and market conditions also play significant roles in determining the shape of its yield curve.

Choice D is incorrect. A high level of solvency implies that ABC Bank has sufficient assets to cover its long-term liabilities which should ideally lead to a lower or at least stable yield curve rather than an elevated one as observed in this case.

Things to Remember

- **Liquidity Risk:** Liquidity risk refers to the risk that a company or bank may not be able

to meet its short-term financial obligations due to an inability to liquidate assets or obtain funding at a reasonable cost. It is important for banks to manage liquidity risk effectively to ensure their stability and solvency.

- **Solvency:** Solvency is the ability of a company or bank to meet its long-term financial obligations. A high level of solvency indicates that the entity has sufficient assets to cover its liabilities, providing a measure of financial stability.
 - **Stress Testing:** Stress testing is a risk management technique used by banks and financial institutions to evaluate their resilience to adverse market conditions or economic shocks. It helps identify potential vulnerabilities and assess the impact of extreme scenarios on the institution's financial health.
 - **Yield Curve:** A yield curve is a graphical representation of the interest rates on debt for a range of maturities. It provides insights into market expectations regarding future interest rates and economic conditions. Changes in the shape of the yield curve can signal shifts in investor sentiment and economic outlook.
-

Q.4063 A qualitative monthly report is prepared and send to the head office group treasury for the banking groups operating across country jurisdictions or multiple subsidiaries for a good understanding of the liquidity position in the respective countries. Among the components of the report, the liquidity gap is captured, what is a liquidity gap in this context?

- A. A mismatch in the supply or demand for a security or the maturity dates of securities
- B. Lower liabilities than assets
- C. The difference between the average maturity of assets and liabilities
- D. None of the above

The correct answer is **A**.

A liquidity gap refers to a mismatch in the supply or demand for a security or the maturity dates of securities. This discrepancy can lead to a cash shortage, which is a critical issue for any financial institution. The liquidity gap is a crucial component of the monthly report as it provides an insight into the liquidity position of the banking groups. Understanding the liquidity gap can help the head office group treasury to make informed decisions regarding the management of liquidity and to mitigate potential risks associated with liquidity mismatches.

Choice B is incorrect. A liquidity gap does not refer to having lower liabilities than assets. In fact, a bank may have more assets than liabilities and still face a liquidity gap if the maturity dates of its assets and liabilities do not align.

Choice C is incorrect. The liquidity gap is not defined as the difference between the average maturity of assets and liabilities. While this could potentially contribute to a liquidity gap, it does not fully capture the concept as it ignores other factors such as cash inflows and outflows.

Choice D is incorrect. As explained above, there are specific definitions for what constitutes a liquidity gap in banking groups operating across multiple jurisdictions or subsidiaries, so 'None of the above' would be an inaccurate choice.

Things to Remember

- **Liquidity Risk:** The risk that a company or bank may be unable to meet its short-term financial obligations due to a lack of liquid assets.
 - **Asset-Liability Management (ALM):** The process of managing the risks that arise due to mismatches between the assets and liabilities of a financial institution.
 - **Net Stable Funding Ratio (NSFR):** A regulatory requirement that aims to ensure that banks maintain a stable funding profile in relation to the composition of their assets and off-balance sheet activities.
 - **Liquidity Coverage Ratio (LCR):** Another regulatory requirement that ensures banks have enough high-quality liquid assets to withstand a significant stress scenario over a 30-day period.
-

Q.4064 A report is based on granular analysis of the firm's marketable asset holdings, prepared every month, and submitted in 15 business days after the month-end. Which of the following reports is most accurately described in the given description?

- A. Daily flows report
- B. Wholesale liabilities report
- C. Liquidity buffer qualifying securities report
- D. Currency analysis report

The correct answer is **C**.

A Liquidity Buffer Qualifying Securities Report is indeed based on a granular analysis of a firm's marketable asset holdings. It is prepared monthly and is typically submitted within 15 business days after the end of the month. This report is crucial for firms as it provides a detailed overview of the liquidity of their assets. It helps firms to understand their financial position better and make informed decisions regarding their asset management. The report includes information about the firm's liquid assets, which are assets that can be quickly converted into cash without affecting their market price. These assets are crucial for a firm's financial stability as they can be used to meet short-term obligations.

Choice A is incorrect. A daily flows report, as the name suggests, is prepared on a daily basis and not monthly. It primarily focuses on the inflow and outflow of cash rather than a detailed examination of marketable asset holdings.

Choice B is incorrect. A wholesale liabilities report mainly deals with the firm's liabilities that are due for payment to its creditors or lenders. It does not provide a detailed examination of a firm's marketable asset holdings.

Choice D is incorrect. A currency analysis report primarily focuses on analyzing different currencies' performance and their impact on the firm's financial position. It does not focus specifically on the firm's marketable asset holdings.

Things to Remember

- Granular analysis involves a detailed examination of individual components or elements within a larger system or structure.
- Marketable assets are financial assets that can be easily bought or sold in the market due to their liquidity.
- Liquidity buffer refers to a reserve of highly liquid assets that can be quickly converted into cash to meet short-term obligations.
- Qualifying securities are assets that meet specific criteria set by regulatory bodies or internal policies to be included in certain reports or calculations.
- Monthly reporting is a common practice in financial institutions to track and analyze performance over regular intervals.

Q.4065 How various specific types of cashflow are treated is an essential question in liquidity reporting. Under what circumstance should a bank treat 50% of deposit funds as long-term funds?

A. If it is evident that they can remain relatively stable over time

- B. If their stability is uncertain
- C. If they contribute to the highest portion of the bank's funding
- D. None of the above

The correct answer is **A**.

A deposit is typically considered a short-term fund. However, regulatory authorities allow for a different classification under certain conditions. If there is evidence that deposits can remain stable over a longer period, then 50% of these deposits can be treated as long-term funds. This is because the stability of these funds over time reduces the liquidity risk for the bank, allowing it to rely on these funds for longer-term obligations. Therefore, in the context of liquidity reporting, these deposits can be considered as long-term funds.

Choice B is incorrect. A banking institution cannot categorize deposits as long-term funds if their stability is uncertain. Long-term funds are typically stable and reliable sources of funding that can be expected to remain in place over a longer time horizon.

Choice C is incorrect. The contribution of the deposits to the bank's overall funding does not determine whether they can be classified as long-term funds. It's the stability and predictability of these funds that matter, not their proportion in the bank's total funding.

Choice D is incorrect. As explained above, a banking institution would be justified in categorizing 50% of its deposit funds as long-term if it is evident that they can remain relatively stable over time (choice A). Therefore, 'None of the above' cannot be correct.

Things to Remember

- **Stable funding:** Stable funding refers to funds that are expected to remain with the bank over a longer period of time, providing a reliable source of liquidity.
- **Liquidity risk:** Liquidity risk is the risk that a bank may not be able to meet its short-term obligations due to a lack of liquid assets or the inability to quickly convert assets into cash.
- **Regulatory requirements:** Regulatory authorities often set guidelines for how banks should classify and report their funding sources to ensure stability and manage liquidity risk.
- **Deposit stability:** The stability of deposits is crucial in determining whether they can be classified as long-term funds, as stable deposits reduce liquidity risk for the bank.

Reading 137: Contingency Funding Planning

Q.4067 The design of a CFP should integrate with the broader risk management frameworks. Which among the following choices is not part of the risk management frameworks under consideration?

- A. Enterprise risk management (ERM)
- B. Business continuity and crisis management
- C. Financial management
- D. Capital management

The correct answer is **C**.

Financial management is not typically considered a part of the risk management frameworks that are integrated with a Contingency Funding Plan (CFP). While financial management is an essential aspect of any organization, it does not directly fall under the risk management frameworks that are considered for integration with a CFP. Financial management primarily deals with the financial resources of an organization, including budgeting, forecasting, and financial reporting. It is more concerned with the efficient management of an organization's financial resources rather than managing risks. However, it is worth noting that financial management is a crucial framework in the long-term council community plan (LTCCP) analysis.

Choice A is incorrect. Enterprise risk management (ERM) is a crucial framework that should be integrated with a CFP. ERM provides an organization-wide approach to identify, assess, manage, and control potential risks. It helps in ensuring the effectiveness and consistency of the CFP by providing a holistic view of all risks.

Choice B is incorrect. Business continuity and crisis management are also essential frameworks to integrate with a CFP. These frameworks help in planning for unexpected events or crises that could disrupt business operations, including financial disruptions. Therefore, they play an important role in designing an effective CFP.

Choice D is incorrect. Capital management should also be considered when integrating with a CFP as it involves managing the firm's capital structure to ensure financial stability and mitigate risks associated with insolvency or bankruptcy.

Things to Remember

- Enterprise Risk Management (ERM) is a comprehensive approach to identifying, assessing, and managing risks across an organization.
- Business Continuity and Crisis Management involve planning and preparing for unexpected events that could disrupt normal business operations.

- Financial Management focuses on managing an organization's financial resources, including budgeting, forecasting, and financial reporting.
 - Capital Management involves managing the firm's capital structure to ensure financial stability and mitigate risks related to insolvency or bankruptcy.
 - Contingency Funding Plan (CFP) is a crucial component of liquidity risk management, ensuring that an organization has adequate funding sources during times of stress or crisis.
-

Q.4068 Roy Jakes, the financial risk manager for Pathway financial institution, is preparing a design for a CFP. As an essential consideration in the plan, he must include appropriate stakeholders. Which among the following members is NOT part of the considered stakeholders?

- A. Internal audit committee
- B. Risk and capital committee
- C. Management committee
- D. Operations and technology

The correct answer is **A**.

Internal audit committee is not a stakeholder when designing a CFP. However, internal audit is involved with periodical reviews on the liquidity stress testing framework and procedures.

B is incorrect: Risk and capital committee is an essential stakeholder responsible for setting the principles and parameters for measuring and assigning risk and capital within the business.

C is incorrect: Different management committees such as the asset-liability committee (ALCO), risk and capital committee, and investment committee are key stakeholders in CFP.

D is incorrect: The operational and technology committee reviews and approves the institution's technology planning and strategy.

Things to Remember

- Internal audit committee plays a crucial role in providing independent assurance on the effectiveness of an organization's risk management, internal control, and governance processes.
- Risk and capital committee is responsible for overseeing the risk management framework, ensuring alignment with the organization's risk appetite, and making decisions related to capital allocation.
- Management committees, such as ALCO and investment committee, are involved in strategic decision-making processes related to financial risk management.
- Operations and technology committee focuses on ensuring the efficiency and effectiveness of operational processes and technology infrastructure within the organization.

Q.4069 In JCL bank, Fatou James is the head of the corporate treasury. Which among the following choices best lists the roles of her committee?

- A. It serves as an advisor and counsel to the management committee and LCT teams
- B. It monitors the ongoing business, funding, risk, and liquidity profile as part of its BAU activities.
- C. It provides recommendations on CFP actions.
- D. It provides oversight of the LCT and does a consultation with the board of directors to monitor the institution's liquidity risk profile

The correct answer is **B**.

The corporate treasury in a banking institution, such as JCL bank, is primarily responsible for monitoring the ongoing business, funding, risk, and liquidity profile as part of its Business As Usual (BAU) activities. This involves keeping a close eye on the bank's financial activities and ensuring that the bank has sufficient liquidity to meet its obligations. The corporate treasury also plays a crucial role in managing the bank's financial risks, which includes credit risk, market risk, and operational risk. By monitoring these aspects, the corporate treasury can help the bank maintain financial stability and avoid potential financial crises. This role is critical in ensuring the bank's long-term sustainability and success.

Choice A is incorrect. While the corporate treasury does provide advice and counsel, it is not its primary responsibility. The main role of the corporate treasury in a banking institution is to manage financial risks and ensure financial stability, which includes monitoring ongoing business, funding risk, and liquidity profile.

Choice C is incorrect. Providing recommendations on Contingency Funding Plan (CFP) actions may be part of the responsibilities of the corporate treasury but it's not its primary duty. The main function revolves around managing financial risks and ensuring stability.

Choice D is incorrect. Although oversight of Liquidity Contingency Teams (LCT) and consultation with the board of directors are important tasks performed by the corporate treasury, they do not constitute its primary responsibilities. The core duties involve managing financial risks including monitoring ongoing business activities, funding risk, and liquidity profile.

Things to Remember

- Corporate Treasury plays a crucial role in managing financial risks such as credit risk, market risk, and operational risk.
- Monitoring ongoing business activities, funding risk, and liquidity profile is a key responsibility of the corporate treasury.
- Contingency Funding Plan (CFP) actions may be part of the responsibilities of the

corporate treasury.

- Liquidity Contingency Teams (LCT) oversight and consultation with the board of directors are important tasks performed by the corporate treasury.
-

Q.4070 An effective CFP should ensure that there are contingency plans in place to cushion the bank when certain events that can potentially impact liquidity happen. Which among the following events does NOT have a negative liquidity impact on a financial institution?

- A. Presence of predatory trade
- B. Unavailability of the Federal Home Loan Bank Funding
- C. Shifting allocation from short-term funding to long-term funding sources
- D. When the intraday debit cap with Fedwire is exceeded

The correct answer is **C**.

Shifting allocation from short-term funding to long-term funding sources is a strategic move that can actually enhance a financial institution's liquidity position. Short-term funding sources, while readily available, can be volatile and subject to sudden changes in the market. On the other hand, long-term funding sources provide a more stable and reliable stream of capital. By shifting the allocation towards long-term funding, the institution can ensure a steady flow of funds over a longer period, thereby reducing the risk of liquidity stress. This strategy also allows the institution to better manage its cash flow and meet its financial obligations in a timely manner. Therefore, this action does not have a negative liquidity impact on the financial institution, but rather, it strengthens its liquidity position.

Choice A is incorrect. The presence of predatory trade can indeed pose a negative impact on a bank's liquidity. Predatory trading practices, such as front running or pump and dump schemes, can lead to significant financial losses for the bank and thereby reduce its available liquid assets.

Choice B is incorrect. The unavailability of the Federal Home Loan Bank Funding could also negatively affect a bank's liquidity. This funding source provides banks with access to secured loans which they can use to meet their short-term liquidity needs. If this source becomes unavailable, it could put strain on the bank's liquidity position.

Choice D is incorrect. Exceeding the intraday debit cap with Fedwire could also have negative implications for a bank's liquidity position. Fedwire is a real-time gross settlement system of central bank money used by Federal Reserve Banks to transfer funds electronically between member institutions. If a bank exceeds its intraday debit cap, it may face penalties or restrictions that could hinder its ability to manage its daily cash flows effectively.

Things to Remember

- Contingency planning is a crucial aspect of liquidity risk management for financial institutions.
 - Short-term funding sources are typically more volatile and can lead to liquidity stress during market disruptions.
 - Long-term funding sources provide stability and reliability in funding, reducing liquidity risk.
 - Front running and pump and dump schemes are examples of predatory trading practices that can impact a bank's liquidity.
 - Federal Home Loan Bank Funding is a source of secured loans that banks use for short-term liquidity needs.
 - Exceeding the intraday debit cap with Fedwire can result in penalties or restrictions that affect a bank's liquidity management.
-

Q.4072 Several market factors can affect the institution's ability to take contingent actions that help in strengthening the institution's liquidity position. Which among the following factors does NOT pose such an effect to institutions?

- A. Shutdown of securitization markets
- B. Deposit runoff
- C. Failure of the counterparties to roll over their deposits
- D. Withdrawal of debit caps

The correct answer is **D**.

The withdrawal of debit caps does not impact an institution's ability to take contingent actions to strengthen its liquidity position. Debit caps are limits set on the amount that can be debited from an account in a given period. Withdrawing these caps simply allows institutions to access more intraday credit, but it does not directly affect their ability to take contingent actions. It is important to note that while this may not be a typical scenario, it is a possibility that institutions must consider in their risk management strategies.

Choice A is incorrect. The shutdown of securitization markets can significantly impact an institution's ability to take contingent actions. Securitization markets provide a source of

liquidity for financial institutions by allowing them to convert assets into cash through the sale of asset-backed securities. If these markets shut down, it could limit the institution's access to liquidity and thus its ability to take contingent actions.

Choice B is incorrect. Deposit runoff, which refers to the withdrawal of deposits by customers, can also affect an institution's liquidity position and its ability to take contingent actions. If a significant number of depositors withdraw their funds, this could lead to a liquidity crunch for the institution.

Choice C is incorrect. The failure of counterparties to roll over their deposits can have a direct impact on an institution's liquidity position and its ability to take contingent actions. When counterparties decide not to renew or "roll over" their deposits at maturity, this reduces the available pool of funds that the financial institution has at its disposal.

Things to Remember

- Contingent actions are strategies or measures that institutions can take in response to changes in market conditions or other factors that may impact their liquidity position.
 - Securitization markets play a crucial role in providing liquidity to financial institutions by allowing them to convert assets into cash through the sale of asset-backed securities.
 - Deposit runoff can lead to a liquidity crunch for institutions if a significant number of depositors withdraw their funds.
 - Counterparties failing to roll over their deposits can directly impact an institution's liquidity position by reducing the available pool of funds.
 - Debit caps are limits set on the amount that can be debited from an account in a given period, and withdrawing these caps can provide institutions with more intraday credit.
-

Q.4073 When designing a CFP, monitoring and escalation is a key component of the CFP framework. In the framework, Institutions are required to define early warning indicators using market and business factors. Which of the following factors is NOT a potential factor considered?

- A. Macro-environment measures
- B. Industry measures
- C. Financial measures
- D. Institution-specific measures

The correct answer is **C**.

Financial measures are not necessarily required by institutions to monitor market trends and trends among their peer group. The term 'financial measures' is quite broad and does not specifically pertain to any particular aspect of an institution's operations or the market. While financial measures can provide valuable insights into an institution's financial health and performance, they are not typically used as early warning indicators in a Contingency Funding Plan (CFP). This is because financial measures often reflect the outcomes of various business and market factors, rather than the factors themselves. Therefore, they may not provide timely or accurate warnings of potential liquidity crises or other financial risks.

Choice A is incorrect. Macro-environment measures are typically considered in the creation of a Contingency Funding Plan (CFP). These measures can include economic indicators, market trends, and geopolitical events that could potentially impact the institution's liquidity position.

Choice B is incorrect. Industry measures are also taken into account when creating a CFP. These can include industry-specific risks, competitive dynamics, regulatory changes and technological advancements that could affect the institution's financial stability.

Choice D is incorrect. Institution-specific measures are crucial in developing a CFP as they directly relate to the institution's own operations and risk profile. These may encompass internal factors such as operational efficiency, financial performance, strategic initiatives and management quality.

Things to Remember

- Contingency Funding Plan (CFP) is a framework that helps institutions to manage liquidity risk by identifying potential stress scenarios and ensuring that they have adequate funding sources to meet their obligations.
- Monitoring and escalation in a CFP involves regularly tracking key indicators and triggers to detect early warning signs of liquidity stress and taking appropriate actions to address them.
- Early warning indicators in a CFP are typically based on a combination of market factors, business factors, and other external influences that could impact an institution's liquidity position.
- Market factors can include macro-environment measures, industry measures, and institution-specific measures that are relevant to the institution's operations and risk profile.
- Business factors such as operational efficiency, financial performance, strategic

initiatives, and management quality are important considerations in developing a CFP.

Q.4074 Early warning indicators (EWIs) should act as warning signs to the management to evaluate how changing market conditions and the institution's business strategy may be impacted. This should prompt the management to act in advance of oncoming market disruptions proactively. Which among the following factors is NOT an EWI encompassing market and business factors?

- A. A spike in the market volatility
- B. An increase in asset quality
- C. Real or perceived negative publicity
- D. Canceling loan commitments and refusing to renew maturing loans

The correct answer is **B**.

An increase in asset quality is not considered an Early Warning Indicator (EWI). EWIs are typically negative or adverse changes in market or business conditions that signal potential risks or disruptions. An increase in asset quality, on the other hand, is generally a positive development for a financial institution. It indicates that the institution's assets are performing well, which can enhance its financial stability and reduce the likelihood of liquidity stress. Therefore, an increase in asset quality does not serve as a warning sign of potential market disruptions or adverse changes in the institution's business strategy.

Choice A is incorrect. A spike in the market volatility can indeed serve as an Early Warning Indicator (EWI). Increased market volatility may signal potential disruptions or changes in the financial markets, which could impact a financial institution's business strategy and performance.

Choice C is incorrect. Real or perceived negative publicity can also be considered an EWI. Negative publicity, whether based on fact or perception, can affect a financial institution's reputation and customer trust, potentially leading to loss of business and revenue.

Choice D is incorrect. Cancelling loan commitments and refusing to renew maturing loans can be seen as an EWI as well. These actions might indicate that the institution is facing liquidity issues or other internal problems that could negatively affect its operations and profitability.

Things to Remember

- Early Warning Indicators (EWIs) are crucial for identifying potential risks and disruptions in the market or business environment.
- EWIs help management to take proactive measures to mitigate risks and adapt to

changing conditions.

- Market volatility can be a significant EWI as it may signal potential disruptions in the financial markets.
 - Negative publicity, whether real or perceived, can impact a financial institution's reputation and should be considered as an EWI.
 - Cancelling loan commitments and refusing to renew maturing loans can indicate internal problems and serve as EWIs.
 - Asset quality improvement is generally a positive development and not typically considered an EWI.
-

Q.4075 An institution must monitor a suite of liquidity health ratios besides reviewing macro-economic and industry measures as part of a CFP framework. These ratios aid in quantifying the impact of liquidity risks and supporting decision making on the CFP actions under consideration. The used capacity to total borrowing capacity is one of the liquidity health ratios. Which of the following accurately describes this measure?

- A. Measures the borrowing capacity available to the institution, based on used capacity relative to the total borrowing capacity
- B. Measures the cushion that liquid assets offer over obligatory funding needs and can be used to track the level of liquid assets available to offset volatile funding
- C. Measures the exposure to credit facilities that may be required at a future date
- D. Measures the funding and borrowing enough to finance the institution's increased lending banking activities

The correct answer is **A**.

The 'used capacity to total borrowing capacity' ratio is a measure of the borrowing capacity available to a financial institution. It is calculated based on the used capacity relative to the total borrowing capacity. This ratio is crucial as it provides insights into the extent to which the institution has utilized its borrowing capacity. A higher ratio indicates that the institution has used a larger portion of its total borrowing capacity, which could potentially signal a higher risk of liquidity issues. Conversely, a lower ratio suggests that the institution has a larger unused borrowing capacity, indicating a lower risk of liquidity problems.

Choice B is incorrect. This choice refers to the liquidity coverage ratio (LCR), which measures

the amount of high-quality liquid assets an institution has to meet its short-term obligations. It does not represent the 'used capacity to total borrowing capacity' ratio.

Choice C is incorrect. This option describes a potential future exposure measure, which estimates potential changes in credit exposure over time due to market movements or other factors. It does not describe the 'used capacity to total borrowing capacity' ratio.

Choice D is incorrect. This choice seems to refer more towards a measure of an institution's ability to finance its lending activities through funding and borrowing, rather than indicating how much of its total borrowing capacity has been used up, as described by the 'used capacity to total borrowing capacity' ratio.

Things to Remember

- **Liquidity Risk Management:** It is essential for financial institutions to manage liquidity risk effectively to ensure they can meet their obligations as they come due.
 - **Liquidity Coverage Ratio (LCR):** This ratio measures a financial institution's ability to meet its short-term obligations with high-quality liquid assets.
 - **Net Stable Funding Ratio (NSFR):** This ratio evaluates the stability of a financial institution's funding profile over a longer time horizon.
 - **Stress Testing:** Financial institutions conduct stress tests to assess their resilience to adverse market conditions and liquidity shocks.
 - **Contingency Funding Plan (CFP):** A CFP outlines strategies for managing liquidity shortfalls in times of stress or crisis.
-

Q.4076 In designing a CFP, institutions develop a series of escalation levels adequately aligned to the scenarios, contingency actions, and liquidity measures. However, there are no guidelines in the number of escalation levels required by CFPs. Which among the following statements is correct about the first level of escalation?

- A. At this level, the institution focuses mainly on survival
- B. It represents the elevated monitoring of market conditions and its effect on the business performance
- C. At this level, the institution has experienced noticeable markets and idiosyncratic events that are adversely affecting its business and liquidity risk profile
- D. None of the above

The correct answer is **B**.

The first level of escalation in a Contingency Funding Plan (CFP) is characterized by an increased level of monitoring of market conditions and their impact on the institution's business segments and performance. This level does not necessarily indicate a crisis but rather a heightened state of vigilance. The institution at this stage is not in survival mode or experiencing adverse market and idiosyncratic events significantly affecting its business and liquidity risk profile. Instead, it is proactively monitoring the market conditions to identify any potential threats to its business performance and take necessary actions in a timely manner. This proactive approach helps the institution to manage its risks effectively and ensure its financial stability.

Choice A is incorrect. The first level of escalation in a CFP does not focus mainly on survival. This stage is more about monitoring and identifying potential risks rather than dealing with immediate threats to the institution's survival.

Choice C is incorrect. At the first level of escalation, the institution has not necessarily experienced noticeable market and idiosyncratic events that are adversely affecting its business and liquidity risk profile. This stage involves elevated monitoring of market conditions, but it does not imply that adverse events have already occurred.

Choice D is incorrect. As explained above, option B correctly describes the characteristics and implications of the first level of escalation in a CFP, so 'None of the above' cannot be correct.

Things to Remember

- Contingency Funding Plan (CFP) is a crucial risk management tool for financial institutions to manage liquidity risk effectively.
- CFPs help institutions to identify potential liquidity stress scenarios, develop contingency actions, and establish liquidity measures to ensure financial stability.
- Escalation levels in a CFP help institutions to categorize the severity of liquidity risk situations and respond accordingly.
- Effective monitoring of market conditions is essential at all levels of escalation to proactively manage liquidity risks.
- Having a well-defined CFP with appropriate escalation levels can enhance an institution's resilience to liquidity shocks and crises.

Q.4077 In designing a CFP, liquidity health measures need to be put into consideration. Which among the following choices most accurately states the significance of these measures as part of

the monitoring and escalation component of the CFP framework?

- A. They indicate the institution's liquidity base and strength
- B. They indicate the institution's management policy
- C. They indicate the institution's liquidity profile
- D. None of the above

The correct answer is **A**.

Liquidity health measures are pivotal in indicating the institution's liquidity base and strength. These measures, such as the proportion of short-term funding to total funding, the deposits-to-loan ratio for depository businesses, and the firm's credit rating, are specifically designed to reflect any deterioration in these metrics that could have a direct and adverse impact on the institution's current and projected liquidity profile and strength. Therefore, these measures serve as a reliable indicator of the institution's liquidity base and strength, making them an essential component of the CFP framework.

Choice B is incorrect. While the institution's management policy may influence its liquidity health measures, it is not the primary reason why these measures are significant in designing a Contingency Funding Plan (CFP). The CFP primarily focuses on the institution's liquidity base and strength, which directly impacts its ability to meet financial obligations during a crisis.

Choice C is incorrect. Although an institution's liquidity profile can provide insights into its overall financial health, it does not specifically indicate why liquidity health measures are crucial in designing a CFP. The main purpose of these measures within a CFP framework is to assess and monitor the institution's liquidity base and strength.

Choice D is incorrect. As explained above, none of the above cannot be correct as option A correctly identifies that liquidity health measures indicate an institution's liquidity base and strength which forms an integral part of monitoring and escalation component of the CFP framework.

Things to Remember

- Liquidity risk management is a key aspect of financial risk management, focusing on the ability of an institution to meet its short-term obligations without incurring unacceptable losses.
- Contingency Funding Plans (CFPs) are designed to ensure that institutions have adequate liquidity to withstand stress events and disruptions in funding sources.
- Liquidity health measures can include metrics such as the liquidity coverage ratio (LCR), net stable funding ratio (NSFR), and cash flow projections under various stress

scenarios.

- Monitoring and escalation components of a CFP involve regular assessment of liquidity health measures to identify potential liquidity risks and trigger appropriate actions to address them.
 - Regulatory bodies like the Basel Committee on Banking Supervision provide guidelines and standards for liquidity risk management to promote stability in the financial system.
-

Q.4078 Which among the following choices is FALSE about scenario and liquidity gap analysis, a key component of the CFP framework?

- A. Institutions should align their CFP stress scenarios to those in its liquidity stress testing framework.
- B. The scenarios should align to those in other frameworks such as the recovery and resolution plans
- C. The liquidity stress testing scenarios should cover both systemic and institution-specific risks
- D. None of the above

The correct answer is **D**.

The statement 'None of the above' is false because all the given choices (A, B, and C) are true. Institutions should align their CFP stress scenarios to those in its liquidity stress testing framework. The scenarios should also align to those in other frameworks such as the recovery and resolution plans. The liquidity stress testing scenarios should cover both systemic and institution-specific risks. Therefore, the assertion that none of the above statements are true is incorrect.

Choice A is incorrect. Institutions should indeed align their Contingency Funding Plan (CFP) stress scenarios with those in its liquidity stress testing framework. This alignment ensures that the institution is prepared for potential liquidity stresses and can respond effectively.

Choice B is incorrect. The statement is true as the scenarios should align with other frameworks such as recovery and resolution plans. This alignment provides a comprehensive view of potential risks and ensures that all aspects of risk management are considered in the planning process.

Choice C is incorrect. The liquidity stress testing scenarios should cover both systemic and

institution-specific risks, which means this statement is also true. By considering both types of risks, institutions can ensure they are prepared for a wide range of potential stress events.

Things to Remember

- Contingency Funding Plan (CFP) framework is a key component of liquidity risk management for financial institutions.
 - Liquidity stress testing is a crucial tool for assessing an institution's ability to withstand liquidity stresses under various scenarios.
 - Systemic risks refer to risks that affect the entire financial system, while institution-specific risks are risks specific to a particular institution.
 - Recovery and resolution plans are frameworks designed to help institutions recover from financial distress or resolve their operations in an orderly manner in case of failure.
 - Aligning stress scenarios across different risk management frameworks helps ensure a comprehensive and integrated approach to risk management.
-

Q.4080 Which of the following components is NOT a part of the CFP framework?

- A. Governance and oversight
- B. Liquidity monitoring
- C. Contingent actions
- D. Monitoring and escalation

The correct answer is **B**.

Liquidity monitoring is not a component of the CFP framework. While it is a critical process in financial management, it stands as a separate entity and is not included in the CFP framework. Liquidity monitoring involves the regular assessment of an organization's liquidity position to ensure that it has sufficient funds to meet its short-term obligations. This process includes monitoring cash flows, analyzing liquidity ratios, and assessing the organization's overall financial health. However, it is not directly incorporated into the CFP framework, which focuses more on contingency planning and risk mitigation strategies.

Choice A is incorrect. Governance and oversight are crucial components of the CFP

framework. They ensure that there is a clear structure for decision-making, accountability, and control in managing financial resources and mitigating risks.

Choice C is incorrect. Contingent actions are also part of the CFP framework. These are predefined actions that an organization plans to take if certain risk events occur or certain conditions are met.

Choice D is incorrect. Monitoring and escalation procedures form an integral part of the CFP framework as well. They help in identifying potential risks early on and escalating them to the appropriate level for timely action.

Things to Remember

- CFP stands for Corporate Financial Planning, which is a framework used by organizations to manage their financial resources and mitigate risks.
 - Governance and oversight ensure clear decision-making, accountability, and control in financial management.
 - Contingent actions are predefined strategies that organizations plan to implement in response to specific risk events.
 - Monitoring and escalation procedures help in early identification of risks and timely actions to address them.
 - Liquidity monitoring involves assessing an organization's ability to meet short-term obligations with available funds.
-

Reading 138: Managing and Pricing Deposit Services

Q.4082 Which of the following statements about deposit pricing methods is **correct**?

- A. Cost-plus deposit pricing encourages banks to determine the costs they incur in labor and, management, and materials, among others, in offering each deposit service.
- B. Relationship pricing makes the customers less concerned about the prices of other competing financial institutions.
- C. Conditional pricing is whereby a depository institution sets up a schedule of fees in which the customer pays a low cost or no fee given that the deposit balance is above some minimum level.
- D. All of the above

The correct answer is **D**.

A is correct: Cost-plus deposit pricing encourages banks to determine the costs they incur in labor and management, and materials, among others, in offering each deposit service. Cost-plus pricing typically calls for a bank to charge deposit service fees enough to cover all the costs of providing the service in addition to a small margin for profit.

B is correct: In relationship pricing, the depository institution prices deposits according to the number of services purchased or utilized. The depositor may be given lower fees or have a part of the cost waived if they have used two or more services. Relationship pricing increases the dependency of customers on the institution which promotes greater customer loyalty. This also makes the customers less concerned about the prices of other competing financial institutions.

C is correct: Depository institutions use conditional pricing as a tool to attract the kind of depositors they want to have as customers. In this case, a depository institution sets up a schedule of fees in which the customer pays a low cost or no fee, given that the deposit balance is above some minimum level but is liable to higher charges if the average balance drops below that minimum level. Therefore, the price paid by customers is conditional on how they use their deposit accounts

Things to Remember

- Deposit pricing methods are important for banks to attract and retain customers by offering competitive pricing structures.
- Cost-plus deposit pricing involves determining the costs incurred in offering each deposit service and adding a margin for profit.
- Relationship pricing ties deposit pricing to the number of services utilized, promoting

customer loyalty.

- Conditional pricing sets fees based on the deposit balance, incentivizing customers to maintain a minimum balance.
 - Deposit pricing strategies can impact customer behavior and loyalty towards the financial institution.
-

Q.4083 XYZ bank determines that its basic checking account costs the bank \$5.00 per month in servicing costs (assume the servicing costs are labor and computer time) and \$3.00 per month in overhead expenses. This account requires a \$500 minimum balance. Additionally, the bank also tries to build a \$0.40 per month profit margin on these accounts. Determine the monthly fee that the bank should charge each customer.

- A. \$5.00
- B. \$7.60
- C. \$8.00
- D. \$8.40

The correct answer is **D**.

Recall that a bank should price every deposit service high enough to recover all or most of the cost of offering that service, using the following cost-plus pricing formula:

Unit price charged the customer for each deposit service
= (Operating expense per unit of deposit service
+ Estimated overhead expense allocated to the deposit service function
+ Planned profit margin from each service unit sold)

In this case,

$$\text{Unit price charged per month} = \$5.00 + \$3.00 + \$0.40 = \$8.40$$

Q.4084 XYZ bank determines that its basic checking account costs the bank \$5.00 per month in servicing costs (assume the servicing costs are labor and computer time) and \$3.00 per month in overhead expenses. This account requires a \$500 minimum balance. Additionally, the bank also tries to build a \$0.40 per month profit margin on these accounts. Further analysis of ABC Savings Bank customer accounts reveals that for each \$120 above the \$500 minimum balance maintained in its checking accounts, the bank saves about 6% in operating expenses with each customer account. For a customer who is consistent in maintaining an average monthly balance of \$860, how much should the bank charge to protect its profit margin?

- A. \$4.25
- B. \$5.00
- C. \$7.00
- D. \$7.5

The correct answer is **D**.

If the bank saves about 6% in operating expenses for each \$120 held in balances above the minimum of \$500, then a customer who maintains an average monthly balance of \$860 saves the bank 18% in operating expenses.

$$\text{New operating expenses} = \$5.00 - (18\% \times \$5.00) = \$4.1$$

The appropriate amount that the bank should charge to protect its profit margin is therefore

$$\$4.1 + 3.00 + 0.40 = \$7.5 \text{ per month}$$

Q.4085 Spike Bank determines from an analysis of its cost-accounting figures that for each \$1,000 minimum-balance checking account it transacts, account processing and other operating costs will average \$10.00 per month. Further, its overhead expenses will run an average of \$4.00 per month. The bank's goal is to achieve a profit margin over these particular costs of 5% of the total monthly expenses. What monthly fee should it charge a customer who opens a checking account?

- A. \$7.00
- B. \$10.00
- C. \$14.04
- D. \$14.70

The correct answer is **D**.

The appropriate formula is:

$$\begin{aligned} \text{Unit price charged per month} \\ &= \text{Operating expense per unit} + \text{Overhead expense per unit} \\ &+ \text{Planned profit margin per unit} \end{aligned}$$

$$\text{Unit price charged per month} = \$10.00 + \$4.00 + (0.05 \times (10.00 + 4.00)) = \$14.70$$

Q.4086 For depository organizations to price their deposits adequately, they must know the cost of the deposits. Different approaches may be used in determining deposit costs. Which one is the most appropriate?

- A. Historical average cost approach
- B. Marginal cost deposit pricing approach
- C. Average percentage yield approach
- D. None of the above

The correct answer is **B**.

The marginal cost deposit pricing approach is a method that focuses on the weighted average cost of new funds raised from various sources that the bank draws upon or plans to draw upon in the current period. This approach is preferred by depository institutions over the historical average cost approach because it takes into account the frequent changes in interest rates, making it a more reliable standard for pricing. The marginal cost deposit pricing approach is

more dynamic and responsive to current market conditions, allowing depository institutions to price their deposits more accurately and competitively.

Choice A is incorrect. The historical average cost approach is not the most suitable method for determining deposit costs. This approach uses past data to calculate the average cost of deposits, which may not accurately reflect current market conditions or future trends. It also does not take into account changes in interest rates or other factors that can significantly impact deposit costs.

Choice C is incorrect. The average percentage yield approach calculates the average yield on all deposits, but it does not provide a clear picture of the actual costs associated with each individual deposit. This method fails to consider variations in deposit types and terms, which can lead to inaccurate pricing and potential losses for depository institutions.

Choice D is incorrect. There are indeed methodologies that are considered suitable for determining deposit costs, hence 'None of the above' is an incorrect choice.

Things to Remember

- Deposit costs are a crucial factor for depository organizations as they directly impact profitability and competitiveness in the market.
 - Understanding the cost of deposits helps depository institutions in setting appropriate pricing strategies to attract and retain customers.
 - Marginal cost deposit pricing approach considers the current market conditions and changes in interest rates, making it a more accurate method for pricing deposits.
 - Depository institutions need to regularly review and adjust their deposit pricing strategies to remain competitive and profitable.
 - Factors such as regulatory requirements, liquidity needs, and risk management considerations also play a role in determining deposit costs.
-

Q.4087 Use the APY formula needed by the Truth in Savings Act for the following calculation. Sundeep Raichura is a customer at Bright Bank. Raichura holds a savings deposit in the bank for a year. The balance in the account stood at \$5,000 for 100 days and \$200 for the remaining days of the year. Assume that Bright Bank paid Raichura \$20.00 in interest earnings for the year, what APY did Raichura receive?

- A. 0.01
- B. 0.0132
- C. 0.0154
- D. 0.0199

The correct answer is **B**.

The right formula is:

$$\text{APY earned} = 100 \left[1 + \left(\frac{\text{Interest earned}}{\text{Average account balance}} \right)^{\frac{365}{\text{Days in period}}} - 1 \right]$$

$$\text{Average account balance} = \frac{(\$5,000 \times 100 \text{ days}) + (\$200 \times 265 \text{ days})}{365 \text{ days}} = \$1,515$$

$$\text{APY} = 100 \left[\left(1 + \frac{\$20.00}{\$1,515} \right)^{\frac{365}{365}} - 1 \right] = 1.32\%$$

Q.4088 Brainstorm Savings Bank posts the following fees schedule for its small business transaction accounts.

Average Monthly Account Balances		
Range	Monthly Maintenance Fee	Charge Per Check
Over \$1,000	\$0	\$0
\$500 – \$1,000	\$2	\$0.10
Less than \$500	\$4	\$0.20

What form of business pricing is this?

- A. Relationship pricing
- B. Conditional pricing
- C. Cost-plus profit pricing
- D. Marginal cost approach

The correct answer is **B**.

Conditional pricing is a pricing strategy where the price of a product or service is determined based on certain conditions. In this case, Brainstorm Savings Bank is using conditional pricing as the fees charged are dependent on the average monthly account balance. The bank is encouraging customers to maintain a higher balance in their accounts by waiving fees for accounts with an average monthly balance over \$1,000. This strategy not only provides the bank with a more stable funding base but also allows it to have more money available for use. The higher fees charged on accounts with less than \$500 could potentially deter small depositors, driving them to other banks.

Choice A is incorrect. Relationship pricing refers to a strategy where the bank offers better rates or lower fees to customers who have multiple relationships with the bank, such as having a checking account, mortgage, and credit card with the same institution. In this case, there is no information provided about multiple relationships influencing the fee structure.

Choice C is incorrect. Cost-plus profit pricing involves setting prices based on covering costs and achieving a certain profit margin. The question does not provide any information about cost coverage or profit margins in determining fees.

Choice D is incorrect. Marginal cost approach involves setting prices based on the additional cost of producing one more unit of output (in this case, providing one more transaction). The fee schedule provided does not appear to be based on marginal costs as it varies by average monthly balance rather than per transaction.

Q.4089 XYZ Bank establishes that it can attract the following amounts of deposits if it agrees to offer new depositors and those rolling over their maturing term deposits the interest rates in the

following table:

Expected Amount of New Deposits	Rate of Interest Offered to Depositors
6,000,000	3.00%
10,000,000	3.25%
12,000,000	3.50%
15,000,000	3.75%
20,000,000	5.50%

Management hopes to invest any new deposits raised in loans yielding 7%. How far should this thrift institution go in raising its deposit interest rate to maximize total profits (excluding operational costs)?

- A. 0.0325
- B. 0.035
- C. 0.0375
- D. 0.055

The correct answer is **C**.

Funds Raised	Average Rate Paid	Total Interest	Marginal Cost	Change in Total Cost	Expected Revenue	Difference Expected less Marginal	Total Additional Profit
6,000,000	3.00%	180,000	180,000	3.00%	7.00%	4.00%	240,000
10,000,000	3.25%	325,000	145,000	3.63%	7.00%	3.38%	375,000
12,000,000	3.50%	420,000	95,000	4.75%	7.00%	2.25%	420,000
15,000,000	3.75%	562,500	142,500	4.75%	7.00%	2.25%	487,500
20,000,000	5.50%	1,100,000	537,500	10.75%	7.00%	-3.75%	300,000

The calculations are as follows:

$$\begin{aligned}\text{Total interest} &= \text{Funds raised} \times \text{Average Rate Paid} \\ &= 6,000,000 \times 3.00\% = 180,000\end{aligned}$$

$$\begin{aligned}\text{Marginal cost} &= \text{Change in total cost} \\ &= \text{New interest rates} \times \text{Total funds raised at new rate} \\ &\quad - \text{Old interest rate} \times \text{Total funds raised at old rate} \\ &= (3.25 \times 10,000,000) - (3.00\% \times 6,000,000) = 145,000\end{aligned}$$

Alternatively, marginal cost is the additional interest earned in new deposit money.

$$\begin{aligned}\text{Marginal cost} &= 325,000 - 180,000 = 145,000 \\ \text{Change in Total Cost} &= \frac{\text{Marginal cost}}{\text{New Funds Raised}} \\ &= \frac{145,000}{6,000,000} = 3.00\%\end{aligned}$$

For the second case,

$$\text{Change in Total Cost} = \frac{145,000}{10,000,000 - 6,000,000} = 3.63\% \text{ and so on.}$$

XYZ Bank should raise its deposit rate to 3.75%, attracting \$15,000,000 in new deposits; because up to then, the marginal revenue rate is higher than the marginal cost rate, and total profits are also rising. At 5.50%, the marginal cost rate is higher than the marginal revenue rate, and overall profits have fallen from a high of \$487,500 back down to \$300,000.

Q.4090 Young Money Savings Bank realizes that its basic transaction account, which requires a \$300 minimum balance, costs this savings bank an average of \$3.00 per month in servicing costs (including labor and computer time) and \$1.50 per month in overhead expenses. The savings bank also tries to build in a \$0.70 per month profit margin on these accounts. What monthly fee should the bank charge each customer?

- A. \$3.00
- B. \$4.50
- C. \$5.00
- D. \$5.20

The correct answer is **D**.

Following the cost-plus-profit approach, the monthly fee should be:

$$\text{Monthly fee} = \$3.00 + \$1.50 + \$0.70 = \$5.20 \text{ per month.}$$

Q.4091 Young Money Savings Bank realizes that its basic transaction account, which requires a \$300 minimum balance, costs this savings bank an average of \$3.00 per month in servicing costs (including labor and computer time) and \$1.50 per month in overhead expenses. The savings bank also tries to build in a \$0.70 per month profit margin on these accounts. Further analysis of customer accounts reveals that for each \$50 above the \$300 minimum in the average balance maintained in its transaction accounts, Young Money Savings Bank saves about 2% in operating expenses with each account. For a customer who consistently holds an average balance of \$400 per month, how much should the bank charge to cushion its profit margin?

- A. \$4.00
- B. \$4.82
- C. \$5.08
- D. \$5.10

The correct answer is C.

If the bank saves about 2% in operating expenses for each \$50 held in balances above the minimum of \$300, then a customer who maintains an average monthly balance of \$400 saves the bank 4% in operating expenses.

That is, $\$400 - \$300 = \frac{\$100}{\$50} = 2 \times 2\% = 4\%$ saving in the operating costs

$$\text{New operating expenses} = \$3.00 - (4\% \times \$3.00) = \$2.88$$

The appropriate amount that the bank should charge to protect its profit margin is therefore

$$\$2.88 + 1.50 + 0.70 = \$5.08 \text{ per month}$$

Note.

Overheads are the expenditure which cannot be conveniently traced to or identified with any particular cost unit, unlike operating expenses such as raw material and labor. That's why we add the \$1.5 overhead as a separate item.

Q.4092 Neha Datta maintains a savings deposit with XYZ Credit Union. In 2019, Datta received \$12.25 in interest earnings from her savings account. Her savings deposit had the following average balance each month:

Month	Account Balance
January	520
February	380
March	440
April	300
May	385
June	470
July	530
August	615
September	750
October	810
November	845
December	530

What was the annual percentage yield (APY) earned on Datta's savings account?

- A. 0.0125
- B. 0.0156
- C. 0.0196
- D. 0.0223

The correct answer is **D**.

$$\text{APY earned} = 100 \left[\left(1 + \frac{\text{Interest earned}}{\text{Average account balance}} \right)^{\frac{365}{\text{Days in period}}} - 1 \right]$$

Month	Account Balance	Number of Days in the Month	AC Balance* Days
January	520	31	16,120
February	380	28	10,640
March	440	31	13,640
April	300	30	9,000
May	385	31	11,935
June	470	30	14,100
July	530	31	16,430
August	615	31	19,065
September	750	30	22,500
October	810	31	25,110
November	845	30	25,350
December	530	31	16,430
Total		365	200,320

$$\text{Average account balance} = \frac{200,320}{365 \text{ days}} = \$548.82$$

$$\text{APY} = 100 \left[\left(1 + \frac{\$12.25}{\$548.82} \right)^{\frac{365}{365}} - 1 \right] = 2.23\%$$

Q.4093 A hypothetical Bank of India quotes an APY of 4.0% on a one-year money market Certificate of Deposit sold to one of the businesses in town. The firm posted a balance of \$3,000 for the first 100 days of the year, \$2,500 over the next 100 days, and \$1,000 for the remainder of the year. How much in total interest earnings did this business customer receive for the year?

- A. 56.7
- B. 60.54
- C. 78.36
- D. 80.46

The correct answer is **C**.

Using the APY formula, we proceed as follows:

$$\text{APY earned} = 100 \left[\left(1 + \frac{\text{Interest earned}}{\text{Average account balance}} \right)^{\frac{365}{\text{Days in period}}} - 1 \right]$$

$$\begin{aligned} \text{Average account balance} \\ = \frac{\$3,000 \times 100 \text{ days} + (\$2,500 \times 100 \text{ days}) + (\$1,000 \times 165 \text{ days})}{365 \text{ days}} = \$1,959 \end{aligned}$$

$$4.0\% = 100 \left[\left(1 + \frac{\text{Interest}}{\$1,959} \right)^{\frac{365}{365}} - 1 \right]$$

$$\text{Interest earned} = \$78.36$$

Q.4094 Fatou Heckerman has been holding \$ 250,000 in Smart-way Bank and another \$250,000 in Pathway Savings Bank. Fatou has learned that the two institutions have merged one month

ago. Fatou reported the case to FDIC to ensure appropriate deposit coverage. How much should be the total protection offered to Fatou given the above scenario?

- A. \$50,000
- B. \$250,000
- C. \$500,000
- D. \$25,000

The correct answer is **C**.

As per the rules of the Federal Deposit Insurance Corporation (FDIC), the standard insurance amount is \$250,000 per depositor, per insured bank, for each account ownership category. This means that in the event of a bank failure, the FDIC insures deposits up to \$250,000.

However, in the case of a merger of two FDIC-insured banks - like in Fatou's situation - the deposits from the assumed bank continue to be insured separately for some time until they find a way to transfer some of the funds to other institutions. This grace period gives a depositor the opportunity to restructure his or her accounts, if necessary

Given that Fatou learned about the merger in December and the banks had merged a month before, it's less than six months from the merger date. Therefore, Fatou's deposits in both banks would still be separately insured. As such, she should be covered for \$250,000 in each bank, for a total of \$500,000.

Things to Remember

- FDIC (Federal Deposit Insurance Corporation) provides deposit insurance to depositors in US banks in case of bank failures.
- The standard insurance amount provided by FDIC is \$250,000 per depositor, per insured bank, for each account ownership category.
- In the event of a bank merger, deposits from the assumed bank continue to be insured separately for a grace period.
- Depositors have the opportunity to restructure their accounts during this grace period to ensure full coverage.

- It's important for depositors to be aware of the FDIC insurance limits and to spread their deposits across different institutions if they have large amounts.

Q.4096 ABC Credit Union is launching a new deposit campaign next week in anticipation of bringing in from \$200 million to \$700 million in new deposit money, which it expects to invest at a 7% yield. The management believes that an offer rate on new deposits of 3% would attract \$200 million in new deposits and rollover funds. To attract \$300 million, the bank would probably be forced to offer 4%. ABC's forecast suggests that \$400 million might be available at 4.25%, \$500 million at 4.50%, \$600 million at 4.75%, and \$700 million at 5.25%. What volume of deposits should the institution try to avoid to ensure that marginal cost does not exceed marginal revenue?

- A. \$300 Million
- B. \$400 Million
- C. \$600 Million
- D. \$700 Million

The correct answer is **D**.

Expected Inflows in Millions	Rate Offered on New Funds	Total Interest Cost	Marginal Interest Cost	Marginal Cost Rate	Marginal Revenue Rate	Expected Difference In Marginal Revenues and Costs	Total Profits Earned
200	3.00%	6.00	6.00	3.00%	7.00%	4.00%	8.00
300	4.00%	12.00	6.00	6.00%	7.00%	1.00%	9.00
400	4.25%	17.00	5.00	5.00%	7.00%	2.00%	11.00
500	4.50%	22.50	5.50	5.50%	7.00%	1.50%	12.50
600	4.75%	28.50	6.00	6.00%	7.00%	1.00%	13.50
700	5.25%	36.75	8.25	8.25%	7.00%	-1.25%	12.25

Total Interest Costs = \$Expected inflows x Rate of new funds%

Marginal Interest Costs = Total interest costs for the current expected inflow - Total interest costs for the previous expected inflow

Marginal Cost Rate = Marginal costs ÷ Increase in the expected inflows

Increase in the Expected Inflows = Current expected inflow - Previous expected inflow

Marginal Revenue Rate = Expected investment yield

Expected Difference in Marginal Revenues and Costs = Marginal revenue rate - Marginal costs rate

Total Profits Earned = (Marginal revenue rate - Rate offered on new funds)% x Expected inflows

From the table above, at \$700 million, the marginal cost rate of 8.25%% is greater than the marginal revenue rate of 7.00%. Therefore, ABC Credit Union should try to avoid \$700 million in new deposits.

Q.5414 Given the information below, what is the correct amount for the bank to charge per ATM visit according to the cost-plus pricing model?

- Profit required per ATM visit: USD 0.08
- The bank's target return on capital: 14%
- Bank's return on equity: 17%
- Bank's cost to income ratio: 39%
- Operating expense per ATM visit: USD 0.33
- Estimated overhead cost allocated per ATM visit: USD 0.42

A. \$0.83

B. \$0.75

C. \$0.97

D. \$0.94

The correct answer is **A**.

Based on a cost-plus pricing formula:

$$\begin{aligned}\text{Unit price charged by bank} &= \text{Operating expense per unit of deposit service} \\ &\quad + \text{Estimated overhead expense allocated to the} \\ &\quad \text{deposit service function} \\ &\quad + \text{Profit margin per unit service sold} \\ &= 0.33 + 0.42 + 0.08 = \$ 0.83\end{aligned}$$

Reading 139: Managing Nondeposit Liabilities

Q.4097 Promise Bank takes out a loan of \$30 million overnight through a repurchase agreement (RP), which is collateralized by Treasury bills. The current RP rate is 4%. Calculate the interest costs that the bank pays due to this borrowing.

- A. \$3,333.33
- B. \$6,657.67
- C. \$8,768.90
- D. \$9,754.78

The correct answer is **A**.

$$\begin{aligned} &\text{Interest cost of RP} \\ &= \frac{\text{Amount borrowed} \times \text{Current RP rate} \times \text{Number of days in RP borrowing}}{360 \text{ days}} \\ &= \$30,000,000 \times 4\% \times \frac{1}{360} = 3,333.33 \end{aligned}$$

Q.4098 Short term notes with maturities of 3 or 4 days to 9 months issued by well-known companies are known as:

- A. Commercial paper
- B. Negotiable CDs
- C. Eurocurrency deposits
- D. None of the above

The correct answer is **A**.

Commercial paper is a type of unsecured, short-term debt instrument issued by corporations, typically for the financing of accounts receivable, inventories and meeting short-term liabilities. Maturities on commercial paper rarely range any longer than 270 days. The debt is usually issued at a discount, reflecting prevailing market interest rates. Commercial paper is an important part of many money market funds. They are issued with a fixed interest rate and sold at a discount from face value. The maturity period of commercial paper typically ranges from overnight to just under 9 months. Since they are issued by well-known companies and are short-term, they are considered relatively low-risk.

Choice B is incorrect. Negotiable CDs are certificates of deposit that can be traded in the secondary market, unlike traditional CDs which are held until maturity. They have longer maturities than commercial papers and are not typically used for short-term financing needs.

Choice C is incorrect. Eurocurrency deposits refer to deposits made in a bank located outside the country of the currency being deposited. These deposits do not necessarily have to be short-term and can range from overnight to several years, making them different from the described financial instrument.

Choice D is incorrect. As explained above, there exists a specific term for these short-term notes - Commercial Paper (Option A), hence 'None of the above' does not apply here.

Things to Remember

- Commercial paper is typically issued at a discount and has maturities ranging from 3 to 270 days.
 - Money market funds often invest in commercial paper due to its short-term nature and relatively low risk.
 - Negotiable CDs are certificates of deposit that can be traded in the secondary market.
 - Eurocurrency deposits are made in a bank located outside the country of the currency being deposited.
 - Commercial paper is an important source of short-term financing for corporations.
-

Q.4099 A bank with immediate reserve needs can borrow from the federal reserve. It can acquire different types of loans, based on its obligations. Gift Bank is a sound financial institution. Its management agrees to borrow a short term loan whose rate is higher than the federal funds rate. Which of the following refers to this type of loan?

- A. Primary credit
- B. Secondary credit
- C. Seasonal credit
- D. None of the above

The correct answer is **A**.

Primary credit is the type of loan that the Federal Reserve provides to financially sound

institutions for short-term needs. The interest rate for primary credit is typically higher than the federal funds rate. This type of credit is usually used to meet unexpected liquidity demands or to fund regular short-term borrowing needs. The Federal Reserve offers this type of credit with minimal administrative procedures, usually on a very short-term basis, typically overnight. The primary credit program is the principal safety valve for ensuring adequate liquidity in the banking system.

Choice B is incorrect. Secondary credit is typically offered to smaller, less financially secure institutions that are in need of short-term liquidity or experiencing severe financial difficulties. This does not align with the scenario presented where Gift Bank is described as a financially stable institution.

Choice C is incorrect. Seasonal credit is designed to assist small depository institutions in managing significant seasonal swings in their loans and deposits, which does not apply to the situation described for Gift Bank.

Choice D is incorrect. The scenario clearly states that Gift Bank has decided to borrow a short-term loan from the Federal Reserve, which contradicts with this option of 'None of the above'.

Things to Remember

- Primary credit is typically provided by the Federal Reserve to financially sound institutions for short-term needs at a higher interest rate than the federal funds rate.
 - Secondary credit is offered to smaller, less financially secure institutions in need of short-term liquidity.
 - Seasonal credit is designed to help small depository institutions manage seasonal fluctuations in loans and deposits.
 - Primary credit is a key tool for ensuring liquidity in the banking system.
-

Q.4100 Trust Bank purchases a 50-day negotiable CD with a \$10 million denomination from Promise Bank, bearing a 5% annual yield. How much interest is the bank obliged to pay at the maturity of the CD?

- A. \$10,456
- B. \$12,567
- C. \$50,678
- D. \$69,444

The correct answer is **D**.

$$\begin{aligned}\text{Interest owed} &= \text{Principal} \times \frac{\text{Days to maturity}}{360 \text{ days}} \times \text{Annual interest rate} \\ \text{Interest owed} &= \$10,000,000 \times \frac{50}{360} \times 5\% = \$69,444\end{aligned}$$

Q.4101 Central Credit Union borrows \$200 million overnight through a repurchase agreement (RP), which is collateralized by Treasury bills. The current RP rate is 5%. What is the interest cost that the bank should pay for this borrowing?

- A. \$27,777.80
- B. \$30,433.76
- C. \$33,333.67
- D. \$46,879.20

The correct answer is **A**.

$$\begin{aligned}\text{Interest cost of RP} &= \frac{\text{Amount borrowed} \times \text{Current RP rate} \times \text{Number of days in RP borrowing}}{360 \text{ days}} \\ &= \$200,000,000 \times 5\% \times \frac{1}{360} = \$27,777.80\end{aligned}$$

Q.4102 Prime Limited is expecting new deposit inflows of \$400 million and deposit withdrawals

of \$600 million next month. The bank has projected that new loan demand will reach \$500 million, and customers with approved credit lines will need \$200 million in cash. The bank will sell \$375 million in securities but plans to add \$100 million in securities to its portfolio. Calculate the projected available funds gap.

- A. \$600
- B. \$775
- C. \$625
- D. \$425

The correct answer is C.

To calculate the projected available funds gap for Prime Limited, we need to sum up all the expected inflows and subtract from that the expected outflows, including loans disbursed and securities sold, while considering the additional securities purchased.

Given.

- New deposit inflows: \$400 million
- Deposit withdrawals: \$600 million
- New loan demand: \$500 million
- Customers with approved credit lines needing cash: \$200 million
- Securities to be sold: \$375 million
- Securities to be added: \$100 million

We can calculate the net expected inflows and outflows:

Total inflows = Deposit inflows + Securities to be sold

$$= 400 + 375 = 775 \text{ million}$$

Total outflows = Deposit withdrawals + New loan demand + Securities to be added + Approved credit lines to be funded

$$= 600 + 500 + 100 + 200 = 1400 \text{ million}$$

Now, we can calculate the projected available funds gap:

Projected available funds gap = Total inflows - Total outflows

$$= 775 - 1400 = -625 \text{ million}$$

Since the result is negative, it indicates that Prime Limited is projected to have a shortfall or gap in available funds of \$625 million. This means they may need to seek additional funding sources or adjust their operations to manage the shortfall.

Q.4103 Suppose that Bluemont bank borrows \$85 million using an RP transaction with a government bond collateral for twelve days, and the current RP rate in the market is 4%. How much would be the bank's total interest cost?

- A. \$113,333.33
- B. \$123,445.90
- C. \$134,670.08
- D. \$167,346.78

The correct answer is **A**.

Interest cost of RP

$$= \frac{\text{Amount borrowed} \times \text{Current RP rate} \times \text{Number of days in RP borrowing}}{360 \text{ days}}$$

$$= \$85,000,000 \times 4\% \times \frac{12}{360} = \$113,333.33$$

Q.4104 Suppose a depository institution promises a 7% annual interest rate to a buyer of a \$320,000, 150 days Negotiable CD. How much will the depositor have at the maturity date?

- A. \$9,345
- B. 14678
- C. \$329,333
- D. \$339,345

The correct answer is **C**.

$$\begin{aligned} &\text{Amount due CD customer} \\ &= \text{Principal} + \text{principal} \times \frac{\text{Days to maturity}}{360 \text{ days}} \times \text{Annual rate of interest} \\ &= \$320,000 + \$320,000 \times \frac{150}{360} \times 7\% = \$329,333 \end{aligned}$$

Q.4105 Suppose that a commercial bank has a new loan request that meets its quality standards of \$660 million; it expects to purchase \$330 million in new Treasury securities being issued on Friday the same week and expects drawings on credit lines from its best corporate customers of \$358 million. Deposits and other customer funds received today total \$456 million, and those expected in the coming week will inject another \$265 million. If the current funding gap is zero, what is the bank's estimated available funds gap (AFG) for the coming week?

- A. \$326
- B. \$423
- C. \$347
- D. \$627

The correct answer is **D**.

Available funds gap (AFG) = (Current and projected loans and investments the lending institution desires to make **LESS** current and expected deposit inflows and other available funds)

$$\text{AFG} = (\$660 + \$330 + \$358) - (\$456 + \$265) = \$1,348 - \$721 = \$627 \text{ million.}$$

Q.4106 Suppose that banks and other lending affiliates within Niche Bank Holding Company have a substantial loan demand from several companies that have a significant expansion project of their facilities before the beginning of the next financial year. The holding company and the bank in charge have a plan to raise \$1 billion in short term funds this week, of which \$0.9 billion will be used to meet these new loan requests. The following table shows the current annual interest rates on alternative sources of funds.

	Market Interest rates	Noninterest cost rates
Federal Funds	2.10%	0.42%
Negotiable CDs	2.25%	0.42%
Eurodollars	2.15%	0.52%
Commercial paper	2.20%	0.67%
Fed Discount Rate	3.10%	0.42%

Calculate the effective cost rates of the Federal Funds and Negotiable CDs, respectively.

- A. 2.10% and 2.25%
- B. 2.80% and 2.97%
- C. 2.97% and 3.19%
- D. 2.80% and 3.91%

The correct answer is **B**.

	Market Interest Rates	Noninterest Cost Rates	Market Interest Cost	Noninterest Cost
Federal Funds	2.10%	0.42%	21,000,000	4,200,000
Negotiable CDs	2.25%	0.42%	22,500,000	4,200,000

$$\begin{aligned}
 &\text{Effective cost rate on deposit and nondeposit sources of funds} \\
 &\quad (\text{Current interest cost on amounts borrowed} \\
 &\quad + \text{Noninterest costs incurred to access these funds}) \\
 &= \frac{\quad}{\text{Net investable funds raised from this source}}
 \end{aligned}$$

Where:

$$\begin{aligned}
 &\text{Current interest cost on amounts borrowed} \\
 &= \text{Prevailing interest rate in the money market} \times \text{Amount of funds borrowed}
 \end{aligned}$$

For Federal Funds,

$$\begin{aligned}\text{market interest cost} &= 2.10\% \times 1,000,000,000 = 21,000,000 \\ \text{Noninterest cost} &= 0.42\% \times 1,000,000,000 = 4,200,000\end{aligned}$$

Thus the effective cost rate of the Federal Funds is $\frac{21,000,000+4,200,000}{900,000,000} = 2.80\%$

Similarly, the effective cost rate of the Negotiable CDs is $\frac{22,500,000+4,200,000}{900,000,000} = 2.97\%$.

Q.4107 Suppose that banks and other lending affiliates within Niche Bank holding company have a substantial loan demand from several companies that have a significant expansion project of their facilities before the beginning of the next financial year. The holding company and the bank in charge have a plan to raise \$1billion in short term funds this week, of which \$0.9 billion will be used to meet these new loan requests. The following table shows the current annual interest rates on alternative sources of funds.

	Market Interest rates	Noninterest cost rates
Federal Funds	2.10%	0.42%
Negotiable CDs	2.25%	0.42%
Eurodollars	2.15%	0.52%
Commercial paper	2.20%	0.67%
Fed Discount Rate	3.10%	0.42%

Assume that you are to advise Niche Bank Holding Company management on the best source of funding to use. Which one would you recommend?

- A. Federal Funds
- B. Negotiable CDs
- C. Eurodollars
- D. Federal Discount Rate

The correct answer is **A**.

Federal Funds is the most cost-effective source of funding for Niche Bank Holding Company. The total cost of borrowing from the Federal Funds market, which includes both the market interest cost and the noninterest cost, is the lowest among all the alternative sources of funds. The market interest cost for Federal Funds is calculated as 2.10% of \$1 billion, which equals \$21 million. The noninterest cost is calculated as 0.42% of \$1 billion, which equals \$4.2 million. Therefore, the total cost of borrowing from the Federal Funds market is \$25.2 million. When this total cost is divided by the amount of funds that will be used to meet the new loan requests (\$0.9 billion), the effective cost rate of borrowing from the Federal Funds market is found to be 2.80%. This is the lowest effective cost rate among all the alternative sources of funds, making Federal

Funds the most cost-effective source of funding for Niche Bank Holding Company.

Choice B is incorrect. Negotiable CDs, while a viable option for raising funds, typically have higher interest rates compared to Federal Funds. This would mean that Niche Bank Holding Company would incur more costs in terms of interest payments if they choose this funding source.

Choice C is incorrect. Eurodollars are deposits denominated in U.S. dollars at banks and bank branches located outside the United States, and thus involve additional risks such as exchange rate risk and country risk which could potentially increase the cost of borrowing for Niche Bank Holding Company.

Choice D is incorrect. The Federal Discount Rate is usually higher than the rate on Federal Funds because it's a penalty rate set above market levels to discourage regular use by banks. Therefore, it would be more expensive for Niche Bank Holding Company to raise funds through this source.

Q.4109 Suppose that ABC Bank Ltd receives \$1,530 million from the following sources. These sources are listed below with their associated costs:

Funding source	Amount	Interest Costs	Noninterest Costs	Additional costs	Total Costs
Money-Market Borrowings	275	3.03%	0.24%	0.24%	3.51%
Time and Savings Deposits	580	2.27%	0.23%	0.60%	3.10%
Checkable deposits	385	0.52%	1.96%	0.80%	3.28%
Stockholders Equity	290	12.75%			12.75%

Assuming that the bank has earning assets worth \$1,240 (i.e., \$1,530 - \$290), calculate the break-even cost rate.

- A. 1.9%
- B. 5.0%
- C. 3.2%
- D. 6.2%

The correct answer is **D**.

$$\text{Break even cost rate} = \frac{\text{Total funding costs}}{\text{Earning assets}}$$

Funding Source	Amount	Interest Cost	Other Cost	Total
Money-Market Borrowings	275	8.33	1.32	9.65
Time and Savings Deposits	580	13.17	4.81	17.98
Checkable deposits	385	2.00	10.63	12.63
Stockholders Equity	290	36.98	—	36.98
Total	1,530	23.50	16.76	77.24

Interest cost = \$Amount x Interest costs %

Other costs = \$Amount x (Non-interest costs + Additional costs)%

Total costs = \$ Interest costs + \$Other costs

$$\text{Break even cost rate} = \frac{77.24}{1,240} = 6.2\%$$

Q.4110 Suppose that Mug Bank, a fictional financial institution in the Netherlands, has an estimated cost of overhead expenses and salaries of \$13 million. The earning assets are worth \$540 million. Suppose further that the total interest cost is \$127 million. Evaluate the bank's break-even cost rate on borrowed funds invested in earning assets.

- A. 26%
- B. 28%
- C. 32%
- D. 40%

The correct answer is **A**.

$$\begin{aligned} &\text{Break-even cost rate on borrowed funds invested in earning assets} \\ &= \frac{\text{Interest cost} + (\text{Other operating costs})}{(\text{Total earning assets})} = \frac{127 + 13}{540} = 26\% \end{aligned}$$

Q.4111 The financial data in the following table belongs to Pathway Bank. Use the data to evaluate the weighted average overall cost of capital for the institution.

Break-even cost	14.50%
The after-tax cost of stock of stockholder's investment	12%
Tax rate	15%
Stockholder's investment	\$260,000
Earning assets	\$340,000

- A. 12.9%
- B. 15.7%
- C. 22.7%
- D. 25.3%

The correct answer is **D**.

The weighted average overall cost of capital is given by:

$$\text{Break-even cost} + \frac{\text{After-tax cost of stockholders' investment}}{(1 - \text{Tax rate})} \times \frac{\text{Stockholders' investment}}{\text{Earning assets}}$$

Which is equivalent to:

$$14.5\% + \frac{12\%}{(1 - 0.15)} \times \frac{\$260,000}{\$340,000} = 25.3\%$$

Thus, 25.3% is the lowest rate of return overall fund-raising costs Pathway Bank can afford to earn on its assets if its equity shareholders invest \$260,000 in the institution.

Q.4113 Which of the following types of Negotiable CDs is issued by foreign banks in the US?

- A. Domestic CDs
- B. Euro CDs
- C. Yankee CDs
- D. Thrift CDs

The correct answer is **C**.

Yankee CDs are a type of Negotiable Certificate of Deposit (CD) that are issued by foreign banks in the United States. These CDs are usually denominated in U.S. dollars. The term 'Yankee' is often used to refer to foreign entities that are operating in the U.S. market. For instance, 'Yankee bonds' are bonds issued by foreign entities in the U.S. Similarly, Yankee CDs are CDs issued by foreign banks in the U.S. These CDs provide an opportunity for foreign banks to raise funds in the U.S. market. They are subject to the same regulations as the CDs issued by domestic banks. Therefore, they offer a similar level of risk and return as domestic CDs.

Choice A is incorrect. Domestic CDs are not issued by foreign banks operating within the United States. Instead, they are issued by domestic banks within their own country.

Choice B is incorrect. Euro CDs are not specific to foreign banks operating in the United States. They refer to certificates of deposit that are issued outside of the currency's home country, but not necessarily in Europe or in US.

Choice D is incorrect. Thrift CDs do not specifically pertain to foreign banks operating within the United States. These types of CDs are typically offered by savings and loan associations or credit unions rather than foreign banking institutions.

Things to Remember

- Yankee CDs are denominated in U.S. dollars and are issued by foreign banks in the United States.
 - Yankee bonds are also issued by foreign entities in the U.S. market.
 - Domestic CDs are issued by domestic banks within their own country.
 - Euro CDs refer to certificates of deposit issued outside of the currency's home country.
 - Thrift CDs are typically offered by savings and loan associations or credit unions.
-

Q.4114 A loan which is unwritten, often uncollateralized agreements usually negotiated over a telephone and is payable in the next day is known as:

- A. Term loan
- B. Overnight loan
- C. Continuing contract
- D. None of the above

The correct answer is **B**.

Overnight loans are a type of short-term loan where the duration of the loan lasts for only one night. They are typically unwritten, often uncollateralized agreements that are usually negotiated over a telephone. The borrower is expected to repay the loan the following day. These loans are commonly used in the interbank market to help banks manage their daily cash flow needs. Banks with surplus funds lend to those in need of short-term cash, and the loan is repaid the next day when the borrowing bank's customers' payments are deposited.

Choice A is incorrect. A term loan is a monetary loan that is repaid in regular payments over a set period of time. Term loans usually last between one and ten years, but may last as long as 30 years in some cases. It does not match the characteristics of the loan described in the question which is expected to be repaid the following day.

Choice C is incorrect. A continuing contract, also known as an evergreen contract or rolling contract, automatically renews at the end of each term until either party gives notice to terminate. This type of contract does not align with the features mentioned in the question where repayment is expected on next day.

Choice D is incorrect. As explained above, there exists a specific term for this type of loan - an overnight loan - making this option invalid.

Things to Remember

- Overnight loans are commonly used in the interbank market to help banks manage their daily cash flow needs.
 - These loans are typically short-term and are repaid the following day.
 - Term loans are repaid over a set period of time with regular payments.
 - Continuing contracts automatically renew at the end of each term until terminated by either party.
-

Q.4115 KBC Bank is considering funding a package of new loans for \$600 million. Money Bank has projected that it must raise \$800 million to have \$600 million available to make the new loans. It expects to raise \$550 million of the total by selling time deposits at an average interest rate of 5%. Noninterest costs from the time of sale deposits add an estimated 2% in operating expenses. KBC Bank expects another \$250 million to come from noninterest-bearing transaction deposits, whose noninterest costs are expected to be 3% of the total amount of these deposits. What is the Bank's projected pooled-funds marginal cost?

	Dollar Amount (\$ millions)	Interest rate	Non interest cost rate	Total interest expenses	Total non-interest expenses
Time deposits	550	5.00%	2.00%	27.50	11.00
Transaction deposits	250	0%	3.00%	0	7.50
Total	800			27.50	18.50

- A. 5.75%
- B. 6.25%
- C. 2.31%
- D. 3.44%

The correct answer is **A**.

$$\begin{aligned} &\text{Projected pooled-funds marginal cost} \\ &= \frac{\text{All expected operating expenses}}{(\text{All new funds expected})} = \frac{27.50 + 18.50}{800} = 5.75\% \end{aligned}$$

Q.4116 Trust Bank purchases a 50-day negotiable CD with a \$10 million denomination from Promise Bank, bearing a 5% annual yield. What is the total amount that Promise Bank will have to pay back to Trust Bank at the end of 50 days?

- A. \$10,000,000
- B. \$10,065,765
- C. \$10,069,444
- D. \$12,086,657

The correct answer is **C**.

$$\begin{aligned} &\text{Amount due CD customer} \\ &= \text{Principal} + \text{Principal} \times \frac{\text{Days to maturity}}{360 \text{ days}} \times \text{Annual interest rate} \\ &= \$10,000,000 + \$10,000,000 \times \frac{50}{360} \times 5\% = \$10,069,444 \end{aligned}$$

Q.4117 KBC Bank is considering funding a package of new loans for \$600 million. The bank has projected that it must raise \$800 million to have \$600 million available to make the new loans. It expects to raise \$550 million of the total by selling time deposits at an average interest rate of 5%. Noninterest costs from the time of sale deposits add an estimated 2% in operating expenses. The bank expects another \$250 million to come from noninterest-bearing transaction deposits, whose noninterest costs are expected to be 3% of the total amount of these deposits. What hurdle rate must the bank achieve in its earning assets?

- A. 5.46%
- B. 7.67%
- C. 9.83%
- D. 10.33%

The correct answer is **B**.

The hurdle rate of return is equivalent to:

$$\frac{\text{All expected operating costs}}{\text{Dollars available to place in earning assets}}$$

	Dollar Amount (\$ millions)	Interest rate	Non interest cost rate	Total interest expenses	Total non-interest expenses
Time deposits	550	5.00%	2.00%	27.50	11.00
Transaction deposits	250	0%	3.00%	0	7.50
Total	800			27.50	18.50

$$= \frac{(27.50 + 18.50)}{600} = 0.0767\%$$

This means that KCB Bank should earn 7.67% and above on average before taxation on all its new funds invested in meeting the expected new funding cost.

Reading 140: Repurchase Agreements and Financing

Q.2239 Sover Bank needs short-term liquidity. It has high-quality US government bonds in its portfolio. In order to fund its liquidity needs, Sover Bank has entered into a repurchase agreement (repo) with GNR Bank. According to the repo agreement, Sover bank has to:

- A. Sell US government bonds to third parties in order to repay any funds borrowed from GNR bank.
- B. Buy US government bonds from GNR Bank.
- C. Pledge US government bonds to receive a loan from GNR Bank.
- D. Sell government bonds to GNR Bank with an obligation to repurchase those bonds.

The correct answer is **D**.

A repurchase agreement, or 'repo', is a form of short-term borrowing for dealers in government securities. In the case of a repo, a dealer sells government securities to investors, usually on an overnight basis, and buys them back the following day at a slightly higher price. That slightly higher price is the overnight interest rate. Thus, repos are essentially collateralized short-term loans, where the dealer (in this case, Sover Bank) sells the securities to the investor (in this case, GNR Bank) with the agreement to repurchase them at a later date. The dealer is raising short-term funds at the expense of the investor's short-term investment needs. The securities serve as collateral for the loan. Therefore, in the given scenario, Sover Bank would sell government bonds to GNR Bank with an obligation to repurchase those bonds, which is exactly what choice D states.

Choice A is incorrect. While selling US government bonds to third parties could potentially generate funds, it does not align with the concept of a repurchase agreement. In a repo, Sover Bank would sell securities to GNR Bank and agree to repurchase them at a later date, not sell them to third parties.

Choice B is incorrect. Buying US government bonds from GNR Bank would require Sover Bank to expend funds, which contradicts its current need for liquidity. This action does not align with the nature of a repurchase agreement where Sover bank should be selling or pledging securities rather than buying more.

Choice C is incorrect. Pledging US government bonds as collateral for a loan from GNR Bank could be an option in some circumstances but it's not typically how repos work. In a standard repo transaction, the borrower (Sover bank) sells securities (US Government Bonds) and agrees to buy them back later at higher price which includes interest on borrowed money.

Things to Remember

- A repurchase agreement (repo) involves the selling of securities with an agreement to repurchase them at a later date.

- Repos are collateralized short-term loans where securities serve as collateral for the borrowed funds.
 - The party selling the securities in a repo is the borrower, while the party buying the securities is the lender.
 - Repos are commonly used by financial institutions to manage short-term liquidity needs.
 - Interest rates in repos are typically higher than traditional loans due to the collateralized nature of the transaction.
-

Q.2240 Bolny Bank has entered into a repo agreement with Kovit Bank. In the first part of the transaction, it receives US government bonds from Kovit Bank. In the second part of the transaction Bolny bank has to:

- A. Return the bonds, which it received as collateral, to Kovit.
- B. Receive money from Kovit.
- C. Sell bonds back to Kovit at their market price on the settlement date.
- D. Sell bonds to Kovit at the fixed price agreed upon in advance.

The correct answer is **D**.

In a repurchase agreement, the seller (in this case, Bolny Bank) agrees to sell a security (in this case, US government bonds) to the buyer (in this case, Kovit Bank) and also agrees to repurchase the same security at a later date at a fixed price. This fixed price is agreed upon in advance when the initial sale takes place. Therefore, in the second part of the transaction, Bolny Bank is required to sell the bonds back to Kovit Bank at the fixed price that was agreed upon in advance. This is the fundamental principle of a repurchase agreement, which is essentially a short-term borrowing for dealers in government securities. The dealer sells the government securities to investors, usually on an overnight basis, and buys them back the following day at a slightly higher price. The slightly higher price represents the interest rate on the loan.

Choice A is incorrect. Bolny Bank would not return the bonds to Kovit Bank without first selling them back as per the terms of the repurchase agreement. The purpose of a repo agreement is for one party to sell securities and agree to repurchase them later, not simply return them.

Choice B is incorrect. In a repo agreement, Bolny Bank would not receive money from Kovit Bank in the second part of this transaction. Instead, it's usually the opposite - Bolny bank would

be paying money back to Kovit bank along with some interest.

Choice C is incorrect. The bonds are not sold back at their market price on the settlement date in a typical repo agreement. Instead, they are sold back at a fixed price agreed upon in advance which includes an implied interest rate.

Things to Remember

- Repurchase Agreement (Repo): A financial transaction in which one party sells a security to another party with a commitment to repurchase the same security at a specified price at a future date.
 - Collateral: The security (in this case, US government bonds) provided by the seller (Bolny Bank) to the buyer (Kovit Bank) to secure the transaction in a repo agreement.
 - Interest Rate: The difference between the selling price and the repurchase price in a repo agreement represents the interest rate on the short-term loan.
 - Settlement Date: The date on which the repurchase of the security occurs in a repo agreement.
 - Counterparty Risk: The risk that the other party in the transaction (Kovit Bank) may default on its obligation to repurchase the security in a repo agreement.
-

Q.2241 Sokolov Bank is going to enter into a repurchase agreement with Branitzky Bank. Sokolov Bank will sell high-quality securities to Branitzky bank in the first part of the transaction. At what price will Branitzkybank sell the aforementioned securities back to Sokolov Bank?

- A. A fixed price agreed upon at the time of entering into the repo agreement.
- B. At the market price of the securities at the time of entering into the repo agreement.
- C. At the market price of the securities at the time of the second transaction.
- D. At the price that should have been agreed upon at the time of the second transaction.

The correct answer is **A**.

In a repurchase agreement, the price at which the securities will be repurchased is determined at the time of entering into the agreement. This price is fixed and does not change regardless of

market fluctuations. The reason for this is to provide certainty to both parties involved in the transaction. The selling party (Sokolov Bank in this case) knows exactly how much it will have to pay to repurchase the securities, and the buying party (Branitzky Bank) knows exactly how much it will receive when the securities are repurchased. This eliminates the risk of price volatility that could potentially result in a loss for either party. Therefore, the price at which Branitzky Bank will sell the securities back to Sokolov Bank is a fixed price agreed upon at the time of entering into the repo agreement.

Choice B is incorrect. The resale price in a repurchase agreement is not determined by the market price of the securities at the time of entering into the repo agreement. It is agreed upon at the outset of the transaction.

Choice C is incorrect. The resale price in a repurchase agreement does not depend on the market price of securities at the time of second transaction. This would introduce unnecessary risk for both parties, as it could result in significant losses if market prices fluctuate unfavorably.

Choice D is incorrect. The resale price cannot be determined or agreed upon at the time of second transaction as it contradicts with basic principle of repurchase agreements where terms are set upfront to mitigate risks associated with future uncertainties.

Things to Remember

- Repurchase agreements (repos) are short-term borrowing arrangements where one party sells securities to another party with an agreement to repurchase them at a later date.
- The party selling the securities (the borrower) pays interest to the party buying the securities (the lender) for the use of the funds.
- Repos are commonly used in the financial markets for short-term funding and liquidity management.
- The difference between the sale price and the repurchase price represents the interest paid on the transaction.
- Repos are typically used by financial institutions such as banks and broker-dealers to manage their short-term funding needs.

Q.2242 Ludska Bank entered into a repo agreement with SKA Bank. Under the agreement, Ludska Bank sold government bonds to SKA Bank as per the following details:

- Face amount: USD 1,000,000,
- Invoice price: USD 1,100,000,
- Repo rate: 4% p.a.,
- Settlement date: 90 days after the conclusion of the agreement

What is going to happen on the date of the settlement, assuming the 360-day convention applies?

- A. Ludska Bank will repurchase the bonds from SKA Bank for USD 1,100,000.
- B. Ludska Bank will repurchase the bonds from SKA Bank for USD 1,000,000.
- C. Ludska Bank will repurchase bonds from SKA Bank for USD 1,101,466.66.
- D. Ludska Bank will repurchase bonds from SKA Bank for USD 1,111,000.

The correct answer is **D**.

The formula for calculation of the repurchase price is as follows:

$$\text{Repurchase price} = \text{Invoice price} \left(1 + \frac{(\text{Repo rate} * \text{Repo term in days})}{360} \right)$$

Therefore,

$$\text{Repurchase price} = 1,100,000 \left(1 + \frac{(4\% * 90)}{360} \right) = 1,111,000$$

Q.2243 Which of the following statements is correct about repurchase agreements?

- A. They can be traded on primary markets.
- B. They are unsecured.
- C. They are classified as money market instruments.
- D. They can have a maturity period of up to 20 years.

The correct answer is C.

Repurchase agreements are indeed classified as money market instruments. Money market instruments are short-term debt instruments that are highly liquid and have low risk. They are used by participants as a means for borrowing and lending in the short term, with maturities that usually range from overnight to just under a year. Repurchase agreements, or repos, are one such instrument. In a repo transaction, a dealer sells government securities to investors, usually on an overnight basis, and buys them back the following day at a slightly higher price. The difference in price is the interest, or repo rate. Repos are typically used to raise short-term capital. They are also used by the central banks to conduct monetary policy operations. The repo market is a crucial component of the financial system as it serves as a source of secured, short-term funding for a variety of participants, including banks, mutual funds, hedge funds and securities dealers.

Choice A is incorrect. Repurchase agreements are typically traded in secondary markets.

Choice B is incorrect. Repurchase agreements are not unsecured; they are backed by collateral, usually in the form of government securities. The borrower sells the securities to the lender with an agreement to repurchase them at a later date at a higher price.

Choice D is incorrect. While repurchase agreements can have varying maturity periods, they do not typically extend up to 20 years. Most repos have a maturity period of one day (overnight) or up to one year.

Things to Remember

- Repurchase agreements are also known as repos.
- They are commonly used by central banks for monetary policy operations.
- Repos are considered low-risk investments due to the collateral backing.
- Participants in a repo transaction include banks, mutual funds, hedge funds, and securities dealers.
- The interest rate in a repo transaction is known as the repo rate.

Q.2246 For a particular high-quality security transaction, the agreement is 'repo' from the point of view of:

- A. The initial security buyer
- B. The initial security seller
- C. The clearinghouse
- D. The Federal Reserve

The correct answer is **B**.

The term 'repo' or repurchase agreement is used from the perspective of the initial security seller. In a repo transaction, the initial security seller sells securities to the buyer with an agreement to repurchase them at a later date. The transaction is essentially a loan, where the seller is the borrower and the buyer is the lender. The securities act as collateral for the loan. The initial seller agrees to buy back the securities at a higher price, and the difference between the selling price and the repurchase price is the interest on the loan. This arrangement allows the initial seller to raise short-term capital while providing the buyer with a secure investment. Therefore, from the perspective of the initial security seller, the transaction is a 'repo'.

Choice A is incorrect. The security buyer in a repo transaction is actually entering into a reverse repo agreement, not a repo. In this case, the buyer is lending money and receiving securities as collateral.

Choice C is incorrect. The clearinghouse does not participate in the transaction as either the buyer or seller of securities but rather facilitates transactions between parties and manages risk.

Choice D is incorrect. The Federal Reserve can participate in repo transactions to implement monetary policy, but it's not always involved in every such transaction. Therefore, it's not correct to say that every repurchase agreement is considered a 'repo' from the perspective of the Federal Reserve.

Things to Remember

- Repurchase agreements (repos) are commonly used in the financial markets for short-term borrowing and lending of securities.
- In a repo transaction, the initial security seller agrees to repurchase the securities at a later date at a slightly higher price, effectively paying interest on the borrowed funds.
- The buyer of the securities in a repo transaction is essentially acting as a lender, providing funds in exchange for the security collateral.
- Repo transactions are important for liquidity management and short-term funding

needs in financial markets.

- The Federal Reserve can use repo transactions as a tool for implementing monetary policy by adjusting the level of reserves in the banking system.
-

Q.2247 Briggs Bank buys US treasury bonds from Roland Bank under a repo agreement. Sometime before the settlement date agreed upon in advance, the market value of the bonds significantly reduces. Which of the following actions would *most likely* be taken by Briggs bank in response to the declining market value? The bank will:

- A. Terminate the agreement.
- B. Demand immediate maturity of the contract so as to avoid further losses that may arise from the bonds losing more of their value.
- C. Do nothing, except hope that the bonds regain their value as the settlement date approaches.
- D. Demand more collateral.

The correct answer is **D**.

Repo agreements, such as the one between Briggs Bank and Roland Bank, are typically subject to margin calls. This means that in a declining market, the borrower of cash (in this case, Roland Bank) is required to provide additional collateral. The purpose of this is to maintain the overall value of the collateral, despite the decrease in the market value of the bonds. Therefore, in response to the declining market value of the U.S. Treasury bonds, Briggs Bank would most likely demand more collateral from Roland Bank. This action would help Briggs Bank mitigate the risk of potential losses due to the decrease in the value of the bonds. It's important to note that the borrower may also withdraw collateral in advancing markets. This flexibility is a key feature of repo agreements and plays a crucial role in managing risk and maintaining the value of collateral.

Choice A is incorrect. Terminating the agreement would not be a feasible option for Briggs Bank as it would result in a breach of contract, which could lead to legal consequences and damage the bank's reputation. Furthermore, termination does not guarantee recovery of losses incurred due to the drop in bond value.

Choice B is incorrect. Demanding immediate maturity of the contract may not be possible as repurchase agreements typically have fixed terms that cannot be altered unilaterally. Moreover, this action might not prevent further losses if bond prices continue to fall.

Choice C is incorrect. Doing nothing and hoping that the bonds regain their value is a passive strategy that exposes Briggs Bank to unnecessary risk. It does not actively manage or mitigate potential losses from declining bond values.

Things to Remember

- Repo agreements involve the sale of securities with an agreement to repurchase them at a later date, effectively serving as a short-term collateralized loan.
 - Margin calls in repo agreements require the borrower to provide additional collateral if the market value of the securities declines.
 - Terminating a repo agreement can have legal and reputational consequences, and may not guarantee recovery of losses.
 - Repo agreements typically have fixed terms that cannot be unilaterally altered.
 - Passively waiting for the market value of securities to recover exposes the party to unnecessary risk.
-

Q.2248 Voynet Bank buys a security from Bornet bBank under a repo agreement. In the period leading up to the settlement date, the market value of the security significantly increases. Assuming a margin call forms part of the agreement, Bornet Bank will most likely:

- A. Terminate the agreement.
- B. Request that the contract matures immediately so that it can sell the mortgage to third parties at a profit.
- C. Request that the amount of collateral be reduced to reflect the increased value.
- D. Do nothing, except wait for maturity of the agreement as scheduled.

The correct answer is **C**.

In a repurchase agreement, or 'repo', two parties agree to sell and repurchase the same security. The seller sells the security to the buyer at a price with a commitment to repurchase the same or another part of the same security at a specified price on a future date. In this case, Voynet Bank (the buyer) and Bornet Bank (the seller) have entered into such an agreement. As the market value of the security increases, Bornet Bank, being the borrower of cash, has the right to request a reduction in the amount of collateral. This is because the increased value of the security provides greater assurance to Voynet Bank about the return of their cash. This is a common practice in repo agreements, where margin calls allow the borrower to supply extra collateral in declining markets and withdraw collateral in advancing markets. Therefore, Bornet Bank is most likely to request that the amount of collateral be reduced to reflect the increased value of the security.

Choice A is incorrect. Terminating the agreement would not be a beneficial action for Bornet Bank. The bank has an obligation to repurchase the security at a later date, and terminating the agreement would likely result in penalties or other negative consequences.

Choice B is incorrect. Requesting that the contract matures immediately so that it can sell the mortgage to third parties at a profit is not typically an option in a repurchase agreement. The terms of these agreements are usually fixed and cannot be altered unilaterally.

Choice D is incorrect. While Bornet Bank could choose to do nothing and wait for maturity of the agreement as scheduled, this would not take advantage of the increased value of the security. By requesting that the amount of collateral be reduced (choice C), Bornet Bank can benefit from this increase in value.

Things to Remember

- **Repurchase Agreement (Repo):** A financial agreement where one party sells a security to another party with a commitment to repurchase the same or another part of the security at a specified price on a future date.
- **Margin Call:** A margin call is a demand from a broker or lender for additional funds or collateral when the value of the securities held as collateral falls below a certain level.
- **Collateral Management:** The process of managing the collateral provided in financial transactions to mitigate credit risk.
- **Market Value:** The price at which an asset would trade in a competitive auction setting.
- **Beneficial Actions in Repo Agreements:** Parties in repo agreements can benefit from margin calls to adjust collateral based on market conditions and take advantage of changes in security values.

Q.2249 After modeling daily cash flows at BBB Bank, its managers strongly believe that the bank will be unable to meet day-to-day funding requirements in about 1 years' time. As such, the bank has to secure long-term funding to avert a possible cash crunch. BBB Bank has high-quality US treasury bonds in its portfolio. Should the bank sell the bonds under a repo agreement so as to secure the required funds?

- A. Yes, because US treasury bonds are highly marketable.
- B. No, because repo financing is usually short-term.
- C. Yes, because the repo market is highly illiquid.

D. No, because repo borrowing has to be backed up by collateral of a similar value.

The correct answer is **B**.

Repo financing is typically short-term in nature. This means that if BBB Bank were to sell its US treasury bonds under a repo agreement, it would have to buy them back in a short period of time, typically a few days or months. This would not provide a long-term solution to the bank's projected cash flow problem. In fact, it could potentially exacerbate the situation. This is because, in addition to repaying the borrowed amount, the bank would also have to pay accrued interest. Therefore, using repo financing as a means to secure long-term funding would not be a viable solution for BBB Bank.

Choice A is incorrect. While it's true that US treasury bonds are highly marketable, this doesn't necessarily mean they should be sold under a repo agreement to secure long-term funding. The high marketability of these bonds makes them a good choice for short-term financing needs, but not necessarily for long-term ones.

Choice C is incorrect. The repo market is actually quite liquid, which makes it suitable for short-term financing needs but not ideal for securing long-term funding as the bank requires in this scenario.

Choice D is incorrect. Although it's true that repo borrowing has to be backed up by collateral of a similar value, this does not make it unsuitable for securing long-term funding. The issue here isn't the value of the collateral but rather the duration of the financing need - repos are typically used for short term financing rather than long term.

Things to Remember

- Repo agreements involve the sale of securities with an agreement to repurchase them at a later date, usually at a slightly higher price, effectively serving as a short-term collateralized loan.
- US treasury bonds are considered high-quality and highly liquid assets, making them attractive collateral for repo agreements.
- Repo financing is commonly used by financial institutions to manage short-term funding needs and liquidity.
- Long-term funding solutions may involve issuing bonds, obtaining long-term loans, or raising capital through equity.
- It is important for banks to carefully assess their funding requirements and choose appropriate financing options based on the duration and nature of their cash flow needs.

Q.2250 During the financial crisis of 2007/2008, repo transactions were partly to blame for the failure of quite a number of firms. This is because:

- A. The repo market was largely unregulated.
- B. Firms borrowed money at relatively high-interest rates using low-quality financial instruments as collateral.
- C. Firms borrowed money at relatively low-interest rates using low-quality financial instruments as collateral.
- D. The US government defaulted on most treasury bonds.

The correct answer is **C**.

During the financial crisis of 2007/2008, many firms borrowed money at relatively low-interest rates using low-quality financial instruments as collateral. This practice was prevalent in the repo market, where borrowers financed lower-quality collateral, like lower-quality corporate bonds and lower-quality mortgage-backed securities, at the relatively low rates and haircuts available. Lenders, on their part, accepted this collateral in exchange for rates somewhat higher than those available when lending on higher-quality collateral. Therefore, in case of counterparty default in a repo agreement, the other party would struggle to sell the securities in the market. Those lucky enough to find a buyer would still incur heavy losses due to decreased market value. This practice contributed significantly to the financial crisis as it increased the risk exposure of firms and made them vulnerable to market fluctuations.

Choice A is incorrect. While it's true that the repo market was not heavily regulated during the financial crisis, this alone does not fully explain the role of repo transactions in contributing to the crisis. The key issue was not just lack of regulation but also how firms were using these transactions.

Choice B is incorrect. This statement is inaccurate because firms were actually borrowing money at relatively low-interest rates, not high ones, using low-quality financial instruments as collateral. This practice increased their risk exposure and contributed to their eventual failure when these low-quality securities significantly dropped in value.

Choice D is incorrect. The US government did not default on most treasury bonds during the financial crisis of 2007/2008. In fact, US Treasury bonds are considered one of the safest investments because they are backed by the full faith and credit of the US government.

Things to Remember

- Repo transactions involve the sale of securities with an agreement to repurchase them at a later date, effectively acting as a short-term collateralized loan.
- Collateral used in repo transactions can vary in quality, with lower-quality collateral

carrying higher risk.

- Haircuts are used in repo transactions to adjust the amount of financing provided based on the quality of the collateral.
 - Counterparty risk is a significant concern in repo transactions, as default by one party can lead to financial losses for the other.
 - Market liquidity and the ability to sell collateral quickly in case of default are crucial factors in repo transactions.
-

Q.2251 Bank Aluvia has an excess liquidity and it has decided to finance another bank under a repo agreement. Why would Aluvia Bank prefer a repo transaction to other market products such as the purchase of stocks?

- A. A repo transaction would guarantee a quick return.
- B. A repo transaction would be faster and easier to execute.
- C. A repo transaction would be risk-free.
- D. A repo-transaction would be more secure.

The correct answer is **D**.

A repo transaction would indeed be more secure. This is primarily due to two factors. First, repo transactions typically have short maturities. This means that the lender (in this case, Bank Aluvia) would get its money back in a relatively short period of time, reducing the duration of exposure to risk. Second, the borrower in a repo transaction is required to provide high-quality collateral. This collateral serves as a form of security for the lender, ensuring that even if the borrower defaults on the loan, the lender can sell the collateral to recover its funds. Therefore, compared to other market products such as stocks, which are subject to market volatility and other risks, repo transactions provide a higher level of security for the lender.

Choice A is incorrect. While a repo transaction can provide a quick return, it does not guarantee it. The speed of return in a repo transaction depends on the terms of the agreement and market conditions.

Choice B is incorrect. Although repo transactions can be executed quickly due to their standardized nature, they are not necessarily faster or easier than other financial instruments. The execution speed and ease depend on various factors such as the complexity of the instrument, market liquidity, and regulatory requirements.

Choice C is incorrect. No financial transaction, including a repo transaction, is completely risk-

free. Repo transactions carry several risks such as counterparty risk (the risk that the other party will not fulfill its obligations), collateral risk (the risk that the collateral will lose value), and liquidity risk (the risk that one cannot exit the position without affecting market prices).

Things to Remember

- **Repo Agreement:** A repurchase agreement (repo) is a form of short-term borrowing for dealers in government securities. In a repo transaction, one party sells securities to another party with a commitment to repurchase the same or similar securities at a specified price at a future date.
 - **Collateral:** Collateral is an asset or property that a borrower offers to a lender as security for a loan. In repo transactions, high-quality collateral is required to mitigate the risk for the lender.
 - **Counterparty Risk:** Counterparty risk is the risk that the other party in a financial transaction will not fulfill its obligations. It is an important consideration in repo transactions.
 - **Liquidity Risk:** Liquidity risk is the risk that an investor cannot exit a position without affecting the market price. It is one of the risks associated with repo transactions.
-

Q.2252 Boulogne Bank sells securities to Betis Bank as part of a repo agreement. Which of the following statements would not be consistent with best market practice under such agreements?

- A. The “buy back” price would be fixed.
- B. The contract would be settled overnight.
- C. Betis Bank would be free to sell the securities back to Boulogne Bank at any time.
- D. The agreement would be legally binding.

The correct answer is **C**.

A repurchase agreement normally has a fixed settlement date.

Things to Remember

- A repurchase agreement, also known as a repo, is a form of short-term borrowing for dealers in government securities. The dealer sells the government securities to investors, usually on an overnight basis, and buys them back the following day.
 - For the party selling the security (and agreeing to repurchase it in the future) it is a repo; for the party on the other end of the transaction (buying the security and agreeing to sell in the future) it is a reverse repurchase agreement.
 - Repurchase agreements are typically used to raise short-term capital. They are also a common tool of central bank open market operations.
 - The 'buy back' price is fixed and includes the original sale price plus an additional amount that represents the interest on the loan.
 - The agreement is legally binding, meaning both parties are obligated to fulfill their respective obligations.
-

Q.2253 Alameda Bank is looking to buy bonds which are “trading special”. Which type of bonds is the bank looking for?

- A. General collateral bonds.
- B. Zero-collateral bonds.
- C. Bonds issued by the government.
- D. Bonds most in demand.

The correct answer is **D**.

Repo trades can be divided into those using general collateral (GC) and those using special collateral or "specials." In the former, the lender of cash is willing to take any particular security, although the broad categories of acceptable securities might be specified with some precision. In specials trading, the lender of cash initiates the repo in order to take possession of a particular security. Bonds most in demand to be borrowed are said to be **trading special**, although any request for specific collateral is a specials trade.

Things to Remember

1. Repo trades can be divided into those using general collateral (GC) and those using special collateral or 'specials'.
 2. In general collateral trading, the lender of cash is willing to accept any particular security, although the broad categories of acceptable securities might be specified with some precision.
 3. In specials trading, the lender of cash initiates the repo in order to take possession of a particular security.
 4. Bonds most in demand to be borrowed are said to be trading special, although any request for specific collateral is a specials trade.
 5. 'Trading special' does not refer to any specific type of bond (e.g., government bonds, zero-collateral bonds), but rather to bonds that are most in demand to be borrowed.
-

Q.2732 Calculate the financing value per \$100 market value of an on-the-run bond if it was issued on March 31st and trades at a special spread of 0.30%. The bond is expected to trade at GC rates after September 30th. Use the actual/360 day convention.

- A. \$0.100
- B. \$0.187
- C. \$0.125
- D. \$0.153

The correct answer is **D**.

The financing value of the bond can be determined by finding its value for the period that it trades at the special spread.

Number of days it is traded at the special spread = 183 days

$$\text{Financing value} = 100 \times \frac{183}{360} \times 0.30\% = \$0.153 \text{ per } \$100$$

Q.2972 Assuming that a counterparty X sells a €250 million face amount of DBR 4's of December 9th, 2014, to a counterparty Y, for settlement on April 1st, 2013, at an invoice price of €280.131 million. At the same time, counterparty X decides to rebuy the €250 million face amount five months later, for settlement on September 1st 2013 at a purchase rate equivalent to the invoice price including interest at a repo rate of 0.31%. Compute the repurchase price.

- A. €250.166 million
- B. €259.187 million
- C. €281.129 million
- D. €280.500 million

The correct answer is **D**.

By applying the actual/360 convention popular for most money market instruments, and using 153 days between April 1st and September 1st.

Therefore:

$$\begin{aligned} \text{€}280,131,000 \left(1 + \frac{(0.0031 \times 153)}{360}\right) &= \text{€}280,500,072 \\ &\approx \text{€}280.5 \text{ Million} \end{aligned}$$

Note: Repurchase agreements use the actual/360 day count convention.

They are classified as money market instruments in the same category as T-bills.

Q.2973 Which of the following is the value of lending \$1,000 of cash at a spread of 0.31% for 133 days?

- A. \$1.145 per \$1,000
- B. \$2.982 per \$1,000
- C. \$1.675 per \$1,000
- D. \$2.442 per \$1,000

The correct answer is **A**.

The value of lending \$1,000 of cash at a spread of 0.31% for 133 days is computed as:

$$\$1,000 \times \frac{(133 \times 0.0031)}{360} = \$1.145$$

= \$1.145 per \$1,000 market value of the bond

Q.2974 Supposing that DBR 4s of face value \$120 million of January 14th, 2030, are to be sold by HJK Bank to a counterparty for settlement for \$159 million. HJK Bank selling the DBR 4s then decides to buy the \$120 million face amount some 122 days later for settlement at a buying price equal to that of the invoice price with a repo rate of 0.26%. At what price can HJK Bank repurchase the bond?

- A. \$159.14 million
- B. \$121.65 million
- C. \$150.11 million
- D. \$121.33 million

The correct answer is **A**.

The repurchase price is:

$$\$159 \left(1 + \left(0.0026 \times \frac{122}{360} \right) \right) = \$159.14 \text{ million}$$

Q.4118 A counterparty K sells a €300 million face amount of DBR 4's of June 10th, 2016, to a counterparty W, for settlement on January 1st, 2015, at an invoice price of €350.25 million. At the same time, counterparty X decides to rebuy the €300 million face amount six months later, for settlement on July 1st, 2015, at a purchase rate equal to the invoice price with interest at a repo rate of 0.26%. Compute the repurchase price.

- A. 350.16 million
- B. 350.56 million
- C. 350.71 million
- D. 350.85 million

The correct answer is **C**.

We apply the actual/360 convention popular for most money market instruments. There are using 181 days between January 1st and July 1st.

$$\begin{aligned} &\text{Repurchase Price} \\ &= \text{Principal} + \left(\text{Principal} \times \text{Repo Rate} \times \frac{\text{Repo Term in Days}}{360 \text{ days}} \right) \\ &\text{Repurchase Price} \\ &= \$350.25 + \left(350.25 \times 0.26\% \times \frac{181}{360} \right) = \$350.71 \text{ Million} \end{aligned}$$

Note: Repurchase agreements use the actual/360 day count convention. They are classified as money market instruments in the same category as T-bills.

Q.4119 Prime Bank sells DBR 4s of face value \$200 million of September 1st, 2040, to a counterparty for settlement for 250 million. Prime Bank then decides to buy the \$200 million face amount after 183 days for settlement at a buying price equal to that of the invoice price with a repo rate of 0.34%. At what price can Prime Bank repurchase the bond?

- A. \$250.43 million
- B. \$250.57 million
- C. \$200.68 million
- D. \$200.97 million

The correct answer is **A**.

$$\begin{aligned}\text{Repurchase Price} &= \text{Principal} \left(1 + \left(\text{Repo Rate} \times \frac{\text{Repo Term in Days}}{360 \text{ days}} \right) \right) \\ \text{Repurchase Price} &= \$250 \left(1 + \left(0.34\% \times \frac{183}{360} \right) \right) = \$250.43 \text{ Million}\end{aligned}$$

Q.4120 Bright Bank lends \$10,000 of cash at a spread of 0.26% for 150 days. Calculate the Bank's lending value.

- A. \$1.034 per \$10,000
- B. \$5.45 per \$10,000
- C. \$10.83 per \$10,000
- D. \$10.68 per \$10,000

The correct answer is **C**.

The value of lending \$100 of cash at a spread of 0.26% for 150 days is calculated as:

$$\$10,000 \times \frac{150 \times 0.26\%}{360} = \$10.83$$

i.e., \$10.83 per \$10,000 market value of the bond.

Q.4121 Calculate the financing value per \$1,000 market value of an on-the-run bond if it was issued on July 1st 2019 and trades at a special spread of 0.50%. The bond is expected to trade at GC rates after October 31st 2019. Use the actual/360 day convention.

- A. \$1.57
- B. \$1.60
- C. \$1.67
- D. \$1.69

The correct answer is **D**.

The financing value of the bond can be determined by finding its value for the period that it trades at the special spread.

Number of days it is traded at the special spread = 122 days

$$\text{Financing value} = 1,000 \times \frac{122}{360} \times 0.50\% = \$1.69 \text{ per } \$1,000$$

Q.4123 Calculate the value of lending \$200 cash at a spread of 0.25% for 150 days.

- A. \$0.208
- B. \$0.304
- C. \$0.308
- D. \$0.205

The correct answer is **A**.

$$\text{Lending value (financial advantage)} = \$200 \times 0.25\% \times \frac{150}{360} = \$0.208$$

Q.4162 The following are characteristics of a Repurchase Agreement. Which one is **NOT**?

- A. It has a specified price
- B. It usually takes a short duration
- C. It is generally considered a safe investment.
- D. It usually takes a long duration.

The correct answer is **D**.

Repurchase Agreements, commonly known as repos, are typically short-term financial instruments. They are often used for overnight borrowing but can extend up to 21 days. The duration of a repo is agreed upon at the initiation of the contract and is typically short to minimize the risk and provide quick liquidity to the borrower. Therefore, stating that a Repurchase Agreement usually takes a long duration is incorrect.

Choice A is incorrect. Repurchase Agreements do indeed have a specified price. This price is agreed upon by both parties at the beginning of the agreement and it represents the amount that will be repaid at the end of the contract term.

Choice B is incorrect. Repurchase Agreements are typically short-term borrowing instruments, often with durations as short as overnight. Therefore, this statement accurately represents a characteristic of a Repurchase Agreement.

Choice C is incorrect. Generally, repos are considered safe investments because they are collateralized loans - i.e., in case of default by the borrower, the lender can sell off the collateral to recover their funds. Hence, this statement correctly describes one aspect of repos' nature in financial markets.

Things to Remember

- Repurchase Agreements (repos) involve the sale of securities with an agreement to repurchase them at a later date.
- Repos are commonly used in money markets for short-term funding needs.
- Collateral is a key component of repos, providing security to the lender in case of default.
- Interest rates in repos are determined by the difference between the sale price and the repurchase price.
- Repos are often used by financial institutions to manage their short-term liquidity needs.

Q.5378 During a presentation on the quarterly performance of his bank, a risk manager highlights the differences between repurchase agreements (repos) and reverse repurchase agreements (reverse repos). Which of the following statements made by the manager is correct?

- A. In a repo, the seller of securities agrees to buy them back at a later date at a slightly lower price.
- B. In a reverse repo, the buyer of securities agrees to sell them back at a later date at a slightly lower price.
- C. Repos are typically used to raise cash, while reverse repos are typically used to invest cash.
- D. Repos are initiated by the buyer of securities, while reverse repos are initiated by the seller of securities.

The correct answer is **C**.

Repos are typically used by financial institutions to raise cash. In a repo transaction, the seller of securities agrees to sell them to the buyer at a specified price and then repurchase them at a later date at a slightly higher price. The securities serve as collateral for the loan. On the other hand, reverse repos are typically used by financial institutions to invest cash.

A is incorrect. In a repo, the seller of securities agrees to repurchase them at a later date at a slightly higher price.

B is incorrect. In a reverse repo, the buyer of securities agrees to sell them back at a later date, but not at a slightly lower price. Instead, the buyer (who is effectively lending money) agrees to sell them back at a slightly higher price. This mirrors a repo but from the perspective of the counterparty.

D is incorrect. This is not always true. While it's true that the buyer usually initiates the transaction by making an offer, both repos and reverse repos can be initiated by either party depending on the market conditions and their individual needs.

Things to Remember

- Repurchase agreements (repos) involve the sale of securities with an agreement to repurchase them at a later date, while reverse repurchase agreements (reverse repos) involve the purchase of securities with an agreement to sell them back at a later date.
- Repos are commonly used by financial institutions to raise short-term funds by using securities as collateral, while reverse repos are used to invest excess cash for short periods.

- The difference in price between the initial sale and repurchase in a repo transaction represents the interest earned by the buyer of the securities.
 - Both repos and reverse repos are important tools for liquidity management and short-term funding in financial markets.
-

Q.5412 Three months after purchasing the bond on its coupon payment date, ABC Bank opts to enter into a repurchase agreement (repo) on the bond to generate immediate liquidity. The bond's and repo's details are shown below.

Notional (GBP)	10,000,000
Annual coupon (paid semiannually)	3.5%
Current price of bond (GBP)	97
Repo interest rate	2%
Repo haircut	3.5%

What is the anticipated cash outflow for ABC Bank when the repo contract reaches its expiration in 6 months?

- A. 9,539,387
- B. 9,624,669
- C. 9,949,300
- D. 9,273,864

The correct answer is **B**.

Cash inflow at the beginning of the repo.

This is the bond's adjusted price (including the fraction of the coupon) minus the haircut.

$$\begin{aligned}
 \text{Initial Cash Inflow} &= (\text{Current price of bond} + \text{Fraction of Coupon}) \times \text{Notional} \times (1 - \text{Repo haircut}) \\
 &= (0.97 + 0.035 \times \frac{1}{2}) \times 10,000,000 \times 0.965 \\
 &= 9,529,375
 \end{aligned}$$

Cash outflows at the end of the repo.

This is the initial cash inflow plus the repo interest for the 6-month period.

$$\begin{aligned}
 \text{Cash Outflow at the End} &= \text{Initial Cash Inflow} \times (1 + \text{Repo interest rate} \times \frac{1}{2}) \\
 &= 9,529,375 \times (1 + 0.02 \times 0.5) \\
 &= 9,624,668.75
 \end{aligned}$$

The anticipated cash outflow for ABC Bank when the repo contract expires in 6 months is GBP 9,624,668.75.

Reading 141: Liquidity Transfer Pricing: A Guide to Better Practice

Q.4201 Calculate the rate charged for contingent liquidity risk of a line of credit assumed to have a limit of \$56million where \$22 million has already been drawn. Further, suppose there is a 38% chance that the customer will draw on the remaining credit and that the cost of term funding assets in the liquidity cushion is 7.5 bps.

- A. 1.73 bps
- B. 2.85bps
- C. 4.40 bps
- D. 2.71 bps

The correct answer is **A**.

$$\text{Rate charged} = \frac{\text{Limit-Drawn Amount}}{\text{Limit}} \times \text{Likelihood of Drawdown} \\ \times \text{Cost of Funding Liquidity Cushion}$$

Hence,

$$\frac{(\$56\text{m} - \$22\text{m})}{\$56\text{m}} \times 0.38 \times 0.00075 = 0.000173 \cong 1.73\text{bps}$$

Q.4202 A line of credit has a limit of \$120 million, in which \$56 million has already been drawn. An assumption is made that customer K will draw on the remaining credit with a 75% probability and the cost of term funding assets in the liquidity cushion at 32bps. Calculate the rate charged for the contingent liquidity risk of the line of credit.

A. 5.14 bps

B. 1.12 bps

C. 12.8 bps

D. 1.28 bps

The correct answer is **C**.

Formula.

$$\frac{\text{Limit-Drawn Amount}}{\text{Limit}} \times \text{Likelihood of Drawdown} \times \text{Cost of Funding Liquidity Cushion}$$

Hence,

$$\frac{(\$120\text{m} - \$56\text{m})}{\$120\text{m}} \times 0.75 \times 0.0032 = 0.00128 \cong 12.8\text{bps}$$

Q.4204 Given that a six-year amortizing loan in recent times has the following annual liquidity premiums: 6, 14, 18, 24, 27, 32. Use the matched-maturity marginal cost of funds approach to calculate the charge of funding liquidity risk of this loan.

- A. 32.00 bps
- B. 42.33 bps
- C. 33.43 bps
- D. 24.33 bps

The correct answer is **D**.

$$\frac{(\text{Term in year 1} \times \text{Term liquidity premium}) + (\text{Term in year 2} \times \text{Term liquidity premium}) + \dots (\text{Term in year n} \times \text{Term liquidity premium})}{\text{Term in year } (1 + 2 + 3 + \dots + n)}$$

Thus, the charge of funding liquidity risk of this loan is

$$\frac{(1 \times 6) + (2 \times 14) + (3 \times 18) + (4 \times 24) + (5 \times 27) + (6 \times 32)}{(1 + 2 + 3 + 4 + 5 + 6)} = 24.3\text{bps}$$

Q.4206 Assume the following term liquidity premiums and the average cost of funds were recorded by a bank at a point before the crisis (Pre-GFC), and more recently.

Pre-GFC and Current Term Liquidity Premiums and Average Cost of Funds

Term in years	1	2	3	4	5
Pre-Global Financial Crisis					
Term liquidity premium	1	3	5	7	10
Average cost of funds	3	3	3	3	3
Current					
Term liquidity premium	5	8	10	18	35
Average cost of funds	10	10	10	10	10

Using the matched-maturity marginal cost of funds approach, calculate the amount of charge that a one-year non-amortizing bullet loan will be charged for funding liquidity risk if it originated pre-crisis and more recently, respectively.

- A. \$1 and \$3
- B. \$1 and \$5

C. \$3 and \$5

D. \$10 and \$35

The correct answer is **B**.

To calculate the charge for funding liquidity risk using the matched-maturity marginal cost of funds approach, we need to look at the term liquidity premium for the corresponding maturity of the loan. In this case, we're dealing with a one-year non-amortizing bullet loan, so we'll focus on the one-year term.

Let's calculate this for both the pre-crisis and current scenarios:

1. Pre-Global Financial Crisis:

For a one-year loan:

Term liquidity premium = 1 basis point

So, the charge for funding liquidity risk = 1 basis point

2. Current:

For a one-year loan:

Term liquidity premium = 5 basis points

So, the charge for funding liquidity risk = 5 basis points

Therefore:

- Pre-crisis, a one-year non-amortizing bullet loan would be charged 1 basis point for funding liquidity risk.
- Currently, a one-year non-amortizing bullet loan would be charged 5 basis points for funding liquidity risk.

We can observe that the charge for funding liquidity risk has increased significantly from the pre-crisis period to the current period, reflecting the increased perception of liquidity risk in the banking system.

Q.4207 Suppose the following term liquidity premiums and the average cost of funds were recorded by a bank at a point before the crisis (Pre-GFC), and more recently.

Pre-GFC and Current Term Liquidity Premiums and Average Cost of Funds

Term in years	1	2	3	4	5
Pre-Global Financial Crisis					
Term liquidity premium	1	3	5	7	10
Average cost of funds	3	3	3	3	3
Current					
Term liquidity premium	5	8	10	18	35
Average cost of funds	10	10	10	10	10

Assume that the principal of the loan was \$5 million. Using the matched-maturity marginal cost of funds approach, calculate the charge that a one-year loan translated to the business unit(s) writing the loans if it originated pre-crisis and more recently, respectively.

- A. \$100 and \$500
- B. \$500 and \$2,500
- C. \$1,000 and \$2,000
- D. \$2,000 and \$2,500

The correct answer is **B**.

From the table, a one-year non-amortizing bullet loan should have received a charge of 1 bp if originated pre-crisis, and 5 bps if arisen more recently. Given the principal of the loan as \$5 million. This should have translated to charges of $\$500(0.0001 \times 5,000,000)$ and $\$2,500(0.0005 \times 5,000,000)$, respectively, to the business unit(s) writing the loans.

Q.4209 A five-year linearly amortizing bullet loan has a principal amount of \$2 million. Think of these as five separate annual loans, each of \$400,000, using a matched-maturity marginal cost of funds approach, calculate the charge that this loan receives using the information provided in the following table:

Pre-GFC and Current Term Liquidity Premiums and Average Cost of Funds

Term in years	1	2	3	4	5
Pre-Global Financial Crisis					
Term liquidity premium	2	6	8	9	12
Average cost of funds	4	4	4	4	4
Difference	-2	2	4	5	8

- A. 3.93bps
- B. 8.93bps
- C. 9.83bps
- D. 398.00bps

The correct answer is **B**.

$$\frac{(\text{Term in year 1} \times \text{Term liquidity premium}) + (\text{Term in year 2} \times \text{Term liquidity premium}) + \dots (\text{Term in year n} \times \text{Term liquidity premium})}{\text{Term in year } (1 + 2 + 3 + \dots + n)}$$

$$\frac{(1 \times 2) + (2 \times 6) + (3 \times 8) + (4 \times 9) + (5 \times 12)}{(1 + 2 + 3 + 4 + 5)} = 8.93\text{bps}$$

Q.4211 There is an \$80 million limit on a line of credit and \$24 million already drawn. Jack is expected to draw the remaining with a 45% probability at the cost of term funding assets in the liquidity cushion at 24bps. Calculate the rate charged for contingent liquidity risk.

- A. 0.756 bps
- B. 7.56 bps
- C. 0.60 bps
- D. 75.6 bps

The correct answer is **B**.

Formula.

$$\text{Rate charged} = \frac{\text{Limit-Drawn Amount}}{\text{Limit}} \times \text{Likelihood of Drawdown} \\ \times \text{Cost of Funding Liquidity Cushion}$$

Hence,

$$\frac{(\$80\text{m} - \$24\text{m})}{\$80\text{m}} \times 0.45 \times 0.0024 = 0.000756 \cong 7.56\text{bps}$$

Q.4213 Use the matched-maturity marginal cost of funds approach to calculate the cost of funding liquidity risk given that a four-year amortizing loan has annual liquidity premiums of 12, 17, 24, 30, respectively.

- A. 23.8bps
- B. 32.8bps
- C. 238bps
- D. 832bps

The correct answer is **A**.

$$\frac{(\text{Term in year 1} \times \text{Term liquidity premium}) + (\text{Term in year 2} \times \text{Term liquidity premium}) + \dots (\text{Term in year n} \times \text{Term liquidity premium})}{\text{Term in year } (1 + 2 + 3 + \dots + n)}$$

$$\frac{(1 \times 12) + (2 \times 17) + (3 \times 24) + (4 \times 30)}{(1 + 2 + 3 + 4)} = 23.8\text{bps}$$

Q.4214 Most banks are moving towards better Liquidity Transfer Pricing policies. Which of the given options is the *most accurate* about some of the following steps banks are taking in the actualization of improved LTP practices?

- A. Banks operating with decentralized funding centers are changing towards wholesale funding controlled centrally via a treasury function.
- B. Banks are coming up with trading book procedures and policies
- C. Banks with small trading book exposures in their business activities want to do away with over-trading behaviors by applying higher funding rates on net funding needs upon breaching certain funding limits.
- D. All of the above

The correct answer is **D**.

All the options provided accurately describe the steps banks are taking to improve their Liquidity Transfer Pricing (LTP) policies.

Option A states that banks operating with decentralized funding centers are changing towards wholesale funding controlled centrally via a treasury function. This is a strategic move aimed at

preventing arbitrage between treasury and business units as well as between the businesses themselves.

Option B mentions that banks are coming up with trading book procedures and policies. This is a crucial step as it involves the development of risk limits and controls for trading activities. These measures are necessary for the proper measuring, monitoring, and assessment of liquidity threats attached to business units and products.

Option C indicates that banks with small trading book exposures in their business activities want to do away with over-trading behaviors by applying higher funding rates on net funding needs upon breaching certain funding limits. This is a strategic move aimed at curbing over-trading behaviors and ensuring financial stability.

Choice A is incorrect. While it is true that banks are moving towards a more centralized funding approach, this does not fully capture the strategic steps being taken to enhance LTP policies. The shift also involves other measures such as implementing trading book procedures and policies, and applying higher funding rates on net funding needs upon breaching certain limits.

Choice B is incorrect. Although the development of trading book procedures and policies is part of the strategic steps being taken by banks to improve their LTP practices, it does not encompass all the measures being implemented. Other steps include centralizing wholesale funding through a treasury function and applying higher funding rates on net funding needs when certain limits are breached.

Choice C is incorrect. The application of higher funding rates on net funding needs upon breaching certain limits is indeed one of the strategies used by banks to curb over-trading behaviors in their business activities with small trading book exposures. However, this alone does not represent all the strategic steps being taken to enhance LTP practices in banking sector.

Things to Remember

- Liquidity Transfer Pricing (LTP) is a mechanism used by banks to allocate the cost of liquidity to different business units based on their liquidity needs and contributions to overall liquidity risk.
- Centralized funding approach helps in better management of liquidity risk by ensuring that funding decisions are made at a central level, reducing the possibility of arbitrage and optimizing liquidity allocation.
- Trading book procedures and policies are essential for managing market risk associated with trading activities, including setting risk limits, monitoring exposures, and ensuring compliance with regulations.
- Over-trading behaviors can lead to excessive risk-taking and liquidity challenges, hence applying higher funding rates on net funding needs can act as a deterrent and promote

responsible trading practices.

- Effective LTP practices are crucial for maintaining financial stability, managing liquidity risk, and aligning funding costs with the underlying risks of different business activities.
-

Q.4215 The following are the cost components of liquidity transfer pricing (LTP). Which of the following is **NOT**?

- A. Liquidity risk costs
- B. Costs of the liquidity cushion
- C. Costs of the liquidity reserves
- D. Expense costs

The correct answer is **D**.

Expense costs are not a component of Liquidity Transfer Pricing (LTP). LTP is a method used by financial institutions to allocate the costs and benefits of liquidity across different business lines. It includes several cost components, such as liquidity risk costs, costs of the liquidity cushion, and costs of the liquidity reserves. However, expense costs, which typically refer to the costs incurred in the course of conducting business, such as salaries, utilities, and rent, are not considered a part of LTP. These costs are usually accounted for separately in the financial institution's overall cost structure.

Choice A is incorrect. Liquidity risk costs are a component of Liquidity Transfer Pricing. These costs arise from the uncertainty and potential loss caused by an institution's inability to meet its obligations as they come due without incurring unacceptable losses.

Choice B is incorrect. The costs of the liquidity cushion are also a part of Liquidity Transfer Pricing. The liquidity cushion refers to the liquid assets held by a financial institution to meet unexpected cash flow needs or market disruptions, and maintaining this cushion incurs certain costs.

Choice C is incorrect. The cost of liquidity reserves is another component of Liquidity Transfer Pricing. Financial institutions need to hold certain reserves as per regulatory requirements, which can be costly as these funds could have been otherwise invested for returns.

Things to Remember

- Liquidity Transfer Pricing (LTP) is a method used by financial institutions to allocate the costs and benefits of liquidity across different business lines.

- Liquidity risk costs arise from the uncertainty and potential loss caused by an institution's inability to meet its obligations without incurring unacceptable losses.
 - The liquidity cushion refers to the liquid assets held by a financial institution to meet unexpected cash flow needs or market disruptions.
 - Costs of liquidity reserves are incurred by financial institutions to meet regulatory requirements for holding certain reserves.
 - Expense costs typically refer to the costs incurred in the course of conducting business, such as salaries, utilities, and rent, and are not considered a part of LTP.
-

Q.5418 A risk analyst is having a conversation with her supervisor about the different approaches to liquidity transfer pricing and the potential effects of each technique. Which of the following claims made by the risk analyst on liquidity transfer price is most likely correct?

- A. The bank's fixed-rate borrowing cost becomes a floating-rate borrowing cost when a matched-maturity marginal cost of funds technique is used.
- B. When management remuneration is based on net income, both an average cost-of-funds approach and a zero-cost-of-funds approach correctly align the maturity of the bank's lending and borrowing activities.
- C. When a bank uses a zero-cost-of-funds strategy, it frequently ends up with long-term, very illiquid assets that are financed by stable, long-term liabilities.
- D. The balance sheet of a bank often experiences the biggest maturity transformation when using an average cost of funds strategy.

The correct answer is **A**.

The bank's fixed rate borrowing cost is converted to floating via a matched-maturity marginal cost of funds liquidity pricing strategy, which also equals the spread over a reference rate. This approach ensures that the cost of funds aligns with the maturity of the bank's assets and liabilities, considering the marginal cost of acquiring additional funds.

B is incorrect. When compensation is based on net income, neither a zero-cost-of-funds approach nor an average cost-of-funds approach will properly align the maturity of the bank's lending and borrowing activities. Management might build an important maturity mismatch between assets and liabilities to boost net income.

C is incorrect. The bank will typically retain long-term, highly illiquid assets matched with

short-term liabilities under a zero-cost liquidity pricing structure.

D is incorrect. The balance sheet maturity transformation for a bank will often be the biggest under a zero-cost-of-funds liquidity pricing system.

Things to Remember

- Liquidity Transfer Pricing (LTP) is a mechanism used by banks to allocate the cost of liquidity to different business units based on their liquidity usage.
 - Matched-maturity marginal cost of funds technique ensures that the cost of funds aligns with the maturity of the bank's assets and liabilities.
 - Zero-cost-of-funds strategy can lead to maturity mismatches between assets and liabilities, potentially increasing risk.
 - Long-term, illiquid assets financed by stable, long-term liabilities can create liquidity risk for a bank.
 - Balance sheet maturity transformation refers to the process of converting short-term liabilities into long-term assets, which can impact a bank's risk profile.
-

Reading 142: The US Dollar Shortage in Global Banking and the International Policy Response

Q.4164 The stretch to which banks invest in one currency and fund in another via F.X. swaps is known as:

- A. Cross currency funding
- B. Funding gap
- C. Carry trading
- D. Hedging

The correct answer is **A**.

This term refers to the practice where banks invest in one currency and fund in another through foreign exchange (F.X.) swaps. This strategy is often employed by banks to take advantage of differences in interest rates between two currencies. For instance, a bank might invest in a currency with a high interest rate while funding in a currency with a lower interest rate. This allows the bank to earn a profit from the interest rate differential. The European and Japanese banking systems have been known to engage in cross-currency funding extensively since 2000, taking on significant net on-balance-sheet positions in foreign currencies, particularly in U.S. dollars.

Choice B is incorrect. A funding gap refers to the difference between a company's available capital and the amount required to carry out its operations or projects. It does not specifically involve investing in one currency and funding in another via foreign exchange (F.X.) swaps.

Choice C is incorrect. Carry trading is a strategy that involves borrowing at a low interest rate and investing in an asset that provides higher returns. While it can involve different currencies, it doesn't necessarily imply using F.X. swaps for investment and funding purposes as described in the question.

Choice D is incorrect. Hedging refers to making an investment to reduce the risk of adverse price movements in an asset, typically by taking an offsetting position in a related security, such as options or futures contracts. It does not directly relate to the strategy of investing in one currency and funding another through F.X. swaps.

Things to Remember

- Foreign Exchange (F.X.) swaps involve the exchange of one currency for another at an agreed-upon exchange rate with an obligation to reverse the transaction at a future date.
- Interest rate differentials between currencies play a crucial role in determining the profitability of cross-currency funding strategies.

- Net on-balance-sheet positions in foreign currencies can expose banks to currency risk and fluctuations in exchange rates.
 - Understanding the concept of basis risk is important when engaging in cross-currency funding, as it refers to the risk that the interest rate differential may not offset the exchange rate risk.
-

Q.4165 The international policy response came into place due to the severe U.S. dollar shortage among banks outside the United States. It brought about the following success. Which of the following choices is the most accurate?

- A. It increased the appetite for foreign currency assets
- B. It led to the U.S. dollar funding gap
- C. It contributed to the reversal of carry trades
- D. It mitigated upward pressure and interbank rate volatility on the U.S. dollar.

The correct answer is **D**.

The international policy response was successful in mitigating upward pressure and interbank rate volatility on the U.S. dollar. During the financial crisis, there was a severe shortage of U.S. dollars among banks outside the United States. This shortage was due to a variety of factors, including an increased appetite for foreign currency assets, a U.S. dollar funding gap, and the reversal of carry trades. The international policy response was designed to address this shortage and stabilize the financial markets. One of the key successes of this response was its ability to mitigate upward pressure and interbank rate volatility on the U.S. dollar. By doing so, it helped to stabilize the value of the U.S. dollar and reduce the risk of further financial instability.

Choice A is incorrect. The international policy response was not aimed at increasing the appetite for foreign currency assets. Instead, it was designed to address the shortage of U.S. dollars among banks outside the United States during the financial crisis.

Choice B is incorrect. The international policy response did not lead to a U.S. dollar funding gap; rather, it was a measure taken to alleviate such a gap that had arisen due to the financial crisis.

Choice C is incorrect. While carry trades may have been affected by various factors during the financial crisis, there's no evidence that suggests that they were directly influenced by this particular international policy response.

Things to Remember

- U.S. Dollar Shortage: During times of financial crisis, there can be a shortage of U.S. dollars among banks outside the United States, leading to various challenges in the financial markets.
 - Foreign Currency Assets: Increased appetite for foreign currency assets can impact currency exchange rates and market stability.
 - U.S. Dollar Funding Gap: A U.S. dollar funding gap occurs when there is a shortage of U.S. dollars available for lending or investment purposes.
 - Carry Trades: Carry trades involve borrowing in a low-interest rate currency to invest in a higher-interest rate currency, and reversals in carry trades can impact market dynamics.
 - Interbank Rate Volatility: Fluctuations in interbank rates can affect borrowing costs for financial institutions and impact overall market stability.
-

Q.4166 In a discussion panel on mitigating the recurrence of the great financial crisis, a delegate mentioned the following causes of the U.S Dollar shortage during the great financial crisis.

- I. Increased appetite for foreign currency assets by non-US investors
- II. Cross currency funding
- III. U.S. dollar funding gap

Which of the above causes are correct?

- A. I and II
- B. I and III
- C. II and III
- D. All of the above

The correct answer is **D**.

During the great financial crisis, all three factors mentioned played a significant role in the shortage of the U.S Dollar.

Firstly, there was an increased demand for foreign currency assets, particularly U.S.

denominated claims on non-bank entities, by investors outside the U.S. This increased demand put pressure on the availability of the U.S Dollar.

Secondly, banks engaged in cross-currency funding, a practice where they invested in one currency and funded in another via foreign exchange swaps. This practice was particularly prevalent in the banking systems of Europe and Japan. Since 2000, these banking systems had taken on significant net on-balance-sheet positions in foreign currencies, especially in U.S. dollars. The associated currency exposures were left hedged off-balance-sheet, leading to a situation where the banks were exposed to foreign currency funding risk. In other words, these banks were at risk that their funding positions could not be rolled over, contributing to the U.S Dollar shortage.

Lastly, the U.S. dollar funding gap also contributed to the shortage. Non-U.S banks, unlike domestic banks, had limited access to a stable base of dollar deposits. This forced them to rely on short-term and potentially more volatile sources of funding, such as commercial paper and loans from other banks, leading to the U.S. dollar funding gap.

Choice A is incorrect. While it is true that the growing demand for foreign currency assets by investors outside the U.S and the practice of cross-currency funding contributed to the shortage of U.S Dollar during the crisis, this choice ignores another significant factor - The existence of a U.S. dollar funding gap. This gap was created due to an imbalance between dollar-denominated liabilities and assets held by financial institutions outside of the United States.

Choice B is incorrect. Although it correctly identifies two factors - The growing demand for foreign currency assets by investors outside the U.S and The existence of a U.S. dollar funding gap, it fails to consider another crucial factor - The practice of cross-currency funding. Cross-currency funding refers to borrowing in one currency (usually low-interest-rate currencies) and investing in another (usually high-interest-rate currencies). During times of financial stress, this can lead to a shortage in certain currencies like what happened with USD during the crisis.

Choice C is incorrect. While it rightly points out two factors - The practice of cross-currency funding and The existence of a U.S. dollar funding gap, it overlooks an important factor -The growing demand for foreign currency assets by investors outside the U.S., which also played a role in causing USD shortage during that period.

Things to Remember

- **Foreign Currency Assets:** Increased demand for foreign currency assets can lead to a shortage of a specific currency, such as the U.S. Dollar, during times of financial crisis.
- **Cross-Currency Funding:** Cross-currency funding involves borrowing in one currency and investing in another, which can create imbalances and funding risks for financial institutions.
- **U.S. Dollar Funding Gap:** The U.S. dollar funding gap refers to the imbalance between dollar-denominated liabilities and assets held by financial institutions outside the United States, leading to a shortage of U.S. Dollars.

- Financial Crisis Impact: During a financial crisis, factors like increased demand for foreign currency assets, cross-currency funding practices, and funding gaps can exacerbate currency shortages and liquidity issues.
-

Q.4167 The banks' international balance sheet expansions since 2000 are highly associated with the U.S. dollar shortage. The stock on banks' foreign claims significantly grew from \$10 trillion at the start of 2000 to \$34 trillion by the end of 2007. By 2001, international claims were at 10% while at the end of 2007, it approached 30%. These significant improvements occurred mainly during the financial innovation period. This innovation period included the following, EXCEPT:

- A. Expansion growth in the hedge fund industry
- B. The introduction of financial structures
- C. The spreading of universal banking
- D. Auctioning of the U.S dollar.

The correct answer is **D**.

Auctioning of the U.S dollar was not a part of the financial innovation period. It was, in fact, an international policy response aimed at minimizing the level of volatile swap spread during the great financial crisis. The auctioning of the U.S dollar was a measure taken to provide liquidity to the market and stabilize the financial system during a period of intense crisis. It was not a part of the financial innovation that led to the expansion of banks' international balance sheets.

Choice A is incorrect. The expansion growth in the hedge fund industry was indeed a part of the financial innovation period. Hedge funds played a significant role in providing liquidity to the market and diversifying investment risks.

Choice B is incorrect. The introduction of financial structures, such as derivatives and structured products, was also a key development during this period. These instruments allowed for better risk management and facilitated capital flow across different sectors.

Choice C is incorrect. The spreading of universal banking was another characteristic feature of this period. Universal banks offer a wide range of financial services under one roof, which contributed to the expansion in international balance sheets.

Things to Remember

- Financial Innovation Period: This period saw significant developments in the financial industry, including the introduction of new financial structures and products.
- Hedge Fund Industry Growth: The expansion of the hedge fund industry played a

crucial role in providing liquidity and diversifying investment risks.

- **Introduction of Financial Structures:** The development of financial structures like derivatives and structured products improved risk management and facilitated capital flow.
 - **Universal Banking:** The spreading of universal banking, offering a wide range of financial services under one roof, contributed to the expansion of international balance sheets.
 - **Market Liquidity:** Liquidity provision is essential for the smooth functioning of financial markets and stability of the financial system.
-

Q.4168 You are a student in a Risk Management class. Your lecturer asks you to explain maturity transformation across banks' balance sheets. Which of the following is correct about maturity transformation across banks' balance sheets?

- A. A maturity mismatch needed to facilitate long term investment projects should allow banks to earn a spread in a negative sloping yield curve surrounding.
- B. Banks do not transfer funds from agents in surplus, demanding short-term deposits to agents in deficit with long-term financing needs.
- C. Banks might be motivated to increase their maturity mismatch excessively and thus, making themselves vulnerable to funding risks related to short-term liabilities roll-over.
- D. It mitigates upward pressure and interbank rate volatility on the U.S dollar

The correct answer is **C**.

Maturity transformation is a fundamental function of banks, where they convert short-term liabilities (like deposits) into long-term assets (like loans). This process allows banks to earn a spread, as long-term interest rates are typically higher than short-term rates. However, this also exposes banks to funding risks, particularly in times of financial stress. If a bank has a large maturity mismatch, meaning its liabilities are much shorter-term than its assets, it may face difficulties in rolling over its short-term liabilities. This could lead to a liquidity crisis if depositors demand their money back and the bank is unable to meet these demands. Therefore, while maturity transformation is a necessary function of banks, excessive maturity mismatch can make banks vulnerable to funding risks.

Choice A is incorrect. While it's true that banks can earn a spread through maturity transformation, this isn't necessarily tied to a negative sloping yield curve. In fact, banks often

face challenges in such an environment because they're borrowing short-term at higher rates and lending long-term at lower rates.

Choice B is incorrect. This statement contradicts the fundamental concept of maturity transformation. Banks indeed act as intermediaries, transferring funds from agents with surplus (who demand short-term deposits) to those in deficit (who require long-term financing).

Choice D is incorrect. Maturity transformation doesn't directly mitigate upward pressure or volatility on the U.S dollar interbank rate. The interbank rate depends on various factors like monetary policy, liquidity in the market etc., not specifically on maturity transformation.

Things to Remember

- Maturity transformation is a key function of banks where they convert short-term liabilities into long-term assets to earn a spread.
 - Excessive maturity mismatch can expose banks to funding risks, especially during times of financial stress.
 - Banks play a crucial role in transferring funds from surplus agents to deficit agents through maturity transformation.
 - Maturity transformation does not directly mitigate upward pressure or volatility on interbank rates.
 - Banks need to carefully manage their maturity profiles to ensure liquidity and stability in their operations.
-

Q.4169 In a class discussion, each student was asked to mention something about the U.S dollar shortage during the great financial crisis. The following statements were recorded from the students' responses.

- I. The dollar shortage refers to a situation where a country has an inadequate supply of the dollar to manage international trade effectively.
- II. During the U.S. dollar shortage, countries had to pay more U.S dollars for imports relative to the U.S dollars received from exports.
- III. The U.S. dollar shortage refers to the mismatches between the maturity, currency, and counterparty of assets and liabilities.

Which of the following alternatives is correct?

- A. I and II
- B. II and III
- C. I, II, and III
- D. None of the above

The correct answer is **A**.

Statements I and II accurately describe the U.S. dollar shortage during the great financial crisis. A dollar shortage is indeed a situation where a country lacks a sufficient supply of the U.S. dollar to effectively manage its international trade. This shortage can occur when a country has to pay more U.S. dollars for its imports than it receives from exports. The U.S. dollar acts as a 'safe haven' for other countries' currencies, and a shortage can have a significant impact on global trade. During the great financial crisis, many countries experienced this shortage, which exacerbated the economic downturn.

Choice B is incorrect. Statement III does not accurately represent the U.S. dollar shortage during the great financial crisis. The U.S. dollar shortage was not a result of mismatches between the maturity, currency, and counterparty of assets and liabilities but rather due to a sudden increase in demand for safe assets like US dollars during times of economic uncertainty.

Choice C is incorrect. As explained above, statement III does not accurately represent the U.S. dollar shortage during the great financial crisis hence combination I, II and III cannot be correct.

Choice D is incorrect. Statements I and II do accurately represent aspects of the U.S. dollar shortage that occurred during the great financial crisis hence 'None of the above' cannot be correct.

Things to Remember

- Safe haven currency: The U.S. dollar is considered a safe haven currency, meaning that investors tend to flock to it during times of economic uncertainty or market turmoil.
- Impact on global trade: A U.S. dollar shortage can have significant implications for global trade, as countries may struggle to pay for imports or service their dollar-denominated debts.
- Foreign exchange reserves: Countries often hold U.S. dollars in their foreign exchange reserves to ensure they have an adequate supply of the currency for international transactions.
- Role of central banks: Central banks can intervene in foreign exchange markets to stabilize their currency's value or address shortages of foreign currencies like the U.S.

dollar.

Q.4170 Malcolm was asked to identify two main challenges related to international lending of last resort solved by international swap arrangements. He gave four responses as follows:

- I. The Federal Reserve and its foreign counterparts were now accorded power of creating whatever amounts of money they choose as compared to global financial institutions administering resources in limited nature.
- II. The auctioning of the U.S dollar led to a minimized level of volatile swap spread.
- III. The swap network does not consist of information issues that can lead to moral dangers.
- IV. It mitigated upward pressure and interbank rate volatility on the U.S. dollar.

Which of the responses correctly identifies the two main challenges?

- A. I and III
- B. II and IV
- C. I and IV
- D. None of the above

The correct answer is **A**.

The two main challenges related to international lending of last resort that are solved by international swap arrangements are indeed the ones identified in responses I and III. The first challenge is the limited nature of resources administered by global financial institutions. This is addressed by the power accorded to the Federal Reserve and its foreign counterparts to create any amount of money they choose. The second challenge is the potential for information issues within the swap network to lead to moral hazards. This is mitigated by the absence of such issues within the swap network. These two benefits directly address the challenges faced in international lending of last resort, making them the correct responses.

Choice B is incorrect. While the auctioning of the U.S dollar (II) can lead to a reduction in the volatility of swap spreads, it is not one of the main challenges that international swap arrangements are designed to address. The mitigation of upward pressure and volatility in the interbank rate on the U.S. dollar (IV) is indeed a benefit, but it does not pair with II as one of the two main challenges addressed by these arrangements.

Choice C is incorrect. The ability of central banks like Federal Reserve and its foreign counterparts to create any amount of money they choose (I) is indeed a key benefit, but it does not pair with IV as one of two main challenges addressed by these arrangements. Although mitigating upward pressure and volatility in interbank rates on U.S. dollar (IV) can be seen as a benefit, it doesn't form part with I as primary issues that international swap arrangements aim to solve.

Choice D is incorrect. As explained above, there are indeed pairs among these benefits that correctly identify two main challenges that international swap arrangements are designed to address.

Things to Remember

- **International Lending of Last Resort:** Refers to the provision of emergency liquidity assistance to financial institutions or countries facing a liquidity crisis in the international financial system.
 - **Swap Arrangements:** Agreements between central banks or monetary authorities to exchange one currency for another at a pre-agreed exchange rate and for a specific period of time.
 - **Global Financial Institutions:** Entities such as the International Monetary Fund (IMF) and World Bank that play a key role in providing financial assistance and stability in the global financial system.
 - **Moral Hazard:** Refers to the risk that one party may act in a riskier manner knowing that it is protected against the consequences of its actions, leading to potential negative outcomes.
 - **Interbank Rate Volatility:** Fluctuations in interest rates at which banks lend to each other in the interbank market, impacting overall liquidity and stability in the financial system.
-

Q.4171 The following statements are about the structure of bank operations in long and short of the banks' global balance sheets. Which one is most definitely TRUE?

- A. Banks management of maturity and currencies are based on a consolidated global entity instead of per office.
- B. Banks actively operating across the world have numerous offices in various countries.
- C. Significant mismatches measured on an office's balanced sheet located in a different office may be offset/hedged off-balance sheet through an on-balance sheet position booked by other offices elsewhere
- D. All of the above

The correct answer is **D**.

All the statements provided in the choices accurately reflect the structure and operations of banks on their global balance sheets. Banks that operate on a global scale have numerous offices in various countries (Choice B). They manage their maturity and currencies based on a consolidated global entity instead of per office (Choice A). This implies that significant mismatches measured on an office's balance sheet located in a different office may be offset/hedged off-balance sheet through an on-balance sheet position booked by other offices elsewhere (Choice C). Therefore, all the statements are true, making Choice D the most accurate answer.

Choice A is incorrect. While it's true that banks manage maturity and currencies on a consolidated global entity basis, this statement alone does not fully capture the complexity of global banking operations. Banks also manage these aspects at the individual office level, taking into account local market conditions and regulations.

Choice B is incorrect. The presence of numerous offices in various countries is a characteristic of globally operating banks, but this statement does not provide any insight into how these banks manage their global balance sheets or handle mismatches in maturity and currencies.

Choice C is incorrect. Although significant mismatches measured on an office's balance sheet may be offset/hedged off-balance sheet through an on-balance sheet position booked by other offices elsewhere, this statement doesn't encompass all the strategies used by banks to manage their global operations. Other methods such as netting agreements or internal transfers are also commonly used.

Things to Remember

- Banks' global balance sheets reflect the overall financial position of the entire bank, taking into account all offices and subsidiaries worldwide.
- Global banks often use sophisticated risk management techniques to mitigate risks associated with currency fluctuations, interest rate changes, and other market variables.
- Regulatory requirements may differ across countries, leading banks to adapt their operations and risk management strategies accordingly.
- Off-balance sheet activities, such as derivatives trading and securitization, play a significant role in shaping a bank's overall risk profile and financial performance.
- Global banks must comply with international standards such as Basel III to ensure the stability and resilience of the global financial system.

Q.4172 Which of the following entails an investor holding a high-yielding currency asset (target asset), which is financed with a low-yielding currency liability (funding liability)?

- A. Carry trade
- B. Cross currency funding
- C. Hedging
- D. Funding gap

The correct answer is **A**.

A carry trade is a strategy that involves borrowing at a low interest rate and investing in an asset that provides a higher rate of return. In the context of foreign exchange markets, a carry trade involves borrowing in a low-yielding currency (funding liability) and investing in a high-yielding currency (target asset). The investor 'carries' the difference between the two rates. Carry trades are typically used by large hedge funds and are a major source of leverage in the foreign exchange market.

Choice B is incorrect. Cross currency funding refers to the process of borrowing in one currency where rates are low and then using that funding to invest in another currency where returns are high. While this may seem similar to the scenario described, it does not involve an investor possessing a target asset that yields high returns. Instead, it involves direct investment in a different currency.

Choice C is incorrect. Hedging is a risk management strategy used by investors to protect against potential losses from fluctuations in exchange rates, interest rates or commodity prices. It does not involve financing an asset with a liability as described in the question.

Choice D is incorrect. The funding gap refers to the difference between available funds (liabilities) and required funds (assets). This concept does not involve investing or financing assets with liabilities as described in the question.

Things to Remember

- Carry trade involves borrowing in a low-yielding currency and investing in a high-yielding currency to profit from the interest rate differential.
- Interest rate parity is an important concept related to carry trades, stating that the forward exchange rate should incorporate the interest rate differential between two currencies.
- Uncovered interest rate parity (UIP) is another concept relevant to carry trades, suggesting that expected exchange rate changes should offset interest rate differentials.

- Leverage is often used in carry trades to amplify returns, but it also increases the risk of losses if the exchange rate moves unfavorably.
-

Q.4173 Why is the U.S. dollar is referred to as a “safe haven”?

- A. It acts as a peg for other countries due to its liquidity.
- B. It has less value relative to other currencies.
- C. It is highly illiquid
- D. The U.S. has weak political systems

The correct answer is **A**.

Some features make a currency a “safe haven” for other currencies. One of them is acting as a peg for other countries since it is highly liquid. Moreover, it is of high value, as compared to other currencies. Finally, the United States has a stable political system. These make the U.S. dollar be known as a safe haven.

B is incorrect: The U.S. dollar is considered to have the most substantial value as compared to other currencies.

C is incorrect: The U.S. dollar has high liquidity. In other words, it is easy to buy or sell.

D is incorrect: The United States has a stable political system; thus, making the U.S. dollar a stable currency against other currencies.

Things to Remember

1. A 'safe haven' currency is one that investors tend to gravitate towards in times of market turbulence or economic uncertainty. This is usually because the currency is perceived as being stable and reliable.
 2. The U.S. dollar is often referred to as a 'safe haven' currency due to its high liquidity and the economic and political stability of the United States.
 3. Liquidity refers to the ease with which an asset or security can be bought or sold in the market without affecting its price. High liquidity is a desirable attribute for any currency as it allows for easy transactions and conversions.
 4. The U.S. dollar's status as the world's primary reserve currency also contributes to its 'safe haven' status. As the primary reserve currency, many international transactions are denominated in U.S. dollars, further increasing its liquidity and stability.
-

Q.4175 The federal reserve may extend a loan using nominal figure collateral of foreign currencies to foreign central banks. Moreover, a bank can access the funds through U.S. dollar auction in the bank's respective jurisdiction/country. Which of the following best describes this?

- A. International lending of last resort
- B. Maturity transformation across banks' balance sheets
- C. Cross-currency funding
- D. Balance sheet expansion

The correct answer is **A**.

This term refers to the practice where a central bank, such as the Federal Reserve, provides financial assistance to foreign central banks in times of financial distress. In this case, the Federal Reserve extends loans using foreign currencies as collateral to foreign central banks. These banks can then access these funds through an auction of U.S. dollars within their own jurisdiction or country. This process allows commercial banks worldwide to access U.S. dollar liquidity, including those that lack sufficient eligible collateral and subsidiaries, as they can directly borrow from the Federal Reserve system.

Choice B is incorrect. Maturity transformation across banks' balance sheets refers to the process where banks borrow funds short-term and lend long-term, thereby transforming short-term liabilities into long-term assets. This concept does not encapsulate the process of extending loans using foreign currencies as collateral to foreign central banks.

Choice C is incorrect. Cross-currency funding refers to a situation where an entity borrows in one currency and then swaps that borrowing into another currency using a cross-currency swap. This term does not accurately describe the process of Federal Reserve extending loans using foreign currencies as collateral.

Choice D is incorrect. Balance sheet expansion generally refers to an increase in a company's assets or decrease in its liabilities on its balance sheet, which can occur due to various reasons such as acquisition of new assets or reduction of liabilities through repayments etc., but it doesn't specifically refer to the process described in the question.

Things to Remember

- International Lending of Last Resort is a concept where central banks provide financial assistance to foreign central banks in times of financial distress.
- Foreign exchange reserves are assets held by central banks in foreign currencies, used to back liabilities on their own issued currency as well as to influence monetary policy.
- U.S. dollar liquidity is the availability of U.S. dollars in the financial system, which is crucial for international trade and financial stability.

- Central banks play a key role in maintaining financial stability and providing liquidity to the banking system during times of crisis.
-

Q.4176 Which of the following correctly defines the U.S. dollar funding gap?

- A. A scenario where non-U.S. banks, have liabilities denominated in U.S. dollars that exceed their U.S. dollar-denominated assets.
- B. An offshore investment fund, typically formed as a private limited partnership that engages in speculation using a credit or borrowed capital.
- C. A scenario where a country lacks a sufficient supply of the U.S. dollar for effective management of international trade.
- D. A situation that involves an investor holding a high-yielding currency asset (target asset), which is financed with a low-yielding currency liability (funding liability).

The correct answer is **A**.

The U.S. dollar funding gap refers to a situation where non-U.S. banks have liabilities denominated in U.S. dollars that exceed their U.S. dollar-denominated assets. This gap arises because many global financial transactions, trade, and investment activities are conducted in U.S. dollars, even by institutions that primarily operate in other currencies. As a result, non-U.S. banks may borrow in U.S. dollars or issue dollar-denominated debt to meet the needs of clients or fund their operations. However, these banks might not hold enough U.S. dollar assets—such as cash reserves, U.S. dollar-denominated loans, or investments—to offset their dollar liabilities.

Choice B is incorrect. An offshore investment fund, typically formed as a private limited partnership that engages in speculation using credit or borrowed capital, does not define the U.S. dollar funding gap. This choice describes an investment strategy rather than a specific financial situation related to the U.S. dollar.

Choice C is incorrect. A scenario where a country lacks a sufficient supply of the U.S. dollar for effective management of international trade does not accurately describe the U.S. dollar funding gap either. While this could be a result of such a gap, it is not its definition.

Choice D is incorrect. A situation that involves an investor holding a high-yielding currency asset (target asset), which is financed with a low-yielding currency liability (funding liability) also fails to define the term correctly as it refers more to carry trade strategy and not specifically to any imbalance in US dollars.

Things to Remember

- U.S. dollar funding gap is a crucial concept in international finance and banking.

- It highlights the mismatch between U.S. dollar assets and liabilities in terms of maturity.
 - Banks and financial institutions need to manage this gap effectively to avoid liquidity issues.
 - Rolling over the funding or finding alternative sources of U.S. dollar funding are common strategies to address this gap.
 - The U.S. dollar's role as a global reserve currency amplifies the importance of understanding and managing the funding gap.
-

Q.4177 A maturity mismatch commonly involves a firm's balance sheet. Which of the following is correct about a maturity mismatch?

- A. It is not visible on a firm's balance sheet and masks its liquidity.
- B. Involves short-term borrowers funding long-term assets
- C. It cannot occur in hedging
- D. The maturity mismatch needed for the facilitation of long-term investment projects and for serving the liquidity needs of the investor should allow banks to earn a spread in a negatively sloped yield curve surrounding.

The correct answer is **B**.

A maturity mismatch typically involves short-term borrowers funding long-term assets. This situation arises when a firm uses short-term liabilities to fund long-term assets. This is a common practice in many financial institutions, particularly banks, where they borrow funds on a short-term basis (such as deposits from customers) and lend on a long-term basis (such as mortgages and business loans). This practice allows the firm to earn a spread between the interest it pays on short-term borrowings and the interest it earns on long-term loans. However, it also exposes the firm to liquidity risk, as the firm may face difficulties in meeting its short-term obligations if it cannot quickly convert its long-term assets into cash.

Choice A is incorrect. While a maturity mismatch may not be immediately apparent on a firm's balance sheet, it does not necessarily mask its liquidity. Liquidity refers to the ability of a firm to meet its short-term obligations, which can be affected by various factors other than maturity mismatch.

Choice C is incorrect. Maturity mismatches can indeed occur in hedging scenarios. For instance, if a company uses short-term futures contracts to hedge against long-term price risks,

this could result in a maturity mismatch.

Choice D is incorrect. This statement incorrectly assumes that banks can always earn a spread in negatively sloped yield curve environments due to maturity mismatches. However, this isn't always the case as the spread depends on various factors including interest rates and market conditions.

Things to Remember

- Liquidity risk: Refers to the risk that a firm may not be able to meet its short-term obligations due to a mismatch between its short-term liabilities and assets.
 - Interest rate risk: Maturity mismatches can expose firms to interest rate risk, where changes in interest rates can impact the profitability of the firm.
 - Funding sources: Firms can use various sources of funding such as equity, debt, and deposits to finance their operations and investments.
 - Yield curve: A graphical representation of interest rates at different maturities, which can impact the profitability of firms with maturity mismatches.
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Reading 143: Covered Interest Rate Parity Lost: Understanding the Cross-Currency Basis

Q.4178 Bright Ltd., a US-based corporation, enters a currency basis swap with Gamble, a British company, in which the original principal amounts are \$300 million and £240 million. That is: At inception, there is an initial principal exchange in which Bright Ltd. pays Gamble \$300 million and receives £240 million. Subsequently, at each interest payment date, Bright Ltd pays Gamble the GBP-Libor rate on £240million and receives the USD-Libor rate on \$300 million. Finally, at maturity, a re-exchange of principals occurs in which Bright Ltd pays £240million in exchange for \$300 million. Suppose the spot exchange rate is $\$1.25 = \pounds 1$ at the time of entering the swap. Assuming Bright Ltd. And Gamble both have AA credit ratings at this time and can access funds at Libor flat, the value of the swap at inception to Bright Ltd is?

- A. Positive
- B. Negative
- C. Zero
- D. Cannot be determined from the above information

The correct answer is **C**.

A cross-currency swap does not involve an initial capital outlay since its initial market value is zero.

Things to Remember

- Cross-currency swaps involve the exchange of cash flows in different currencies.
- These swaps help companies manage their exposure to foreign exchange rate risk.
- The notional amounts in a cross-currency swap are exchanged at the beginning and end of the contract.
- The value of a cross-currency swap at inception is typically zero, assuming no arbitrage opportunities.
- Credit ratings of the counterparties play a crucial role in determining the terms and pricing of the swap.

Q.4180 Suppose the market EUR/USD spot exchange rate is 1.3536 with a one-year forward rate of 1.3280. The market consists of a risk-free rate of 4% USD and 6% EUR per annum, respectively. Calculate the ratio of returns.

A. 0.1097

B. 0.9811

C. 12.45

D. 109.7

The correct answer is **B**.

Covered interest rate parity is checked using the formula:

$$\frac{F}{S} = \frac{1 + r}{1 + r^*}$$

$$\Rightarrow \text{The ratio of returns} = \frac{1 + \text{Quote currency interest rate}}{1 + \text{Base currency interest rate}}$$

In this case, the quote currency rate is the US dollar interest rate, while the base currency rate is the EUR interest rate.

$$= \frac{1 + 4\%}{1 + 6\%} = 0.9811$$

Q.4181 Suppose the market USD/EUR spot exchange rate is 1.3536 EUR with a one-year forward rate of 1.3280 EUR. The market consists of a risk-free rate of 4% USD and 6% EUR, respectively. Calculate the ratio of forward rate to spot rate.

- A. 0.9811
- B. 1.0193
- C. 0.0189
- D. 1.5467

The correct answer is **A**.

$$\text{The ratio of forward to spot rate} = \frac{F}{S} = \frac{1.3280}{1.3536} = 0.9811$$

Q.4184 Counterparty A wants to carry out a cross-currency swap where it will exchange the sterling pound for the U.S. dollar with counterparty B. Today, the U.S Libor rate is 1.0% while the sterling pound Libor rate is -2.6%. A dollar shortage occurs, and counterparty B quotes a basis of -40 bps. Compute the cost of the swap for counterparty A. (Assume that both parties pay interest on the currency they receive.)

- A. 4.0%
- B. 2.0%
- C. 2.6%
- D. 3.2%

The correct answer is **A**.

$$\text{Cost of the swap} = 1.0\% + 2.6\% + 0.4\% = 4.0\%$$

Note: The Sterling pound Libor is negative (-2.6%); thus, counterparty B will also have to pay for it, hence the reason why we add it.

Detailed Explanation

Counterparty A is exchanging pounds for dollars. In other words, it's giving out pounds to receive dollars. Both parties pay interest on the currency they receive (because they have technically borrowed that currency). In our case, A should pay the dollar LIBOR (1%) plus the cross-currency basis (-0.4%). Because B has "borrowed" the pound from A, it should pay the pound LIBOR (-2.6%) to A. But here's the problem: this rate is negative, and just like in any other scenario where the interest rate is negative, **it would be the lender and not the borrower, who pays the interest**. Counterparty A would have to pay pound interest to B.

At the end of the day, therefore, the total theoretical cost of the Pound/USD currency swap to A is $1\% + 0.4\% + 2.6\% = 4.0\%$

If the pound LIBOR was +2.6%, party A would have actually received interest on the pound. A's cost would have been $1\% + 0.4\% - 2.6\% = -1.2\%$; it would receive a net rate of interest of 1.2% from B.

Q.4185 Company A wants to transform \$60 million floating rate debt into a fixed rate GBP loan. On trade date, Company A exchanges \$60 million with Company B in return for 48 million GBP. Calculate the GBP/USD exchange rate if their agreement is set to expire in 10 years.

- A. 0.5
- B. 0.86
- C. 0.8
- D. 1.25

The correct answer is **D**.

$$\text{GBP/USD} = \frac{60}{48} = 1.25$$

Note that,

The way Foreign exchange is quoted is always interpreted as; you exchange many units of the denominator currency for 1 unit of the given numerator currency. The rates are different when you quote GBP/USD or USD/GBP.

For example, in this case, USD/GBP=0.8.

However, the general idea here is to quote GBP/USD instead of USD/GBP.

Q.4186 Assume that two parties enter a five -year FX currency swap for 100 million. Party, A is a US firm, while the other, B, is a European firm. The EUR/USD exchange rate is \$1.25 by the time the swap is arranged. In other words, the dollar is EUR 0.8. The start of the swap is December 31st, 2019. On this date, A pays B \$100m, while B pays A EUR 80m. Assume that the agreed dollar interest rate is 6%, while the euro interest rate is 3.5%. Calculate the amount that company A pays annually to company B.

- A. €800,000
- B. €900,000
- C. €1,900,000
- D. €2,800,000

The correct answer is **D**.

Every year, company A pays $€80,000,000 \times 3.5\% = €2,800,000$ to Company B.

Things to Remember

- FX Currency Swap: A financial agreement between two parties to exchange a specified amount of one currency for another at a predetermined exchange rate on a future date.
 - Exchange Rate: The rate at which one currency can be exchanged for another currency.
 - Interest Rate Parity: The theory that the interest rate differential between two countries is equal to the differential between the forward exchange rate and the spot exchange rate.
 - Notional Principal: The amount on which the interest rate is calculated in a swap agreement, but not actually exchanged.
 - Counterparty Risk: The risk that one party in a swap agreement will default on its obligations.
-

Q.4187 Assume that a U.S. firm A wants to acquire Japanese yen while a Japanese firm, B, is aiming to borrow U.S. dollars (USD). Suppose that A wants to borrow ¥100 million. The Japanese firm needs \$50 million. The two firms enter a USD/JPY swap with an exchange rate of 2. Calculate the quarterly interest amount that A pays to B, given that A pays 4% on ¥100 million annually, and the interest rates remain unchanged.

- A. ¥250,000
- B. ¥500,000
- C. ¥1,000,000
- D. ¥1,500,000

The correct answer is **C**.

Let's calculate the quarterly interest amount that Firm A pays to Firm B:

Given Data:

- Firm A borrows ¥100 million from Firm B.
- Firm B borrows \$50 million from Firm A.
- The exchange rate is $\text{USD/JPY} = 2$, meaning $1 \text{ USD} = 2 \text{ JPY}$.
- Firm A pays 4% annual interest on ¥100 million.

Step 1: Calculate the annual interest payment in JPY.

$$\text{Annual Interest Payment (JPY)} = 0.04 \times 100,000,000 = \text{¥}4,000,000$$

Step 2: Convert the annual interest payment to a quarterly interest payment.

$$\text{Quarterly Interest Payment (JPY)} = \frac{\text{¥}4,000,000}{4} = \text{¥}1,000,000$$

Step 3: Understanding the interest payment from A to B:

Firm A pays interest in JPY to Firm B, as they have borrowed ¥100 million. The quarterly interest payment by Firm A to Firm B is therefore ¥1,000,000.

Final Answer:

The quarterly interest amount that Firm A pays to Firm B is **¥1,000,000**.

Q.4188 An investor borrows \$2,000 and converts the funds into British pounds, allowing him to purchase a British bond. If the purchased bond has a yield of 8% while the equivalent U.S. bond yield is 5%, calculate the interest rate differential if the exchange rate between dollars and pounds remains constant.

- A. 3.0%
- B. 5.0%
- C. 6.5%
- D. 13.0%

The correct answer is **A**.

An interest rate differential is simply the difference in the interest rate between two currencies. This differential in the cash money markets should equal the differential between the forward and spot exchange rates. Otherwise, arbitrageurs could make a seemingly riskless profit.

In this case, the interest rate differential = $8\% - 5\% = 3\%$

Things to Remember

- Interest rate differentials reflect the difference in interest rates between two currencies and play a crucial role in the foreign exchange market.
 - Arbitrage opportunities arise when interest rate differentials do not align with the forward and spot exchange rates.
 - Investors often engage in carry trades based on interest rate differentials to profit from the variances in interest rates between currencies.
 - Exchange rate movements can impact the overall returns of investments made based on interest rate differentials.
-

Q.4189 Calculate the 1-year Euro interest rate that will satisfy the covered interest rate parity if the EUR/USD spot rate is \$1.32 and the U.S dollar has a 1-year interest rate of 6% and the EUR/USD forward rate is 1.22.

- A. 10.24%
- B. 29.32%
- C. 12.42%
- D. 14.69%

The correct answer is **D**.

$$\begin{aligned}F_{\text{EUR/USD}} &= S \frac{(1 + R_{\text{USD}})}{(1 + R_{\text{EUR}})} \\1.22 &= 1.32 \frac{(1.06)}{(1 + R_{\text{EUR}})} \\ \Rightarrow R_{\text{EUR}} &= \frac{1.32 \times 1.06}{1.22} - 1 \\ &= 0.14688\end{aligned}$$

Thus, the one-year Euro interest rate is 14.69%

Q.4190 A Japanese company X wants to compute the one-year forward USD/JPY rate, with a spot dollar selling at 132 USD/JPY. The JPY per annum interest rate is equal to 4%, and the annual interest rate for USD is 11%. Calculate the one-year forward exchange rate USD/JPY.

- A. 0.14
- B. 123.68
- C. 136.87
- D. 140.88

The correct answer is **B**.

According to covered interest rate parity, the interest rate differential between two countries is equal to the differential between the forward exchange rate and the spot exchange rate.

The forward exchange rate is given by:

$$\begin{aligned} F &= S \times \left(\frac{1 + R_{JPY}}{1 + R_{USD}} \right)^T \\ &= 132 \times \left(\frac{1.04}{1.11} \right)^1 \\ &= 123.6758 \end{aligned}$$

Q.4191 FX swaps and Currency Swaps are derivative instruments utilized in the hedging of foreign currency exposures. Which of the following correctly points out the similarities between the two?

- A. Both instruments involve the exchange of principal and interest payments in different currencies.
- B. Both instruments are used primarily to manage long-term interest rate risk.
- C. Both instruments involve an exchange of currencies.
- D. Both instruments are commonly utilized by central banks to manage their currency reserves.

The correct answer is **C**.

Both instruments involve an exchange of currencies. However, the key distinction is that FX swaps involve an agreement to reverse the transaction at a later date, whereas currency swaps do not necessarily involve a reversal. This similarity captures the essence of the derivative

nature of both instruments and their involvement in the exchange of currencies.

A is incorrect. While currency swaps involve the exchange of principal and interest payments in different currencies, FX swaps only involve the exchange of currencies without the exchange of interest payments.

B is incorrect. Currency swaps are used to manage long-term interest rate risk, but FX swaps are primarily used for short-term foreign exchange exposure management, not long-term interest rate risk.

D is incorrect. While central banks may use both FX swaps and currency swaps for various purposes, it is not a common practice for central banks to use currency swaps to manage their currency reserves. FX swaps are more commonly used for such purposes.

Things to Remember

- FX swaps involve the exchange of currencies for a specific period of time, typically ranging from a few days to a year, while currency swaps involve the exchange of principal and interest payments in different currencies over a longer period, often several years.
- FX swaps are commonly used by multinational corporations to hedge against foreign exchange rate risk, while currency swaps are often used by entities looking to manage both interest rate and foreign exchange risk simultaneously.
- Both FX swaps and currency swaps are over-the-counter (OTC) derivatives, meaning they are customized contracts traded directly between two parties rather than on an exchange.
- Counterparty risk is an important consideration in both FX swaps and currency swaps, as the creditworthiness of the parties involved can impact the effectiveness of the hedging strategy.
- Regulatory requirements and accounting treatment may differ for FX swaps and currency swaps, so it is important for users of these instruments to be aware of the relevant guidelines and standards.

Q.4192 A hypothetical condition where the correlation between interest rates, spot and forwards currency value of two countries are equal is referred to as:

- A. FX swaps
- B. Cross-currency swaps
- C. Covered Interest Rate Parity
- D. Mark-to-market

The correct answer is **C**.

Covered Interest Rate Parity (CIP) is the term that describes the hypothetical condition where the correlation between interest rates, spot, and forward currency values of two countries are equal. CIP is a no-arbitrage condition in financial economics that states that the nominal interest rates in two different countries should be equal, once the foreign exchange risk is hedged. This condition is expressed mathematically as:

$$\frac{F}{S} = \frac{1 + r}{1 + r^*}$$

Where: S is the spot exchange rate in US dollar per foreign currency, F is the corresponding forward exchange rate, r is the US dollar interest rate, and r* is the foreign currency interest rate. This condition is a cornerstone of the foreign exchange markets and is used to prevent potential arbitrage opportunities.

Choice A is incorrect. FX swaps are a financial instrument used to hedge against foreign exchange risk, not a theoretical condition that stipulates an equal correlation between the interest rates, spot, and forward currency values of two different countries.

Choice B is incorrect. Cross-currency swaps are agreements in which two parties exchange principal and interest in one currency for the same in another currency. They do not represent a theoretical condition correlating interest rates, spot and forward currency values.

Choice D is incorrect. Mark-to-market refers to the accounting practice of valuing an asset according to its current market price rather than its book value. It does not describe any correlation between interest rates, spot and forward currency values of different countries.

Q.4193 Assume that the GBP/USD spot exchange rate is 1.500 (i.e., 1 GBP = 1.500 USD). Further assume that the 180-day USD LIBOR rate stands at 1.2% per annum, and the 180-day GBP LIBOR rate is 2.0% per annum. Compute the 180-day forward GBP/USD exchange rate.

A. 1.4882

B. 1.4292

C. 1.4276

D. 1.4941

The correct answer is **D**.

$$\begin{aligned}
 F &= S \times \frac{1 + r \times T}{1 + r^* \times T} \\
 &= 1.500 \left(\frac{1 + 0.012 \times \frac{180}{360}}{1 + 0.02 \times \frac{180}{360}} \right) \\
 &\approx 1.49405
 \end{aligned}$$

Alternatively,

$$\begin{aligned}
 F &= S \times \left(\frac{1 + r}{1 + r^*} \right)^T \\
 &= 1.500 \left[\left(\frac{1 + 0.012}{1 + 0.02} \right)^{\frac{180}{360}} \right] \\
 &\approx 1.494106
 \end{aligned}$$

Q.4194 Assuming a forward rate of \$0.40/JPY, a one-year interest rate for the U.S. currency of 5%, and a spot rate of \$0.5/JPY, calculate the current one-year JPY interest rate that will satisfy the covered interest rate parity.

- A. 14.35%
- B. 16.53%
- C. 31.25%
- D. 41.53%

The correct answer is **C**.

$$\begin{aligned}\frac{F}{S} &= \frac{1 + r}{1 + r^*} \\ r^* &= (1 + r) \frac{S}{F} - 1 \\ &= ((1 + 0.05) \times \frac{0.50}{0.40}) - 1 = 0.3125\end{aligned}$$

JPY interest rate 31.25%

Q.5413 John Paul, the chief investment officer at Herald Advisors, is particularly concerned about the cross-currency basis since the Global Financial Crisis. Which of the following is the most suitable explanation for the continued presence of a positive cross-currency basis between two currencies?

- A. Regulatory changes have allowed for an expansion in speculative proprietary trading activities conducted by US banks.
- B. Given the overall decrease in global interest rates, the inclusion of a liquidity risk cost in swap pricing is no longer deemed necessary.
- C. The basis increasingly incorporates the financing costs incurred by arbitrageurs to maintain their positions.
- D. As arbitrage positions typically mitigate counterparty risks, the expenses associated with credit value adjustments have risen.

The correct answer is **C**.

Following the Global Financial Crisis (GFC), there have been structural changes in how market

participants assess market, credit, counterparty, and liquidity risks, which have imposed stricter limits on arbitrage opportunities. Consequently, arbitrage now incurs a persistent balance sheet cost. The cost associated with financing offsetting positions for all arbitrage participants is now being factored into the FX swap basis.

A is incorrect. Regulatory changes, such as the implementation of the Volcker rule, have imposed restrictions on speculative proprietary trading activities for US banks since the Global Financial Crisis (GFC).

B is incorrect. Market participants must incorporate liquidity risk costs into their transactions. While overall funding costs may have decreased, it does not imply that a charge for liquidity risk is no longer necessary.

D is incorrect. The incorporation of credit value adjustments into pricing has accounted for the elimination of counterparty risks in arbitrage positions.

Things to Remember

- Global Financial Crisis (GFC) had a significant impact on financial markets and led to structural changes in how market participants assess risks.
 - Arbitrage opportunities have become more limited post-GFC due to stricter limits on market, credit, counterparty, and liquidity risks.
 - Regulatory changes, such as the Volcker rule, have restricted speculative proprietary trading activities for US banks.
 - Market participants need to consider liquidity risk costs in their transactions, even with the overall decrease in global interest rates.
 - Credit value adjustments are important in pricing to account for counterparty risks in arbitrage positions.
-

Reading 144: Risk Management for Changing Interest Rates: Asset-Liability Management and Duration Techniques

Q.4228 Bright Bank posted the following financial entries:

Interest revenues	\$83
Interest costs	\$72
Total earning assets	\$942

Calculate the bank's net interest margin.

- A. 1.17%
- B. 1.56%
- C. 2.25%
- D. 4.78%

The correct answer is **A**.

$$\begin{aligned}\text{Net interest margin (NIM)} &= \frac{\text{Net interest income}}{\text{Total earning assets}} \\ \text{Net Interest Margin} &= \frac{\$83 - \$72}{\$942} = 0.011677 \approx 1.17\%\end{aligned}$$

Q.4229 National Bank has a cumulative gap for the coming year of +\$128 million. The interest rates are expected to decrease by one and a half percentage points. Calculate the expected change in the Bank's net interest income.

- A. -1.23
- B. -1.92
- C. 1.89
- D. 128

The correct answer is **B**.

$$\begin{aligned}\text{Change in net interest income} &= \text{Total change in interest rate (Percentage points)} \\ &\times \text{Size of cumulative gap in dollars.}\end{aligned}$$

Thus,

$$\text{Expected change in net interest income} = \$128 \times (-0.015) = -1.92$$

Q.4230 Mutual Friends Bank has interest-sensitive assets worth \$470 million and interest-sensitive liabilities of \$876 million. Calculate its dollar interest-sensitive gap.

- A. -\$406m
- B. -\$470m
- C. -\$876m
- D. -\$1,346m

The correct answer is **A**.

$$\begin{aligned}\text{Dollar interest sensitive gap (IS GAP)} &= \text{Interest sensitive assets (IS GAP)} \\ &- \text{Interest sensitive liabilities (ISL)} \\ &= \$470 - \$876 = -\$406\text{m}\end{aligned}$$

Q.4231 Mutual Friends Bank has interest-sensitive assets worth \$470 million and interest-sensitive liabilities of \$876 million and the total value of assets is \$1000 million. Calculate its relative interest-sensitive gap ratio.

- A. -30.00%
- B. -40.87%
- C. -40.60%
- D. -50.32%

The correct answer is **C**.

$$\text{Relative Gap} = \frac{\text{ISGap}}{\text{Total Assets}}$$

$$\frac{(\$470 - \$876)}{\$1000} \approx -0.406 \text{ or } -40.60\%$$

Q.4232 Davidson Bank registered a net interest margin of 2.45% in its previous financial report, with total interest revenues of \$100 million and total interest expenses worth \$64 million. Compute the volume of the earning assets held by the bank.

- A. \$845.67 million
- B. \$1,045.23 million
- C. \$1,236.87 million
- D. \$1,469.39 million

The correct answer is **D**.

$$\text{Net interest margin} = \frac{\text{Net interest income}}{\text{Total earning assets}}$$

$$0.0245 = \frac{(100 - 64) \text{ million}}{\text{Total earning assets}}$$

$$\Rightarrow \text{Total earning assets} = \frac{(100 - 64) \text{ million}}{0.0245}$$

$$= \$1,469.39 \text{ million}$$

Q.4233 Banks find it challenging to forecast interest rate changes on an individual level. Which of the following correctly identifies factors as to why this happens?

- A. Interest rates are determined by the numerous investors trading in the credit market. A group of banks or an individual cannot set the interest rates.
- B. Each market rate of interest has various components – the risk-free interest rate plus different risk premia. A change in any of these rate components can cause interest rates to change.
- C. Bankers need to have perfect timing to be able to identify the changes in interest rates. That is, to maximize their predictions, they must have an insight into when the changes will take place.
- D. All of the above.

The correct answer is **D**.

The complexity of forecasting interest rate changes at an individual level in the banking sector is due to a combination of factors. Firstly, interest rates are not determined by a single bank or a group of banks. Instead, they are influenced by the actions of numerous investors trading in the credit market. This makes it impossible for a single entity to control or predict changes in interest rates. Secondly, each market interest rate is composed of various components, including the risk-free interest rate and different risk premia. A change in any of these components can cause interest rates to fluctuate, adding another layer of complexity to the forecasting process. Lastly, for accurate forecasting, bankers need to have perfect timing to identify changes in interest rates. They need to have insight into when these changes will occur to maximize their predictions. Therefore, all the options (A, B, and C) are correct, making choice D the correct answer.

Things to Remember

- Interest rate determination by numerous investors in the credit market
 - Components of market interest rates -risk-free interest rate and risk premia
 - Timing and insight required for accurate interest rate forecasting
 - Market expectations and economic indicators influencing interest rate changes
 - Yield curve analysis and its impact on interest rate forecasting
-

Q.4234 The purchase price of a government bond is \$750. It pays \$150 per year in interest for 4 years when it matures. If the redemption value of this bond is \$1,000, calculate its yield to maturity if it is purchased today for \$750.

- A. 12.00%
- B. 13.23%
- C. 24.87%
- D. 25.73%

The correct answer is **D**.

$$\$750 = \frac{\$150}{(1 + \text{YTM})^1} + \frac{\$150}{(1 + \text{YTM})^2} + \frac{\$150}{(1 + \text{YTM})^3} + \frac{\$150}{(1 + \text{YTM})^4} + \frac{\$1000}{(1 + \text{YTM})^4}$$

At YTM of 26%, the bond's price is \$744.78, while at 25%, its bond price is 763.84. Therefore, the actual YTM is between 25% and 26%.

By linear interpolation, we calculate YTM as follows:

$$\text{YTM} = 0.25 + \frac{763.84 - 750}{763.84 - 744.78} \times 0.01 = 0.2573 \approx 25.73\%$$

Q.4235 Suppose that Millennial National bank consists of an average asset duration of 2.53 years and an average liability duration of 0.65 years. Its assets size is a total of \$367 million, and the volume of its liabilities is \$219 million. Suppose that its original interest rates were 5% before rising to 8.5%. Compute Millennial National Bank's estimated leverage adjusted the duration gap and the change in its net worth as a result of the rise in interest rates.

- A. +1.34 years, -\$10.34million
- B. +2.14 years, -\$12.56million
- C. +1.34 years, -\$13.10million
- D. +2.14 years, -\$26.21million

The correct answer is **D**.

Leverage Adjusted Duration Gap

$$= \text{Dollar-weighted duration of Assets Portfolio} - \text{Dollar-weighted Duration of Liability Portfolio} \\ \times \frac{\text{Total Liabilities}}{\text{Total Assets}}$$

Millennial National Bank's duration gap is:

$$2.53 - 0.65 \times \frac{219}{367} = +2.14 \text{ years}$$

The change in net worth:

$$\begin{aligned} \text{Change in value of net worth} &= -DA \times \frac{\Delta r}{(1+r)} \times A - [-DL \times \frac{\Delta r}{(1+r)} \times L] \\ &= -2.53 \times \frac{+0.035}{(1+0.05)} \times 367 - [-0.65 \times \frac{+0.035}{(1+0.05)} \times 219] \\ &= -\$26.21 \text{ million} \end{aligned}$$

This means that Millennial National Bank's net worth will drop by approximately \$26.21 million if the interest increases by 3.5%.

Q.4236 A financial firm is seeking to have itself hedged against interest rate fluctuations by the use of duration-gap analysis. How can the firm management get to know when it is fully hedged through this approach?

- A. When fully hedged, it will have a zero-duration gap position.

- B. When it can track the variations between the interest earned on loans and the interest paid on deposits
- C. When it can get a clear comparison of the amount of liabilities and assets in each period in the interest rate sensitivity gap table, enabling it to get an estimated perspective of the interest rates risk of the firm's balance sheet being assessed.
- D. When it has higher interest-rate sensitive liabilities relative to its interest-rate sensitive assets.

The correct answer is **A**.

Duration-gap analysis is a financial risk management tool used by firms to hedge against the risk of interest rate fluctuations. The 'duration' in this context refers to the sensitivity of a financial instrument's price to changes in interest rates. When a firm is fully hedged using duration-gap analysis, it means that the dollar-weighted duration of the assets in its portfolio matches the dollar-weighted duration of its liabilities. In other words, the firm has a zero duration-gap position. This is because the interest rate risk on the assets is perfectly offset by the interest rate risk on the liabilities, thereby neutralizing the overall interest rate risk of the firm. Therefore, a zero duration-gap position is the most accurate indicator that a firm has achieved a fully hedged position using duration-gap analysis.

Choice B is incorrect. While tracking the variations between the interest earned on loans and the interest paid on deposits can provide some insight into a firm's financial health, it does not necessarily indicate a fully hedged position through duration-gap analysis. This is because duration-gap analysis specifically focuses on managing interest rate risk by aligning the durations of assets and liabilities, which isn't directly related to the difference in interest earned and paid.

Choice C is incorrect. Although comparing amounts of liabilities and assets in each period in an interest rate sensitivity gap table can help estimate perspective of a firm's balance sheet's interest rates risk, it doesn't signify that a fully hedged position has been achieved through duration-gap analysis. A fully hedged position using this method would require matching durations of assets and liabilities to ensure that changes in interest rates affect both equally.

Choice D is incorrect. Having higher interest-rate sensitive liabilities relative to its interest-rate sensitive assets does not indicate a fully hedged position using duration-gap analysis. In fact, this could expose the firm to greater risk if there are unfavorable movements in market rates as their cost (liabilities) may increase more than their income (assets). A zero-duration gap signifies that changes in market rates will have equal impact on both sides leading to no net effect.

Things to Remember

- **Duration:** Duration measures the sensitivity of a bond's price to changes in interest rates. It helps in assessing the interest rate risk of a bond or a portfolio.
- **Duration-Weighted:** When calculating the duration of a portfolio, the weights of

individual assets are considered based on their market values.

- Interest Rate Risk: The risk that changes in interest rates will affect the value of investments or assets.
- Duration Matching: Aligning the durations of assets and liabilities to mitigate interest rate risk.
- Duration Gap: The difference between the duration of assets and the duration of liabilities in a portfolio.

It's important for firms to actively manage their interest rate risk through tools like duration

- gap analysis to ensure financial stability and mitigate potential losses due to interest rate fluctuations.
-

Q.4237 Suppose that a hypothetical bank has an average yield on rate-sensitive and fixed assets of 5% and 10%, respectively. Additionally, the bank has rate-sensitive and non-rate-sensitive liabilities cost of 4% and 6%, respectively. During the coming week, the bank holds \$542 million in rate-sensitive assets and \$156 million in rate-sensitive liabilities. Assume that the asset total is \$1,212. Further, suppose that these annualized interest rates remain steady. Assuming that equity is zero, calculate the firm's net interest income on an annualized basis.

- A. 20.0 million
- B. 23.4 million
- C. 24.5 million
- D. 30.0 million

The correct answer is **C**.

$$\begin{aligned}
 \text{Net interest income} &= \text{Total interest income} - \text{Total interest cost} \\
 &= [\text{Average interest yield on rate-sensitive assets} \times \text{Volume of rate-sensitive assets} \\
 &\quad + \text{Average interest yield on fixed (non-rate-sensitive) assets} \times \text{Volume of fixed assets}] \\
 &\quad - [\text{Average interest cost on rate-sensitive liabilities} \times \text{Volume of interest-sensitive liabilities} \\
 &\quad - \text{Average interest cost on fixed (non-rate-sensitive) liabilities} \times \text{Volume of fixed} \\
 &\quad \text{(non-rate-sensitive) liabilities}] \\
 &= 0.05 \times \$542 + 0.10 \times [\$1,212 - \$542] - 0.04 \times \$156 - 0.06 \times [\$1,212 - \$156] \\
 &= \$24.5 \text{ million}
 \end{aligned}$$

Note.

The balance sheet is based on the fundamental equation:

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

In this case, we've been told there's no equity, which means that:

$$\text{Assets} = \text{Liabilities}$$

Q.4238 The interest-sensitive gap approach is not enough to fully hedge a financial firm from interest rates risks, thus the need for duration-gap management. The information below identifies some of the disadvantages of duration-gap management, which one is NOT?

- A. Duration matching (immunization) can be costly and time-consuming

- B. Immunization is a dynamic problem such that the duration of liabilities and assets changes as they near their maturity
- C. The interest-sensitive gap fails to put into consideration the implications of interest rate fluctuations on the positions of equity
- D. Duration is not accurate for significant interest rate changes unless convexity is taken into account

The correct answer is **C**.

The statement 'The interest-sensitive gap fails to put into consideration the implications of interest rate fluctuations on the positions of equity' is not a disadvantage of duration-gap management, but rather a limitation of the interest-sensitive gap approach. The duration gap management approach, in contrast, does take into account the effects of interest rate fluctuations on equity positions. This is because duration-gap management involves adjusting the duration of assets and liabilities to match, thereby reducing the sensitivity of the firm's equity to changes in interest rates. Therefore, this choice does not accurately represent a disadvantage of duration-gap management.

Choice A is incorrect. Duration matching, also known as immunization, can indeed be costly and time-consuming. This is because it requires constant rebalancing of the portfolio to maintain the duration match as interest rates change and as assets and liabilities move closer to their maturity dates.

Choice B is incorrect. Immunization is a dynamic problem in that the duration of liabilities and assets changes as they near their maturity. This means that even if a firm has perfectly matched its asset and liability durations at one point in time, it will need to continually adjust its portfolio to maintain this balance due to changes in market conditions or as these instruments approach their maturity dates.

Choice D is incorrect. Duration alone does not accurately measure interest rate risk for significant interest rate changes unless convexity is taken into account. Convexity measures the sensitivity of a bond's price change given a change in interest rates, which becomes more pronounced when there are large swings in rates. Therefore, ignoring convexity could lead to underestimation or overestimation of potential losses or gains from interest rate movements.

Things to Remember

- Duration: Measures the sensitivity of the price of a bond or portfolio to changes in interest rates.
- Convexity: Measures the curvature of the price-yield relationship of a bond and helps refine the accuracy of duration in measuring interest rate risk.
- Immunization: A strategy used to protect a portfolio from interest rate risk by matching the duration of assets with liabilities.

- Rebalancing: The process of adjusting the portfolio to maintain the desired duration match, which can be time-consuming and costly.
 - Interest-sensitive gap: Measures the difference between rate-sensitive assets and rate-sensitive liabilities to assess interest rate risk exposure.
-

Q.4239 Suppose that a treasury bill has a face value of \$100 set for payment at maturity. Its purchase price is \$88. If the security is to mature in 120 days, calculate the interest rate measured using the bank discount rate.

- A. 18%
- B. 20%
- C. 24%
- D. 36%

The correct answer is **D**.

$$DR = \frac{(100 - 88)}{100} \times \frac{360}{120} = 0.36 = 36\%$$

Q.4240 Assume that a treasury bill has a face value of \$100 set for payment at maturity. Its purchase price is \$88. If the security is to mature in 120 days, calculate the YTM equivalent yield of this bill.

- A. 36.00%
- B. 40.91%
- C. 50.86%
- D. 41.48%

The correct answer is **D**.

$$\begin{aligned}\text{YTM equivalent yield} &= \frac{(100 - \text{Purchase Price})}{\text{Purchase Price}} \times \frac{365}{\text{Days of maturity}} \\ &= \frac{(100 - 88)}{88} \times \frac{365}{120} = 41.48\%\end{aligned}$$

Note that we use 360 days when calculating discount rate and 365 days when calculating YTM.

Q.4241 Pride Bank's interest-sensitive assets have a value of \$500, and its interest-sensitive liabilities are worth \$300. Calculate the interest-sensitive ratio of the bank.

- A. 1.67
- B. 2.54
- C. 2.67
- D. 3.84

The correct answer is **A**.

$$\begin{aligned}\text{Interest Sensitive Ratio (ISR)} &= \frac{\text{ISA}}{\text{ISL}} \\ &= \frac{\$500}{\$300} = 1.67\end{aligned}$$

Q.4243 Assuming that Green House National bank holds liabilities and assets with the following average duration and dollar amount as given below:

Asset Composition	Avg. Duration (Years)	Dollar Amount (M)	Liability Composition	Avg. Duration (Years)	Dollar Amount (M)
Investment-grade bonds	7.5	80	Deposits	2.3	\$376
Commercial loans	4.23	423	Non-deposit borrowings	0.12	\$25
Consumer loans	3.2	\$145			
Total Assets		\$648	Total Liabilities		\$401

Calculate the dollar-weighted duration of the bank's liability portfolio and asset portfolio; thus, calculate the leverage-adjusted duration gap, respectively.

- A. 2.16 years, 4.40 years, 3.06 years
- B. 3.06 years, 4.40 years, 2.16 years
- C. 4.40 years, 2.16 years, 3.06 years
- D. 0.12 years, 3.2 years, 4.23 years

The correct answer is **A**.

$$D_A = 7.5 \times \frac{80}{648} + 4.23 \times \frac{423}{648} + 3.2 \times \frac{145}{648} = 4.40 \text{ years}$$

$$D_L = 2.3 \times \frac{376}{401} + 0.12 \times \frac{25}{401} = 2.16 \text{ years}$$

Thus, the leverage-adjusted duration gap is

$$D_A - D_L \times \frac{\text{Total Liabilities}}{\text{Total Assets}}$$

Leverage-adjusted duration Gap

$$4.40 - 2.16 \times \frac{401}{648} = 3.06 \text{ years}$$

Q.4244 A Canadian Treasury bill is available for purchase this week with a price of \$89.84 based on a par value of \$100, with its maturity indicated as 124 days. Calculate the discounted rate of this treasury bill.

- A. 2.950 %
- B. 29.500%
- C. 32.950%
- D. 295.000%

The correct answer is **B**.

$$\begin{aligned}
 DR &= \left(\frac{100 - \text{Purchase price on loan or security}}{100} \right) \\
 &\times \frac{360}{\text{Number of days to maturity}} \\
 &= \left(\frac{100 - 89.84}{100} \right) \times \frac{360}{124} = 0.2949 \approx 29.5\%
 \end{aligned}$$

Q.5391 Thomas Peters, the Chief Risk Officer at ABC Bank, is assessing the bank's net worth sensitivity to a potential interest rate change. By conducting duration analysis, Mr. Peters evaluates the impact of a 200-basis point change in interest rates on the bank's net worth. Which of the following statements accurately represents the conclusion made by the CRO?

- A. A non-prepayable loan has a lower duration compared to a prepayable mortgage loan, making it less sensitive to overall interest rate changes.
- B. Assets and liabilities exhibit higher sensitivity to changes in interest rates When the coupon is lower, the duration is higher, or the overall level of interest rates is lower.
- C. The percentage change in the value of an asset or liability caused by a 200-basis point change in interest rates is directly related to its duration and remains unaffected by the corresponding percentage change in interest rates.
- D. Paying a cash dividend of USD 50 million using cash will have no impact on the overall asset duration of the bank.

The correct answer is **B**.

The sensitivity of asset and liability values to changes in interest rates is higher for instruments with lower coupons, higher durations, and lower overall interest rates. When the coupon is

lower, it means the interest payments on the instrument are lower relative to its face value, resulting in a higher duration. A higher duration implies that the instrument's price is more sensitive to changes in interest rates.

A is incorrect. A prepayable mortgage often has a shorter duration than a comparable non-prepayable investment and is less vulnerable to changes in the market's average interest rate.

C is incorrect. The percentage change in an asset or liability's value caused by a 100-basis point change in interest rates is directly proportional to the asset or liability's duration and also depends on the same percentage change in interest rates.

D is incorrect. The bank's overall asset duration will be impacted by using USD 50 million of cash to pay the bank's cash dividend payment, as the bank's overall asset level would decline as a result of the cash dividend payment.

Things to Remember

- Duration analysis is a risk management technique used to measure the sensitivity of a financial instrument's price to changes in interest rates.
 - Duration is a measure of the weighted average time it takes for the present value of cash flows from a bond or other fixed-income investment to repay its price.
 - Duration helps in assessing the interest rate risk associated with an investment or a portfolio.
 - Lower coupon bonds tend to have higher durations and are more sensitive to changes in interest rates.
 - Duration is a key concept in asset-liability management and helps in managing interest rate risk effectively.
-

Q.5404 The senior management of ABC Bank is discussing ways of protecting the bank against adverse interest rate changes. One of the managers is explaining to the other senior managers the bank's use of the net interest margin. Given the information below on ABC Bank, the Bank's net interest margin (NIM) is closest to?

Cash & cash equivalents	\$93,000,000
Interest income from loans and investments	\$143,000,000
Shareholder's equity	\$300,000,000
Interest expense on deposits & borrowed funds	\$128,000,000
Total earning assets	\$950,000,000

- A. 1.58%
- B. 5.00%
- C. 11.37%
- D. 42.94%

The correct answer is **A**.

$$\text{NIM} = \frac{(\text{Interest income from loans \& investments} - \text{Interest expense on deposits \& borrow})}{(\text{Total earning assets})}$$

$$\text{NIM} = \frac{(\$143\text{M} - \$128\text{M})}{\$950\text{M}} = 0.0158 \approx 1.58\%$$

Q.5415 ABC Bank conducts a scenario analysis to assess the potential impact of the recent interest rate hikes implemented by the US Federal Reserve, aiming to manage inflation. The analysis includes a 300 basis points increase in interest rates, along with a duration analysis. The following presents ABC Bank's balance sheet and the duration of its individual assets and liabilities.

Assets	Amount (USD million)	Duration (years)
Cash	75	0
Federal funds loans	130	1.0
Governments securities and mortgages	160	4.0
Loans and leases	240	2.0
Total assets	605	
Liabilities		
Interest-bearing marketable deposits	300	0.5
Other borrowings	270	3.0
Total liabilities	570	
Shareholder's Equity	35	
Total liabilities & Shareholder's equity	605	

Which of the following statements best captures the manager's evaluation of this stress scenario at the 3% current interest rate level?

- A. An increase of 3% in interest rates will result in a decrease of USD 16.32 million in the bank's net worth.
- B. An increase of 3% in interest rates will result in an increase of USD 16.32 million in the bank's net worth.
- C. An increase of 3% in interest rates will result in an increase of USD 56.49 million in the bank's net worth.
- D. An increase of 3% in interest rates will result in a decrease of USD 8.44 million in the bank's net worth.

The correct answer is **D**.

The duration of the bank's assets and liabilities is:

$$D_{\text{Assets}} = \frac{(75 \times 0 + 130 \times 1 + 160 \times 4 + 240 \times 2)}{605} = \frac{1,250}{605} = 2.066$$

$$D_{\text{Liabilities}} = \frac{(300 \times 0.5 + 270 \times 3)}{570} = 1.6842$$

The effect of a 3% increase in interest rates can be calculated as:

$$\frac{-(2.066 \times 0.03 \times 605)}{1.03} - \frac{(1.6842 \times 0.03 \times 570)}{1.03} \approx -8.44$$

This implies that an increase of 3% in interest rates will result in a decrease of USD 8.44 million in the bank's net worth.

Q.5420 John Doe is a treasury analyst at ABC Bank. John has been tasked with calculating the bank's discount rate.

Purchase price	\$95
Required rate of return	12%
Face value	\$100
Yield to maturity	8%
Matures in	85 days

Using the information provided above on a money market security, what is the discount rate of the bank?

- A. 6.8%
- B. 15.0%
- C. 21.2%
- D. 11.4%

The correct answer is **C**.

$$\begin{aligned}
 \text{Discount rate} &= \frac{(100 - \text{Purchase price on security})}{100} \times \frac{360}{\text{Number of days to maturity}} \\
 &= \frac{(100 - 95)}{100} \times \frac{360}{85} \\
 &= 0.05 \times 4.24 = 0.2117 \approx 21.2\%
 \end{aligned}$$

Q.5421 ABC Bank has an average asset duration of four years, an average liability duration of three years, total liabilities amounting to \$150 million, and total assets valued at \$180 million. Initially, interest rates stood at 9 percent, but suddenly, they increased to 13 percent. What is the bank's new net worth after the increase in interest rates?

- A. -\$26.42 million
- B. -\$16.51 million
- C. -\$9.91 million
- D. -\$1.10 million

The correct answer is C.

$$\begin{aligned} &\text{Bank's new net worth} \\ &= [-\text{Average duration of assets} \times \frac{\text{Change in interest rate}}{(1 + \text{Original discount rate})} \\ &\quad \times \text{Total assets}] \\ &\quad - [-\text{Average duration liabilities} \times \frac{\text{Change in interest rate}}{(1 + \text{Original discount rate})} \\ &\quad \times \text{Total liabilities}] \\ &= [-4 \times \frac{0.04}{(1 + 0.09)} \times \$180 \text{ million}] - [-3 \times \frac{0.04}{(1 + 0.09)} \times \$150 \text{ million}] \\ &= -26.42 - (-16.51) = -\$9.908 \text{ million} \end{aligned}$$

Reading 145: Illiquid Assets

Q.2426 Jude Wang, a portfolio manager, works at an investment management firm. She has been recently asked to manage the endowment fund of a large university. She meets the head of the university to discuss finer details about the fund, including the composition of the endowment fund. The university head informs her that the university plans to withdraw a substantial amount of the fund in next two years and use the money to establish a state-of-the-art financial laboratory. He also adds that the university board desires to keep the transaction cost as low as possible. After the discussion, Wang meets her colleague, Jane Jones, with whom she explores the asset classes that can possibly be included in the endowment fund. According to Jones, Wang should consider including private equity and real estate in the endowment fund as these asset classes have low correlation with stocks and bonds and can provide a better diversification effect to the overall portfolio. Jones goes further to opine that private equity and real estate would make it easier for Wang to generate an excess return compared to stocks and bonds. Which of the following does not support the idea of including private equity and real estate in the portfolio?

- A. It would provide a better diversification effect to the portfolio.
- B. It would help Wang to generate an excess return.
- C. The university needs a substantial amount of funds in next two years for the laboratory.
- D. It would improve the portfolio because of the assets' low correlation with stocks and bonds.

The correct answer is **C**.

Private equity and real estate fall under the category of illiquid assets. Large trades of illiquid assets results in substantial market movement which may in turn trigger adverse price movements. Thus, an investor trying to cash in on illiquid assets in a short time period may have to suffer significant losses due to adverse price movement. Barring a change of plans by the university, a substantial amount of money is needed in next two years – and that's where the problem lies. If the fund invests in illiquid assets, divesting such assets may force the institution to accept a lower market price or incur substantial transaction costs. Therefore, investment in illiquid assets must be avoided.

Things to Remember

- Illiquid assets are assets that cannot be easily sold or converted into cash without a significant loss in value.
- Private equity investments involve investing directly in private companies or through private equity funds, which typically have a longer investment horizon and are illiquid

in nature.

- Real estate investments also tend to be illiquid, as they require time to sell and convert into cash.
 - Transaction costs associated with buying and selling illiquid assets can be high, especially when large amounts need to be traded in a short period.
 - Short-term liquidity needs should be matched with liquid assets to avoid the risk of having to sell illiquid assets at a loss.
-

Q.2427 Jude Wang, a portfolio manager, works at an investment management firm. She has been recently asked to manage the endowment fund of a large university. She meets the head of the university to discuss finer details about the fund, including the composition of the endowment fund. The university head informs her that the university plans to withdraw a substantial amount of the fund in next two years and use the money to establish a state-of-the-art financial laboratory. He also adds that the university board desires to keep the transaction cost as low as possible. After the discussion, Wang meets her colleague, Jane Jones, with whom she explores the asset classes that can possibly be included in the endowment fund. According to Jones, Wang should consider including private equity and real estate in the endowment fund as these asset classes have low correlation with stocks and bonds and can provide a better diversification effect to the overall portfolio. Jones goes further to opine that private equity and real estate would make it easier for Wang to generate an excess return compared to stocks and bonds.

Which of the following statements is correct?

- A. Jones may include private equity and real estate since they will reduce transaction costs.
- B. Jones must not include private equity and real estate because they may increase transaction costs.
- C. The inclusion of private equity and real estate in the portfolio will have no effect on transaction costs.
- D. Private equity and real estate will make the portfolio more diversified and reduce transaction costs.

The correct answer is **B**.

Private equity and real estate are considered illiquid assets. Illiquid assets are those that cannot be easily bought or sold without a substantial change in price. These assets are characterized by information asymmetry, which means that the information about these assets is not readily

available or is unevenly distributed among market participants. As a result, the cost of obtaining information about such assets is high. This increases the transaction costs associated with these assets. Therefore, including private equity and real estate in the endowment fund may increase transaction costs, contrary to the university board's desire to keep these costs low. Furthermore, the university's plan to withdraw a substantial amount from the fund in the next two years may be hindered by the illiquidity of these assets, as selling them quickly could result in a significant loss.

Choice A is incorrect. While private equity and real estate can provide diversification benefits, they do not necessarily reduce transaction costs. In fact, these asset classes often involve higher transaction costs due to their illiquid nature and the complexity of the transactions involved.

Choice C is incorrect. The inclusion of private equity and real estate in the portfolio will likely increase transaction costs, not have no effect on them. These asset classes are typically less liquid than stocks and bonds, which can lead to higher transaction costs.

Choice D is incorrect. Although private equity and real estate can enhance portfolio diversification, they do not inherently reduce transaction costs. As mentioned earlier, these asset classes often come with higher transaction costs due to their illiquidity and complex nature of transactions.

Things to Remember

- Illiquid assets are assets that cannot be easily bought or sold without a substantial change in price.
- Information asymmetry is when information about assets is not readily available or is unevenly distributed among market participants.
- Illiquid assets like private equity and real estate typically have higher transaction costs due to their nature.
- Diversification benefits can be achieved by including asset classes with low correlation with stocks and bonds.
- Portfolio managers need to consider the liquidity of assets when managing funds that may require withdrawals in the near future.

Q.2428 The bias associated with returns being observed only when the underlying asset values are high is best known as:

- A. Selection bias
- B. Survivorship bias

C. Infrequent trading

D. None of the above

The correct answer is **A**.

Selection bias is a type of distortion that can occur in financial and economic data. It arises when the returns are only observed or recorded when the underlying asset values are high. This can lead to an overestimation of the average return and an underestimation of the risk. For instance, properties are often sold when their values are high, and companies are typically made public when stock values are high in buyout funds. This selective observation can skew the data and create a bias that can impact investment decisions and risk assessments.

Choice B is incorrect. Survivorship bias refers to the tendency of remaining funds in a market to have better past performance, primarily because funds with poor performance are likely to be withdrawn from the market. This does not align with the scenario described in the question where returns are only recorded when asset values are high.

Choice C is incorrect. Infrequent trading refers to a situation where assets are not traded on a regular basis, which can lead to stale or outdated prices and thus distort return calculations. However, this concept does not directly relate to recording returns only when asset values are high.

Choice D is incorrect. The scenario described in the question clearly represents an instance of selection bias, hence it cannot be 'None of the above'.

Things to Remember

- Survivorship bias: Refers to the tendency of remaining funds in a market to have better past performance, primarily because funds with poor performance are likely to be withdrawn from the market.
- Infrequent trading: Situation where assets are not traded on a regular basis, leading to stale or outdated prices and distorting return calculations.
- Random sampling: A method for selecting a subset of individuals from a larger population, where each individual has an equal probability of being chosen.
- Confounding variables: Factors that can influence both the dependent variable (returns) and independent variable (asset values), leading to spurious correlations.
- Measurement error: Inaccuracies in the recording or measurement of data, which can introduce bias and affect the reliability of results.

Q.2430 Sam Kesler, a portfolio manager, is trying to select stocks for his portfolio. The investment policy statement of the fund indicates that the portfolio must consist of only liquid stocks. Kesler has the following investment options:

Category	Stock	Information
Mid Cap	X	Going by the latest trading information available, the stock could be sold at \$12.30 and purchased at \$12.32.
Mid Cap	Y	The majority shareholding of the company rests with a family. The daily turnover of the shares equals 100 stocks only.
Large Cap	Z	The stock was last traded 2 days ago.
Small Cap	A	Going by the latest trading information available, the stock could be sold at \$4.65 and purchased at \$8.25.

Which of the following statements is correct?

- A. Stock Y must not be selected due to its large family holding.
- B. Stock Y must not be selected due to its daily turnover.
- C. Stock Y must be selected due to its large family holding.
- D. Stock Y must not be selected as it belongs to the Mid Cap category.

The correct answer is **B**.

The daily turnover of Stock Y is only 100 stocks, which indicates low liquidity. A stock with high liquidity would have a higher daily turnover, making it easier to buy or sell without causing a significant change in its price. Therefore, the low daily turnover of Stock Y does not make it suitable for inclusion in Kesler's portfolio.

Choices A and C are incorrect. Family holdings often indicate stability in the management and direction of a company but do not necessarily affect the decision-making in terms of liquidity.

Choice D is incorrect. The Mid Cap category itself is not a disqualifier for liquidity; many mid-cap stocks are quite liquid.

Q.2431 Sam Kesler, a portfolio manager, is trying to select stocks for his portfolio. The investment policy statement of the fund indicates that the portfolio must consist of only liquid stocks. Kesler has the following investment options:

Category	Stock	Information
Mid Cap	X	Going by the latest trading information available, the stock could be sold at \$12.30 and purchased at \$12.32.
Mid Cap	Y	The majority shareholding of the company rests with a family. The daily turnover of the shares equals 100 stocks only.
Large Cap	Z	The stock was last traded 2 days ago.
Small Cap	A	Going by the latest trading information available, the stock could be sold at \$4.65 and purchased at \$8.25.

The stock(s) which must be included in the portfolio is (are):

- A. X and Y
- B. X only
- C. Y only
- D. X and A

The correct answer is **B**.

The correct answer is that only Stock X should be included in the portfolio. This is because Stock X exhibits the highest liquidity among the four options. Liquidity in the stock market is determined by the ease with which a stock can be bought or sold without causing a significant change in its price. One of the key indicators of a stock's liquidity is the bid-ask spread, which is the difference between the highest price that a buyer is willing to pay for a stock (the bid) and the lowest price at which a seller is willing to sell a stock (the ask). A smaller bid-ask spread indicates higher liquidity because it means that the stock can be bought or sold at a price close to the market price. In the case of Stock X, the bid-ask spread is very small ($\$12.32 - \$12.30 = \$0.02$), indicating high liquidity. Therefore, Stock X meets the requirement stipulated in the fund's investment policy statement and should be included in the portfolio.

Choice A is incorrect. While stock X is a viable option due to its narrow bid-ask spread indicating high liquidity, stock Y is not a good choice. The majority of the shares are held by a single family and the daily turnover is very low, which suggests that the stock may not be easily tradable and hence lacks liquidity.

Choice C is incorrect. As explained above, Stock Y has low liquidity due to concentrated ownership and low daily turnover.

Choice D is incorrect. Although Stock X can be included in the portfolio due to its high liquidity, Stock A should not be considered because of its wide bid-ask spread ($\$4.65 - \8.25), which indicates lower liquidity.

Q.2432 Sam Kesler, a portfolio manager, is trying to select stocks for his portfolio. The investment policy statement of the fund indicates that the portfolio must consist of only liquid stocks. Kesler has the following investment options:

Category	Stock	Information
Mid Cap	X	Going by the latest trading information available, the stock could be sold at \$12.30 and purchased at \$12.32.
Mid Cap	Y	The majority shareholding of the company rests with a family. The daily turnover of the shares equals 100 stocks only.
Large Cap	Z	The stock was last traded 2 days ago.
Small Cap	A	Going by the latest trading information available, the stock could be sold at \$4.65 and purchased at \$8.25.

The majority shareholding of Y decreases the:

- A. Profitability of the company.
- B. Number of shares available for trading.
- C. Transaction costs.
- D. Both the number of shares available for trading and the transaction costs.

The correct answer is **B**.

The majority shareholding of stock 'Y' by a single family significantly reduces the number of shares available for trading. In the context of stock markets, liquidity refers to the ability to quickly buy or sell a particular security without causing a significant change in its price. A stock is considered liquid if there are enough shares available in the market for trading. When a single entity, such as a family, holds the majority of a company's shares, the number of shares available for trading in the market decreases. This reduction in available shares makes the stock less liquid, as there are fewer shares that can be bought or sold quickly without impacting the stock's price. Therefore, the majority shareholding of stock 'Y' decreases the number of shares available for trading, making it less suitable for inclusion in a portfolio that requires liquid stocks.

Choice A is incorrect. The majority shareholding of a company does not directly affect the profitability of the company. Profitability is determined by factors such as revenue, costs, and operational efficiency, which are independent of who holds the majority shares.

Choice C is incorrect. While transaction costs can be influenced by liquidity and trading volume, they are not directly affected by who holds the majority shares in a company. Transaction costs are typically determined by factors such as brokerage fees and bid-ask spreads.

Choice D is incorrect. As explained above, while the number of shares available for trading can be influenced by who holds the majority shares in a company (as it may limit supply), transaction costs are not directly affected by this factor.

Q.2433 Avery Mason, a financial analyst, is investigating the performance of a mutual fund based on the information on returns provided in the following table:

Fund	Historical Return	Fund Status
1	6%	Active
2	-12%	Closed due to acquisition
3	8%	Active
4	-8%	Closed due to poor performance
5	10%	Active

Mason only considers the funds that are still active for the analysis. The results of his analysis are exposed to:

- A. Selection bias
- B. Survivorship bias
- C. Infrequent sampling
- D. None of the above

The correct answer is **B**.

Survivorship bias refers to the analytical error of focusing only on the entities that have 'survived' a particular process and overlooking those that did not because of their lack of visibility. In the context of financial analysis, survivorship bias can significantly skew the results of an analysis if it only considers the funds that are still active and ignores the ones that have been closed. This is because the performance of the closed funds could have been poor, and excluding them from the analysis could lead to an overestimation of the overall performance of the mutual funds.

In this case, Mason's analysis is subject to survivorship bias because he only considers the active funds for his analysis. By doing so, he excludes the funds that have been closed due to acquisition or poor performance. This could lead to an overestimation of the average return of the mutual funds, as the closed funds, which might have had lower returns, are not considered in the calculation. Therefore, the results of Mason's analysis are not representative of the overall performance of all the funds, leading to a biased conclusion.

Choice A is incorrect. Selection bias refers to the bias introduced by the selection of individuals, groups or data for analysis in such a way that proper randomization is not achieved, thereby ensuring that the sample obtained is not representative of the population intended to be analyzed. In this case, Mason's decision to include only active funds does not constitute selection bias because he has a valid reason for his choice - he wants to analyze funds that are currently operational.

Choice C is incorrect. Infrequent sampling refers to a situation where data points are collected at intervals that are too wide, leading to potential inaccuracies in analysis due to missing fluctuations within those intervals. This type of bias does not apply here as there's no

information suggesting infrequent or irregular sampling of return data.

Choice D is incorrect. As explained above, Mason's analysis would most likely be subject to survivorship bias since he only includes active (or surviving) funds in his study and excludes those which have been closed.

Q.2434 A portfolio manager is analyzing two mutual funds using different time periods. The manager calculates the following volatilities:

	Fund X	Fund Y
Return frequency	Monthly	Weekly
Volatility	0.25	0.15

Which of the following statements is most likely correct?

- A. Fund X is riskier.
- B. Fund Y is riskier.
- C. The two funds are equally risky
- D. The riskiness cannot be determined from volatility alone.

The correct answer is **B**.

To properly compare the risk between Fund X and Fund Y, we need to convert their volatilities to the same time period using the square root of time rule.

The square root of time rule states that volatility scales with the square root of time:

$$\sigma_2 = \sigma_1 \times \sqrt{\frac{t_2}{t_1}}$$

Let's convert both to annualized volatility:

For Fund X (monthly volatility = 0.25):
Monthly to annual:

$$0.25 \times \sqrt{12} = 0.25 \times 3.464 = 0.866$$

For Fund Y (weekly volatility = 0.15):
Weekly to annual:

$$0.15 \times \sqrt{52} = 0.15 \times 7.211 = 1.082$$

After proper annualization, Fund Y's volatility (1.082) is higher than Fund X's (0.866), which means Fund Y is actually riskier.

Q.2435 Which of the following is one of the processes usually employed to account for infrequent trading?

- A. Unsmoothing
- B. Noise reduction
- C. Autocorrelation
- D. Filtering

The correct answer is **A**.

To account for the infrequent trading bias, we need to unsmoothing the returns. In the process, the noise is added back to the reported returns to uncover the true return.

Things to Remember

- Infrequent trading bias can lead to inaccurate estimates of volatility and other risk measures.
 - Unsmoothing returns involves adjusting for the impact of infrequent trading on reported returns.
 - Noise reduction techniques aim to remove random fluctuations in data to reveal underlying trends or patterns.
 - Autocorrelation refers to the correlation of a variable with itself over different time lags.
 - Filtering techniques are used to extract specific components or signals from a dataset while removing unwanted noise.
-

Q.2436 Alco Bank is contemplating creating an index to value artistic work. As art works are not actively traded, the bank decides to create an index from data reported by its members. The members include the major auction firms and art advisors situated in the city.

Sam Park is looking to invest in the art work space to diversify his portfolio. In order to make an informed decision, he decides to use the index created by the bank to calculate the expected return.

The returns as indicated by the bank's index exhibit:

- A. Autocorrelation
- B. Unsmoothing
- C. Auto regression
- D. Filtering

The correct answer is **A**.

Autocorrelation refers to a situation where current values are influenced by previously recorded values. In the context of an illiquid asset such as art, there is limited trading data available. As a result, managers often rely on the most recent and most comparable sales data to estimate the current value of such an asset. They may also use past appraised values, whether estimated or perceived. In these circumstances, the final value assigned to the asset will exhibit strong autocorrelation with the past values used. These values will constitute the index. Consequently, the asset's returns computed based on the index will also show autocorrelation with previous returns. This is because the index, and by extension the returns, are influenced by past values, creating a correlation between current and past values. This characteristic of autocorrelation is particularly prevalent in illiquid markets such as the art market, where trading data is sparse and valuations are often based on past sales or appraisals.

Choice B is incorrect. Unsmoothing refers to the process of adjusting smoothed returns to reflect their true volatility. In this case, the bank's index is not smoothing or adjusting the returns, but rather compiling data from its members to evaluate the value of artistic works.

Choice C is incorrect. Autoregression is a type of time series model that uses observations from previous time points as input for a regression equation to predict the value at the next time point. This concept does not apply here because there's no mention in the question about using past data points for predicting future values in constructing this art index.

Choice D is incorrect. Filtering refers to a technique used in signal processing where an unwanted part of a signal (such as random noise) is removed, or desired part (such as certain component frequencies) are extracted by altering aspects of its spectrum. This concept doesn't apply here because there's no indication that Alco Bank plans on removing or extracting any specific parts from their data while creating this art index.

Things to Remember

Additional important concepts related to the question:

- Illiquid assets: Assets that cannot be easily bought or sold, often resulting in limited trading data and valuation challenges.
- Index creation: The process of compiling data from various sources to create a benchmark for a particular market or asset class.
- Valuation methods: Techniques used to estimate the value of assets, especially in the

absence of active trading data.

- Time series analysis: Statistical techniques used to analyze data points collected at successive and equally spaced time intervals.
-

Q.2437 Alco Bank is contemplating creating an index to value artistic work. As art works are not actively traded, the bank decides to create an index from data reported by its members. The members include major auction firms and art advisors situated in the city.

Sam Park is looking to invest in the artworks space to diversify his portfolio. In order to make an informed decision, he decides to use the index created by the bank to calculate the expected return.

The index created by Alco Bank is an example of a (an):

- A. True index
- B. Appraisal index
- C. Equal-weighted index
- D. Transaction index

The correct answer is **B**.

An appraisal index is a type of index that is constructed using valuation estimates provided by experts, rather than actual transaction prices. In the context of the question, Alco Bank is creating an index to value artistic works using data reported by its members, which include major auction firms and art advisors. These members are likely to provide their expert opinions or appraisals on the value of the artworks, rather than actual transaction prices, as artworks are not actively traded. Therefore, the index created by Alco Bank is an appraisal index. This type of index is often used in markets where assets are not frequently traded, such as the art market, real estate market, and private equity market. It allows investors like Sam Park to estimate the expected return on their investment in these illiquid markets.

Choice A is incorrect. A true index is a type of index that reflects the actual performance of the assets it represents. In this case, Alco Bank is not creating a true index because it's using data provided by its members rather than actual transaction prices from the open market.

Choice C is incorrect. An equal-weighted index assigns the same weight to all components in the portfolio, regardless of their market value or other characteristics. This scenario does not provide any information suggesting that Alco Bank plans to assign equal weights to all artworks in its proposed index.

Choice D is incorrect. A transaction index uses actual transaction prices for its calculation, which would require frequent trading of artworks on an open market - something that isn't

happening here as stated in the question.

Things to Remember

- Appraisal index is constructed using valuation estimates provided by experts, rather than actual transaction prices.
 - True index reflects the actual performance of the assets it represents.
 - Equal-weighted index assigns the same weight to all components in the portfolio.
 - Transaction index uses actual transaction prices for its calculation.
 - Illiquid markets like art market, real estate market, and private equity market often use appraisal indexes.
-

Q.2439 Michael Bay is thinking to include real estate in his portfolio to generate excess returns and to diversify his portfolio. He studies a report prepared by a big consultancy firm on the real estate sector. The report outlines the future prospects of the sector and also contains the list of recent transactions carried out and the return generated.

	Date	Return
Building A	12/2/2016	12.00%
Building B	13/06/2016	11.60%
Building C	12/1/2016	13.00%
Building D	4/9/2016	11.00%
Building E	9/10/2016	14.00%

The report also mentions that there has been an increase in real estate transactions in the recent times due to increases in land prices. Furthermore, it adds that the number of transactions increased from 3 in 2015 to 28 in 2016, indicating the changing fortunes of real estate.

Based on the returns contained in the report and by computing the corresponding market return, Bay tries to calculate the beta (β) and alpha (α - also known as excess return) of the real estate sector. β turns out to be 0.65 while α stands at 3.45%. Thus, Bay concludes that the real estate sector is less risky than the market, and an excess return of 3.45% can be easily generated by investing in real estate. Which of the following statements is accurate?

- A. Bay has overestimated the values of α and β .
- B. Bay has overestimated the value of α but the value of β is correctly estimated.
- C. Bay has overestimated the value of β but the value of α is correctly estimated.
- D. Bay has overestimated the value of α and understated the value of β .

The correct answer is **D**.

Due to selection bias, the returns were observed when the underlying asset values were high. Therefore, the selection bias overestimates the value of α (excess return). The higher observed returns result in an underestimation of β . The security market line, when plotted, gives a flatter line instead of a steeper one.

Q.2440 Michael Bay is thinking to include real estate in his portfolio to generate excess returns and to diversify his portfolio. He studies a report prepared by a big consultancy firm on the real estate sector. The report outlines the future prospects of the sector and also contains the list of recent transactions carried out and the return generated.

	Date	Return
Building A	12/2/2016	12.00%
Building B	13/06/2016	11.60%
Building C	12/1/2016	13.00%
Building D	4/9/2016	11.00%
Building E	9/10/2016	14.00%

The report also mentions that there has been an increase in real estate transactions in the recent times due to increases in land prices. Furthermore, it adds that the number of transactions increased from 3 in 2015 to 28 in 2016, indicating the changing fortunes of real estate. Based on the returns contained in the report and by computing the corresponding market return, Bay tries to calculate the beta (β) and alpha (α - also known as excess return) of the real estate sector. β turns out to be 0.65 while α stands at 3.45%. Thus, Bay concludes that the real estate sector is less risky than the market, and an excess return of 3.45% can be easily generated by investing in real estate. The estimates of α and β can be made more robust by:

- A. Decreasing the number of observations.
- B. Using a multifactor risk model.
- C. Using daily returns.
- D. None of the above.

The correct answer is **B**.

Using a multifactor risk model can indeed make the estimates of alpha (α) and beta (β) more robust. The alpha (α) of an investment indicates the excess return that the investment generates over the expected return, given its beta (β) and the expected market return. Beta (β), on the other hand, measures the investment's sensitivity to market movements. A beta (β) of 1 indicates that the investment's price will move with the market, while a beta (β) less than 1 indicates that the investment will be less volatile than the market. A beta (β) greater than 1 indicates that the investment's price will be more volatile than the market. Therefore, by using a multifactor risk model, one can account for multiple risk factors that could affect the returns of the real estate sector, rather than just considering the market portfolio as the sole risk factor. This can provide a more comprehensive and accurate estimation of the risk and return characteristics of the real estate sector, thereby making the estimates of alpha (α) and beta (β) more robust.

Choice A is incorrect. Decreasing the number of observations would not make the estimates of alpha and beta more robust. In fact, it would likely lead to less reliable estimates due to a smaller sample size. More data points generally provide a better representation of the population, leading to more accurate and reliable results.

Choice C is incorrect. Using daily returns might increase the number of observations, but it may not necessarily improve the robustness of alpha and beta estimates. Daily returns can be very volatile and may introduce noise into the data, which could distort true underlying patterns or trends.

Choice D is incorrect. As explained above, both decreasing the number of observations (choice A) and using daily returns (choice C) are unlikely to improve the robustness of alpha and beta estimates in this context.

Q.2758 People tend to overstate expected returns and understate the risk of illiquid assets. This is due to certain biases that affect their judgment. These biases include all of the following, except the:

- A. Survivorship bias
- B. Infrequent sampling bias
- C. Selection bias
- D. Availability bias

The correct answer is **D**.

Availability bias refers to the tendency of individuals to base their decisions on information that is readily available to them, rather than seeking out all possible information. This bias can lead to skewed perceptions and decisions, as the information that is most readily available may not be the most accurate or comprehensive. In the context of illiquid assets, availability bias might lead an investor to overestimate the returns or underestimate the risks based on the limited information they have. However, it does not directly cause investors to overstate the expected returns and understate the risk of illiquid assets like the other biases listed. Therefore, availability bias is the correct answer.

Choice A is incorrect. Survivorship bias can contribute to the overestimation of returns and underestimation of risks associated with illiquid assets. This is because this bias involves focusing on the assets that have 'survived' or performed well in the past, while ignoring those that have failed or performed poorly. As a result, investors may overestimate the expected returns and underestimate the risks associated with these surviving assets.

Choice B is incorrect. Infrequent sampling bias also contributes to this issue as it refers to a situation where investors do not frequently update their information about an asset's performance due to its illiquidity. This can lead them to overlook recent changes in market conditions or other relevant factors, thereby causing them to overestimate expected returns and underestimate potential risks.

Choice C is incorrect. Selection bias refers to a situation where investors selectively focus on certain information while ignoring other relevant data when making investment decisions. In the context of illiquid assets, this could mean focusing only on positive performance data while ignoring negative indicators, leading again to an overestimation of returns and underestimation of risk.

Things to Remember

- Survivorship bias: This bias occurs when only successful or surviving assets are considered, leading to an overestimation of returns and an underestimation of risks.
 - Infrequent sampling bias: Investors may not update their information frequently due to illiquidity, causing them to overlook changes in market conditions and leading to inaccurate estimations of returns and risks.
 - Selection bias: Investors selectively focus on certain information while ignoring other relevant data, potentially leading to an overestimation of returns and an underestimation of risks.
-

Q.3024 In 2012, a taxonomy of how illiquidity comes about in the market was given by Vayanos and Wang. Which of the following is not a source of market illiquidity according to Vayanos and Wang (2012)?

- A. Costs of transaction
- B. Symmetric information
- C. The effect of prices
- D. Search frictions

The correct answer is **B**.

Symmetric information is not a source of market illiquidity according to the taxonomy proposed by Vayanos and Wang in 2012. In the context of financial markets, symmetric information refers to a situation where all market participants have equal access to relevant information. This scenario promotes market liquidity as it reduces the risk of information asymmetry, where one party has superior information compared to others. In a market characterized by symmetric information, investors are more likely to engage in trading activities as the fear of being exploited due to lack of information is minimized. Therefore, symmetric information encourages market liquidity rather than causing market illiquidity.

Choice A is incorrect. Transaction costs are indeed a cause of market illiquidity according to Vayanos and Wang's classification system. High transaction costs can deter investors from trading, thereby reducing liquidity in the market.

Choice C is incorrect. The effect of prices on liquidity is a recognized factor in Vayanos and Wang's taxonomy. If prices are volatile or uncertain, this can lead to decreased trading activity and thus lower liquidity.

Choice D is incorrect. Search frictions are also considered a cause of market illiquidity by

Vayanos and Wang's classification system because they make it more difficult for buyers and sellers to find each other, which reduces the number of transactions taking place in the marketplace.

Things to Remember

- Market liquidity refers to the ease with which an asset can be bought or sold in the market without causing a significant change in its price.
 - Illiquidity in the market can lead to wider bid-ask spreads, increased price volatility, and higher transaction costs.
 - Transaction costs include brokerage fees, taxes, and other expenses incurred when buying or selling an asset.
 - Information asymmetry occurs when one party in a transaction has more or better information than the other party, leading to potential market inefficiencies.
 - Search frictions refer to obstacles that prevent buyers and sellers from easily finding each other in the market, reducing trading activity and liquidity.
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Q.3025 Expected returns can be overstated and the risk of illiquid assets understated due to some key biases. Survivorship bias is one of these biases. How does survivorship bias arise?

- A. The survivorship bias is the result of infrequent trading where risk estimates like volatility, correlations and betas get too low when calculated using past data.
- B. The survivorship bias results from the tendency of returns to be only observed as the underlying values of the asset get high as compared to other assets.
- C. The survivorship bias results from the tendency of poorly performing funds to stop their activities.
- D. All the above.

The correct answer is **C**.

The tendency of poorly performing funds to stop reporting leads to the survivorship bias. Ultimately, most of these funds will fail, although their failures are rarely counted. Therefore, as compared to the reported data, the true illiquid asset returns are worse due to this fact.

Things to Remember

- Survivorship bias is a common issue in financial analysis where only successful entities or funds are included in the analysis, leading to an overestimation of performance.
 - It is important to account for survivorship bias when analyzing historical data or making investment decisions to avoid making inaccurate assumptions about returns and risks.
 - Other biases that can affect expected returns and risk assessments include selection bias, look-ahead bias, and confirmation bias.
 - Illiquid assets pose unique challenges in terms of valuation, risk assessment, and liquidity risk management due to their limited marketability and potential for mispricing.
 - Proper due diligence and risk assessment processes are essential when dealing with illiquid assets to mitigate the impact of biases and uncertainties.
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Q.5417 Which of the following correctly describes reporting bias in the context of illiquid assets?

- A. It arises because only funds that have survived are included in performance data.
- B. It occurs when firms with poor performance opt not to report their returns due to voluntary reporting practices.
- C. It results from infrequent trading, which limits the availability of precise performance data.
- D. It is caused by the inclusion of backfilled data that inflates historical returns.

The correct answer is **B**.

Reporting bias occurs because reporting is often voluntary for firms dealing with illiquid assets. Firms with poor performance may choose not to report their results, leading to an overstatement of the average returns in the dataset. This selective underreporting skews the reported performance upward, as only better-performing funds are likely to disclose their results.

A is incorrect. This describes survivorship bias , not reporting bias. Survivorship bias arises when only funds still in operation (i.e., those that have "survived") are included, excluding defunct or underperforming funds.

C is incorrect. This refers to challenges caused by infrequent trading of illiquid assets, which affects the precision of performance metrics like beta and correlation.

D is incorrect. This describes backfill bias , which occurs when historical data is added selectively after a fund's inception, potentially inflating past performance.
